All New Matrix Groups: Formulas, Derivations, and Proofs

A consolidated proof anthology organized around established tools

Lambda Distributions workspace

21 July 2026 — integrated with the 14 July matrix-group paper

Abstract

This volume is the proof-level companion to *All New Matrix Groups and Formulas*. It does not merely list the final identities: it begins with the common tools that produce them and then bundles the complete proof-and-verification notes for every family, in the same family order as the formula compendium. The source notes are included without abridgment so that hypotheses, finite-rank qualifications, corrections, computational checks, and references remain attached to the claims they support. Superseded flat exports and byte-level duplicate copies are omitted; their expanded canonical editions appear once in the sections below.

This revised edition treats the foundational identities of *Matrix group* Λ*-distributions* (draft of 14 July 2026) as imported tools rather than reproving them in each family. It corrects the normalized image/kernel convention, separates earlier theorems from new extensions, adds explicit stable-range and central-charge hypotheses, and moves unresolved proof obligations to a final provisional part.

How to use this volume. Start with the derivation toolkit below to identify the relevant reduction: Reynolds/Molien averaging, cycle index, Fourier projection, Weyl constant term, central-character selection, or a two-sided balanced series. Then follow the table of contents to the family-specific note, where the reduction is specialized and independently verified. Exact finite formulas, stable formulas, limiting laws, and numerical diagnostics are deliberately kept distinct.

Conventions and claim labels

Object	Convention used throughout this edition
Representation	A specified continuous finite-dimensional complex representation $\rho : G \to GL(V)$; the represented image $\rho(G)$ is part of the data.
Averaging	Normalized counting measure for finite groups and normalized Haar probability for compact groups. Non-Haar laws are always named explicitly.
σ-MGF	The coefficientwise formal series $\mathrm{SMG}_{G,V} := \mathbb{E}[\mathrm{Exp}_\sigma(X_{G,V}h_1)]$, equivalently $\int_G \prod_i \det(I - t_i\rho(g))^{-1}\,dg$.
Coefficient	For a partition $\tau = (\tau_1 \geq \cdots \geq \tau_s > 0)$, $[m_\tau]\,\mathrm{SMG}_{G,V} = \dim(\bigotimes_i \mathrm{Sym}^{\tau_i} V)^G$ under Haar averaging; this is imported from Lemma 3.3.1 of the 14 July paper.
Image versus kernel	Normalized Haar/counting averages depend only on the represented image. Kernels matter only for unnormalized sums or non-Haar pushforwards.
One-sided versus balanced	Scalar-rich representations often have trivial one-sided series. The balanced series pairs V with V^* and is graded by central charge.
Validity labels	*Exact*: all stated finite parameters. *Stable*: a proved threshold for a fixed coefficient. *Limit*: a specified topology and normalization. *Conditional*: an explicitly named missing structural input. *Diagnostic*: finite computation only.

Global correction. If $H = \rho(G)$ and the average is normalized Haar/counting measure, then integration over G equals integration over H. Earlier warnings that coincident matrices must be counted with "abstract multiplicity" have been removed.

Contents

Part I

Derivation toolkit: how the formulas are obtained

1 Imported tools from *Matrix group Λ-distributions*

We use the following results from M. Bertucci, S. Van Blarcom, B. Glancy, S. Howe, T. Madden, B. Miller, S. Pal, G. Papagrigoriou, N. Raikman, and T. Tran, *Matrix group Λ-distributions*, draft of 14 July 2026. They are not reproved in the family notes below.

Let G be finite or compact, let V be a specified finite-dimensional complex representation, and put

$$W_\tau(V) = \bigotimes_{a=1}^{\ell(\tau)} \mathrm{Sym}^{\tau_a} V.$$

Throughout,

$$\boxed{\mathrm{SMG}_{G,V} := \mathbb{E}[\mathrm{Exp}_\sigma(X_{G,V} h_1)]}$$

(or $\mathrm{SMG}_G := \mathbb{E}[\mathrm{Exp}_\sigma(X_G h_1)]$ when the matrix realization is understood). Thus SMG is only a compact abbreviation for the displayed expectation.

Theorem 1 (Imported coefficient identity; Lemma 3.3.1)**.** *For normalized Haar or counting measure,*

$$\boxed{\mathrm{SMG}_{G,V} = \sum_\tau \dim W_\tau(V)^G \, m_\tau, \qquad [m_\tau]\,\mathrm{SMG}_{G,V} = \dim W_\tau(V)^G.}$$

Theorem 2 (Imported generalized Molien identity; Lemma 3.3.2)**.** *Coefficientwise as a formal symmetric series,*

$$\boxed{\mathrm{SMG}_{G,V} = \int_G \prod_{i\geq 1} \det(I - t_i\rho(g))^{-1}\, dg.}$$

For a finite group the integral is the normalized sum over G.

Theorem 3 (Imported permutation-orbit identity; Lemma 3.3.5)**.** *If $V = \mathbb{C}[X]$ is a permutation representation, then*

$$[m_\tau]\,\mathrm{SMG}_{G,X} = \#\left(G\backslash \prod_a \mathrm{Sym}^{\tau_a} X\right).$$

Theorem 4 (Imported independence criterion; Theorem C)**.** *The power-sum trace variables are independent precisely when the σ-MGF factors into one-variable moment-generating factors in the coordinates p_r/r. The analogous statement for cycle-counting functions uses the falling σ-MGF.*

Theorem A and Theorem B of the same paper are cited rather than reproved for the cyclic, dihedral, Pauli, alternating, and classical Weyl cases. Later sections retain only a correction, an exact finite-rank strengthening, a more general representation, or an independent verification. In particular, the normalized one-dimensional cyclic coefficient is 1, not n, when $n \mid |\tau|$; that section is explicitly presented as a correction and extension.

1.1 Proof pipeline after importing the tools

A family-specific proof now begins at the first structural reduction: weights, cycles, conjugacy classes, a maximal torus, a normal subgroup, or a representation-theoretic decomposition. It then supplies a genuinely independent coefficient extraction and states the exact range of validity. The Reynolds projection and determinant expansion are not repeated merely to identify $[m_\tau]$ with an invariant dimension.

Theorem 5 (Normalized image invariance)**.** *Let $\rho : G \to H := \rho(G)$ be a continuous surjective homomorphism of finite or compact groups. The pushforward of normalized Haar measure on G is normalized Haar measure on H. Consequently*

$$\mathrm{SMG}_{G,V} = \mathrm{SMG}_{H,V}, \qquad W^G = W^H$$

for every module W obtained functorially from V.

Proof. The pushforward is a left-invariant probability measure on H, hence is Haar measure by uniqueness. The fixed spaces agree because the two actions have the same image. □

Intuition. A kernel repeats every image matrix uniformly in a normalized finite average; Haar pushforward is the compact analogue. Central extensions still matter when their central scalars act nontrivially in the chosen linear lift.

2 Plethystic exponentials and trace coordinates

The logarithmic determinant expansion

$$-\log\det(I - tA) = \sum_{r\ge 1} \frac{t^r}{r} \operatorname{Tr}(A^r)$$

rewrites every Molien integrand as

$$\prod_i \det(I - t_i A)^{-1} = \exp\left(\sum_{r\ge 1} \frac{p_r(\mathbf{t})}{r} \operatorname{Tr}(A^r) \right). \tag{1}$$

This is the bridge between eigenvalue calculations and symmetric functions. It explains why a permutation cycle of length r contributes $\operatorname{Exp}_\sigma(p_r)$, why wreath products exponentiate connected cycles, and why Schur-functor problems reduce to polynomial substitutions in trace variables.

Two expansions are repeatedly useful:

$$\operatorname{Exp}_\sigma(F) = \exp\left(\sum_{r\ge 1} \frac{1}{r} p_r[F] \right), \qquad h_d = \sum_{\lambda \vdash d} \frac{p_\lambda}{z_\lambda}.$$

The first builds multisets from connected components; the second changes from power sums (natural for cycles and traces) to complete symmetric functions (natural for symmetric powers).

3 Finite Fourier filters and zero-weight counting

Suppose $A = \prod_{\ell=1}^{r} C_{m_\ell}$ is abelian and $V = \bigoplus_{j=1}^{d} \mathbb{C}_{w_j}$ is decomposed into one-dimensional weight spaces. A monomial in $W_\tau(V)$ is described by occupancies $q_{bj} \ge 0$ with row sums $\sum_j q_{bj} = \tau_b$. Its total A-weight is

$$\left(\sum_{b,j} w_{\ell j} q_{bj} \right)_{\ell=1}^{r}.$$

It is invariant exactly when every component is zero modulo m_ℓ. Equivalently, use character orthogonality

$$\frac{1}{|A|} \sum_{a\in A} \chi(a) = \begin{cases} 1, & \chi = 1, \\ 0, & \chi \ne 1. \end{cases}$$

This derives the simultaneous-congruence formula and, after conditioning on a permutation part, the dual-code filter for monomial groups. The same idea produces scalar-center selection rules: if ζI lies in the group, a one-sided coefficient vanishes unless $\zeta^{|\tau|} = 1$ for every allowed ζ.

Theorem 6 (Normal-abelian Fourier–Reynolds reduction)**.** *Let G be finite, let $A \triangleleft G$ be abelian, put $Q = G/A$, and let W be a complex G-module. With $P_A = |A|^{-1} \sum_{a\in A} \rho_W(a)$ and any section $s : Q \to G$,*

$$P_G = \frac{1}{|Q|} \sum_{q\in Q} \rho_W(s(q))P_A, \qquad \dim W^G = \frac{1}{|Q|} \sum_{q\in Q} \mathrm{Tr}\big(\rho_W(s(q)) \mid W^A\big).$$

The expression is independent of the section.

Proof. Normality makes P_A commute with G. Expanding the displayed average gives $|G|^{-1} \sum_{g\in G} \rho_W(g)$. On W^A the subgroup A acts trivially, so changing a section representative does not change the trace. □

This single reduction contains the finite-abelian, code-monomial, $G(r, p, n)$, generalized-dihedral, and several Pauli/extraspecial filters as special cases.

4 Cycle indices, fixed points, and Möbius inversion

If G permutes a finite set X and $c_d(g; X)$ is the number of d-cycles, then a cycle contributes the determinant factor $1 - t^d$. Hence

$$\boxed{\mathrm{SMG}_{G,X} = \frac{1}{|G|} \sum_{g\in G} \prod_{i,d} (1 - t_i^d)^{-c_d(g;X)}.} \tag{2}$$

The coefficient has an independent orbit interpretation:

$$[m_\tau]\,\mathrm{SMG}_{G,X} = \#\left(G\backslash \prod_j \mathrm{Sym}^{\tau_j} X\right).$$

This is often the cleanest verification route, since it avoids eigenvalues entirely.

When cycles are difficult to enumerate directly but fixed points are easy, use

$$|\,\mathrm{Fix}_X(g^r)| = \sum_{d|r} d\, c_d(g; X), \qquad c_d(g; X) = \frac{1}{d} \sum_{r|d} \mu(d/r)|\,\mathrm{Fix}_X(g^r)|.$$

This fixed-points-to-cycles transform is the principal tool for subset, Hamming, Grassmann, polar, affine, and finite Lie-type actions. The geometry enters only in the fixed-point count: solve a linear system, count invariant subspaces, or impose an isotropic/singular predicate.

5 Wreath products and the block-cycle determinant

Let $H \le GL(V)$ and consider $H \wr S_n$ on $V^{\oplus n}$. Around a permutation cycle $C = (i_1 \cdots i_r)$, multiply the labels in cyclic order, $h_C = h_{i_r} \cdots h_{i_1}$. Direct block elimination gives

$$\det(I - tg_C) = \det(I - t^r h_C).$$

Different cycles are independent. The exponential formula for labelled cycles therefore yields

$$\boxed{\sum_{n\ge 0} u^n\, \mathrm{SMG}_{H\wr S_n}(\mathbf{t}) = \exp\left(\sum_{r\ge 1} \frac{u^r}{r}\, \mathrm{SMG}_H(t_1^r, t_2^r, \ldots)\right).} \tag{3}$$

This is also the recursive mechanism behind rooted-tree automorphism groups: isomorphic branches are permuted, so every vertex layer contributes a wreath product.

6 Conjugacy-class and spectral compression

If the integrand in (2) is a class function, replace the element sum by

$$\frac{1}{|G|}\sum_{C}|C|\prod_{i}\det(I-t_i\rho(g_C))^{-1}.$$

The proof task becomes a classification of spectra. This produces the exact formulas for cyclic, dihedral, dicyclic, polyhedral, Pauli/extraspecial, sporadic frame-shape, and many braid-image examples. A reliable derivation records three pieces separately:

- the number of elements in each spectral sector;
- the eigenvalue multiset in the represented module; and
- the represented image and scalar center. By Theorem 5, normalized Haar/counting averages may be taken directly over the image; the kernel affects only unnormalized or non-Haar weights.

For a regular representation, an element of order d acts as $|G|/d$ disjoint d-cycles. Grouping only by element order immediately gives

$$\mathrm{SMG}_{G,\mathrm{Reg}}=\sum_{d\geq 1}\frac{N_d(G)}{|G|}\,\mathrm{Exp}_\sigma\left(\frac{|G|}{d}p_d\right).$$

7 Weyl integration and constant-term extraction

For compact connected G, choose a maximal torus T, Weyl group W, and positive roots Φ^+. If the weights of V are μ with multiplicities $m(\mu)$, Weyl integration converts the Haar integral into

$$\boxed{\mathrm{SMG}_{G,V}=\frac{1}{|W|}\,\mathrm{CT}_z\left[\prod_{\alpha\in\Phi^+}(1-z^\alpha)(1-z^{-\alpha})\prod_{i,\mu}(1-t_iz^\mu)^{-m(\mu)}\right].}\tag{4}$$

The constant term imposes weight zero; the Weyl denominator performs the alternating projection required to pass from torus weights to G-invariants. This is the exact engine for compact classical groups, exceptional groups, and maximal compact corrections. Low-degree checks can be made independently by constructing invariant bilinear, alternating, cubic, or quartic tensors and by computing Lie-algebra kernels.

8 Schur functors and plethystic pushforward

For a partition $\lambda\vdash d$, the Frobenius character formula gives the trace of a Schur-functor representation:

$$\boxed{\mathrm{Tr}(S_\lambda(g)^r)=s_\lambda[X(g^r)]=\sum_{\alpha\vdash d}\frac{\chi^\lambda(\alpha)}{z_\alpha}\prod_{a\in\alpha}\mathrm{Tr}(g^{ra}).}\tag{5}$$

Substitute (5) into (1). In stable orthogonal and symplectic rank, the base invariant series are respectively $\mathrm{Exp}_\sigma(h_2)$ and $\mathrm{Exp}_\sigma(e_2)$, so the Schur-functor coefficients become Hall inner products against $h_\tau[s_\lambda]$. A genuinely independent finite check constructs $S_\lambda(V)$ with Young symmetrizers and averages the resulting matrices.

9 Central characters and balanced two-sided series

A continuous scalar center often forces every positive-degree holomorphic invariant to vanish. For example, if $U(1)$ acts on V with weight one, then $z\in U(1)$ acts on $W_\tau(V)$ by $z^{|\tau|}$, so the one-sided series is just 1. The correct nontrivial object pairs V and V^*:

$$b_{\alpha,\beta}=\dim\left(\mathrm{Sym}^\alpha V\otimes\mathrm{Sym}^\beta V^*\right)^G,\qquad b_{\alpha,\beta}=0\text{ unless }|\alpha|=|\beta|.$$

The corresponding integrand contains both $\det(I - s_i\rho(g))^{-1}$ and $\det(I - t_j\rho(g)^{-*})^{-1}$. This correction is essential for Clifford groups with all scalars, local unitary tensor products, unitary Schur functors, and conjugation on endomorphisms. Frame potentials arise as the multilinear diagonal coefficients $\mathbb{E}|\operatorname{Tr} g|^{2k}$.

Theorem 7 (Central-charge lattice filter)**.** *Let a central subgroup Z act on modules $V_1, \ldots, V_s$ by characters $\omega_1, \ldots, \omega_s$. Then*

$$\left(\bigotimes_{j=1}^{s} \operatorname{Sym}^{d_j} V_j\right)^{G} = 0 \quad \textit{unless} \quad \prod_{j=1}^{s} \omega_j(z)^{d_j} = 1 \quad (z \in Z).$$

For V and V^* this is the charge condition $|\alpha| - |\beta|$. Exact charge zero is invariant under arbitrary projective rephasing; a finite congruence sector records a chosen scalar lift.

Proof. Every vector in the displayed homogeneous component is an eigenvector for Z with the stated product character. A G-fixed vector must in particular be fixed by Z. □

10 Nonuniform measures, convolution, and heat flow

For a probability measure ν that is not Haar, averaging is generally not a projection. The coefficient is

$$[m_\tau]\,\mathrm{SMG}_{\nu,V} = \operatorname{Tr}\left(\int_G \rho_{W_\tau}(g)\, d\nu(g)\right),$$

so it need not be a nonnegative integer. For a finite random walk $\nu = \kappa^{*r}$, Fourier transforms turn convolution into a power: $\widehat{\nu}(W) = \widehat{\kappa}(W)^r$. For compact heat kernels, irreducible Fourier modes are attenuated by their Casimir eigenvalues. On $U(1)$ this is the explicit weight filter

$$[m_\tau]\,\mathrm{SMG}_t = \sum_{w \in \mathbb{Z}} M_\tau(w) e^{-w^2 t/2},$$

which interpolates from total dimension at $t = 0$ to Haar invariant dimension as $t \to \infty$.

11 Central extensions, projective representations, and scalar filters

The representation, not merely the abstract group, is part of the input. For a central extension $1 \to Z \to \widetilde{G} \to G \to 1$ and an honest representation V of $\widetilde{G}$, a central element acting by $\omega(z)$ acts on $W_\tau(V)$ by $\omega(z)^{|\tau|}$. Hence the one-sided coefficient vanishes unless that central character is trivial. If the lift is only projective, the intrinsic replacement is either the adjoint series or the balanced two-sided series: scalar phases cancel in $V \otimes V^*$. This distinction is essential for spin covers, Weil lifts, Clifford groups, braid images, and sporadic covers.

12 Frame shapes and lattice automorphism groups

For a finite-order integral matrix g with frame shape $\prod_{d \geq 1} d^{k_d(g)}$, one has

$$\det(I - tg) = \prod_{d \geq 1} (1 - t^d)^{k_d(g)}, \qquad \mathrm{SMG}_{G,V} = \frac{1}{|G|} \sum_g \prod_{i,d} (1 - t_i^d)^{-k_d(g)}.$$

Power traces and frame exponents determine one another by Möbius inversion. This gives an exact class-compressed route for root lattices, repeated orthogonal sums, tensor products, and isolated lattices, while keeping rank-tower limits separate from finite exceptional examples.

13 Multiplicative wreath actions and tensor-induced modules

The additive block representation of $H \wr S_n$ is governed by the block-cycle determinant above. Product actions and tensor-induced modules behave differently: their fixed points or characters multiply over coordinate cycles. For $g = (h_1, \ldots, h_n; \pi)$, the data needed from a cycle C are its length and the conjugacy class of the cycle product h_C. Powering splits C into $\gcd(|C|, r)$ cycles and powers the corresponding cycle product. This class-power rule is the common engine for product actions, twisted wreath actions, Fock modules, and multipartition representations.

14 Fast asymptotic diagnostics

Before asserting a coefficientwise limit, inspect the first orbit/moment coefficients. In a permutation module,

$$[m_{(1,1)}]\,\mathrm{SMG}_{G,X} = \#(G\backslash X^2).$$

For a general complex module, $[m_{(1,1)}] = \dim(V \otimes V)^G$; it measures self-duality rather than the usual frame potential. The balanced second moment is $\mathbb{E}|\operatorname{Tr}\rho(g)|^2 = \dim(V \otimes V^*)^G$. Fourth moments often expose spinor or tensor-growth obstructions. If one fixed coefficient diverges, the raw sigma-MGF has no finite coefficientwise limit. When all bounded-degree data depend on finitely many cycle counts, Poisson or compound-Poisson limits are plausible; otherwise the correct statement may require normalization, fixed-tail shapes, fixed level, or no universal limit.

15 Stable range, coefficientwise limits, and honest claim labels

Three statements that look similar must not be conflated:

- an *exact finite-rank identity*, valid for a specified n;
- a *stable-range identity*, whose coefficient of total degree d agrees once n exceeds a proved threshold; and
- a *distributional or coefficientwise limit*, which may have no finite stage at which equality holds.

Stability is usually proved by showing that degree-d data only sees at most d labels, blocks, or trace modes. It can fail immediately when geometry creates new low-degree orbits, as for complete flags or the binary-cube action. Poisson cycle limits and Gaussian trace limits are therefore stated as limits, not silently promoted to finite formulas.

Proposition 1 (Integer convergence implies eventual stability)**.** *For finite or compact Haar models, if a fixed coefficient $a_n = [m_\tau]\,\mathrm{SMG}_{G_n,V_n}$ converges to a finite real limit, then a_n is eventually constant and the limit is a nonnegative integer.*

Proof. Each a_n is an invariant-space dimension. A convergent integer sequence is eventually contained in an interval of length less than one. □

Theorem 8 (Bounded-support orbit stability)**.** *Suppose every degree-D configuration is contained in a supported subobject of size or rank at most $R(D)$, and suppose that for $n \geq N(R)$ every isomorphism between two supported configurations of rank at most R extends to an element of G_n. Then every coefficient of total degree D in the permutation σ-MGF is independent of n for $n \geq N(R(D))$.*

Proof. The coefficient counts orbits of configurations. Bounded support leaves only finitely many abstract support types, and the extension hypothesis makes ambient equivalence exactly support isomorphism. □

A support bound alone is not a stability proof; the extension property must be verified for each family.

16 Verification hierarchy and proof-audit checklist

The bundled notes use four distinct levels of evidence. These labels are not interchanged:

1. **Proof:** a symbolic argument valid for every parameter in the stated range.
2. **Independent exact verification:** a second exact computation based on a different mathematical object, such as orbit enumeration versus a cycle formula, modular weight counting versus complex matrix averaging, or a Lie-algebra kernel versus a character formula.
3. **Cross-check with different implementation failure modes:** two computations share some mathematical input, such as an eigenvalue list or class table, but use different coefficient algorithms.
4. **Numerical diagnostic:** floating-point evaluation, seeded Haar sampling, or evaluation of an identity at finitely many parameter values.

Every reported result should state which category it belongs to. Finite testing is never substituted for an all-parameter proof. When a rank or nullity is obtained numerically, the certificate reports the threshold, the largest singular or eigenvalue classified as zero, and the smallest one classified as positive. If no clear spectral gap is present, the result is labelled inconclusive rather than rounded to a rank.

Fast routing guide. Diagonal abelian action: use a Fourier zero-weight filter. Permutation action: count fixed points, invert to cycles, then use the cycle index. Block permutation: use the wreath determinant. Finite matrix group: compress by spectral classes. Compact connected group: use Weyl constant terms. Continuous scalar center: switch to a balanced two-sided series. Non-Haar law: compute the averaging operator rather than an invariant dimension.

Part II

Finite abelian, monomial, dihedral, and wreath families

Standing convention. The notation $\mathrm{SMG}_{G,V}$ means $\mathbb{E}[\mathrm{Exp}_\sigma(X_{G,V}h_1)]$. The coefficient identity, generalized Molien formula, and permutation-orbit formula are imported Tools 1–3; each section starts at the first family-specific reduction.

17 Finite abelian zero-weight theorem

Abstract

For a finite abelian group acting diagonally, the coefficient of the monomial symmetric function m_τ in the sigma moment-generating function is the number of monomial basis vectors whose total character is trivial. This note proves the resulting simultaneous-congruence formula, derives the diagonal Molien average and finite constant-term form, and records a direct matrix test. The accompanying marimo notebook constructs the represented group, builds the full tensor product of symmetric powers, forms the Reynolds projector, and compares its rank with an exact lattice count and a numerical character average.

17.1 Setup and the represented matrix group

Let

$$A = C_{m_1} \times \cdots \times C_{m_r}, \qquad M = \widehat{A} \cong \mathbb{Z}/m_1\mathbb{Z} \oplus \cdots \oplus \mathbb{Z}/m_r\mathbb{Z}.$$

Every complex representation of A splits into one-dimensional characters. Choose a weight basis and write

$$V = \bigoplus_{j=1}^{d} \mathbb{C}_{w_j}, \qquad w_j = (w_{1j}, \ldots, w_{rj}) \in M.$$

The weights are the columns of an integer matrix $W = (w_{\ell j}) \in \mathbb{Z}^{r\times d}$, understood row by row modulo m_ℓ. For $a = (a_1, \ldots, a_r)$, the corresponding matrix is

$$\rho(a) = \mathrm{diag}_{1\le j\le d}\left(\exp\left(2\pi i \sum_{\ell=1}^{r} \frac{a_\ell w_{\ell j}}{m_\ell}\right)\right). \tag{6}$$

Thus the concrete matrix group is obtained by letting $0 \le a_\ell < m_\ell$ in (6). If ρ has a kernel, several abstract elements give the same matrix; every image matrix has exactly $|\ker\rho|$ preimages. Hence the normalized average over A equals the normalized average over $\rho(A)$ by Theorem 5.

For a partition or finite tuple $\tau = (\tau_1, \ldots, \tau_s)$, put

$$T_\tau(V) = \bigotimes_{b=1}^{s} \mathrm{Sym}^{\tau_b} V.$$

The empty tuple gives $T_\varnothing(V) = \mathbb{C}$.

17.2 The simultaneous-congruence formula

Theorem 1 (Zero-weight formula)**.** *With the notation above,*

$$\dim T_\tau(V)^A = \#\left\{ (q_{bj}) \in \mathbb{Z}_{\geq 0}^{s\times d} : \begin{array}{lll} \displaystyle\sum_{j=1}^{d} q_{bj} = \tau_b & & (1 \leq b \leq s), \\ \displaystyle\sum_{b=1}^{s}\sum_{j=1}^{d} w_{\ell j} q_{bj} \equiv 0 & \pmod{m_\ell} & (1 \leq \ell \leq r) \end{array} \right\}. \tag{7}$$

Equivalently, for $q_b = (q_{b1}, \ldots, q_{bd})^T$ *and* $D = \operatorname{diag}(m_1, \ldots, m_r)$*,*

$$\dim T_\tau(V)^A = \#\left\{(q_1, \ldots, q_s) : q_b \in \mathbb{Z}_{\geq 0}^d, \quad \mathbf{1}^T q_b = \tau_b, \quad W(q_1 + \cdots + q_s) \in D\mathbb{Z}^r\right\}. \tag{8}$$

Proof. The monomials

$$e_1^{q_{b1}} \cdots e_d^{q_{bd}}, \qquad q_{bj} \geq 0, \quad \sum_j q_{bj} = \tau_b,$$

form a basis of $\operatorname{Sym}^{\tau_b} V$. Tensoring these bases over b gives a basis of $T_\tau(V)$ indexed by the arrays in the first line of (7). Since e_j has character w_j, the basis vector indexed by (q_{bj}) has total character

$$\sum_{b=1}^{s}\sum_{j=1}^{d} q_{bj} w_j \in M.$$

It is fixed by every element of A exactly when that character is zero. In coordinates this is precisely the second line of (7). Because the displayed monomial basis is a weight basis, the invariant subspace is spanned by exactly those basis vectors of weight zero. Counting them proves (7); collecting the q_b and writing divisibility by each m_ℓ as membership in $D\mathbb{Z}^r$ gives (8). □

17.3 Fourier form of the imported sigma-MGF

Tools 1 and 2 give, without a new Reynolds–Molien proof,

$$\mathbb{E}_A[\operatorname{Exp}_\sigma(X_A h_1)] = \frac{1}{|A|}\sum_{a\in A}\prod_{i\geq 1}\prod_{j=1}^{d} \frac{1}{1 - t_i \chi_{w_j}(a)}, \tag{9}$$

$$[m_\tau]\,\mathbb{E}_A[\operatorname{Exp}_\sigma(X_A h_1)] = \frac{1}{|A|}\sum_{a\in A}\prod_{b=1}^{s} h_{\tau_b}\big(\chi_{w_1}(a), \ldots, \chi_{w_d}(a)\big) = \dim T_\tau(V)^A. \tag{10}$$

The family-specific step is to evaluate this imported average by finite Fourier projection. Work in the group algebra with $z_\ell^{m_\ell} = 1$ and put $z^{w_j} = \prod_\ell z_\ell^{w_{\ell j}}$. If CT_A extracts the coefficient of the identity character, then Work in the group algebra with $z_\ell^{m_\ell} = 1$ and put $z^{w_j} = \prod_\ell z_\ell^{w_{\ell j}}$. If CT_A extracts the coefficient of the identity character, then

$$\mathbb{E}_A[\operatorname{Exp}_\sigma(X_A h_1)] = \mathrm{CT}_A \prod_{i\geq 1}\prod_{j=1}^{d}(1 - t_i z^{w_j})^{-1}, \tag{11}$$

$$[m_\tau]\,\mathbb{E}_A[\operatorname{Exp}_\sigma(X_A h_1)] = \mathrm{CT}_A \prod_{b=1}^{s} h_{\tau_b}(z^{w_1}, \ldots, z^{w_d}). \tag{12}$$

Character orthogonality identifies this operator with the finite Fourier projector

$$\mathrm{CT}_A\, F = \frac{1}{|A|}\sum_{a\in A} F\big(\zeta_{m_1}^{a_1}, \ldots, \zeta_{m_r}^{a_r}\big).$$

17.4 Verification strategy

The accompanying notebook uses three routes with different failure modes.

1. **Direct represented matrices.** It constructs all $|A| = \prod_\ell m_\ell$ matrices in (6). It first checks the identity, unitarity, and homomorphism laws numerically for every pair of represented elements. In the monomial basis, the diagonal of $\mathrm{Sym}^n(\rho(a))$ is indexed by weak compositions q of n and has entry $\prod_j \rho(a)_{jj}^{q_j}$. Kronecker products give the full matrices on $T_\tau(V)$. Their average is P_A; its numerical rank and trace give the target quantity directly. The residual $\|P_A^2 - P_A\|_F$ checks that the average is a projection.
2. **Exact zero-weight enumeration.** It enumerates the same weak compositions using integer arithmetic only and counts those for which every row of $W(q_1 + \cdots + q_s)$ is divisible by the corresponding m_ℓ. This is a literal implementation of (8); no complex roots of unity are used.
3. **Character/Molien average.** It computes

$$\frac{1}{|A|} \sum_{a \in A} \prod_b \mathrm{tr}\big(\mathrm{Sym}^{\tau_b}(\rho(a))\big)$$

in floating-point complex arithmetic. The result must be within 10^{-8} of the exact count. This route avoids the large Kronecker matrices while still testing the Fourier projection independently of the modular test.

A case passes when the projector rank, its rounded trace, the exact count, and the rounded Molien average agree, and both the numerical discrepancy and idempotence residual are below 10^{-8}.

17.5 Representative results and size choice

The default case is

$$A = C_3 \times C_4, \qquad W = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \end{pmatrix}, \qquad d = 3, \qquad \tau = (3, 2).$$

Here $|A| = 12$ and

$$\dim T_{(3,2)}(V) = \binom{3+3-1}{3}\binom{2+3-1}{2} = 10 \cdot 6 = 60.$$

The notebook constructs twelve 60×60 target matrices. It obtains

$$\operatorname{rank} P_A = 2, \qquad \text{zero-weight count} = 2, \qquad \text{Molien average} = 2 + 3.8 \times 10^{-15} i,$$

with maximum displayed numerical discrepancy 8.21×10^{-15}.

The automated suite broadens the coverage as follows.

Representation	$\lvert A\rvert$	d	largest τ	max. target dim.	checks
C_2, trivial plus sign	2	2	$(3, 2)$	12	5
C_4, nonfaithful	4	3	$(2, 1)$	18	4
$C_2 \times C_2$, all characters	4	4	$(2, 1)$	40	4
$C_3 \times C_4$, mixed weights	12	3	$(3, 2)$	60	5
$C_2 \times C_3$, repeated weights	6	4	$(2, 2)$	100	4
				Total	22

All 22 checks pass. The cases include the empty partition, trivial weights, repeated weights, a nonfaithful representation, one and two cyclic factors, and tensor products with repeated and unequal degrees.

In general the target dimension is

$$N = \prod_{b=1}^{s} \binom{\tau_b + d - 1}{\tau_b}. \tag{13}$$

Dense direct projection requires $O(N^2)$ memory and approximately $O(|A|N^2)$ arithmetic, while literal congruence enumeration visits N monomial basis vectors. The notebook caps the dense target dimension at $N = 1000$ and uses $N = 60$ by default. For larger dimensions, the Molien trace average or an exact convolution of residue distributions on $\prod_\ell \mathbb{Z}/m_\ell\mathbb{Z}$ is the appropriate replacement for a dense projector.

17.6 Reproduction

From the repository root, launch the notebook with

```
.venv/bin/marimo edit \
  marimo_notebooks/finite_abelian.py
```

The controls accept comma-separated moduli and partition parts. Rows of W are comma-separated and separated from one another by semicolons. For example, the default matrix is entered as

```
1, 0, 2; 0, 1, 3
```

The automated table is evaluated whenever the notebook runs.

18 Cyclic-weight invariant distributions: correction and extension

Abstract

This note verifies the homogeneous-basis moments associated with a finite cyclic representation. The answer depends on the multiset of weights of the chosen representation, not only on the abstract cyclic group. We prove the weight-congruence formula, derive its refined Molien series, specialize to the cyclic permutation representation, and describe independent exact and numerical tests implemented in the accompanying marimo notebook.

18.1 Setup

Let $C_m = \mathbb{Z}/m\mathbb{Z} = \langle R\rangle$ act on $V = \mathbb{C}^d$. Since $R^m = I$ and the polynomial $x^m - 1$ has distinct roots over $\mathbb{C}$, the operator R is diagonalizable. Fix an eigenbasis $u_1, \dots, u_d$ such that

$$Ru_j = \omega^{a_j} u_j, \qquad \omega = e^{2\pi i/m}, \qquad a_j \in \mathbb{Z}/m\mathbb{Z}.$$

The multiset $(a_1, \dots, a_d)$ is part of the representation data.

For a partition $\tau = (\tau_1, \dots, \tau_s)$, define

$$W_\tau = \mathrm{Sym}^{\tau_1}(V) \otimes \cdots \otimes \mathrm{Sym}^{\tau_s}(V).$$

18.2 Invariant-count formula

Theorem 1. *For the cyclic representation above,*

$$\dim(W_\tau^{C_m}) = \#\left\{ (q_{i,j}) \in \mathbb{Z}_{\geq 0}^{s\times d} : \begin{array}{l} \sum_{j=1}^d q_{i,j} = \tau_i \quad (1 \leq i \leq s), \\ \sum_{i=1}^s \sum_{j=1}^d a_j q_{i,j} \equiv 0 \pmod m \end{array} \right\}.$$

Proof. A monomial basis of $\mathrm{Sym}^{\tau_i}(V)$ is indexed by the weak compositions $(q_{i,1}, \dots, q_{i,d})$ of τ_i. Hence a basis vector of W_τ has the form

$$b_q = \bigotimes_{i=1}^s u_1^{q_{i,1}} \cdots u_d^{q_{i,d}}.$$

Applying R gives

$$Rb_q = \omega^{\sum_{i,j} a_j q_{i,j}} b_q.$$

Thus every b_q is an eigenvector, and it is fixed by the generator precisely when its exponent is zero modulo m. A vector is fixed by C_m if and only if it is fixed by R. Counting the basis vectors with eigenvalue 1 proves the claim. ☐

By Tool 1, $[m_\tau]\, \mathbb{E}[\mathrm{Exp}_\sigma(X_{C_m} h_1)] = \dim(W_\tau^{C_m})$. Thus Theorem 1 is the representation-specific coefficient formula; it extends the one-dimensional cyclic calculation of Theorem A(1) in the 14 July paper.

18.3 Fourier evaluation of the imported sigma-MGF

Tool 2 gives directly

$$\mathbb{E}[\mathrm{Exp}_\sigma(X_{C_m} h_1)]\,(T) = \mathbb{E}[\mathrm{Exp}_\sigma(X_{C_m} h_1)(T)] = \frac{1}{m} \sum_{k=0}^{m-1} \prod_{r\geq 1} \prod_{j=1}^d \frac{1}{1 - \omega^{ka_j} t_r}, \tag{14}$$

$$\dim(W_\tau^{C_m}) = \frac{1}{m} \sum_{k=0}^{m-1} \prod_{i=1}^s h_{\tau_i}(\omega^{ka_1}, \dots, \omega^{ka_d}). \tag{15}$$

The only new reduction here is the roots-of-unity filter

$$\frac{1}{m} \sum_{k=0}^{m-1} \omega^{kL} = \begin{cases} 1, & L \equiv 0 \pmod m, \\ 0, & L \not\equiv 0 \pmod m, \end{cases}$$

which turns the imported average into the congruence count of Theorem 1. In particular, extracting one monomial-symmetric coefficient gives the proposed formula itself:

$$\boxed{[m_\tau]\,\mathbb{E}[\mathrm{Exp}_\sigma(X_{C_m}h_1)] = \mu_{X_{C_m}}(h_\tau) = \dim(W_\tau^{C_m}) = \#\left\{(q_{i,j}) : \sum_j q_{i,j} = \tau_i,\ \sum_{i,j} a_j q_{i,j} \equiv 0 \pmod{m}\right\}}\,.$$

All infinite products above are interpreted coefficientwise as formal symmetric series, so no analytic convergence assertion is required.

18.4 Cyclic permutation representation

For the cyclic shift on $\mathbb{C}^n$,

$$R(e_j) = e_{j+1}\ (j < n), \qquad R(e_n) = e_1,$$

the Fourier eigenbasis has weights $0, 1, \ldots, n-1$. Thus, with $m = d = n$,

$$\dim(W_\tau^{C_n}) = \#\left\{(q_{i,a}) : \sum_{a=0}^{n-1} q_{i,a} = \tau_i, \quad \sum_{i=1}^{s}\sum_{a=0}^{n-1} a q_{i,a} \equiv 0 \pmod{n}\right\}.$$

For $d = 1$ and weight $a_1 = 1$, every row is forced to be $q_{i,1} = \tau_i$, so the value is 1 when $|\tau| \equiv 0 \pmod{m}$ and 0 otherwise. The surviving value is 1, not m, because the group average is normalized by $1/m$.

18.5 Verification strategy and tested range

The accompanying notebook uses three computational paths.

1. **Direct enumeration.** Generate every weak composition of each τ_i into d parts, form their Cartesian product, and count arrays whose total weight is zero modulo m.
2. **Exact residue convolution.** For each block, count monomials in each residue class of $\mathbb{Z}/m\mathbb{Z}$ and cyclically convolve the block distributions. The residue-zero entry is the answer. This path uses integer arithmetic.
3. **Molien average.** For every k, compute each $h_{\tau_i}(\omega^{ka_1}, \ldots, \omega^{ka_d})$ as the coefficient of z^{τ_i} in $\prod_j (1 - \omega^{ka_j} z)^{-1}$, multiply across blocks, and average over k. This path uses complex arithmetic and is compared with the exact integer count.

The automated suite checks all $2 \le m \le 8$, all partitions of total degree at most 7 with at most three blocks, and four representation families at each order: the faithful one-dimensional character, the m-dimensional cyclic shift, a repeated zero-weight representation, and a nonuniform repeated-weight pattern. These cases include dimensions from 1 through 8 and deliberately test representations that are not determined by the group order alone.

Executed result. All 868 generated cases passed. The exact direct count and exact residue convolution agreed in every case. The largest absolute discrepancy between the numerical Molien average and the exact integer answer was 2.2×10^{-12}.

Remark 1. *Finite testing is evidence, not a substitute for Theorem 1. Its value is to detect normalization errors, mishandling of repeated weights, incorrect tensor-block multiplication, and mistakes in extracting complete homogeneous coefficients.*

18.6 Conclusion

The proposed weight-congruence formula is correct for arbitrary complex representations of C_m. The cyclic-shift specialization follows by using the full set of residues as weights. The refined Molien expression is also correct when read coefficientwise, and normalized expectation is essential in the one-dimensional specialization.

19 Restricted monomial groups: dual-code filter

Outcome. The general dual-code formula and the specialized $G(r, p, n)$ coefficient formula were compared with explicit matrix averages in six representative cases. All 180 coefficients through total degree six agree. A direct Reynolds projector for $G(2, 2, 3)$ on $\mathrm{Sym}^2(V) \otimes V$ has rank and trace 1, with idempotence error 3.21×10^{-16}.

19.1 Reduction to the code-specific calculation

By Tools 1 and 2,

$$\mathrm{SMG}_{G,V} = \mathbb{E}[\mathrm{Exp}_\sigma(X_{G,V} h_1)], \qquad [m_\tau]\,\mathrm{SMG}_{G,V} = \dim W_\tau(V)^G.$$

The new work begins with the cycle spectrum of a code-constrained monomial matrix and the resulting dual-code filter.

19.2 Code-constrained monomial groups

Fix $\zeta = e^{2\pi i/r}$, an H-stable additive code $A \le (\mathbb{Z}/r\mathbb{Z})^n$, and

$$G(A, H) = D(A) \rtimes H, \qquad D(A) = \{\mathrm{diag}(\zeta^{x_1}, \dots, \zeta^{x_n}) : x \in A\}.$$

We use the convention $g = D(x)P_\pi$. If C is a cycle of π, the restriction of g to its coordinate subspace is a weighted cyclic shift. Only the full traversal contributes to its characteristic polynomial, hence

$$\det(I - tg) = \prod_{C \in \mathrm{Cyc}(\pi)} \left(1 - \zeta^{x_C} t^{|C|}\right), \qquad x_C = \sum_{i \in C} x_i. \tag{3}$$

Substitution in (1) proves the finite exact formula

$$\mathrm{SMG}_{A,H}(\mathbf{t}) = \frac{1}{|H||A|} \sum_{\pi \in H} \sum_{x \in A} \prod_{a,C} \left(1 - \zeta^{x_C} t_a^{|C|}\right)^{-1}. \tag{4}$$

19.2.1 The dual-code filter

Expand each factor geometrically and let $m_{C,a} \ge 0$ be its selected multiplicity. Put $M_C = \sum_a m_{C,a}$ and set $e_i = M_C$ for $i \in C$. The phase of a term is $\zeta^{x \cdot e}$. For

$$A^\perp = \{e \in (\mathbb{Z}/r)^n : x \cdot e = 0 \pmod r \text{ for all } x \in A\},$$

finite character orthogonality gives

$$\frac{1}{|A|} \sum_{x \in A} \zeta^{x \cdot e} = \mathbf{1}_{e \in A^\perp}.$$

Therefore the coefficient indexed by $\tau = (\tau_1, \dots, \tau_\ell)$ is

$$\frac{1}{|H|} \sum_{\pi \in H} \# \left\{ (m_{C,a}) : \begin{array}{l} \sum_C |C| m_{C,a} = \tau_a \quad \text{for every } a, \\ e(\pi, m) \in A^\perp \end{array} \right\}. \tag{5}$$

Equation (5) is the implemented general formula. Notice that it contains no matrix eigenvalue computation and no average over diagonal matrices.

For a residue $s \in \mathbb{Z}/r$, the equivalent root-of-unity filter is

$$\Phi_{d,s}(\mathbf{t}) = \frac{1}{r} \sum_{\ell=0}^{r-1} \zeta^{-\ell s} \prod_a (1 - \zeta^\ell t_a^d)^{-1}.$$

It collects those monomials for which $\sum_a m_a \equiv s \pmod r$.

19.3 Specialization to $G(r, p, n)$

Assume $p \mid r$ and

$$A_{r,p,n} = \{x \in (\mathbb{Z}/r)^n : \sum_i x_i = 0 \pmod p\}.$$

The vectors $e_i - e_j \in A_{r,p,n}$ force a dual vector to have equal coordinates. The vectors $pe_i \in A_{r,p,n}$ then force the common coordinate to be a multiple of r/p. Thus

$$A_{r,p,n}^{\perp} = \{q(r/p)(1, \dots, 1) : 0 \le q < p\}. \tag{6}$$

The cycle-index theorem now yields

$$\mathrm{SMG}_{G(r,p,n)}(\mathbf{t}) = \sum_{q=0}^{p-1} [u^n] \exp\left(\sum_{d \ge 1} \frac{u^d}{d} \Phi_{d,qr/p}(\mathbf{t})\right). \tag{7}$$

An equivalent coefficient algorithm regards each coordinate as carrying a label $(m_1, \dots, m_\ell)$ whose coordinate sum has residue qr/p. An S_n-orbit is a multiset of n labels. Bounded dynamic programming over label multiplicities extracts exactly the coefficient of (7); this is the second independent formula route in the software.

With only $t_1 = t$ nonzero, coefficient extraction reduces to

$$\mathrm{SMG}_{G(r,p,n)}(t, 0, \dots) = \frac{1}{(1 - t^{nr/p}) \prod_{j=1}^{n-1} (1 - t^{rj})}. \tag{8}$$

For $W(D_n) = G(2, 2, n)$, the second residue sector requires every coordinate to carry odd total multiplicity. It cannot occur below degree n. In particular, it does not create a degree-one fixed vector; this is the key zero-coordinate correction in the proposed discussion.

19.4 Independent verification design

1. Construct every matrix $D(x)P_\pi$. Check the predicted order $r^n n!/p$ in the $G(r, p, n)$ cases.
2. For every partition τ of every total degree from 0 through 6, compute h_{τ_a} recursively from the eigenvalues of each matrix and average $\prod_a h_{\tau_a}$ over the group.
3. Independently evaluate (5) for a general code case or the multiset coefficient of (7) for $G(r, p, n)$.
4. For a selected target, explicitly embed normalized occupancy bases of symmetric powers in tensor powers, form the full Reynolds matrix, and compare its numerical rank and trace with the coefficient.

These routes share only the mathematical input (r, p, n) or (A, H). The formula route does not call the matrix-average code.

19.5 Representative cases and results

There are 30 partitions through total degree six. The following suite includes an $H = C_3$ code group not covered by the full symmetric-group specialization, the unrestricted wreath products, the D_n parity sector, and two different nontrivial divisors p of r.

Case	$\lvert G\rvert$	$\dim V$	degree	checks	result
even-sign code $\rtimes C_3$	12	3	0–6	30	**PASS**
$G(2, 1, 2)$	8	2	0–6	30	**PASS**
$G(2, 2, 3) = W(D_3)$	24	3	0–6	30	**PASS**
$G(3, 1, 2)$	18	2	0–6	30	**PASS**
$G(4, 2, 2)$	16	2	0–6	30	**PASS**
$G(4, 4, 2)$	8	2	0–6	30	**PASS**

Selected matched dimensions illustrate that the test is multivariate, not only the ordinary Molien specialization.

Case	(2)	(3)	$(2,1)$	$(3,2)$
even-sign code $\rtimes C_3$	1	1	1	4
$G(2,1,2)$	1	0	0	0
$G(2,2,3)$	1	1	1	3
$G(3,1,2)$	0	1	1	0
$G(4,2,2)$	0	0	0	0
$G(4,4,2)$	1	0	0	0

For $G(2,2,3)$ and $\tau = (2,1)$, the target dimension is $\binom{4}{2}\binom{3}{1} = 18$. Direct averaging of the eighteen-dimensional target matrices produced

$$\operatorname{rank} P = 1, \qquad \operatorname{tr} P = 0.9999999999999998, \qquad \|P^2 - P\|_F = 3.21 \times 10^{-16}.$$

This agrees with both coefficient routes.

19.6 Reproducibility and interpretation

Run the command-line suite from the repository root with

```
.venv/bin/python -m for_this_guy.matrix_group_formula_verification.restricted_monomial.verification
```

or open `marimo_notebooks/restricted_monomial_verification.py` with marimo. Matrix averages are accepted only when their imaginary part and distance from the nearest integer are below 3×10^{-7}. The observed residuals are much smaller; the displayed projector error is at machine precision.

Finite tests do not replace the preceding proof. Their role is to detect indexing errors, wrong permutation conventions, missing residue sectors, and incorrect treatment of zero-labeled coordinates. Within the stated small dimensions, no such discrepancy remains.

20 Exact finite-n formula for $G(r,p,n)$

Result. For $p \mid r$ and $n \geq 1$, the auxiliary variable u records the number of coordinates exactly:

$$\boxed{\mathbb{E}\left[\mathrm{Exp}_\sigma\left(X_{G(r,p,n)}h_1\right)\right] = [u^n]\sum_{q=0}^{p-1}\mathrm{Exp}_\sigma\left(u\sum_{j\geq 0}h_{qr/p+jr}(\mathbf{t})\right).}$$

The supplied implementation constructs the matrix group, averages the target character directly, and compares every partition through a chosen degree with an independent bounded-multiset coefficient extraction.

20.1 Imported convention and the finite-size marker

For $V = \mathbb{C}^n$, Tools 1 and 2 identify

$$\mathrm{SMG}_{G(r,p,n)} = \mathbb{E}\left[\mathrm{Exp}_\sigma\left(X_{G(r,p,n)}h_1\right)\right], \qquad [m_\tau]\,\mathrm{SMG}_{G(r,p,n)} = \dim(\mathrm{Sym}^\tau V)^{G(r,p,n)}.$$

The additional variable u records the number of coordinates, with

$$\mathrm{Exp}_\sigma\left(u\sum_{\mathbf{v}\in L}\mathbf{t}^{\mathbf{v}}\right) = \prod_{\mathbf{v}\in L}(1-u\mathbf{t}^{\mathbf{v}})^{-1}.$$

Thus the theorem below is an exact finite-n strengthening, not a reproof of the imported Molien identity.

20.2 The monomial group and exact theorem

Choose a primitive rth root ω. The group is

$$G(r,p,n) = \left\{(\omega^{b_1},\dots,\omega^{b_n};\sigma)\in\mu_r^n\rtimes S_n : \sum_i b_i \equiv 0 \pmod{p}\right\}, \qquad p\mid r,$$

acting by $e_i \mapsto \omega^{b_i}e_{\sigma(i)}$. Its order is $r^n n!/p$. Set

$$a_q = \frac{qr}{p}, \qquad H_q^{(r,p)} = \sum_{j\geq 0}h_{a_q+jr}, \qquad 0\leq q<p.$$

Theorem. For every $n \geq 1$,

$$\mathbb{E}\left[\mathrm{Exp}_\sigma\left(X_{G(r,p,n)}h_1\right)\right] = [u^n]\sum_{q=0}^{p-1}\mathrm{Exp}_\sigma\left(uH_q^{(r,p)}\right) \tag{1}$$

as a complete formal symmetric series. Thus (1) has no stable-range qualification.

20.3 Proof

Fix $\tau = (\tau_1,\dots,\tau_s)$. A monomial basis vector of $\mathrm{Sym}^\tau V$ assigns to coordinate i a label

$$\mathbf{v}_i = (v_i(1),\dots,v_i(s)) \in \mathbb{Z}_{\geq 0}^s, \qquad \sum_{i=1}^n \mathbf{v}_i = \tau.$$

Write $d_i = |\mathbf{v}_i| = \sum_j v_i(j)$. The diagonal element with exponent vector $\mathbf{b}$ acts by $\omega^{\sum_i d_ib_i}$.

The annihilator condition

The character is trivial for every $\mathbf{b}$ satisfying $\sum_i b_i \equiv 0 \pmod p$ if and only if there is a single $q \in \{0, \dots, p-1\}$ such that

$$d_1 \equiv \cdots \equiv d_n \equiv \frac{qr}{p} \pmod r. \tag{2}$$

Indeed, the allowed vector $e_i - e_j$ forces $d_i \equiv d_j \pmod r$. Calling the common residue c, the allowed vector pe_i forces $pc \equiv 0 \pmod r$, so $c = qr/p$ modulo r. Conversely, if (2) holds, then

$$\sum_i d_i b_i \equiv \frac{qr}{p} \sum_i b_i \equiv 0 \pmod r.$$

This proves both necessity and sufficiency, including $n = 1$.

Permutation orbits become multisets

After diagonal invariance is imposed, S_n permutes the labels $\mathbf{v}_1, \dots, \mathbf{v}_n$. The invariant dimension is therefore the number of label multisets. For a fixed q, their generating function is

$$F_q(u, \mathbf{t}) = \prod_{\substack{\mathbf{v} \in \mathbb{Z}^s_{\geq 0} \\ |\mathbf{v}| \equiv a_q \pmod r}} \frac{1}{1 - u\mathbf{t}^{\mathbf{v}}}.$$

Here u counts labels, and $\mathbf{t}$ records their componentwise sum. Since

$$\sum_{|\mathbf{v}|=d} \mathbf{t}^{\mathbf{v}} = h_d(\mathbf{t}),$$

we have $F_q = \mathrm{Exp}_\sigma(uH_q^{(r,p)})$. The residue classes for distinct q are disjoint, so summing $[u^n\mathbf{t}^\tau]\, F_q$ proves (1).

20.4 Types B/C and D

For $G(2,1,n) = W(B_n/C_n)$ only the even branch occurs:

$$\mathrm{SMG}_{B_n/C_n} = [u^n]\, \mathrm{Exp}_\sigma(u(h_0 + h_2 + h_4 + \cdots))\,. \tag{3}$$

The factor contributed by $h_0 = 1$ is $(1-u)^{-1}$; it supplies unused, or zero-labelled, coordinates.

For $G(2,2,n) = W(D_n)$ both common residues occur:

$$\mathrm{SMG}_{D_n} = [u^n]\, \{\mathrm{Exp}_\sigma(u(h_0 + h_2 + h_4 + \cdots)) + \mathrm{Exp}_\sigma(u(h_1 + h_3 + h_5 + \cdots))\}\,. \tag{4}$$

The second branch has no zero label: all n coordinates must have odd positive degree. Consequently it first contributes in total degree n. This explains why deleting u gives a false degree-one term for $n \geq 2$.

More generally, the first $q > 0$ correction has every coordinate of degree at least r/p, hence first appears in degree nr/p. Before that threshold, the $q = 0$ branch has enough zero labels to stabilize, giving

$$\mathrm{SMG}_{G(r,p,n)} \equiv \mathrm{Exp}_\sigma(h_r + h_{2r} + \cdots) \quad \text{in total degrees } d < nr/p. \tag{5}$$

20.5 Independent verification strategy

The notebook does not verify (1) by evaluating its right-hand side twice. It uses two computational paths.

Direct matrix average. For every $\sigma \in S_n$ and exponent vector $\mathbf{b} \in \{0, \dots, r-1\}^n$ with $\sum b_i \equiv 0 \pmod p$, it constructs the matrix

$$g_{\sigma(i),i} = \omega^{b_i}, \qquad g_{k,i} = 0 \quad (k \neq \sigma(i)).$$

For eigenvalues $\lambda_1, \dots, \lambda_n$, the character of $\operatorname{Sym}^d V$ is computed from

$$\sum_{d \geq 0} \chi_{\operatorname{Sym}^d V}(g) z^d = \prod_{i=1}^{n} (1 - \lambda_i z)^{-1}.$$

The observed integer is

$$D_{\text{direct}}(\tau) = \frac{1}{|G|} \sum_{g \in G} \prod_j \chi_{\operatorname{Sym}^{\tau_j} V}(g). \tag{6}$$

Bounded multiset extraction. For each q, the code lists only labels $0 \leq v(j) \leq \tau_j$ with $|\mathbf{v}| \equiv qr/p \pmod r$. It processes each label type once, selecting multiplicities from 0 through n. A dynamic program indexed by "labels used" and "componentwise degree" extracts $[u^n \mathbf{t}^\tau]\, F_q$ using exact Python integers. This route never constructs or diagonalizes a group matrix.

Floating-point arithmetic appears only in roots of unity and eigenvalues on the direct side. The group average is required to lie within $2 \cdot 10^{-8}$ of a real integer before comparison; otherwise the verifier raises an error instead of silently rounding.

20.6 Representative results

Every partition τ of every total degree $0 \leq |\tau| \leq 6$ was tested, for 30 coefficients per row of the table.

Group	Ambient dim.	Order	Coefficients	Result
$G(2,1,2) = W(B_2/C_2)$	2	8	30	pass
$G(2,2,4) = W(D_4)$	4	192	30	pass
$G(3,1,2)$	2	18	30	pass
$G(4,2,2)$	2	16	30	pass
$G(4,4,2)$	2	8	30	pass

These cases exercise the zero-label mechanism, the odd D_n branch, roots of unity of orders 3 and 4, and several nontrivial values of p. They are finite checks rather than a proof, but they are structurally independent and probe exactly the finite-n issue addressed by u.

20.7 Reproduction

Open the marimo notebook `marimo_notebooks/g_r_p_n.py` and press *Construct matrices and verify*. The standalone engine is `verification.py`; running it from the repository root executes the five representative cases and exits nonzero on any discrepancy.

21 Exact finite-n wreath-product formula

Result. Let a finite matrix group $H \le \mathrm{GL}(V)$ act blockwise on $V^{\oplus n}$, with S_n permuting the blocks. Then

$$\boxed{\sum_{n\ge 0} u^n \,\mathrm{SMG}_{H\wr S_n}(\mathbf{t}) = \exp\left(\sum_{\ell\ge 1} \frac{u^\ell}{\ell}\,\mathrm{SMG}_H(t_1^\ell, t_2^\ell, \ldots)\right).}$$

The notebook verifies fixed-n coefficients by comparing explicit block-monomial matrix averages with an exact rational cycle-index recurrence.

21.1 Set-up

Let $H \le \mathrm{GL}(V)$ be finite, with $d = \dim V$. The wreath product $H \wr S_n = H^n \rtimes S_n$ acts on $V^{\oplus n} \cong V \otimes \mathbb{C}^n$. For any finite matrix group K define

$$\mathrm{SMG}_K(\mathbf{t}) = \frac{1}{|K|}\sum_{k\in K}\prod_{i\ge 1}\det(I - t_i k)^{-1}. \tag{1}$$

If $\tau = (\tau_1, \ldots, \tau_s)$, then

$$[m_\tau]\,\mathrm{SMG}_K = \dim\left(\bigotimes_{j=1}^{s} \mathrm{Sym}^{\tau_j} W\right)^K, \tag{2}$$

where W is the representation space of K.

For $\ell \ge 1$ set

$$A_\ell(\mathbf{t}) = \frac{1}{|H|}\sum_{h\in H}\prod_{i\ge 1}\det(I - t_i^\ell h)^{-1} = \mathrm{SMG}_H(t_1^\ell, t_2^\ell, \ldots). \tag{3}$$

21.2 Exact theorem

Theorem. As a formal power series in u and the variables $\mathbf{t}$,

$$\sum_{n\ge 0} u^n\,\mathrm{SMG}_{H\wr S_n}(\mathbf{t}) = \exp\left(\sum_{\ell\ge 1}\frac{u^\ell}{\ell}A_\ell(\mathbf{t})\right). \tag{4}$$

Consequently, at a prescribed number of blocks,

$$\mathrm{SMG}_{H\wr S_n}(\mathbf{t}) = [u^n]\exp\left(\sum_{\ell\ge 1}\frac{u^\ell}{\ell}\,\mathrm{SMG}_H(t_1^\ell, t_2^\ell, \ldots)\right). \tag{5}$$

No limit in n and no stable-range assumption is used.

21.3 Cycle-by-cycle proof

Write $g = (h_1, \ldots, h_n; \sigma)$. For a cycle $c = (i_1\, i_2\, \cdots\, i_\ell)$ of σ, define its ordered cycle product

$$h_c = h_{i_\ell}\cdots h_{i_2} h_{i_1}.$$

On the corresponding ℓ summands of $V^{\oplus n}$, the block action gives the standard determinant identity

$$\det(I - z g_c) = \det(I - z^\ell h_c). \tag{6}$$

One way to see (6) is to solve the first $\ell - 1$ block equations of $(I - z g_c)x = 0$ successively; closing the cycle leaves $(I - z^\ell h_c)x_{i_1} = 0$. Thus the two determinants have the same zeros with multiplicity. Both have degree at most

ℓd and constant term one, so their ratio is the constant one. Equivalently, the identity follows by block Gaussian elimination over $\mathbb{C}[z]$.

Since distinct cycles act on disjoint blocks, (6) yields

$$\prod_{i\geq 1}\det(I-t_i g)^{-1}=\prod_{c}\prod_{i\geq 1}\det(I-t_i^{|c|}h_c)^{-1}. \tag{7}$$

For fixed σ, averaging uniformly over $(h_1,\dots,h_n)\in H^n$ makes the cycle products for distinct cycles independent and uniform in H. This follows because, after all but one h_i on a cycle are chosen, multiplication by the remaining uniform factor makes h_c uniform. If $c_\ell(\sigma)$ is the number of ℓ-cycles, the conditional average of (7) is therefore

$$\prod_{\ell\geq 1}A_\ell(\mathbf{t})^{c_\ell(\sigma)}. \tag{8}$$

Finally apply the cycle-index identity

$$\sum_{n\geq 0}\frac{u^n}{n!}\sum_{\sigma\in S_n}\prod_{\ell\geq 1}x_\ell^{c_\ell(\sigma)}=\exp\left(\sum_{\ell\geq 1}\frac{x_\ell u^\ell}{\ell}\right). \tag{9}$$

Substitution of $x_\ell=A_\ell$ proves (4).

21.4 Roots of unity as a consistency check

For the one-dimensional representation $H=\mu_r$,

$$\mathrm{SMG}_{\mu_r}(\mathbf{t})=\frac{1}{r}\sum_{\zeta^r=1}\prod_i(1-\zeta t_i)^{-1}=\sum_{j\geq 0}h_{rj}(\mathbf{t}), \tag{10}$$

by the roots-of-unity filter. Substituting (10) in (5) recovers the exact $G(r,1,n)=\mu_r\wr S_n$ formula. Thus the monomial-root case is the one-dimensional specialization of the general block theorem.

21.5 How the notebook verifies the theorem

The two sides are implemented through different data structures and algorithms.

Direct block-matrix construction

For each $\sigma\in S_n$ and $(h_1,\dots,h_n)\in H^n$, the code creates an $nd\times nd$ matrix whose only nonzero block in block column i is

$$g_{\sigma(i),i}=h_i. \tag{11}$$

It constructs all $|H|^n n!$ matrices. For each matrix, complete symmetric characters are generated by

$$\sum_{a\geq 0}\chi_{\mathrm{Sym}^a(V^{\oplus n})}(g)z^a=\prod_{j=1}^{nd}(1-\lambda_j z)^{-1}, \tag{12}$$

and the direct target is the normalized average of products of these characters, as in (2).

Exact cycle-index recurrence

Write the right side of (4) as $F(u)=\sum_{n\geq 0}F_n u^n$. Logarithmic differentiation gives the useful fixed-n recurrence

$$nF_n=\sum_{\ell=1}^{n}A_\ell F_{n-\ell},\qquad F_0=1. \tag{13}$$

This is the notebook's main coefficient tool. For a target multi-degree τ, it stores only exponents bounded componentwise by τ. The coefficient of $\mathbf{t}^\alpha$ in A_ℓ is zero unless every component of α is divisible by ℓ; when divisible, it is the direct base-group coefficient at α/ℓ. Truncated polynomial convolution and (13) are performed with exact rational numbers. The final denominator must cancel to one, or the verifier raises an error.

This recurrence is useful beyond testing: it avoids enumerating partitions of permutation cycle types and computes one finite-n answer without expanding an unbounded exponential.

Numerical guardrail

Only the direct eigenvalue calculation uses floating point. Each normalized average must lie within $3 \cdot 10^{-8}$ of a real integer. No comparison is reported if that integrality check fails. The formula side remains exact.

21.6 Representative results

All partitions through total degree five were checked: 19 coefficients for each case.

Base representation $H \curvearrowright V$	$\lvert H\rvert$	$\dim V$	n	$\lvert H \wr S_n\rvert$	Result
C_2 scalar on $\mathbb{C}$	2	1	3	48	pass
C_3 scalar on $\mathbb{C}$	3	1	2	18	pass
C_2 reflection on $\mathbb{C}^2$	2	2	2	8	pass
S_3 standard on $\mathbb{C}^2$	6	2	2	72	pass

The last two cases test genuinely block-valued matrices. The C_2 reflection representation is reducible, while the standard S_3 representation is two-dimensional, irreducible, and has a nonabelian base group. Agreement in these cases tests content not covered by the scalar roots-of-unity specialization.

22 σ-MGF for arbitrary wreath products

Abstract

For a finite matrix group $\rho : H \to GL(V)$, this note proves the exact generating function for the block-monomial representation of $H \wr S_n$ on $V^{\oplus n}$ and its coefficientwise stable limit $\mathrm{Exp}_\sigma(\mathrm{SMG}_H - 1)$. A precise degreewise interpretation explains the marked compound-Poisson language. The companion Marimo notebook constructs all wreath-product matrices and compares their generalized Molien coefficients with two independent exact recurrences. Tests for the sign representation of C_2 and the rational two-dimensional representation of C_3 checked 38 finite coefficients, 19 stable coefficients, and every small cycle-product determinant identity; all passed exactly.

22.1 Setup and target formula

Let $\rho : H \to GL(V)$ be a finite matrix group, $\dim V = d$, and let

$$G_n = H \wr S_n = H^n \rtimes S_n$$

act on $V_n = V^{\oplus n}$ by block-monomial matrices. Write

$$\mathrm{SMG}_{H,n}(\mathbf{t}) = \frac{1}{|H|^n n!} \sum_{g \in H \wr S_n} \prod_{a \geq 1} \det(I - t_a g \mid V_n)^{-1}$$

and

$$\mathrm{SMG}_H(\mathbf{t}) = \frac{1}{|H|} \sum_{h \in H} \prod_{a \geq 1} \det(I - t_a h \mid V)^{-1}.$$

For the Adams substitution $\psi_r F(\mathbf{t}) = F(t_1^r, t_2^r, \ldots)$, the claims under test are

$$\boxed{\sum_{n \geq 0} \mathrm{SMG}_{H,n}(\mathbf{t}) u^n = \exp\left(\sum_{r \geq 1} \frac{u^r}{r} \psi_r \, \mathrm{SMG}_H(\mathbf{t}) \right).} \tag{16}$$

$$\boxed{\mathrm{SMG}_{H,\infty} = \mathrm{Exp}_\sigma(\mathrm{SMG}_H - 1) = \exp\left(\sum_{r \geq 1} \frac{\psi_r \, \mathrm{SMG}_H - 1}{r} \right).} \tag{17}$$

22.2 Cycle-product determinant lemma

Write $g = (h_1, \ldots, h_n; \pi)$. If $C = (i_1 \ i_2 \ \cdots \ i_r)$ is a cycle of π, define

$$h_C = h_{i_r} \cdots h_{i_2} h_{i_1}.$$

On $V_C = V_{i_1} \oplus \cdots \oplus V_{i_r}$, applying g exactly r times returns a vector to its original block and applies h_C. If λ is an eigenvalue of h_C, the corresponding block-cycle eigenvalues are the r roots of $z^r = \lambda$. Their determinant factor is

$$\prod_{z^r = \lambda} (1 - tz) = 1 - t^r \lambda.$$

Multiplying over the eigenvalues of h_C proves

$$\boxed{\det(I - tg \mid V_C) = \det(I - t^r h_C \mid V).} \tag{18}$$

The checker tests (18) without eigenvalues: it constructs both integer matrices and recovers each determinant polynomial from exact traces via Newton identities. Every labeling of cycles of lengths $1, 2, 3$ is tested for both base groups.

22.3 Proof of the exact generating function

Fix $\pi \in S_n$. On an r-cycle, the product map $H^r \to H$ has exactly $|H|^{r-1}$ preimages for each $h \in H$. Thus the cycle product is uniform in H, and disjoint cycles have independent products. By (18), an r-cycle contributes the average

$$B_r(\mathbf{t}) = \frac{1}{|H|} \sum_{h\in H} \prod_a \det(I - t_a^r h)^{-1} = \psi_r \operatorname{SMG}_H(\mathbf{t}).$$

It follows that

$$\operatorname{SMG}_{H,n} = \frac{1}{n!} \sum_{\pi\in S_n} \prod_{C\in \operatorname{Cyc}(\pi)} B_{|C|}.$$

Substitution in the symmetric-group cycle-index identity

$$\sum_{n\geq 0} \frac{u^n}{n!} \sum_{\pi\in S_n} \prod_{r\geq 1} x_r^{c_r(\pi)} = \exp\left(\sum_{r\geq 1} \frac{x_r u^r}{r}\right)$$

proves (16).

22.4 Coefficient recurrence used by the checker

Write the right side of (16) as $F(u) = \sum_{n\geq 0} F_n u^n$. Logarithmic differentiation in u gives

$$\boxed{nF_n = \sum_{r=1}^{n} (\psi_r \operatorname{SMG}_H) F_{n-r}, \qquad F_0 = 1.} \tag{19}$$

For $\tau = (\tau_1, \ldots, \tau_\ell)$ the code represents a truncated multivariate series by an exact dictionary indexed by $0 \leq \alpha_i \leq \tau_i$. The coefficient of $\mathbf{t}^\alpha$ in $\psi_r \operatorname{SMG}_H$ vanishes unless every α_i is divisible by r; in the divisible case it is the direct base-group average

$$\frac{1}{|H|} \sum_{h\in H} \prod_i \chi_{\operatorname{Sym}^{\alpha_i/r} V}(h).$$

This gives an exact rational implementation of (19); all denominators cancel in the final invariant dimensions.

The direct side does not use this recurrence. It constructs every $nd \times nd$ block-monomial matrix and computes its complete symmetric characters from

$$mh_m(g) = \sum_{k=1}^{m} \operatorname{tr}(g^k) h_{m-k}(g), \qquad h_0 = 1. \tag{20}$$

It then averages $\prod_i h_{\tau_i}(g)$ over all $|H|^n n!$ matrices.

22.5 Stable range and an independent stable extractor

Separate the constant term in (16):

$$\sum_{n\geq 0} \operatorname{SMG}_{H,n} u^n = \frac{1}{1-u} \exp\left(\sum_{r\geq 1} \frac{u^r}{r} (\psi_r \operatorname{SMG}_H - 1)\right).$$

Every nonconstant monomial in $\psi_r \operatorname{SMG}_H - 1$ has total $\mathbf{t}$-degree at least r. Products preserve the inequality "u-degree at most $\mathbf{t}$-degree." Therefore a coefficient of total degree k in the second factor is a polynomial in u of degree at most k. Multiplication by $(1-u)^{-1}$ makes its u^n coefficient constant once $n \geq k$. Setting $u = 1$ in the finite polynomial gives (17), and hence

$$\boxed{\operatorname{SMG}_{H,n} \equiv \operatorname{Exp}_\sigma(\operatorname{SMG}_H - 1) \pmod{\deg > n}.}$$

The stable checker does not call the finite recurrence at a large proxy n. If $L = \sum_{r\geq 1}(\psi_r \operatorname{SMG}_H - 1)/r$ and $F = e^L$, the Euler operator $D = \sum_i t_i \partial_{t_i}$ gives the multivariate recurrence

$$|\alpha| F_\alpha = \sum_{0<\beta\leq\alpha} |\beta| L_\beta F_{\alpha-\beta}, \qquad L_\beta = \sum_{r|\gcd(\beta)} \frac{1}{r} [\mathbf{t}^{\beta/r}] \operatorname{SMG}_H . \tag{21}$$

This is a useful general-purpose coefficient extractor for plethystic exponentials: it is finite, exact, and independent of the outer block count.

22.6 Marked compound-Poisson interpretation

Let $\mathcal{C}(H)$ be the conjugacy classes of H and choose $h_C \in C$. Put

$$F_{r,C}(\mathbf{t}) = \prod_a \det(I - t_a^r h_C)^{-1}.$$

Then the logarithm of the stable series is

$$\log \operatorname{SMG}_{H,\infty} = \sum_{r\geq 1} \sum_{C\in\mathcal{C}(H)} \frac{|C|}{r|H|} \big(F_{r,C} - 1\big).$$

This is the probability-generating functional of independent marks

$$N_{r,C} \sim \operatorname{Poisson}\left(\frac{|C|}{r|H|}\right).$$

The precise interpretation is degreewise: for a coefficient of total degree k, only $r \leq k$ can contribute. Although the total intensity summed over all r diverges, every fixed-degree calculation sees a finite marked process. This is exactly the local topology in which cycle counts of uniform permutations converge to independent Poisson variables.

22.7 Representations tested

Two exact integral models exercise different dimensions.

1. $C_2 \to GL_1(\mathbb{Q})$, with matrices $[1]$ and $[-1]$. Here a base coefficient survives precisely when its total degree is even.
2. $C_3 \to GL_2(\mathbb{Q})$, generated by

   $$R = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}, \qquad R^3 = I.$$

   This is the rational form of the two nontrivial complex characters together; it tests a genuinely two-dimensional block and non-scalar determinant.

The largest enumerations construct all $2^4 4! = 384$ matrices of dimension 4 and all $3^3 3! = 162$ matrices of dimension 6.

22.8 Exact verification evidence

For each representation, every partition through total degree 5 is checked. The table shows selected rows; "stable" is asserted only in the proved range $|\tau| \le n$.

Base H	n, τ	direct	finite formula	stable	result
C_2 sign	$4, (1)$	0	0	0	pass
C_2 sign	$4, (2)$	1	1	1	pass
C_2 sign	$4, (1,1)$	1	1	1	pass
C_2 sign	$4, (4)$	2	2	2	pass
C_3 rational	$3, (2)$	1	1	1	pass
C_3 rational	$3, (1,1)$	2	2	2	pass
C_3 rational	$3, (3)$	2	2	2	pass
C_3 rational	$3, (2,1)$	2	2	2	pass
C_3 rational	$3, (4)$	2	2	—	pass

Across the complete suite there are 38 direct-matrix versus finite-formula comparisons and 19 direct-matrix versus stable-formula comparisons. In addition, 53 cycle-product determinant identities are checked: $2+4+8=14$ labelings for C_2 and $3+9+27=39$ for C_3; all pass. The implementation uses integers and `Fraction` throughout, so "pass" means exact equality.

Reproducibility and limitations. The Marimo notebook exposes the base representation, block count, and degree cutoff, displays an actual block-monomial matrix, and marks every coefficient. The standalone module runs the fixed suite and exits nonzero on failure. The proof is valid for any finite matrix group over characteristic zero. The included constructor focuses on two small faithful integral representations so exhaustive matrix averaging remains immediate; adding a new base group requires only its exact matrices.

Conclusion. The explicit matrix averages agree with the exact cycle-product generating function and, in the uniform sufficient range $n \ge |\tau|$, with the independently extracted stable plethystic exponential. This bound is sharp as a uniform degree bound, although a fixed coefficient may stabilize earlier. The marked compound-Poisson description is therefore supported both formally and computationally.

23 Multiplicative wreath representations

Canonical testing artifacts.
Exact backend: `lambda_distributions/dists/multiplicative_wreath_sigma_mgfs/verification.py`.
Regression: `lambda_distributions/dists/multiplicative_wreath_sigma_mgfs/test_verification.py`.
Marimo: `marimo_notebooks/multiplicative_wreath.py`.

Scope and outcome

Let $G_n = H \wr S_n$. We prove the wreath class-power theorem and use it to derive exact σ-MGFs for product actions on Ω^n and tensor-induced modules $V^{\otimes n}$. Their tensor moments are multiset counts, which normally diverge polynomially; diagonal coset and twisted regular actions give related polynomial or exponential rare-event laws. We also record the graded bosonic and fermionic Fock formulas, fixed-tail multipartition limits, growing-multipartition obstructions, varying-base criteria, and the contrasting compound-Poisson law for additive block modules. An executable marimo notebook constructs the relevant permutation and signed-monomial matrix groups in dimensions up to 36. Direct orbit and fixed-space computations agree with the proposed formulas in 13,183 exact checks.

23.1 Common target and wreath class powers

Let H be a finite group and $G_n = H^n \rtimes S_n$. For a complex G_n-module W and a partition $\tau = (\tau_1, \ldots, \tau_s)$, put

$$a_\tau(G_n, W) = \dim \left(\bigotimes_{j=1}^{s} \mathrm{Sym}^{\tau_j} W \right)^{G_n}, \qquad \mathrm{SMG}_{G_n,W}(\mathbf{t}) = \sum_\tau a_\tau m_\tau(\mathbf{t}).$$

Reynolds averaging and $\sum_{d\ge 0} \mathrm{Tr}(\mathrm{Sym}^d A) t^d = \det(I - tA)^{-1}$ give

$$\mathrm{SMG}_{G_n,W} = \frac{1}{|G_n|} \sum_{g\in G_n} \exp\left(\sum_{r\ge 1} \frac{p_r(\mathbf{t})}{r} \chi_W(g^r) \right). \tag{22}$$

Let $\mathcal{C}(H)$ be the conjugacy classes of H. A wreath type is a partition-valued function $\boldsymbol{\mu} : C \mapsto \mu(C)$ of total size n. A part ℓ of $\mu(C)$ denotes an ℓ-cycle of the top permutation whose ordered label product lies in C. For $c \in C$, set $z_C = |C_H(c)|$, and set

$$z_\lambda = \prod_{j\ge 1} j^{m_j(\lambda)} m_j(\lambda)!, \qquad Z_{\boldsymbol{\mu}} = \prod_{C\in\mathcal{C}(H)} z_{\mu(C)} z_C^{\ell(\mu(C))}.$$

Theorem 1 (wreath class-power theorem)**.** *The proportion of elements of type $\boldsymbol{\mu}$ is $Z_{\boldsymbol{\mu}}^{-1}$. If an ℓ-cycle has cycle product c and $d = \gcd(\ell, r)$, then the rth power has d cycles of length ℓ/d, each with cycle product conjugate to $c^{r/d}$. Denote the transformed type by $\boldsymbol{\mu}^{[r]}$. Then*

$$\boxed{\mathrm{SMG}_{G_n,W} = \sum_{\boldsymbol{\mu}\vdash_H n} \frac{1}{Z_{\boldsymbol{\mu}}} \exp\left[\sum_{r\ge 1} \frac{p_r}{r} \chi_W(\boldsymbol{\mu}^{[r]}) \right].} \tag{23}$$

Proof. Fix the top cycle structure and one cycle of length ℓ. After choosing $\ell - 1$ labels freely, the last label has exactly $|C|$ choices that place the cycle product in C. Combining this with the ordinary permutation cycle-index count gives the denominator $z_{\mu(C)} z_C^{\ell(\mu(C))}$; multiplication over C gives $Z_{\boldsymbol{\mu}}$. Advancing by r positions on an ℓ-cycle has $d = \gcd(\ell, r)$ orbits, each of length ℓ/d. Traversing one new orbit winds r/d times around the old cycle, so its label product is conjugate to $c^{r/d}$. Substitution in (22) proves (23). ☐

Remark 1 (orientation)**.** *Changing the starting point of a top cycle conjugates its label product. Thus all formulas are independent of left-versus-right cycle notation once the product is recorded only up to conjugacy.*

23.2 Product-action permutation groups

Suppose H acts on a finite set Ω and G_n acts on $X_n = \Omega^n$. Write $g = (h_1, \dots, h_n; \pi)$. For a cycle Q of π, let a_Q be its ordered label product and put $d_Q(r) = \gcd(|Q|, r)$ and $f_s(a) = |\operatorname{Fix}_\Omega(a^s)|$.

Theorem 2 (product-action fixed powers and σ-MGF)**.** *For every $r \geq 1$,*

$$\boxed{F_r(g) := |\operatorname{Fix}_{\Omega^n}(g^r)| = \prod_{Q \in \operatorname{Cyc}(\pi)} f_{r/d_Q(r)}(a_Q)^{d_Q(r)}.} \tag{24}$$

If $c_j(g)$ is the number of j-cycles on X_n, then

$$c_j(g) = \frac{1}{j} \sum_{r|j} \mu(j/r) F_r(g), \tag{25}$$

and therefore

$$\boxed{\mathrm{SMG}_n^{\mathrm{prod}} = \sum_{\boldsymbol{\mu} \vdash_H n} \frac{1}{Z_{\boldsymbol{\mu}}} \prod_{i,j \geq 1} (1 - t_i^j)^{-c_j(\boldsymbol{\mu})}.} \tag{26}$$

Proof. The rth power splits Q into $d_Q(r)$ coordinate cycles. Going around one new cycle applies an element conjugate to $a_Q^{r/d_Q(r)}$. Each new cycle may independently receive any point fixed by that element, proving (24). Since $F_r(g) = \sum_{j|r} j c_j(g)$, Möbius inversion gives (25). The represented permutation matrix has determinant $\prod_j (1 - t^j)^{c_j(g)}$; Reynolds averaging and Theorem 1 give (26). □

Define $R_s(H, \Omega) = \#(H \backslash \Omega^s)$.

Theorem 3 (exact product moments)**.**

$$\boxed{[m_{(1^s)}] \, \mathrm{SMG}_n^{\mathrm{prod}} = \binom{n + R_s(H, \Omega) - 1}{R_s(H, \Omega) - 1}.} \tag{27}$$

Proof. An ordered s-tuple of words is an $s \times n$ array. The base group H^n replaces each column by its H-orbit in Ω^s, leaving R_s column types. The top group forgets their order. The orbits are therefore multisets of size n from R_s types, counted by (27). The orbit count equals the invariant dimension in the permutation module. □

If the base action is nontrivial and transitive, then $R_2 \geq 2$, so the pair coefficient grows as $n^{R_2 - 1}/(R_2 - 1)!$ and no finite coefficientwise limit exists. If δ_H is the derangement proportion of the base action, a fixed word exists exactly when every top-cycle product is a nonderangement. Since the cycle products are independent uniform elements of H conditional on π,

$$\Pr(F_1 > 0) = \mathbb{E}(1 - \delta_H)^{K(\pi)} = \frac{\Gamma(n + 1 - \delta_H)}{\Gamma(1 - \delta_H)\Gamma(n + 1)} \sim \frac{n^{-\delta_H}}{\Gamma(1 - \delta_H)}. \tag{28}$$

Thus $F_1 \to 0$ in probability while transitivity gives $\mathbb{E} F_1 = 1$ and higher moments grow polynomially.

23.3 Tensor-induced signed matrix groups

Let V be an H-module with character χ, and let $T_n(V) = V^{\otimes n}$ with the usual label action and permutation of tensor factors.

Theorem 4 (tensor cycle character and moments)**.** *For $g = (h_1, \dots, h_n; \pi)$,*

$$\chi_{T_n(V)}(g) = \prod_{Q \in \operatorname{Cyc}(\pi)} \chi(a_Q), \tag{29}$$

$$\chi_{T_n(V)}(g^r) = \prod_Q \chi(a_Q^{r/d_Q(r)})^{d_Q(r)}. \tag{30}$$

Consequently,

$$\boxed{\mathrm{SMG}_n^{\otimes} = \sum_{\boldsymbol{\mu} \vdash_H n} \frac{1}{Z_{\boldsymbol{\mu}}} \exp\left[\sum_{r \geq 1} \frac{p_r}{r} \prod_{C,\ell} \chi(c^{r/d_{\ell,r}})^{d_{\ell,r} m_\ell(\mu(C))}\right].} \tag{31}$$

If $a_s(H, V) = \dim(V^{\otimes s})^H$, *then*

$$\boxed{[m_{(1^s)}]\,\mathrm{SMG}_n^{\otimes} = \begin{cases} \binom{n+a_s-1}{a_s-1}, & a_s > 0, \\ 0, & a_s = 0. \end{cases}} \tag{32}$$

Proof. Contracting tensor indices around one top cycle produces the trace of its label product, proving (29); Theorem 1 gives (30). Substitution in (23) proves (31). For the moment, rearrange tensor factors:

$$(V^{\otimes n})^{\otimes s} \cong (V^{\otimes s})^{\otimes n}.$$

Taking H^n-invariants gives $((V^{\otimes s})^H)^{\otimes n}$; taking the remaining S_n-invariants gives $\mathrm{Sym}^n((V^{\otimes s})^H)$. Its dimension is (32). ☐

If $a_s \geq 2$, the coefficient is asymptotic to $n^{a_s-1}/(a_s-1)!$. If V is self-dual irreducible and $\dim V > 1$, then $a_2 = 1$ and

$$a_4 = \dim \mathrm{End}_H(V \otimes V) \geq 2,$$

because $V \otimes V = \mathrm{Sym}^2 V \oplus \Lambda^2 V$ has two nonzero invariant summands. Hence the fourth tensor moment is at least $n+1$. A one-dimensional character θ of order q is exceptional: $a_s = 1$ when $q \mid s$ and $a_s = 0$ otherwise, so these obstruction coefficients remain bounded.

23.4 Bosonic and fermionic Fock modules

For a finite G-module W, introduce the energy-graded modules

$$\mathcal{F}^+(W) = \bigoplus_{d \geq 0} q^d \,\mathrm{Sym}^d W, \qquad \mathcal{F}^-(W) = \bigoplus_{d \geq 0} q^d \Lambda^d W.$$

The graded σ-MGF applies the outer symmetric-power construction to these graded modules.

Proposition 1 (graded Fock formulas)**.**

$$\mathrm{SMG}^+_{G,W}(q; \mathbf{t}) = \frac{1}{|G|} \sum_{g \in G} \exp\left[\sum_{r \geq 1} \frac{p_r}{r} \det(I - q^r g^r \mid W)^{-1}\right], \tag{33}$$

$$\mathrm{SMG}^-_{G,W}(q; \mathbf{t}) = \frac{1}{|G|} \sum_{g \in G} \exp\left[\sum_{r \geq 1} \frac{p_r}{r} \det(I + q^r g^r \mid W)\right]. \tag{34}$$

Removing the vacuum replaces the determinant inverse in (33) *by that inverse minus* 1.

Proof. For each r, the graded characters are

$$\sum_{d \geq 0} q^{dr}\, \mathrm{Tr}(g^r \mid \mathrm{Sym}^d W) = \det(I - q^r g^r \mid W)^{-1},$$

and

$$\sum_{d \geq 0} q^{dr}\, \mathrm{Tr}(g^r \mid \Lambda^d W) = \det(I + q^r g^r \mid W).$$

Insert these identities into the plethystic form of Reynolds averaging. ☐

For the block module $W_n = V^{\oplus n}$, every total-energy-E monomial has support on at most E blocks. Therefore

$$[q^E m_\tau]\,\mathrm{SMG}^{\pm}_{H \wr S_n, W_n} \quad \text{is independent of } n \text{ for } n \geq E. \tag{35}$$

The grading is essential: at $q = 1$ the bosonic Fock space is infinite-dimensional and ungraded coefficients are generally infinite.

23.5 Irreducible multipartitions

Irreducibles of $H \wr S_n$ are indexed by multipartitions $\boldsymbol{\lambda} = (\lambda^\rho)_{\rho \in \mathrm{Irr}(H)}$. The wreath Frobenius characteristic sends their characters to

$$\chi^{\boldsymbol{\lambda}} \longleftrightarrow \prod_{\rho \in \mathrm{Irr}(H)} s_{\lambda^\rho}(x^{(\rho)}). \tag{36}$$

Theorem 1 immediately gives

$$\boxed{\mathrm{SMG}_{\boldsymbol{\lambda}} = \sum_{\boldsymbol{\mu} \vdash_H n} \frac{1}{Z_{\boldsymbol{\mu}}} \exp\left[\sum_{r \geq 1} \frac{p_r}{r} \chi^{\boldsymbol{\lambda}}(\boldsymbol{\mu}^{[r]})\right].} \tag{37}$$

For a fixed-tail family, only the trivial-color component acquires a long first row and all other boxes form a fixed multipartition $\boldsymbol{\alpha}$. Assume a wreath character-polynomial theorem supplies a polynomial $Q_{\boldsymbol{\alpha}}(X_{\ell,C})$ and an explicit threshold $N_{\boldsymbol{\alpha}}(D)$ valid for all power evaluations needed through total outer degree D. This is the additional input not proved in this section. Under the uniform measure,

$$X_{\ell,C} \Longrightarrow Z_{\ell,C}, \qquad Z_{\ell,C} \sim \mathrm{Poisson}\left(\frac{1}{\ell |C_H(c)|}\right), \tag{38}$$

independently. Define

$$Z^{[r]}_{j,D} = \sum_{\substack{\ell, C:\ \ell/\gcd(\ell,r)=j \\ c^{r/\gcd(\ell,r)} \in D}} \gcd(\ell, r) Z_{\ell,C}. \tag{39}$$

For each fixed degree D and $n \geq N_{\boldsymbol{\alpha}}(D)$, the character-polynomial substitution is valid; marked-cycle moment convergence then yields the conditional marked-Poisson-chaos law

$$\boxed{\mathrm{SMG}_{\boldsymbol{\alpha},\infty} = \mathbb{E} \exp\left[\sum_{r \geq 1} \frac{p_r}{r} Q_{\boldsymbol{\alpha}}(\boldsymbol{Z}^{[r]})\right].} \tag{40}$$

Status: Conditional. Equation (40) is proved once the stated wreath character-polynomial theorem and threshold $N_{\boldsymbol{\alpha}}(D)$ are supplied. A proof obligation is collected in the final provisional part.

There is no corresponding universal raw limit for growing multipartitions. If a self-dual $W_{\boldsymbol{\lambda}}$ satisfies

$$W_{\boldsymbol{\lambda}} \otimes W_{\boldsymbol{\lambda}} = \bigoplus_{\boldsymbol{\nu}} g^H_{\boldsymbol{\lambda},\boldsymbol{\lambda},\boldsymbol{\nu}} W_{\boldsymbol{\nu}},$$

then

$$[m_{(1,1,1,1)}]\,\mathrm{SMG} = \dim \mathrm{End}_{H \wr S_n}(W_{\boldsymbol{\lambda}}^{\otimes 2}) = \sum_{\boldsymbol{\nu}} (g^H_{\boldsymbol{\lambda},\boldsymbol{\lambda},\boldsymbol{\nu}})^2. \tag{41}$$

Growth of this sum obstructs convergence.

23.6 Diagonal coset matrix groups

Let $N_n = H^n$, $\Delta H = \{(h, \dots, h) : h \in H\}$, and let N_n act on $X_n = N_n/\Delta H$. For a class C and $c \in C$, write $z_C = |C_H(c)|$.

Theorem 5 (diagonal fixed powers and moments)**.** *For $g = (g_1, \dots, g_n)$,*

$$F_r(g) = \begin{cases} z_C^{n-1}, & g_1^r, \dots, g_n^r \in C, \\ 0, & \textit{otherwise.} \end{cases} \tag{42}$$

If $X(g) = F_1(g)$, then

$$\boxed{\mathbb{E} X(g)^s = \sum_{C \in \mathcal{C}(H)} z_C^{(s-1)n-s}.} \tag{43}$$

In particular, $[m_{(1,1)}] = \sum_C z_C^{n-2} \geq |H|^{n-2}$.

Proof. A coset $x\Delta H$ is fixed by g^r exactly when the components of $x^{-1}g^r x$ are equal. If the component powers do not lie in one class, there is no solution. If they lie in C, choose the common conjugate and then choose a conjugator in each coordinate; division by $|\Delta H|$ leaves z_C^{n-1} cosets. The event $g_1, \dots, g_n \in C$ has probability z_C^{-n}, and the fixed-point value there is z_C^{n-1}. Summing the sth powers proves (43). $\square$

The identity class alone gives exponential second-moment growth. Meanwhile $\Pr(X > 0) = \sum_C z_C^{-n} \to 0$ for nontrivial H, although $\mathbb{E}X = 1$.

23.7 Twisted-wreath regular matrix groups

Let $N_n = H^n$, let $P_n \leq \mathrm{Aut}(N_n)$, and let $G_n = N_n \rtimes P_n$ act affinely on N_n by $x^{(a,p)} = a\,p(x)$. Put

$$a_r = a\,p(a)\cdots p^{r-1}(a), \qquad L_{p^r}(x) = x\,p^r(x)^{-1}.$$

Theorem 6 (Lang-map criterion and moments)**.**

$$F_r(a,p) = \begin{cases} |\operatorname{Fix}_{N_n}(p^r)|, & a_r \in \operatorname{im} L_{p^r}, \\ 0, & \textit{otherwise}. \end{cases} \tag{44}$$

Conditioned on p, F_1 equals $|\operatorname{Fix}_{N_n}(p)|$ with probability its reciprocal, and otherwise equals zero. Hence

$$\boxed{[m_{(1^s)}] = \frac{1}{|P_n|}\sum_{p\in P_n} |\operatorname{Fix}_{N_n}(p)|^{s-1} = \#(P_n\backslash N_n^{s-1}).} \tag{45}$$

If $P_n = S_n$ only permutes coordinates, then

$$\boxed{[m_{(1^s)}] = \binom{n + |H|^{s-1} - 1}{|H|^{s-1} - 1}.} \tag{46}$$

Proof. The fixed equation $a_r p^r(x) = x$ is equivalent to $a_r = L_{p^r}(x)$. Every nonempty fiber of the Lang map is a coset of $\operatorname{Fix}_{N_n}(p^r)$, proving (44) and the conditional spike law. Averaging its sth power gives the middle expression in (45); Burnside's lemma gives the orbit count. Under coordinate permutations, N_n^{s-1} is an array of n elements from H^{s-1} modulo column permutation, hence a multiset of size n from $|H|^{s-1}$ types. This proves (46). $\square$

For coordinate permutations,

$$\Pr(F_1 > 0) = \frac{(1/|H|)_n}{n!} \sim \frac{n^{1/|H|-1}}{\Gamma(1/|H|)},$$

again giving vanishing fixed points in probability with divergent moments.

23.8 Varying base groups and the additive contrast

Let H_N act on Ω_N and V_N, and put $G_{N,k} = H_N \wr S_k$. Define

$$R_{N,s} = \#(H_N\backslash\Omega_N^s), \qquad a_{N,s} = \dim(V_N^{\otimes s})^{H_N}.$$

Theorems 3 and 4 give the exact two-parameter laws

$$[m_{(1^s)}]\,\mathrm{SMG}_{G_{N,k},\Omega_N^k} = \binom{k + R_{N,s} - 1}{R_{N,s} - 1}, \tag{47}$$

$$[m_{(1^s)}]\,\mathrm{SMG}_{G_{N,k},V_N^{\otimes k}} = \begin{cases} \binom{k+a_{N,s}-1}{a_{N,s}-1}, & a_{N,s} > 0, \\ 0, & a_{N,s} = 0. \end{cases} \tag{48}$$

These binomial coefficients govern every joint-growth path.

The direct-sum block module behaves differently. If $A_N(\mathbf{t}) = \mathrm{SMG}_{H_N,V_N}(\mathbf{t})$, the labelled-cycle exponential formula gives

$$\sum_{k\geq 0} u^k\,\mathrm{SMG}_{H_N\wr S_k,V_N^{\oplus k}} = \exp\left[\sum_{\ell\geq 1}\frac{u^\ell}{\ell}A_N(t_1^\ell,t_2^\ell,\ldots)\right]. \tag{49}$$

Separating its constant term $(1-u)^{-1}$ shows coefficientwise that

$$\lim_{k\to\infty}\mathrm{SMG}_{H_N\wr S_k,V_N^{\oplus k}} = \exp\left[\sum_{\ell\geq 1}\frac{A_N(t_1^\ell,t_2^\ell,\ldots)-1}{\ell}\right]. \tag{50}$$

Thus the additive block representation has a compound-Poisson limit, while (47) and (48) usually diverge.

Corollary 1 (multiplicative wreath obstruction theorem)**.** *For product and tensor-induced wreath representations, the coefficient $[m_{(1^s)}]$ is a multiset count $\binom{n+r_s-1}{r_s-1}$. If $r_s\geq 2$, it grows as $n^{r_s-1}/(r_s-1)!$, so the raw σ-MGF has no finite coefficientwise limit.*

23.9 Exact matrix verification

23.9.1 Independent computational routes

The companion marimo notebook and `verification.py` use the following audit protocol.

1. Construct every group element with the semidirect-product law. Store a permutation matrix by its unique nonzero row in each column; store a signed monomial matrix by that row and a sign. This is the complete matrix, not a character-only surrogate.
2. Form matrix powers directly and compare their fixed points or traces with the structural cycle-product, class-power, diagonal-centralizer, or Lang map formula.
3. Compute $\bigotimes_j \mathrm{Sym}^{\tau_j} W$ on its actual monomial basis. Permutation modules use literal configuration orbits. Signed modules propagate signs around each basis orbit: an orbit contributes one invariant unless a stabilizer sends a basis vector to its negative.
4. Independently compute the same coefficient by the full Reynolds–Molien average. Moment cases receive a third comparison with the closed binomial or centralizer formula.

All checks use integers or exact rational numbers.

23.9.2 Wreath types and product actions

group	order	wreath types	power checks	weight checks
$C_2\wr S_3$	48	10	288	10
$S_3\wr S_2$	72	9	432	9

For $S_3 \curvearrowright \{0,1,2\}$, direct product-action matrices give:

n	matrix dimension	s	R_s	direct/Burnside	formula
2	9	2	2	3	3
2	9	3	5	15	15
3	27	2	2	4	4
3	27	3	5	35	35

The program additionally checks all first six powers of all $72+1296$ product-action matrices, for 8,208 fixed-power comparisons.

23.9.3 Tensor-induced and fixed-tail signed groups

Take $H = C_2$ and $V = \mathbf{1} \oplus \mathrm{sgn}$. Then $a_s = 2^{s-1}$. The tensor-induced results are:

group	dim.	$a_{(1)}$	$a_{(2)}$	$a_{(1,1)}$	$a_{(2,1)}$	$a_{(1^4)}$
$C_2 \wr S_2$	4	1	3	3	7	36
$C_2 \wr S_3$	8	1	4	4	13	120

Every entry is simultaneously a direct signed-fixed-space count and a Molien coefficient; the moment entries also equal (32). Traces of six powers give another 336 checks.

For the fixed-tail bipartition $((n-1),(1))$ of $C_2 \wr S_n$, the natural signed module has character polynomial $X_{1,+} - X_{1,-}$. For $n = 2, 3, 4$, both direct and class-formula routes give

n	(1)	(2)	$(1,1)$	(3)	$(2,1)$	(1^4)
2	0	1	1	0	0	4
3	0	1	1	0	0	4
4	0	1	1	0	0	4

The matrix trace also agrees with the power-transformed character polynomial in 1,760 element-power comparisons. Its limiting character is the Skellam variable $Z_+ - Z_-$ with independent $Z_\pm \sim \mathrm{Poisson}(1/2)$, concretely illustrating (40).

23.9.4 Diagonal, twisted, Fock, and varying-base checks

For the diagonal coset actions, matrix averages and (43) agree:

action	dimension	$\mathbb{E}X$	$\mathbb{E}X^2$	$\mathbb{E}X^3$
$S_3^2 \curvearrowright S_3^2/\Delta S_3$	6	1	3	11
$S_3^3 \curvearrowright S_3^3/\Delta S_3$	36	1	11	251

All first six powers of all 252 matrices give 1,512 further checks. For $S_3^2 \rtimes S_2 \curvearrowright S_3^2$, the direct/formula moments are $1, 21, 666$ for $s = 1, 2, 3$; all 432 fixed-power checks satisfy the Lang-map criterion, and the direct pair-orbit count is 21.

For C_2 on $\mathbf{1} \oplus \mathrm{sgn}$, explicit bosonic monomials give invariant dimensions $1, 1, 2, 2, 3, 3, 4$ in energies 0 through 6. Explicit exterior bases give $1, 1, 0, 0, 0, 0, 0$. These values and all unaveraged power characters agree with (33) and (34) in 94 checks. Regular bases C_2, C_3, with $k = 2, 3$ and $s = 2, 3$, give eight further exact specializations of (47)–(48).

Verification result. The executable suite reports **PASS: 13,183 exact checks**. This includes group construction, semidirect powers, class weights, fixed points, traces, orbit counts, signed fixed spaces, Molien coefficients, moment theorems, and graded Fock characters. The asymptotic claims are deductions from the proved exact formulas; they are not extrapolations from the small cases.

23.10 Remarks and asymptotic classification

family	controlling statistic	behavior
$V^{\oplus n}$	cyclewise determinants	compound-Poisson limit
Ω^n	$R_s = \#(H\backslash\Omega^s)$	polynomial divergence if $R_s \geq 2$
$V^{\otimes n}$	$a_s = \dim(V^{\otimes s})^H$	polynomial divergence if $a_s \geq 2$
bosonic/fermionic Fock	$\det(I \mp q^r g^r)^{\mp 1}$	fixed-energy stability
fixed-tail multipartition	marked character polynomial	marked-Poisson chaos
growing multipartition	wreath Kronecker multiplicities	no universal raw limit
$H^n/\Delta H$	common conjugacy class	exponential moment divergence
$H^n \rtimes P_n \curvearrowright H^n$	$\lvert \mathrm{Fix}_{H^n}(p)\rvert$	Bernoulli-spike rare events
varying $H_N \wr S_k$	$R_{N,s}$ or $a_{N,s}$	exact two-parameter criterion

The computational examples are deliberately modest: their role is to test the representation, conventions, power operation, and coefficient extraction without relying on asymptotic numerics. The formulas themselves are valid at every finite size covered by their hypotheses. The small groups already display the structural distinction: cyclewise sums exponentiate into stable laws; cyclewise products create rare, large contributions that force moments to grow.

23.11 Reproducibility and scope

The standalone source, compiled PDF, marimo notebook, exact backend, and focused tests live together in `new_dists/multiplicative_wreath_sigma_mgfs`. From the repository root, the complete audit is reproduced with

```
.venv/bin/python -m new_dists.multiplicative_wreath_sigma_mgfs.verification,
```

and the interactive laboratory is opened with

```
.venv/bin/marimo run marimo_notebooks/multiplicative_wreath.py.
```

The PDF is compiled directly from its colocated LaTeX source. No combined anthology source or PDF is modified by this package.

The exhaustive computations certify the listed finite instances and test the implementation of each structural reduction. Universality comes from the proofs: Reynolds averaging, wreath class powers, orbit–multiset bijections, and the Lang-map fiber argument. Statements about growing multipartitions remain criteria rather than unconditional convergence theorems, because the relevant wreath Kronecker multiplicities depend on the chosen family.

References

[1] I. G. Macdonald-style wreath characteristic background is developed in *Wreath Product Symmetric Functions*, arXiv:0809.2439.

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24 Generalized-dihedral master theorem

Abstract

Let $G = \mathrm{Dih}(A) = A \rtimes C_2$ for a finite abelian group A, where the nontrivial element acts on A by inversion. This note classifies the complex irreducible representations of G and proves a refined Molien formula for an arbitrary finite-dimensional representation. The formula is then checked by an independent full-matrix computation: the accompanying marimo notebook constructs every represented group element, constructs the action on tensor products of symmetric powers, and compares the rank of the Reynolds projector with coefficient extraction from the claimed formula. All 33 representative checks passed, covering cyclic and noncyclic A, odd and even orders, one- and two-dimensional irreducible blocks, repeated blocks, and representation dimensions from 2 through 5.

24.1 Setup and imported target

Let A be finite abelian and

$$G = \mathrm{Dih}(A) = A \rtimes \langle s\rangle, \qquad s^2 = 1, \qquad sas = a^{-1}.$$

Thus $G = A \sqcup As$ and the usual dihedral group is the case $A = C_n$. For a specified $\rho : G \to GL(V)$, put

$$\mathrm{SMG}_{G,V} := \mathbb{E}[\mathrm{Exp}_\sigma(X_{G,V}h_1)] = \frac{1}{|G|}\sum_{g\in G}\prod_{r\geq 1}\det(I - t_r\rho(g))^{-1}. \tag{51}$$

By Tool 1,

$$\mathrm{SMG}_{G,V} = \sum_\tau \dim(W_\tau^G)m_\tau. \tag{52}$$

These are imported. The new work begins with the irreducible decomposition and the two coset spectra.

24.2 Classification of the irreducibles

Write $\widehat{A} = \mathrm{Hom}(A, \mathbb{C}^\times)$. Conjugation by s acts on $\widehat{A}$ by $\chi \mapsto \chi^{-1}$.

Theorem 1 (Irreducibles of $\mathrm{Dih}(A)$)**.** *The irreducible complex representations of $G = \mathrm{Dih}(A)$ are as follows.*

(i) *If $\eta \in \widehat{A}$ satisfies $\eta^2 = 1$, it has exactly two one-dimensional extensions*

$$L_{\eta,\epsilon}(a) = \eta(a), \qquad L_{\eta,\epsilon}(s) = \epsilon, \qquad \epsilon \in \{+1, -1\}.$$

(ii) *If $\chi \neq \chi^{-1}$, then $\rho_\chi = \mathrm{Ind}_A^G\,\chi$ is irreducible of dimension two. In a suitable basis,*

$$\rho_\chi(a) = \begin{pmatrix}\chi(a) & 0\\ 0 & \chi(a)^{-1}\end{pmatrix}, \qquad \rho_\chi(s) = \begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}.$$

Moreover $\rho_\chi \cong \rho_{\chi^{-1}}$, and these are the only duplications in this list.

Proof. Restrict an irreducible G-module U to A. Since A is finite abelian over $\mathbb{C}$, the restriction is a direct sum of weight spaces

$$U|_A = \bigoplus_{\chi\in\widehat{A}} U_\chi.$$

For $u \in U_\chi$ and $a \in A$,

$$a(su) = s(sas)u = s(a^{-1}u) = \chi(a)^{-1}su,$$

so $sU_\chi = U_{\chi^{-1}}$.

Suppose first that $\chi \neq \chi^{-1}$ occurs. For nonzero $u \in U_\chi$, the span of u and su is G-stable: A acts through the two weights χ, χ^{-1}, while s exchanges the two lines. Irreducibility forces this span to be all of U. In the basis (u, su) the matrices are those displayed above. The two distinct A-weights show that this module is irreducible, and exchanging the basis vectors identifies ρ_χ with $\rho_{\chi^{-1}}$.

Suppose instead that every occurring weight satisfies $\eta = \eta^{-1}$. Then s preserves each U_η, so irreducibility forces a single weight. On that space A acts by scalars and $s^2 = 1$. The operator s is diagonalizable with eigenvalues ± 1, and each eigenspace is G-stable. Thus irreducibility forces dimension one and gives $L_{\eta,+}$ or $L_{\eta,-}$. Conversely, the relation $sas = a^{-1}$ in a one-dimensional representation requires $\eta(a) = \eta(a)^{-1}$ for every a, exactly the condition $\eta^2 = 1$. □

Choose a set Ω containing one representative from each unordered pair $\{\chi, \chi^{-1}\}$ with $\chi \neq \chi^{-1}$. Complete reducibility and Theorem 1 give the unique multiplicity data

$$V \cong \bigoplus_{\substack{\eta^2=1\\ \epsilon=\pm 1}} L_{\eta,\epsilon}^{\oplus c_{\eta,\epsilon}} \oplus \bigoplus_{\chi\in\Omega} \rho_\chi^{\oplus m_\chi}. \tag{53}$$

Put

$$c_\eta = c_{\eta,+} + c_{\eta,-}, \qquad M = \sum_{\chi\in\Omega} m_\chi.$$

24.3 The master formula

Theorem 2 (Generalized-dihedral refined Molien formula)**.** *For the representation (53),*

$$\boxed{\mathrm{SMG}_{G,V} = \frac{1}{2}\mathcal{R}_V + \frac{1}{2}\,\mathrm{SMG}_V,}$$

where

$$\mathcal{R}_V = \frac{1}{|A|}\sum_{a\in A}\prod_{r\geq 1}\left[\prod_{\eta^2=1}(1-\eta(a)t_r)^{-c_\eta}\prod_{\chi\in\Omega}\frac{1}{(1-\chi(a)t_r)^{m_\chi}(1-\chi(a)^{-1}t_r)^{m_\chi}}\right], \tag{54}$$

$$\mathrm{SMG}_V = \frac{1}{|A|}\sum_{a\in A}\prod_{r\geq 1}\left[(1-t_r^2)^{-M}\prod_{\substack{\eta^2=1\\ \epsilon=\pm 1}}(1-\epsilon\eta(a)t_r)^{-c_{\eta,\epsilon}}\right]. \tag{55}$$

Consequently, for every partition τ,

$$[m_\tau]\,\mathrm{SMG}_{G,V} = \dim\left(\bigotimes_b \mathrm{Sym}^{\tau_b} V\right)^G.$$

Proof. Split the normalized sum (51) across $G = A \sqcup As$. This produces the factor $1/2$ and the two normalized A-averages.

For $a \in A$, each copy of $L_{\eta,\epsilon}$ has eigenvalue $\eta(a)$; the sign ϵ does not enter on A. Each copy of ρ_χ has eigenvalues $\chi(a)$ and $\chi(a)^{-1}$. Multiplying $(1 - t_r\lambda)^{-1}$ over this eigenvalue multiset proves (54).

For $as \in As$, a copy of $L_{\eta,\epsilon}$ has eigenvalue $\epsilon\eta(a)$. On a copy of ρ_χ,

$$\rho_\chi(as) = \begin{pmatrix} 0 & \chi(a) \\ \chi(a)^{-1} & 0 \end{pmatrix}.$$

This matrix squares to I and has trace zero, hence eigenvalues $+1$ and -1, independently of a. Its determinant factor is therefore

$$\det(I - t_r\rho_\chi(as))^{-1} = \frac{1}{(1-t_r)(1+t_r)} = (1-t_r^2)^{-1}.$$

There are M such blocks. Multiplication over the one- and two-dimensional summands proves (55). The coefficient statement follows from (52). □

Remark 1 (Image invariance)**.** *The formula may be averaged either over the abstract group or uniformly over the represented image. The two normalized averages agree by Theorem 5; only unnormalized sums or non-Haar laws can distinguish the kernel.*

24.4 Coefficientwise verification method

For X with eigenvalue multiset $\Lambda(X)$, define

$$H_q(X) := [z^q] \prod_{\lambda\in\Lambda(X)} (1-\lambda z)^{-1} = h_q(\Lambda(X)).$$

Extracting the monomial of type τ from Theorem 2 gives the finite prediction

$$\begin{aligned} F_\tau = \frac{1}{2|A|}\sum_{a\in A}\Big\{ &\prod_b h_{\tau_b}\Big(\{\eta(a)^{[c_\eta]}\}_{\eta^2=1}, \{\chi(a)^{[m_\chi]}, (\chi(a)^{-1})^{[m_\chi]}\}_{\chi\in\Omega}\Big) \\ &+ \prod_b h_{\tau_b}\Big(\{(\epsilon\eta(a))^{[c_{\eta,\epsilon}]}\}_{\eta,\epsilon}, \{1^{[M]}, (-1)^{[M]}\}\Big)\Big\}. \end{aligned} \tag{56}$$

Here $x^{[m]}$ denotes m repetitions of x. The notebook evaluates each h_q by truncated coefficient multiplication in $\prod_\lambda (1-\lambda z)^{-1}$.

The independent direct route does not diagonalize reflection matrices and does not use (56). It performs the following steps.

1. Represent $A = C_{n_1}\times\cdots\times C_{n_r}$ by residue vectors and construct the matrices $\rho(a)$ and $\rho(as)$ block by block.
2. Verify $s^2 = I$ and $s\rho(a)s = \rho(a^{-1})$ on generators of A.
3. In the weak-composition monomial basis, expand the image of every degree-q monomial to construct the full matrix $\mathrm{Sym}^q(\rho(g))$.
4. Form
$$\rho_{W_\tau}(g) = \bigotimes_b \mathrm{Sym}^{\tau_b}(\rho(g)) \quad\text{and}\quad P_\tau = \frac{1}{|G|}\sum_{g\in G}\rho_{W_\tau}(g).$$
5. Compare $\operatorname{rank} P_\tau$ and $\operatorname{tr} P_\tau$ with F_τ, while also checking $P_\tau^2 = P_\tau$.

Thus agreement is stronger than comparing two evaluations of the same determinant expression.

24.5 Representative cases and executed results

The direct matrix dimension is

$$\dim W_\tau = \prod_b \binom{\tau_b + d - 1}{\tau_b}, \qquad d = \dim V.$$

The tested cases were selected to keep this dimension moderate while exercising each structural feature of the theorem.

Representation	$\lvert G\rvert$	$\dim V$	checks	largest $\dim W_\tau$
D_5, one standard two-dimensional block	10	2	6	9
D_6, trivial and order-two 1D blocks plus one 2D block	12	4	7	100
$\mathrm{Dih}(C_2\times C_2)$, self-inverse characters only	8	5	6	75
$\mathrm{Dih}(C_3\times C_4)$, mixed 1D and 2D blocks	24	4	7	100
D_5, multiplicity two on the same 2D block	10	4	7	100
Total			**33**	**100**

The partitions include the empty partition, single blocks of degrees $1,2,3$, repeated tensor blocks such as $(1,1)$ and $(2,2)$, and the mixed partition $(2,1)$. The noncyclic examples test character weights with several moduli. The even cyclic and elementary 2-group examples test the extra self-inverse characters. The last example tests the exponent M and repeated two-dimensional summands.

Executed result. All 33 checks passed. Across the full suite, the maximum of the group-relation error, Reynolds-projector idempotence error, trace/formula discrepancy, and distance of the formula from its nearest integer was 4.48×10^{-15}. The largest explicitly materialized target was 100×100.

Remark 2. *Finite testing is not a proof of the theorem. Its role is to expose common implementation and transcription errors: a missing factor* $1/2$*, omission of one coset, incorrect normalization when passing to a represented image, incorrect* ϵ *signs, failure to pair* χ *with* χ^{-1}*, or replacing the reflection contribution* $(1-t_r^2)^{-M}$ *by an a-dependent expression.*

24.6 Conclusion

The proposed classification and master formula are correct for every finite-dimensional complex representation of $\mathrm{Dih}(A)$ with A finite abelian. The proof reduces the representation theory to inversion orbits in $\widehat{A}$ and then reads the determinant factors from the two cosets A and As. The matrix verification agrees coefficientwise with this formula in 33 small but structurally varied cases, including reasonable representation dimensions and full symmetric-power projectors.

25 Exact sigma-MGFs for dicyclic groups

Abstract

For the dicyclic group Dic_m of order $4m$, this note proves the proposed generalized Molien formulas for every complex representation, specializes them to the two-dimensional irreducibles ρ_k, and derives the stated exact coefficient count. It also checks direct sums and the regular representation. The accompanying marimo notebook constructs the matrices themselves and compares the rank of an explicit Reynolds projector with three independent formula evaluations. All representative cases passed to numerical error below 2.6×10^{-15}.

25.1 Imported finite-group reduction

The determinant and coefficient formulas are Tools 2 and 1. The dicyclic argument starts with the representation classification and the spectra on the two cosets.

25.2 Representations of Dic_m

Use

$$\mathrm{Dic}_m = \langle a, x : a^{2m} = 1,\ x^2 = a^m,\ xax^{-1} = a^{-1} \rangle, \qquad \zeta = e^{\pi i/m}.$$

Its elements are a^j and xa^j for $0 \le j < 2m$. The four one-dimensional representations are

$$L_{\alpha,\beta} : \quad a \mapsto \alpha, \quad x \mapsto \beta, \qquad \alpha^2 = 1, \quad \beta^2 = \alpha^m.$$

For $1 \le k \le m-1$, the two-dimensional representations are

$$\rho_k(a) = \begin{pmatrix} \zeta^k & 0 \\ 0 & \zeta^{-k} \end{pmatrix}, \qquad \rho_k(x) = \begin{pmatrix} 0 & (-1)^k \\ 1 & 0 \end{pmatrix}.$$

The displayed matrices satisfy all three defining relations. Their two a-weights are distinct and are interchanged by x, hence ρ_k is irreducible. The representations ρ_k, $1 \le k \le m-1$, are pairwise inequivalent: $\chi_k(a) = \zeta^k + \zeta^{-k} = 2\cos(\pi k/m)$, and these values are distinct because cosine is strictly decreasing on $(0, \pi)$. They are inequivalent to the one-dimensional representations by dimension. The four one-dimensional characters are pairwise distinct by their values on a and x. Therefore the displayed irreducibles are pairwise inequivalent; now $4 \cdot 1^2 + (m-1) \cdot 2^2 = 4m$ proves that the list is complete.

Write an arbitrary representation as

$$V \cong \bigoplus_{\alpha,\beta} L_{\alpha,\beta}^{\oplus c_{\alpha,\beta}} \oplus \bigoplus_{k=1}^{m-1} \rho_k^{\oplus d_k}.$$

On a^j, the displayed summands have eigenvalues α^j and ζ^{kj}, ζ^{-kj}. On xa^j, they have eigenvalue $\beta\alpha^j$ and a pair whose square is $(-1)^k$. Thus

$$\begin{aligned} \mathrm{SMG}_{\mathrm{Dic}_m,V} = {} & \frac{1}{4m} \sum_{j=0}^{2m-1} \prod_i \left[\prod_{\alpha,\beta} (1 - \alpha^j t_i)^{-c_{\alpha,\beta}} \prod_k \frac{1}{(1 - \zeta^{kj} t_i)^{d_k} (1 - \zeta^{-kj} t_i)^{d_k}} \right] \\ & + \frac{1}{4m} \sum_{j=0}^{2m-1} \prod_i \left[\prod_{\alpha,\beta} (1 - \beta\alpha^j t_i)^{-c_{\alpha,\beta}} \prod_k (1 - (-1)^k t_i^2)^{-d_k} \right]. \end{aligned}$$

This proves the arbitrary-representation formula directly from the spectrum.

25.3 A single two-dimensional irreducible

For $V = \rho_k$ the preceding expression becomes

$$\boxed{\mathrm{SMG}_{\mathrm{Dic}_m,\rho_k} = \frac{1}{4m} \sum_{j=0}^{2m-1} \prod_i \frac{1}{(1 - \zeta^{kj} t_i)(1 - \zeta^{-kj} t_i)} + \frac{1}{2} \prod_i \frac{1}{1 - (-1)^k t_i^2}.}$$

Let $q_k = 2m/\gcd(2m,k)$. In the monomial basis $u^{\tau_b - r_b}v^{r_b}$ of $\mathrm{Sym}^{\tau_b} V$, the a-weight is $\zeta^{k(\tau_b - 2r_b)}$. Hence the cyclic-subgroup average selects exactly

$$N_{k,\tau} = \#\left\{(r_b) : 0 \le r_b \le \tau_b, \quad q_k \mid |\tau| - 2\sum_b r_b\right\}.$$

For a coset element the eigenvalues are $q, -q$, where $q^2 = (-1)^k$, and

$$h_d(q,-q) = \begin{cases} q^d, & d \text{ even}, \\ 0, & d \text{ odd}. \end{cases}$$

Half the group lies in each part, giving the exact integer formula

$$\boxed{[m_\tau]\,\mathrm{SMG}_{\mathrm{Dic}_m,\rho_k} = \frac{1}{2}N_{k,\tau} + \frac{1}{2}\mathbf{1}_{\text{all } \tau_b \text{ even}}(-1)^{k|\tau|/2}.}$$

For $k = 1$, the coset factor is $(1 + t_i^2)^{-1}$, recording that lifted reflections square to $-I$.

For $V = \bigoplus_k \rho_k^{\oplus d_k}$, the identical argument yields half a zero-weight lattice count and half $\prod_b B_{\tau_b}$, where

$$B_d = [x^d] \prod_{k=1}^{m-1} (1 - (-1)^k x^2)^{-d_k}.$$

25.4 Regular representation

An element of order d acts on the regular representation as $4m/d$ cycles of length d. There are $\varphi(d)$ elements of order d in $\langle a \rangle$, while every xa^j has order four. Therefore

$$\boxed{\mathrm{SMG}_{\mathrm{Dic}_m,\mathrm{reg}} = \frac{1}{4m}\sum_{d|2m} \varphi(d)\,\mathrm{Exp}_\sigma\left(\frac{4m}{d}p_d\right) + \frac{1}{2}\mathrm{Exp}_\sigma(mp_4).}$$

25.5 Verification strategy and results

The notebook uses an orthonormal occupancy basis for each symmetric power. It embeds that basis in $V^{\otimes q}$, computes $\mathrm{Sym}^q(\rho(g))$ by compression, forms the tensor product W_τ, and then forms the full Reynolds projector. This route does not use eigenvalues or the coefficient formula. It is compared with (i) a character average, (ii) the congruence count above, and (iii) coefficient extraction from the two spectral sums. A safety cap prevents accidental construction of oversized tensor spaces.

Case	$\dim W_\tau$	rank	formula	error
$\mathrm{Dic}_2, \rho_1, \tau = (2)$	3	0	0	$6.8 \cdot 10^{-17}$
$\mathrm{Dic}_3, \rho_1, \tau = (2,2)$	9	2	2	$6.9 \cdot 10^{-16}$
$\mathrm{Dic}_4, \rho_2, \tau = (3,1)$	8	2	2	$3.4 \cdot 10^{-16}$
$\mathrm{Dic}_5, \rho_3, \tau = (4)$	5	1	1	$5.5 \cdot 10^{-16}$
$\mathrm{Dic}_3, L_{1,-1} \oplus L_{-1,i} \oplus \rho_2, \tau = (2,1)$	40	4	4	$< 2.6 \cdot 10^{-15}$
$\mathrm{Dic}_4, \rho_1 \oplus \rho_2, \tau = (2,1)$	40	2	2	$< 10^{-14}$
$\mathrm{Dic}_2, \mathrm{reg}, \tau = (4)$	330	44	44	0

Here "error" is the projector idempotence residual or, for the two added direct-sum rows, a conservative bound including relation and formula residuals. The dimensions 2, 4, and 8 exercise irreducible, reducible, and regular representations without hiding the computation behind large numerical spaces.

Conclusion. Every tested route agrees. More importantly, the proof identifies why the formula must hold: the determinant series gives symmetric-power characters, and averaging those characters is exactly the trace/rank of the invariant projector. The dicyclic sign is forced by $(xa^j)^2 = a^m$.

26 Regular representations of finite groups

Abstract

Using the imported coefficient and Molien identities, we derive the element-order formula for the regular σ-MGF and then test the regular representations of C_5, D_4 (order 8), S_3, and A_4. Every group is constructed as a concrete left-regular matrix group. Coefficients are computed by an explicit Reynolds projector and compared with an independent order-count formula; the full rational functions are also compared at safe numerical specializations. All reported cases agree to at least 10^{-14}.

26.1 Imported reduction

We use Tools 1 and 2. The family-specific datum is that an element of order d acts on the regular basis as $|G|/d$ cycles of length d.

26.2 The regular representation

Let $R_G = \mathbb{C}[G]$ with the left action. If g has order d, left multiplication partitions G into $|G|/d$ cycles of length d. Hence

$$\det(I - t\rho_{R_G}(g)) = (1 - t^d)^{|G|/d}.$$

Writing $N_d(G) = \#\{g \in G : \mathrm{ord}(g) = d\}$ gives

$$\boxed{\mathbb{E}[\mathrm{Exp}_\sigma(X_{G,R_G}h_1)] = \sum_{d\geq 1} \frac{N_d(G)}{|G|} \mathrm{Exp}_\sigma\left(\frac{|G|}{d}p_d\right).} \tag{57}$$

Equivalently, at finitely many scalar variables,

$$\mathbb{E}[\mathrm{Exp}_\sigma(X_{G,R_G}h_1)]\,(t_1,\ldots,t_s) = \frac{1}{|G|}\sum_d N_d(G)\prod_{i=1}^{s}(1-t_i^d)^{-|G|/d}.$$

Thus the regular σ-MGF depends only on the element-order distribution.

The principal families follow immediately:

$$\begin{aligned}
\mathbb{E}[\mathrm{Exp}_\sigma(X_{C_n,R}h_1)] &= \frac{1}{n}\sum_{d|n}\varphi(d)\,\mathrm{Exp}_\sigma\left(\frac{n}{d}p_d\right),\\
\mathbb{E}[\mathrm{Exp}_\sigma(X_{D_n,R}h_1)] &= \frac{1}{2n}\left[\sum_{d|n}\varphi(d)\,\mathrm{Exp}_\sigma\left(\frac{2n}{d}p_d\right) + n\,\mathrm{Exp}_\sigma(np_2)\right], \qquad |D_n| = 2n,\\
\mathbb{E}[\mathrm{Exp}_\sigma(X_{S_n,R}h_1)] &= \sum_{\lambda\vdash n}\frac{1}{z_\lambda}\mathrm{Exp}_\sigma\left(\frac{n!}{o(\lambda)}p_{o(\lambda)}\right),\\
\mathbb{E}[\mathrm{Exp}_\sigma(X_{A_n,R}h_1)] &= 2\sum_{\substack{\lambda\vdash n\\ n-\ell(\lambda)\ meven}}\frac{1}{z_\lambda}\mathrm{Exp}_\sigma\left(\frac{n!/2}{o(\lambda)}p_{o(\lambda)}\right),
\end{aligned}$$

where $z_\lambda = \prod_j j^{m_j} m_j!$ and $o(\lambda) = \mathrm{lcm}\{j : m_j > 0\}$.

26.3 Independent verification method

The implementation uses two routes.

Direct matrix route. Index the standard basis by $x \in G$ and set $(P_g)_{gx,x} = 1$. On $W_\tau = \bigotimes_a \mathrm{Sym}^{\tau_a} R_G$ form

$$\Pi_\tau = \frac{1}{|G|} \sum_{g \in G} \bigotimes_a \mathrm{Sym}^{\tau_a}(P_g).$$

Because Π_τ is the Reynolds projector, $\operatorname{rank} \Pi_\tau = \dim(W_\tau)^G$. The code separately checks $\|\Pi_\tau^2 - \Pi_\tau\|$ and averages the symmetric-power characters.

Order-formula route. Element orders are computed in the abstract multiplication table. For an element of order d, the coefficient of t^k in $(1-t^d)^{-|G|/d}$ is zero unless $d \mid k$, and otherwise is $\binom{|G|/d+k/d-1}{k/d}$. Multiplication over the parts of τ and averaging with $N_d(G)$ extracts the coefficient in (57). This route never uses the Reynolds matrix.

As an untruncated check, both determinant expressions are evaluated at $(t_1, t_2) = (0.07, -0.04)$. These small values are far from all poles.

26.4 Representative results

case	$\dim R_G$	$\dim W_\tau$	projector	formula	max. error
C_5, $\tau = (2,1)$	5	75	15	15	$9.4 \cdot 10^{-16}$
D_4, $\tau = (2)$	8	36	7	7	$5.0 \cdot 10^{-16}$
S_3, $\tau = (1,1)$	6	36	6	6	$4.1 \cdot 10^{-16}$
A_4, $\tau = (2)$	12	78	8	8	$3.4 \cdot 10^{-16}$

Here "max. error" is the larger of the determinant-specialization error and the Reynolds idempotence error. Character averages equal the same displayed integers in every row.

26.5 Dimension choice and a stability diagnostic

The test dimensions $5, 8, 6, 12$ are large enough to include several element orders but keep the largest explicit target at dimension 78. This makes a full matrix projector, rather than only a character computation, practical. The implementation caps interactive symmetric-power targets to prevent an accidental exponential allocation.

Finally, the regular family cannot have a finite coefficientwise limit when $|G| \to \infty$: directly,

$$[m_{(1,1)}]\mathbb{E}[\mathrm{Exp}_\sigma(X_{G,R_G} h_1)] = \dim(R_G \otimes R_G)^G = |G|.$$

This is a proof-level obstruction, not merely a numerical observation.

Reproducibility

The canonical marimo file is `marimo_notebooks/regular_representations.py`. The shared engine, README, and pytest command remain at the package root.

27 Representation variants: exterior powers and regular modules

Outcome. Two structurally different substitutions into the finite-group master formula were tested. For Λ^2 of the standard three-dimensional S_4 representation, explicit compound matrices agree with the pairwise-eigenvalue formula. For the regular representation of S_3, explicit 6×6 permutation matrices agree with the element-order formula. All 38 multivariate coefficients through total degree five pass, as do direct Reynolds projector checks.

27.1 Imported master formula

The universal finite-group identities are Tools 1 and 2. This section tests two nontrivial substitutions into those tools: an exterior power and a regular module.

27.2 Exterior powers

If $g|_V$ has eigenvalues $\lambda_1, \dots, \lambda_n$, then the wedge vectors $v_{i_1} \wedge \cdots \wedge v_{i_k}$ show that $g|_{\Lambda^k V}$ has eigenvalues

$$\lambda_I = \prod_{i \in I} \lambda_i, \qquad |I| = k.$$

Therefore

$$\mathrm{SMG}_{G,\Lambda^k V}(\mathbf{t}) = \frac{1}{|G|} \sum_{g \in G} \prod_{a \geq 1} \prod_{|I|=k} (1 - t_a \lambda_I(g))^{-1}. \tag{3}$$

This proof needs no assumption that G is a reflection group.

27.2.1 Concrete test: Λ^2 of the S_4 reflection representation

Start from the permutation representation of S_4 on $\mathbb{C}^4$ and restrict it to

$$V = \{(x_1, x_2, x_3, x_4) : x_1 + x_2 + x_3 + x_4 = 0\}.$$

Numerical QR orthonormalization supplies a 4×3 basis matrix B, so the explicit standard matrices are $B^* P_\pi B$. The direct exterior-square matrices are constructed without diagonalizing: in the ordered basis $e_i \wedge e_j$, their entries are the 2×2 minors

$$(\Lambda^2 g)_{(k,l),(i,j)} = g_{ki} g_{lj} - g_{kj} g_{li}. \tag{4}$$

The formula route instead diagonalizes only the original 3×3 standard matrices and forms the three pairwise eigenvalue products required by (3). Thus an error in the compound-matrix construction is not reproduced in the formula route.

27.3 The regular representation

Let G act on its basis by left multiplication. If g has order $o(g)$, then its permutation of the $|G|$ basis elements has $|G|/o(g)$ cycles, each of length $o(g)$. Hence

$$\det(I - tg|_{\mathrm{Reg}}) = (1 - t^{o(g)})^{|G|/o(g)}. \tag{5}$$

If N_m is the number of elements of order m, substitution into (1) gives

$$\mathrm{SMG}_{G,\mathrm{Reg}}(\mathbf{t}) = \sum_m \frac{N_m}{|G|} \prod_{a \geq 1} (1 - t_a^m)^{-|G|/m}. \tag{6}$$

For coefficient extraction, use

$$[t^d](1 - t^m)^{-c} = \begin{cases} \binom{c+d/m-1}{d/m}, & m \mid d, \\ 0, & m \nmid d. \end{cases}$$

For S_3, the order counts are

$$N_1 = 1, \qquad N_2 = 3, \qquad N_3 = 2. \tag{7}$$

Equations (6)–(7) therefore give an integer-arithmetic formula for every multivariate coefficient. The direct route constructs all six left-regular 6×6 permutation matrices and never consults these order counts.

27.4 Verification protocol

For each representation and every partition τ through total degree five:

1. Construct all representation matrices explicitly.
2. Recursively compute h_d from each matrix's eigenvalues and average $\prod_a h_{\tau_a}$ over the group.
3. Compare with the representation-specific formula: pairwise spectra in (3) for S_4, or binomial coefficient extraction from (6) for S_3.
4. Embed the normalized occupancy basis of $\mathrm{Sym}^2(W)$ in $W^{\otimes 2}$, form the explicit Reynolds matrix, and compare rank with the coefficient.

27.5 Results

Representation	$\lvert G \rvert$	$\dim W$	degree	checks	result
S_4 on $\Lambda^2(V_{\mathrm{std}})$	24	3	0–5	19	**PASS**
S_3 on Reg	6	6	0–5	19	**PASS**

Selected matched dimensions are

Representation	(1)	(2)	(3)	$(2,1)$	$(2,2)$	$(3,2)$
$\Lambda^2(V_{\mathrm{std}})$ of S_4	0	1	0	0	3	1
Reg of S_3	1	5	10	21	78	196

The large mixed coefficients in the regular case provide a useful stress test: agreement is not caused merely by both routes rounding small numbers to zero.

For the exterior case, $\mathrm{Sym}^2(W)$ has dimension six and the Reynolds projector has rank 1, trace 0.9999999999999992, and idempotence error 8.31×10^{-16}. For the regular case, $\mathrm{Sym}^2(W)$ has dimension 21; the projector has rank 5, trace 4.999999999999999, and idempotence error 3.72×10^{-16}. These ranks agree with the (2) coefficient in each row of the table.

27.6 What these tests establish

The exterior test exercises a nonlinear eigenvalue transformation and a coordinate-level compound-matrix construction. The regular test exercises a combinatorial compression from individual group elements to element-order counts. Together they show that the master formula is representation-agnostic in precisely the claimed way: only the spectra on the chosen representation change.

The tests are finite and numerical; the all-degree statements (3) and (6) follow from the preceding eigenvalue and cycle proofs. Numerical acceptance requires the direct average to be within 3×10^{-7} of an integer. The projector residuals are at machine precision.

27.7 Reproducibility

Run from the repository root:

```
PYTHONPATH=for_this_guy/matrix_group_formula_verification \
  .venv/bin/python \
  for_this_guy/matrix_group_formula_verification/\
  representation_variants/verification.py
```

The marimo notebook allows the maximum degree to be changed interactively and displays every coefficient comparison and both projector diagnostics.

Part III

Symmetric, alternating, cube, and permutation actions

28 Exact finite-n sigma-MGF for S_n

Abstract

For the permutation representation $V = \mathbb{C}^n$, this note proves the exact cycle-type formula for $\mathbb{E}[\mathrm{Exp}_\sigma(X_{S_n} h_1)]$, identifies its coefficients with vector partitions, and explains the stable range. The accompanying marimo notebook constructs all permutation matrices and compares a full Reynolds-projector rank with both formulas. All representative tests passed.

28.1 Imported coefficient interpretation

The coefficient identity is Tool 1; its orbit form is Tool 3. The new content is the exact finite-n cycle-type formula, the vector-partition size constraint, and stabilization.

28.2 Exact cycle-type formula

Write a cycle type as $\lambda = (1^{m_1} 2^{m_2} \cdots) \vdash n$ and set

$$z_\lambda = \prod_{d \geq 1} d^{m_d} m_d!.$$

A d-cycle has all dth roots of unity as eigenvalues, so $\det(I - tP_d) = 1 - t^d$. A permutation of type λ therefore contributes

$$\prod_i \prod_{d \geq 1} (1 - t_i^d)^{-m_d(\lambda)}.$$

The conjugacy class has size $n!/z_\lambda$, proving

$$\boxed{\mathbb{E}[\mathrm{Exp}_\sigma(X_{S_n} h_1)] = \sum_{\lambda \vdash n} \frac{1}{z_\lambda} \prod_i \prod_{d \geq 1} (1 - t_i^d)^{-m_d(\lambda)}.}$$

This statement is exact for every n and in every degree; it is not a stable approximation.

28.3 Exact coefficient formula

A monomial-basis vector of W_τ is encoded by a labeling

$$\ell : \{1, \ldots, n\} \longrightarrow \mathbb{Z}^s_{\geq 0}, \qquad \sum_{j=1}^n \ell(j) = \tau$$

coordinatewise. Two labelings lie in the same S_n-orbit exactly when the multiplicities $a_\mathbf{v} = \#\{j : \ell(j) = \mathbf{v}\}$ agree. Since the dimension of the invariant space of a permutation module equals its number of orbits,

$$\boxed{[m_\tau]\mathbb{E}[\mathrm{Exp}_\sigma(X_{S_n} h_1)] = \#\left\{(a_\mathbf{v})_{\mathbf{v} \geq 0} : a_\mathbf{v} \geq 0, \quad \sum_\mathbf{v} a_\mathbf{v} = n, \quad \sum_\mathbf{v} a_\mathbf{v} \mathbf{v} = \tau\right\}.}$$

Equivalently, this is the number of vector partitions of τ into at most n nonzero parts; zero parts fill the remaining slots.

Marking the number of labels with u gives

$$\boxed{\sum_{n\geq 0} u^n \mathbb{E}[\mathrm{Exp}_\sigma(X_{S_n} h_1)] = \prod_{\mathbf{v}\in\mathbb{Z}_{\geq 0}^{(\mathbb{N})}} (1 - ut^{\mathbf{v}})^{-1}.}$$

The exponential formula for permutation cycles gives the equivalent form

$$\sum_{n\geq 0} u^n \mathbb{E}[\mathrm{Exp}_\sigma(X_{S_n} h_1)] = \exp\left(\sum_{d\geq 1} \frac{u^d}{d} \prod_i \frac{1}{1 - t_i^d}\right).$$

28.4 Stable range

Separate the zero label. Every nonzero vector has total degree at least one, so a vector partition of τ uses at most $|\tau|$ nonzero parts. If $n \geq |\tau|$, zeros can always be added and the coefficient stabilizes:

$$[m_\tau]\mathbb{E}[\mathrm{Exp}_\sigma(X_{S_n} h_1)] = [m_\tau] \prod_{\mathbf{v}\neq 0} (1 - t^{\mathbf{v}})^{-1}.$$

When $n < |\tau|$, the exact finite-n formula simply removes decompositions requiring too many nonzero parts.

28.5 Verification method and evidence

For each $g \in S_n$, the notebook creates its permutation matrix, compresses $g^{\otimes q}$ to an orthonormal occupancy basis of $\mathrm{Sym}^q(V)$, tensors the selected blocks, and averages the resulting matrices. The projector rank is compared independently with the symmetric-power character average, cycle-type coefficient extraction, and a dynamic-programming vector-partition count.

Case	$\dim W_\tau$	rank	character	cycle	orbit
$S_2, \tau = (3)$	4	2	2	2	2
$S_3, \tau = (2,1)$	18	4	4	4	4
$S_4, \tau = (2,2)$	100	9	9	9	9
$S_5, \tau = (2,1)$	75	4	4	4	4

The largest projector here is 100×100 and averages 24 explicitly constructed matrices. Idempotence residuals range from 2.2×10^{-16} to 1.1×10^{-15}. These cases cover degree above the stable threshold, multiblock tensor products, and two different values of n for the same τ.

Conclusion. The cycle-index and orbit formulas are two descriptions of the same invariant dimension, and the explicit matrix calculations agree with both in every tested case.

29 Exact finite-n sigma-MGF for A_n

Abstract

The asymptotic agreement with S_n is imported from Theorem B; for $n \geq 2$, this section proves the exact cycle-type formula for the permutation representation of A_n, derives the correction to the S_n vector-partition count from orbit splitting, and proves equality with S_n in degree at most $n-2$. Explicit even permutation matrices and Reynolds projectors verify the formula, including a boundary case with a nonzero splitting correction.

29.1 From S_n to A_n

For any class function F on S_n, the indicator of an even permutation gives

$$\mathbb{E}_{A_n}[F] = \frac{1}{n!} \sum_{g \in S_n} (1 + \operatorname{sgn}(g)) F(g).$$

By Tool 1, the coefficient of m_τ is $\dim(\bigotimes_b \operatorname{Sym}^{\tau_b}(\mathbb{C}^n))^{A_n}$. No new Reynolds argument is needed.

If $\lambda \vdash n$, then

$$\operatorname{sgn}(\lambda) = (-1)^{n-\ell(\lambda)}, \qquad z_\lambda = \prod_{d \geq 1} d^{m_d(\lambda)} m_d(\lambda)!.$$

Combining the sign indicator with the determinant of each permutation cycle proves the exact expression

$$\boxed{\mathbb{E}[\operatorname{Exp}_\sigma(X_{A_n} h_1)] = \sum_{\lambda \vdash n} \frac{1 + (-1)^{n-\ell(\lambda)}}{z_\lambda} \prod_i \prod_{d \geq 1} (1 - t_i^d)^{-m_d(\lambda)}.}$$

Only even cycle types remain, each with coefficient $2/z_\lambda$.

29.2 Generating function

Set

$$G_+(u) = \prod_{\mathbf{v} \geq 0} (1 - u t^{\mathbf{v}})^{-1}, \qquad G_-(u) = \prod_{\mathbf{v} \geq 0} (1 + u t^{\mathbf{v}}).$$

The first product records multisets of labels. The second records sets of distinct labels and is the sign-weighted cycle-index contribution. Hence, for $n \geq 2$,

$$\boxed{\mathbb{E}[\operatorname{Exp}_\sigma(X_{A_n} h_1)] = [u^n]\big(G_+(u) + G_-(u)\big).}$$

Equivalently, applying the exponential formula to both factors gives

$$\mathbb{E}[\operatorname{Exp}_\sigma(X_{A_n} h_1)] = [u^n]\Bigg[\exp\left(\sum_{d \geq 1} \frac{u^d}{d} \prod_i \frac{1}{1 - t_i^d}\right) + \exp\left(\sum_{d \geq 1} \frac{(-1)^{d-1} u^d}{d} \prod_i \frac{1}{1 - t_i^d}\right)\Bigg].$$

29.3 Exact orbit interpretation

An S_n-orbit of monomial-basis labelings is determined by multiplicities $(a_{\mathbf{v}})_{\mathbf{v} \geq 0}$. Its stabilizer is $\prod_{\mathbf{v}} S_{a_{\mathbf{v}}}$. If some $a_{\mathbf{v}} \geq 2$, the stabilizer contains the odd transposition of two equally labeled points, so the S_n-orbit remains a single A_n-orbit. If every $a_{\mathbf{v}}$ is zero or one, the stabilizer lies in A_n, and the orbit splits into two A_n-orbits. Thus

$$[m_\tau]\mathbb{E}[\operatorname{Exp}_\sigma(X_{A_n} h_1)] = \#\Big\{(a_{\mathbf{v}}) \geq 0 : \sum a_{\mathbf{v}} = n,\ \sum a_{\mathbf{v}} \mathbf{v} = \tau\Big\} + \#\Big\{(a_{\mathbf{v}}) \in \{0,1\} : \sum a_{\mathbf{v}} = n,\ \sum a_{\mathbf{v}} \mathbf{v} = \tau\Big\}.$$

The first term is the S_n coefficient. The correction counts sets of distinct nonzero vectors summing to τ, with either n such vectors or $n-1$ such vectors and one zero vector.

29.4 Stable agreement with S_n

Any labeling of total degree $|\tau|$ has at most $|\tau|$ nonzero labels. If $|\tau| \le n-2$, at least two labels are zero. Their transposition is odd and lies in the stabilizer, so no S_n-orbit can split. Therefore

$$\boxed{[m_\tau]\mathbb{E}[\mathrm{Exp}_\sigma(X_{A_n}h_1)] = [m_\tau]\,\mathrm{SMG}_{S_n} \quad \text{whenever } |\tau| \le n-2.}$$

Outside that range the distinct-label term is the exact correction, not merely an error estimate.

29.5 Verification method and evidence

The notebook enumerates even permutations, constructs their permutation matrices, builds the full matrices on each symmetric tensor block, and averages them to obtain the A_n Reynolds projector. Its rank is compared independently with the character average, the even-cycle-type formula, and the S_n orbit count plus a distinct-label dynamic-programming correction.

Case	$\dim W_\tau$	rank	cycle	S_n	corr.	total
$A_3, \tau = (2)$	6	2	2	2	0	2
$A_3, \tau = (3)$	10	4	4	3	1	4
$A_4, \tau = (2,1)$	40	4	4	4	0	4
$A_5, \tau = (2,2)$	225	9	9	9	0	9
$A_5, \tau = (3,1)$	175	7	7	7	0	7

The $A_3, \tau = (3)$ row deliberately lies outside the stable range and confirms that one S_3 orbit splits, increasing the invariant dimension from three to four. The remaining rows test multiblock targets and larger projectors. Idempotence residuals are at most 1.1×10^{-15}.

Conclusion. The sign-weighted cycle formula and orbit-splitting formula agree exactly, and both match direct matrix invariant dimensions in all representative cases.

30 Selected representations of S_n

Abstract

We verify the σ-MGF formulas for the sign and standard representations of S_n, the symmetric/exterior/tensor character constructions, and the induced representation $\mathrm{Ind}_{C_3}^{S_3}\omega$. Matrices are constructed element by element. Reynolds ranks, character averages, cycle formulas, and decorated coset-cycle determinants agree in all small cases.

30.1 Sign representation

For $n \geq 2$, exactly half of S_n has sign $+1$ and half has sign -1. Therefore

$$\boxed{\mathrm{SMG}_{S_n,\mathrm{sgn}}(t_\bullet) = \frac{1}{2}\left[\prod_i (1-t_i)^{-1} + \prod_i (1+t_i)^{-1}\right].} \tag{58}$$

The coefficient of m_τ is 1 when $|\tau|$ is even and 0 when it is odd. This follows either by expanding (58), or because $\bigotimes_a \mathrm{Sym}^{\tau_a}(\mathrm{sgn}) = \mathrm{sgn}^{|\tau|}$.

30.2 The standard representation

Let $M_n = \mathbb{C}^n$ be the natural permutation representation and $V_n = S^{(n-1,1)}$. Since $M_n = \mathbf{1} \oplus V_n$, for every permutation g,

$$\prod_i \det(I - t_i g|_{V_n})^{-1} = \prod_i (1-t_i) \prod_i \det(I - t_i g|_{M_n})^{-1}.$$

If g has cycle type λ, then $\det(I - tg|_{M_n}) = \prod_{d\in\lambda}(1-t^d)$. Hence the exact finite-n formula is

$$\boxed{\mathrm{SMG}_{S_n,V_n}(t_1,\ldots,t_s) = \sum_{\lambda\vdash n} \frac{1}{z_\lambda} \prod_{i=1}^{s} \left[(1-t_i)\prod_{d\in\lambda}(1-t_i^d)^{-1}\right].} \tag{59}$$

The natural-action coefficient counts S_n-orbits on tuples of multisets. Multiplication by $\prod_i(1-t_i)$ gives the independent coefficient rule

$$[t_1^{\tau_1}\cdots t_s^{\tau_s}]\,\mathrm{SMG}_{S_n,V_n} = \sum_{J\subseteq[s]} (-1)^{|J|} N_n(\tau - \mathbf{1}_J),$$

where $N_n(\alpha)$ is the corresponding vector-partition count and terms with negative coordinates vanish. Once total degree is at most n, the familiar stable form follows:

$$\mathrm{SMG}_{S_n,V_n} \equiv \mathrm{Exp}_\sigma(h_2 + h_3 + \cdots) \pmod{\deg > n}.$$

Concrete matrices. The code uses the basis $e_i - e_n$ for $1 \leq i < n$. If P_g is the natural permutation matrix and $B = [e_1 - e_n \ \cdots \ e_{n-1} - e_n]$, the standard matrix is the coordinate matrix of $P_g B$. This produces exact integer matrices of dimension $n-1$. Although this basis is not orthonormal, the rank and idempotence of the Reynolds operator are basis-invariant.

30.3 Symmetric, exterior, and tensor constructions

For any representation V and $g \in G$, the cycle-index expansions give

$$\chi_{\mathrm{Sym}^k V}(g) = \sum_{\mu\vdash k} \frac{1}{z_\mu} \prod_{r\geq 1} \chi_V(g^r)^{m_r(\mu)},$$

$$\chi_{\wedge^k V}(g) = \sum_{\mu\vdash k} \frac{(-1)^{k-\ell(\mu)}}{z_\mu} \prod_{r\geq 1} \chi_V(g^r)^{m_r(\mu)},$$

$$\chi_{V^{\otimes k}}(g) = \chi_V(g)^k.$$

For $k = 2$ these reduce to

$$\chi_{\mathrm{Sym}^2 V}(g) = \frac{\chi(g)^2 + \chi(g^2)}{2}, \qquad \chi_{\wedge^2 V}(g) = \frac{\chi(g)^2 - \chi(g^2)}{2}.$$

The implementation constructs the symmetric-square matrices on an occupancy basis, the exterior square in the two-dimensional S_3 case, and tensor characters. Maximum discrepancies are below 10^{-15}.

30.4 An induced representation and the coset-cycle proof

Let $H \leq G$, let W be an H-representation, and choose representatives x for G/H. If g has a coset cycle C of length ℓ_C beginning at xH, define $h_C = x^{-1} g^{\ell_C} x \in H$.

Proposition 1. *For $V = \operatorname{Ind}_H^G W$,*

$$\det(I - t\rho_V(g)) = \prod_{C \subseteq G/H} \det(I - t^{\ell_C} \rho_W(h_C)).$$

Proof. Along one coset cycle, g cyclically moves ℓ_C copies of W. After one lap the map on the starting copy is $\rho_W(h_C)$. An eigenvalue α of this monodromy produces all roots of $z^{\ell_C} = \alpha$, whose determinant factor is $1 - \alpha t^{\ell_C}$. Multiply over eigenvalues and cycles. □

The test takes $G = S_3$, $H = A_3 = C_3$, and $\omega((123)) = e^{2\pi i/3}$. Two left-coset representatives give explicit 2×2 monomial matrices. The resulting induced representation is the standard irreducible. Its direct determinant average is compared with the coset-cycle product, including the complex monodromy scalars.

30.5 Representative results

case	rep. dim.	target dim.	projector	formula	max. error
S_4 sign, (2)	1	1	1	1	$2.3 \cdot 10^{-16}$
S_4 sign, (3)	1	1	0	0	$2.3 \cdot 10^{-16}$
S_3 standard, $(2, 1)$	2	6	1	1	$3.5 \cdot 10^{-16}$
S_4 standard, $(2, 2)$	3	36	3	3	$2.5 \cdot 10^{-15}$
S_5 standard, $(2, 1)$	4	40	1	1	$1.3 \cdot 10^{-15}$
$\mathrm{Ind}_{C_3}^{S_3} \omega$, $(2, 1)$	2	6	1	1	$7.1 \cdot 10^{-16}$

The determinant tests use $(t_1, t_2) = (0.11, -0.06)$ for sign/standard and $(0.09, -0.04)$ for the induced example. These tests exercise the entire rational expression, while the projector tests certify selected exact coefficients.

Reproducibility

Open `marimo_notebooks/symmetric_representations.py` with marimo. The shared pytest suite reruns every row and the derived-character checks.

31 Specht, Foulkes, and free-Lie modules

Scope. This section collects the non-permutation S_n-modules that were formerly grouped with spin covers and rooted trees. The common Frobenius–Molien formula is followed by fixed-tail Specht limits, Foulkes modules, and multilinear free-Lie modules. Exact finite formulas are kept separate from the low-degree coefficients that obstruct raw limits.

31.1 The common target and generalized Molien theorem

Let G be finite, let $\rho : G \to GL(V)$ be a complex representation with character χ_V, and let $\tau = (\tau_1, \dots, \tau_s)$. Put

$$W_\tau(V) = \bigotimes_{j=1}^{s} \mathrm{Sym}^{\tau_j} V, \qquad a_\tau(G, V) = \dim W_\tau(V)^G.$$

The σ-MGF is the symmetric series

$$\mathrm{SMG}_{G,V} = \sum_\tau a_\tau(G, V) m_\tau.$$

Theorem 1 (Imported generalized Molien identity)**.** *For every finite group and finite-dimensional complex representation,*

$$\boxed{\mathrm{SMG}_{G,V} = \frac{1}{|G|} \sum_{g\in G} \exp\left(\sum_{r\geq 1} \frac{p_r}{r} \chi_V(g^r) \right) = \sum_\tau \dim W_\tau(V)^G m_\tau.} \tag{60}$$

Equivalently, in an alphabet $\mathbf{t} = (t_1, t_2, \dots)$,

$$\mathrm{SMG}_{G,V}(\mathbf{t}) = \frac{1}{|G|} \sum_{g\in G} \prod_i \det(I - t_i\rho(g))^{-1}. \tag{61}$$

Source. This is Tools 1 and 2, together with the standard logarithmic determinant expansion. It is quoted here only to fix the notation used in the representation-specific formulas below. □

Thus a character table without power maps is not sufficient: the exponent uses $\chi_V(g^r)$ for every r. For a permutation g of cycle type $\mu = 1^{m_1}2^{m_2}\cdots$, the rth power has type $\mu^{[r]}$ satisfying

$$\boxed{m_j(\mu^{[r]}) = \sum_{\substack{d\mid r\\ (j,r/d)=1}} d\, m_{jd}(\mu).} \tag{62}$$

Indeed, a cycle of length jd splits under the rth power into d cycles of length j precisely when $d = (jd, r)$, which is equivalent to the displayed divisibility and coprimality conditions.

31.2 Ordinary symmetric-group modules

31.2.1 Frobenius master formula

For an S_n-module V_n with Frobenius characteristic $f_n = \mathrm{ch}(V_n)$, the standard identity $\chi_{V_n}(\mu) = z_\mu[p_\mu]f_n$ and Theorem 1 give

$$\boxed{\mathrm{SMG}_{V_n} = \sum_{\mu\vdash n} \frac{1}{z_\mu} \exp\left(\sum_{r\geq 1} \frac{p_r}{r} z_{\mu^{[r]}}[p_{\mu^{[r]}}]f_n \right).} \tag{63}$$

For a Specht module S^λ, this specializes to

$$\boxed{\mathrm{SMG}_{S_n,S^\lambda} = \sum_{\mu\vdash n} \frac{1}{z_\mu} \exp\left(\sum_{r\geq 1} \frac{p_r}{r} \chi^\lambda(\mu^{[r]}) \right).} \tag{64}$$

The formula is exact for every n and λ; character evaluation can be performed by Murnaghan–Nakayama or any equivalent character algorithm.

Every irreducible S_n-module is self-dual and admits an invariant symmetric bilinear form. Therefore, for a nontrivial irreducible,

$$[m_{(1)}] = 0, \qquad [m_{(2)}] = [m_{(1,1)}] = 1.$$

The fourth multilinear coefficient already sees Kronecker complexity:

$$[m_{(1,1,1)}] = g(\lambda,\lambda,\lambda), \qquad [m_{(1,1,1,1)}] = \sum_{\nu\vdash n} g(\lambda,\lambda,\nu)^2.$$

Consequently (64) is a complete exact formula, but not a claim that arbitrary Kronecker multiplicities have a simpler closed form.

31.2.2 Hooks, two rows, and the matrix audit

Let $C_j(\sigma)$ count j-cycles. Since $S^{(n-k,1^k)} \cong \bigwedge^k S^{(n-1,1)}$, its character is

$$\boxed{Q_{1^k}(C) = [u^k]\frac{1}{1+u}\prod_{j\geq 1}(1-(-u)^j)^{C_j}.} \tag{65}$$

This follows by deleting the trivial eigenline from the natural permutation module and taking the generating function of exterior-power characters.

For $n \geq 2k$, let

$$A_k(C) = [u^k]\prod_{j\geq 1}(1+u^j)^{C_j}.$$

This is the number of fixed k-subsets. Young's rule gives

$$\mathbb{C}\left[\binom{[n]}{k}\right] \cong \bigoplus_{j=0}^{k} S^{(n-j,j)},$$

and subtraction of consecutive characters proves

$$\boxed{\chi^{(n-k,k)} = A_k - A_{k-1}.} \tag{66}$$

The verifier constructs the standard module on the integral basis $e_i - e_n$, takes exterior minors for the hook cases, and realizes $S^{(n-2,2)}$ as the vertex-sum-zero subspace of the two-subset permutation module. It checks the traces of powers $1 \leq r \leq 4$ for every matrix against (65) or (66). It then constructs $W_\tau(V)$ for $\tau = (1), (2), (1,1), (3)$, finds the common generator-fixed space, and compares its dimension with the full Molien average.

module	dim.	$[m_1]$	$[m_2]$	$[m_{11}]$	$[m_3]$
$S^{(3,1)}$ of S_4	3	0	1	1	1
$S^{(3,1,1)}$ of S_5	6	0	1	1	0
$S^{(2,2)}$ of S_4	2	0	1	1	1
$S^{(3,2)}$ of S_5	5	0	1	1	1

All 1,152 character-power comparisons and all 16 fixed-space/Molien comparisons pass. Numerical rank residuals are at most 1.3×10^{-15}; the matrices and character calculations themselves have integral entries.

31.2.3 Fixed-tail asymptotics and the boundary of the claim

Fix a partition λ and put $\lambda[n] = (n - |\lambda|, \lambda_1, \lambda_2, \ldots)$. For $n \geq |\lambda| + \lambda_1$, the partition is defined and its irreducible character is given by a fixed character polynomial $Q_\lambda(C_1, C_2, \ldots)$. Let independent $Z_j \sim \text{Poisson}(1/j)$ and define

$$Z_j^{[r]} = \sum_{\substack{d \mid r \\ (j, r/d)=1}} dZ_{jd}. \tag{67}$$

Then, coefficientwise in bounded total degree,

$$\boxed{\text{SMG}_{\lambda,\infty} = \mathbb{E}\exp\left(\sum_{r\geq 1} \frac{p_r}{r} Q_\lambda(Z_1^{[r]}, Z_2^{[r]}, \ldots)\right).} \tag{68}$$

Proof. In total degree at most D, the exponential uses only $r \leq D$ and finitely many cycle variables in each character polynomial. Formula (62) expresses the cycle counts of σ^r in those of σ. The finite collection of cycle counts of a uniform permutation converges jointly, with all fixed moments, to the independent Z_j. Substitution and moment convergence prove (68). □

This is a polynomial Poisson-chaos law, not generally an ordinary Poisson law. It applies to fixed hooks and fixed two-row tails through (65)–(66). It does not apply to rectangles or diagrams with two growing dimensions. For those shapes (64) remains exact, while normalized character asymptotics alone do not imply a raw σ-MGF limit; boundedness of the fourth coefficient is already a necessary test.

31.3 Foulkes permutation modules

Let

$$H^{(a^b)} = \text{Ind}_{S_a \wr S_b}^{S_{ab}} \mathbf{1}.$$

It is the permutation module on partitions of $[ab]$ into b unlabeled blocks of size a, and its Frobenius characteristic is

$$\text{ch}\, H^{(a^b)} = h_b[h_a].$$

Substitution in (63) proves

$$\boxed{\text{SMG}_{H^{(a^b)}} = \sum_{\mu \vdash ab} \frac{1}{z_\mu} \exp\left(\sum_{r\geq 1} \frac{p_r}{r} z_{\mu^{[r]}}[p_{\mu^{[r]}}] h_b[h_a]\right).} \tag{69}$$

The pair coefficient has a concrete orbit interpretation. Given two block partitions, label their blocks temporarily and set $A_{ij} = |B_i \cap C_j|$. Every row and column sum is a. Relabeling the two block systems permutes rows and columns, and two pairs lie in the same S_{ab}-orbit exactly when the tables are so related. Therefore

$$\boxed{[m_{(1,1)}]\, \text{SMG}_{H^{(a^b)}} = \#\{A \in \mathbb{Z}_{\geq 0}^{b\times b} : \text{all margins are } a\}/(S_b \times S_b).} \tag{70}$$

Theorem 2 (Foulkes pair-growth obstruction)**.** *Let $c_{a,b}$ denote the coefficient in (70).*

(i) For fixed $a \geq 2$, $c_{a,b} \geq p(b)$, where $p(b)$ is the partition number.

(ii) For fixed $b \geq 2$, $c_{a,b} \geq \lfloor a/2 \rfloor + 1$.

Consequently the raw Foulkes σ-MGF has no finite coefficientwise limit in either standard regime $b \to \infty$ with $a \geq 2$ fixed or $a \to \infty$ with $b \geq 2$ fixed.

Proof. For a partition $b = s_1 + \cdots + s_r$, take a block diagonal matrix with a 1×1 block $[a]$ for $s_i = 1$ and, for $s_i \geq 2$, the block $(a-1)I_{s_i} + P_{s_i}$, where P_{s_i} is a cyclic permutation matrix. The sizes of the connected components of the weighted bipartite support are invariant under independent row and column permutations and recover the partition of b. This proves (i).

For (ii), use the 2×2 block $\begin{pmatrix} j & a-j \\ a-j & j \end{pmatrix}$ for $0 \le j \le \lfloor a/2 \rfloor$, together with $b-2$ blocks $[a]$. The unique nontrivial weighted component and its edge weights distinguish these values of j up to the already imposed symmetry $j \leftrightarrow a-j$. ☐

The verification lists the block partitions, constructs every zero-one permutation matrix, and independently expands

$$h_b[h_a] = \sum_{\nu \vdash b} \frac{1}{z_\nu} \prod_{r \in \nu} \left(\sum_{\lambda \vdash a} \frac{p_{r\lambda}}{z_\lambda} \right).$$

Thus every matrix trace is compared with a power-sum coefficient, not with a second traversal of the same action. The results are

module	dim.	$[m_1]$	$[m_2]$	$[m_{11}]$	$[m_3]$	$[m_{21}]$
$H^{(2^2)}$ of S_4	3	1	2	2	3	4
$H^{(2^3)}$ of S_6	15	1	3	3	8	12

All 744 character comparisons and ten direct-orbit/Molien comparisons pass; the contingency-table counts are respectively 2 and 3. The preceding theorem, rather than the finite computations, proves divergence in the two standard Foulkes regimes.

31.4 The multilinear free-Lie representation

The degree-n multilinear free-Lie module has Frobenius characteristic

$$\boxed{L_n = \operatorname{ch} \operatorname{Lie}_n = \frac{1}{n} \sum_{d \mid n} \mu(d) p_d^{n/d}.} \tag{71}$$

Formula (63) therefore yields

$$\boxed{\operatorname{SMG}_{\operatorname{Lie}_n} = \sum_{\mu \vdash n} \frac{1}{z_\mu} \exp\left(\sum_{r \ge 1} \frac{p_r}{r} z_{\mu^{[r]}} [p_{\mu^{[r]}}] L_n \right).} \tag{72}$$

Power sums are orthogonal with $\langle p_\lambda, p_\lambda \rangle = z_\lambda$. Distinct divisors in (71) give distinct cycle types, hence

$$\boxed{[m_{(1,1)}] \operatorname{SMG}_{\operatorname{Lie}_n} = \langle L_n, L_n \rangle = \frac{1}{n^2} \sum_{d \mid n} \mu(d)^2 d^{n/d} (n/d)!.} \tag{73}$$

The $d = 1$ term gives the lower bound $n!/n^2$, so the raw series cannot have a coefficientwise limit as $n \to \infty$. A sign twist, such as that appearing in top partition-lattice homology, leaves this norm unchanged.

For an independent matrix construction, fix x_n and use the bracket basis

$$[\cdots[[x_n, x_{\sigma(1)}], x_{\sigma(2)}], \ldots, x_{\sigma(n-1)}], \qquad \sigma \in S_{n-1}.$$

Expanding brackets in the multilinear associative word basis gives integral matrices for relabeling by S_n. For $n = 3, 4$, all 30 matrix characters match (71). Direct fixed spaces and Molien averages give

module	dim.	$[m_1]$	$[m_2]$	$[m_{11}]$	$[m_3]$
Lie_3	2	0	1	1	1
Lie_4	6	0	2	2	2

The pair values $1, 2$ agree exactly with (73).

31.5 Verification and asymptotic scope

The matrix/fixed-space route constructs integral Specht and free-Lie matrices, while the formula route uses power maps, character polynomials, power-sum plethysm, and the Witt characteristic. Foulkes coefficients are also checked by literal orbit enumeration. The relevant conclusions are:

family	exact finite formula	asymptotic status
fixed-tail Specht	(64), (68)	explicit polynomial Poisson chaos
hooks and two rows	(65), (66)	explicit in the fixed-tail regime
growing diagrams	(64)	normalized character theory alone does not control raw moments
Foulkes	(69), (70)	pair-orbit growth obstructs standard raw limits
Lie_n	(72), (73)	$[m_{11}] \geq n!/n^2$, hence no raw coefficientwise limit

32 Subset actions and the universal permutation method

Abstract

We prove and test the fixed-point, Mobius-inversion, determinant, and orbit formulas for the permutation representation on k-subsets. Every induced permutation matrix is constructed for small S_n and A_n. Direct Reynolds ranks agree with the independent cycle formula in target spaces up to dimension 550. The symmetric-group quadratic coefficient $k+1$ is checked at its boundary, and the alternating-group orbit-splitting boundary is tested separately.

32.1 Imported permutation tools

For $V = \mathbb{C}[\Omega]$, Tools 2 and 3 give

$$\mathrm{SMG}_{G,V}(t_\bullet) = \frac{1}{|G|}\sum_{g\in G}\prod_{i,d\geq 1}(1-t_i^d)^{-c_d^\Omega(g)}, \tag{74}$$

$$[m_\tau]\,\mathrm{SMG}_{G,V} = \#\left(G\backslash\prod_a \mathrm{Sym}^{\tau_a}(\Omega)\right). \tag{75}$$

These identities are quoted, not reproved. The new calculation begins with the conversion from fixed points of powers to induced cycles.

32.2 Recovering cycles from fixed points

For any permutation q of finite order,

$$\#\,\mathrm{Fix}(q^r) = \sum_{d|r} d\, c_d(q).$$

Mobius inversion gives

$$\boxed{c_d(q) = \frac{1}{d}\sum_{r|d}\mu(d/r)\#\,\mathrm{Fix}(q^r).} \tag{76}$$

This identity is exact: the code checks divisibility by d before accepting a cycle count.

Now take $G = S_n$ or A_n and $\Omega = \binom{[n]}{k}$. A k-subset fixed by σ^r is a union of cycles of σ^r. If $X_j(\sigma^r)$ is its number of j-cycles, then

$$\boxed{\#\,\mathrm{Fix}_{\binom{[n]}{k}}(\sigma^r) = [z^k]\prod_{j\geq 1}(1+z^j)^{X_j(\sigma^r)}.} \tag{77}$$

Equations (74), (76), and (77) are the proposed exact finite formula.

32.3 Stability and the first nontrivial coefficients

For fixed k and τ, a tuple of $|\tau|$ many k-subsets uses at most $k|\tau|$ labels. Deleting isolated labels identifies the stable orbits with finite colored k-uniform multihypergraphs. Consequently

$$[m_\tau]\,\mathrm{SMG}_{S_n,\mathbb{C}[\binom{[n]}{k}]} \quad \text{stabilizes for} \quad n \geq k|\tau|.$$

For A_n, the same coefficients agree once two unused labels are available, so $n \geq k|\tau|+2$ is sufficient: compose any odd relabeling with a transposition of the unused labels.

For S_n, when $n \geq 2k$, an ordered or unordered pair of k-subsets is classified by $|A\cap B| \in \{0,1,\ldots,k\}$. Thus

$$[m_{(2)}]\,\mathrm{SMG} = [m_{(1,1)}]\,\mathrm{SMG} = k+1 \quad \text{for } S_n \text{ and } n\geq 2k. \tag{78}$$

For A_n, the same conclusion is guaranteed once two unused labels are available, so $n \geq 2k+2$ is sufficient. At $n = 2k$ and $n = 2k+1$, an S_n orbit may split and must be checked separately. For example, A_4 acting on 2-subsets has three orbits on unordered pairs but four orbits on ordered pairs: the ordered adjacent-pair class splits into two A_4-orbits. Thus

$$[m_{(2)}]\,\mathrm{SMG} = 3, \qquad [m_{(1,1)}]\,\mathrm{SMG} = 4 \quad \text{for } (n,k) = (4,2).$$

The regression suite checks the symmetric boundary and the alternating cases A_4, A_5, A_6 separately.

32.4 Four independent computational layers

1. Enumerate S_n (or its even subgroup) as image-tuples and construct its action on the lexicographically ordered k-subsets.
2. Build the full Reynolds projector on $\bigotimes_a \mathrm{Sym}^{\tau_a} \mathbb{C}[\binom{[n]}{k}]$ and take its numerical rank. Its idempotence residual is recorded.
3. Without examining the induced matrix cycles, compute (77), apply (76), and extract the coefficient of (74) using integer dynamic programming.
4. Evaluate both full determinant expressions at $(t_1, t_2) = (0.08, -0.03)$, safely inside the common disk of convergence.

The symmetric-power occupancy basis has dimension $\binom{\dim V + d - 1}{d}$. The chosen cases therefore exercise nontrivial spaces while keeping full projectors feasible.

32.5 Representative results

case	rep. dim.	target dim.	projector	formula	max. error
S_4 on 2-subsets, (2)	6	21	3	3	$2.4 \cdot 10^{-16}$
S_5 on 2-subsets, $(1,1)$	10	100	3	3	$2.6 \cdot 10^{-16}$
S_5 on 2-subsets, $(2,1)$	10	550	11	11	$2.2 \cdot 10^{-15}$
A_5 on 2-subsets, (2)	10	55	3	3	$2.6 \cdot 10^{-16}$
A_4 on 2-subsets, $(1,1)$	6	36	4	4	$< 10^{-14}$
A_5 on 2-subsets, $(1,1)$	10	100	3	3	$< 10^{-14}$
A_6 on 2-subsets, $(1,1)$	15	225	3	3	$< 10^{-14}$

The character average equals the same integer in every row. The 550 dimensional case is a useful stress test: the exact cycle calculation predicts 11, and the independently assembled projector has rank 11.

Reproducibility

The interactive file is `marimo_notebooks/permutation_actions.py`. It exposes n, k, τ, and the choice S_n/A_n, with conservative control ranges. The shared pytest suite also checks the stability boundary.

33 Johnson-scheme permutation representations

Abstract

For the permutation representation on the k-subsets of $[v]$, we prove the cycle-factor and orbit formulas, derive the fixed-subset formula, treat the middle-dimensional complementation coset, and verify all coefficients through total degree three in three representative finite cases. The computation uses the actual permutation matrices and does not assume the desired cycle formula.

33.1 Imported permutation tools

Let $c_\ell(g)$ count the ℓ-cycles of g on X. Tools 2 and 3 give

$$\mathbb{E}\big[\mathrm{Exp}_\sigma\big(X_{G,\mathbb{C}[X]}h_1\big)\big] = \frac{1}{|G|}\sum_{g\in G}\prod_{i\geq 1}\prod_{\ell\geq 1}(1-t_i^\ell)^{-c_\ell(g)}, \tag{79}$$

$$[m_\tau]\,\mathbb{E}\big[\mathrm{Exp}_\sigma\big(X_{G,\mathbb{C}[X]}h_1\big)\big] = \#\big(G\backslash(\mathrm{Sym}^{\tau_1} X\times\cdots\times\mathrm{Sym}^{\tau_s} X)\big). \tag{80}$$

The Johnson-specific work begins with the fixed-subset calculation below.

33.2 Johnson fixed points and complementation

Put $X = \binom{[v]}{k}$. If $\sigma\in S_v$ has a_j cycles of length j, then a j-cycle splits under σ^r into $\gcd(j,r)$ cycles of length $j/\gcd(j,r)$. A fixed subset is exactly a union of these cycles; hence

$$F_r^J(\sigma) = [u^k]\prod_{j\geq 1}\Big(1+u^{j/\gcd(j,r)}\Big)^{a_j\gcd(j,r)}. \tag{81}$$

The vertex-cycle counts follow from

$$c_\ell(\sigma) = \frac{1}{\ell}\sum_{d\mid\ell}\mu(\ell/d)F_d^J(\sigma). \tag{82}$$

When $v=2k$ and $k>1$, complementation $T(A)=A^c$ commutes with S_{2k} and supplies the additional coset in the full Johnson-scheme automorphism group. For odd r, a subset fixed by σ^rT must alternate membership around every cycle of σ^r. Thus

$$F_r(\sigma T) = \begin{cases} F_r^J(\sigma), & r \text{ even},\\ 2^{c(\sigma^r)}, & r \text{ odd and every cycle of } \sigma^r \text{ is even},\\ 0, & \text{otherwise}.\end{cases} \tag{83}$$

Every alternating choice has size exactly half of v, so no further size constraint is missing. Equations (79)–(83) prove the claimed formula, including the middle case.

33.3 Verification design

For each group element, the code explicitly builds its $|X|\times|X|$ permutation matrix P. It computes $h_d(P)=\mathrm{tr}(\mathrm{Sym}^d P)$ from the matrix traces by Newton's recurrence

$$dh_d(P) = \sum_{r=1}^{d}\mathrm{tr}(P^r)h_{d-r}(P), \qquad h_0 = 1.$$

It averages $\prod_j h_{\tau_j}(P)$ over the group. Independently, it extracts the same coefficient from the cycle factors in (79) and counts the orbits in (80) by explicit set traversal. Finally, it evaluates (81) or (83) for every required power and checks that (82) reproduces every actual cycle count.

33.4 Exact results

The entries list the common value in the order $\tau = (1), (2), (1,1), (3), (2,1), (1,1,1)$.

Action	$\lvert G\rvert$	$\lvert X\rvert$	common coefficient vector
Aut $J(4,2)$ (including T)	48	6	$(1,3,3,5,8,11)$
S_5 on 2-subsets	120	10	$(1,3,3,7,11,15)$
S_5 on 1-subsets	120	5	$(1,2,2,3,4,5)$

All 18 triple comparisons are exact, and all fixed-point reconstructions pass.

At $(t_1, t_2) = (0.037, 0.061)$ the direct determinant average and cycle product for Aut $J(4,2)$ coincide at

$$1.1237420509116285 \qquad \text{(reported error 0)}.$$

The pair values also give $[m_{(2)}] = [m_{(1,1)}] = \min(k, v-k) + 1$, as predicted.

33.5 Reproduction

From the repository root run

```
marimo run marimo_notebooks/johnson_scheme.py
```

or run the family module directly with Python. The notebook displays every matrix/cycle/orbit comparison rather than only the summary above.

34 Hamming-scheme permutation representations

Abstract

We prove and test the generalized Molien formula for the full Hamming graph automorphism group $S_q \wr S_D = S_q^D \rtimes S_D$ acting on Q^D. The tests construct every wreath-product element, not merely pure coordinate permutations, and compare matrix, cycle, and orbit calculations through total degree three.

34.1 Imported permutation tools

Tools 2 and 3 give

$$\mathrm{SMG}_{G,X}(\mathbf{t}) = \frac{1}{|G|} \sum_{g\in G} \prod_{i,\ell} (1 - t_i^{\ell})^{-c_\ell(g)}, \tag{84}$$

$$[m_\tau]\,\mathrm{SMG}_{G,X} = \#\left(G\backslash \prod_j \mathrm{Sym}^{\tau_j}(X)\right). \tag{85}$$

They are quoted here only to fix notation. The family-specific step is the fixed-word formula for a full Hamming wreath-product element.

34.2 Fixed words in the full wreath product

Write $g = (\sigma_1, \dots, \sigma_D; \pi)$. For a fixed power g^r, follow a coordinate around a cycle $C = (i_1, \dots, i_s)$ of its coordinate permutation. The successive alphabet relabelings compose to a permutation $h_{C,r} \in S_q$. A fixed word is determined on C by one symbol fixed by $h_{C,r}$, and different coordinate cycles are independent. Therefore

$$F_r^H(g) = \prod_{C\in \mathrm{Cyc}(\pi^r)} |\operatorname{Fix}_Q(h_{C,r})|. \tag{86}$$

The exact vertex-cycle counts are then

$$c_\ell^H(g) = \frac{1}{\ell} \sum_{d|\ell} \mu(\ell/d) F_d^H(g). \tag{87}$$

Substitution in (84) proves the proposed Hamming formula. For pure coordinate permutations the label products are identities, reducing (86) to $q^{c(\pi^r)}$.

Two ordered pairs of words are equivalent exactly when they have equal Hamming distance. Hence vertex transitivity and the distances $0, \dots, D$ give

$$[m_{(1)}] = 1, \qquad [m_{(2)}] = [m_{(1,1)}] = D + 1.$$

34.3 Independent computational checks

The implementation explicitly constructs every word permutation and its $q^D \times q^D$ matrix P. It computes complete symmetric characters from actual matrix traces,

$$d h_d(P) = \sum_{r=1}^{d} \operatorname{tr}(P^r) h_{d-r}(P), \quad h_0 = 1,$$

and averages $\prod_j h_{\tau_j}(P)$. A second route extracts coefficients from (84); a third explicitly traverses the orbits in (85). Separately, the coordinate maps and alphabet labels are powered by composition, (86) is evaluated, and (87) is required to reproduce the actual vertex cycles for every group element.

34.4 Exact results

Values are ordered by $\tau = (1), (2), (1,1), (3), (2,1), (1,1,1)$.

Action	$\lvert G\rvert$	$\lvert X\rvert$	common coefficient vector
$S_2 \wr S_2$ on Q^2	8	4	$(1, 3, 3, 4, 7, 10)$
$S_2 \wr S_3$ on Q^3	48	8	$(1, 4, 4, 7, 13, 20)$
$S_3 \wr S_2$ on Q^2	72	9	$(1, 3, 3, 7, 11, 15)$

All 18 matrix/cycle/orbit comparisons and every fixed-word reconstruction pass. For $(q, D) = (3, 2)$ at $(t_1, t_2) = (0.037, 0.061)$, the determinant and cycle averages are 1.1254339873575003 and 1.1254339873575000; the absolute error is 2.22×10^{-16}.

34.5 Reproduction

Run

```
marimo run marimo_notebooks/hamming_scheme.py
```

from the repository root. The table in the notebook exposes every exact comparison and the global fixed-point reconstruction result.

35 Grassmann-scheme actions

Abstract

For $GL_N(q)$ acting on the k-subspaces of $\mathbb{F}_q^N$, we prove the permutation Molien formula and verify its invariant-subspace input by direct prime-field matrix construction. Three representative cases test points, hyperplanes, and a nontrivial scalar kernel.

35.1 General formula

Let $X = \mathrm{Gr}_q(N, k)$ and $V = \mathbb{C}[X]$. If $c_\ell(g)$ is the number of ℓ-cycles of g on X, the cycle-block determinant identity gives

$$\mathrm{SMG}_{G,X}(\mathbf{t}) = \frac{1}{|G|} \sum_{g\in G} \prod_{i,\ell} (1 - t_i^\ell)^{-c_\ell(g)}. \tag{88}$$

The monomial basis of $\mathrm{Sym}^d(V)$ is indexed by d-multisets of subspaces, so Burnside's lemma also proves

$$[m_\tau]\,\mathrm{SMG}_{G,X} = \#\left(G\backslash \prod_j \mathrm{Sym}^{\tau_j}(X)\right). \tag{89}$$

For a linear matrix g, define

$$F_r^{\mathrm{Gr}}(g) = \#\{U \le \mathbb{F}_q^N : \dim U = k,\ g^r U = U\}. \tag{90}$$

Every ℓ-cycle contributes its ℓ vertices to F_r exactly when $\ell \mid r$. Möbius inversion therefore yields

$$c_\ell^{\mathrm{Gr}}(g) = \frac{1}{\ell} \sum_{d\mid\ell} \mu(\ell/d) F_d^{\mathrm{Gr}}(g). \tag{91}$$

Together, (88), (90), and (91) prove the claimed exact formula. Interpreting the ambient space as an $\mathbb{F}_q[x]$-module with x acting as g^r identifies (90) with the stated invariant-submodule problem.

Pair orbits are classified by $\dim(U \cap W)$, and so, with $s = \min(k, N - k)$,

$$[m_{(1)}] = 1, \qquad [m_{(2)}] = [m_{(1,1)}] = s + 1.$$

35.2 Verification strategy

All matrices in $GL_N(q)$ are enumerated over the prime field. Every k-subspace is generated and stored in unique reduced-row-echelon form. Matrix multiplication then gives both the induced vertex permutation and the independent invariance test in (90). Scalar matrices are deduplicated before averaging, so the average is over the actual permutation image; this is equivalent to averaging over $GL_N(q)$ because the scalar kernel has constant size.

The target coefficient is computed from actual permutation matrices using Newton's trace recurrence for $h_d(P) = \mathrm{tr}(\mathrm{Sym}^d P)$. The same coefficient is independently extracted from (88) and counted as the explicit orbit number in (89). Finally, invariant-subspace counts for powers of every matrix representative must reconstruct every cycle through (91).

35.3 Exact results

The coefficient vector is ordered by $\tau = (1), (2), (1, 1), (3), (2, 1), (1, 1, 1)$.

Action	$\lvert GL\rvert$	$\lvert G$ image$\rvert$	$\lvert X\rvert$	common vector
$\mathrm{Gr}_2(3, 1)$	168	168	7	$(1, 2, 2, 4, 5, 6)$
$\mathrm{Gr}_2(3, 2)$	168	168	7	$(1, 2, 2, 4, 5, 6)$
$\mathrm{Gr}_3(2, 1)$	48	24	4	$(1, 2, 2, 3, 4, 5)$

All 18 exact triple comparisons and all invariant-subspace reconstructions pass. The final row explicitly checks the scalar kernel. In the first case, the direct determinant and cycle averages at $(0.037, 0.061)$ both equal 1.1152638116688691 (reported error 0).

35.4 Reproduction

Run

```
marimo run marimo_notebooks/grassmann_scheme.py
```

from the repository root. The implementation is intentionally restricted to prime fields for transparency; the theorem itself is not so restricted.

36 Polar-scheme permutation representations

Abstract

We prove the fixed-polar-subspace version of the permutation Molien formula and test two nontrivial actions of the explicitly constructed matrix group $Sp_4(2)$: isotropic points and maximal totally isotropic 2-spaces. The latter is a rank-two dual-polar scheme.

36.1 Cycle and orbit formulas

Let a finite isometry group G act on a finite family X of totally isotropic or singular subspaces, and put $V = \mathbb{C}[X]$. If $c_\ell(g)$ counts ℓ-cycles on X, the determinant of an ℓ-cycle block is $1 - t^\ell$, so

$$\mathrm{SMG}_{G,X}(\mathbf{t}) = \frac{1}{|G|} \sum_{g \in G} \prod_{i,\ell} (1 - t_i^\ell)^{-c_\ell(g)}. \tag{92}$$

Since the monomial basis of $\mathrm{Sym}^d(V)$ is indexed by d-multisets in X, Burnside's lemma independently gives

$$[m_\tau]\,\mathrm{SMG}_{G,X} = \#\left(G\backslash \prod_j \mathrm{Sym}^{\tau_j}(X) \right). \tag{93}$$

Define the geometric fixed-point count

$$F_r^{\mathrm{pol}}(g) = \#\{U \in X : g^r U = U\}. \tag{94}$$

The universal relation $F_r = \sum_{\ell | r} \ell c_\ell$ and Möbius inversion yield

$$c_\ell^{\mathrm{pol}}(g) = \frac{1}{\ell} \sum_{d|\ell} \mu(\ell/d) F_d^{\mathrm{pol}}(g). \tag{95}$$

Equations (92)–(95) prove the proposed formula. Relative to the Grassmann case, the only additional predicate in (94) is total isotropy or singularity.

For maximal isotropic subspaces in Witt rank r, pair orbits are classified by $r - \dim(U \cap W)$. Consequently

$$[m_{(1)}] = 1, \qquad [m_{(2)}] = [m_{(1,1)}] = r + 1.$$

36.2 Concrete symplectic construction

On $\mathbb{F}_2^4$ use the nondegenerate alternating form

$$\langle x, y \rangle = x_1 y_2 + x_2 y_1 + x_3 y_4 + x_4 y_3.$$

The code enumerates all 20,160 matrices in $GL_4(2)$ and retains exactly the 720 satisfying $\langle gx, gy \rangle = \langle x, y \rangle$ on basis vectors. It separately enumerates reduced-row-echelon subspaces. There are 15 one-spaces (all isotropic for an alternating form) and 15 two-spaces on which the form vanishes identically. Thus both induced representations have dimension 15, but their vertices are different geometric objects.

For every group matrix and every needed power, the program acts on the stored bases and counts invariant members of X directly. Equation (95) must reproduce the cycles of the separately constructed vertex permutation.

36.3 Three independent coefficient routes

For the actual 15×15 permutation matrix P, complete symmetric characters are computed without cycle data from

$$d h_d(P) = \sum_{a=1}^{d} \mathrm{tr}(P^a) h_{d-a}(P), \qquad h_0 = 1.$$

The matrix-group target is the average of $\prod_j h_{\tau_j}(P)$. A second calculation extracts the coefficient from (92); a third explicitly enumerates the configuration orbits in (93). Agreement of these routes tests the representation, the proposed formula, and its geometric interpretation separately.

36.4 Exact results

Values are ordered by $\tau = (1), (2), (1, 1), (3), (2, 1), (1, 1, 1)$.

Action	$\lvert G\rvert$	$\lvert X\rvert$	common coefficient vector
$Sp_4(2)$ on isotropic points	720	15	$(1, 3, 3, 8, 12, 16)$
$Sp_4(2)$ on maximal isotropic 2-spaces	720	15	$(1, 3, 3, 8, 12, 16)$

All 12 matrix/cycle/orbit comparisons and all geometric fixed-point reconstructions pass. The dual-polar pair value is $3 = r + 1$ for $r = 2$. For the maximal action at $(t_1, t_2) = (0.037, 0.061)$, both numerical averages equal 1.1263214575673286 (reported error 0).

36.5 Scope and reproduction

The exact proof applies equally to alternating, Hermitian, and quadratic forms. The executable representative is symplectic because $Sp_4(2)$ is large enough to exhibit all rank-two pair relations while remaining fully enumerable and auditable.

Run

```
marimo run marimo_notebooks/polar_scheme.py
```

from the repository root to inspect every comparison.

37 Association-scheme permutation groups

Canonical testing artifacts.
Exact backend: `lambda_distributions/dists/association_scheme_sigma_mgfs/verification.py`.
Regression: `lambda_distributions/dists/association_scheme_sigma_mgfs/test_verification.py`.
Marimo: `marimo_notebooks/association_schemes.py`.

Scope and outcome

For every finite automorphism group $G \le \mathrm{Sym}(\Omega)$, the imported permutation tools show that the complete σ-MGF on $\mathbb{C}[\Omega]$ is determined by the power-fixed-point data $F_r(g) = |\operatorname{Fix}_\Omega(g^r)|$. We then prove explicit fixed-point templates for homogeneous spaces, affine translation actions, and wreath-product actions. These specialize to Johnson, Hamming, Grassmann, form, polar, building, graph, Doob, folded-cube, and halved-cube families. Stable ranges and degree-two obstructions are kept separate from exact finite formulas. An accompanying marimo notebook constructs representative matrix groups and compares direct matrix traces, cycle products, and literal configuration orbits in 99 exact coefficient checks. Every check passes.

37.1 The universal power-fixed-point theorem

Let a finite group G act on a finite set Ω, let $V = \mathbb{C}[\Omega]$, and let P_g denote the resulting permutation matrix. For a composition or partition $\tau = (\tau_1, \dots, \tau_s)$ put

$$W_\tau(V) = \bigotimes_{j=1}^{s} \mathrm{Sym}^{\tau_j} V, \qquad a_\tau(G,\Omega) = \dim W_\tau(V)^G.$$

Our convention is

$$\mathrm{SMG}_{G,\Omega}(\mathbf{t}) = \mathbb{E}_{g\in G} \operatorname{Exp}_\sigma(X_V(g)h_1) = \sum_\tau a_\tau(G,\Omega) m_\tau(\mathbf{t}).$$

For $g \in G$ define

$$F_r(g) = |\operatorname{Fix}_\Omega(g^r)|, \qquad c_d(g) = \#\{d\text{-cycles of } g \text{ on } \Omega\}.$$

Theorem 1 (Association-scheme power-fixed-point theorem)**.** *For every finite permutation action,*

$$F_r(g) = \sum_{d\mid r} d\, c_d(g), \tag{96}$$

$$c_d(g) = \frac{1}{d} \sum_{r\mid d} \mu(d/r) F_r(g), \tag{97}$$

$$\mathrm{SMG}_{G,\Omega}(\mathbf{t}) = \frac{1}{|G|} \sum_{g\in G} \prod_{i\ge 1} \prod_{d\ge 1} (1 - t_i^d)^{-c_d(g)}, \tag{98}$$

$$[m_\tau]\, \mathrm{SMG}_{G,\Omega} = \# \left(G \backslash \prod_j \mathrm{Sym}^{\tau_j} \Omega \right). \tag{99}$$

Thus the complete σ-MGF is determined by the collection of numbers $F_r(g)$.

Imported identities and the fixed-point reduction. The sigma-MGF and orbit formulas are Tools 2 and 3. A point on a d-cycle is fixed by g^r exactly when $d \mid r$, proving (96); arithmetic Möbius inversion gives (97). Substituting the recovered cycle counts into the imported permutation formula proves (98). No Reynolds–Molien or orbit identity is reproved. □

Remark 1 (What is tested directly)**.** *The verification stores P_g exactly by the location of the single nonzero entry in each column. It computes symmetric-power characters from the actual matrix traces using Newton's recurrence*

$$d\, h_d(P_g) = \sum_{r=1}^{d} \mathrm{Tr}(P_g^r) h_{d-r}(P_g), \qquad h_0 = 1.$$

This matrix route, the determinant cycle-factor route, and literal configuration orbits are separate computations of the target coefficient.

37.2 Three reusable fixed-point templates

37.2.1 Homogeneous and parabolic actions

Let $\Omega = G/H$ with left multiplication.

Proposition 1 (Coset fixed points)**.** *For $x \in G$,*

$$\boxed{|\operatorname{Fix}_{G/H}(x)| = \frac{|C_G(x)|\,|x^G \cap H|}{|H|}.} \tag{100}$$

Proof. The coset yH is fixed precisely when $y^{-1}xy \in H$. For each element of $x^G \cap H$, exactly $|C_G(x)|$ choices of y conjugate x to it. Dividing the number of such y by $|H|$ accounts for the representatives of a coset. Apply this to $x = g^r$, then use (97)–(98). ☐

This covers Grassmannians and other parabolic quotients, dual-polar spaces, generalized polygons from BN-pairs, and building residues.

37.2.2 Affine translation actions

Let A be a finite additive group, $H \le \operatorname{Aut}(A)$, and $G = A \rtimes H$ act by $x \mapsto b + h(x)$. Define

$$b_r = (I + h + \cdots + h^{r-1})b.$$

Proposition 2 (Affine kernel/image formula)**.** *For $g = (b, h)$,*

$$\boxed{F_r(b,h) = \begin{cases} |\ker(I - h^r)|, & b_r \in \operatorname{im}(I - h^r), \\ 0, & b_r \notin \operatorname{im}(I - h^r). \end{cases}} \tag{101}$$

Proof. Induction gives $g^r(x) = b_r + h^r(x)$. Thus x is fixed exactly when $(I - h^r)x = b_r$. A fiber of the homomorphism $I - h^r$ is empty outside its image and is a coset of its kernel otherwise. ☐

37.2.3 Product and wreath actions

Let $H \curvearrowright \Gamma$ and let $H \wr S_d = H^d \rtimes S_d$ act on Γ^d. Write $g = (h_1, \ldots, h_d; \pi)$. For a cycle $C = (i_1 \ldots i_\ell)$ of π, let

$$a_C = h_{i_\ell} \cdots h_{i_1}, \qquad e_C(r) = \gcd(\ell, r), \qquad f_s(a) = |\operatorname{Fix}_\Gamma(a^s)|.$$

Proposition 3 (Marked-cycle product formula)**.**

$$\boxed{F_r(g) = \prod_{C \in \operatorname{Cyc}(\pi)} f_{r/e_C(r)}(a_C)^{e_C(r)}.} \tag{102}$$

Proof. Under π^r, an ℓ-cycle splits into $e = \gcd(\ell, r)$ coordinate cycles. Transport once around any one has monodromy conjugate to $a_C^{r/e}$. Each of the e cycles admits $f_{r/e}(a_C)$ choices. Different cycles of π use disjoint coordinates, so their counts multiply. ☐

37.3 Johnson schemes

Let $\Omega = \binom{[v]}{k}$ and $G = S_v$. Suppose σ has cycle type $\lambda = 1^{m_1}2^{m_2}\cdots$. A j-cycle of σ splits under σ^r into $e = \gcd(j,r)$ cycles of length j/e. A fixed k-subset is a union of these cycles, hence

$$\boxed{F_r^{J(v,k)}(\lambda) = [u^k]\prod_{j\ge 1}\left(1+u^{j/\gcd(j,r)}\right)^{m_j\gcd(j,r)}.} \tag{103}$$

With $c_d^J(\lambda) = d^{-1}\sum_{r|d}\mu(d/r)F_r^J(\lambda)$ and $z_\lambda = \prod_j j^{m_j}m_j!$,

$$\boxed{\mathrm{SMG}_{J(v,k)} = \sum_{\lambda\vdash v}\frac{1}{z_\lambda}\prod_{i,d\ge 1}(1-t_i^d)^{-c_d^J(\lambda)}.} \tag{104}$$

For $D = |\tau|$, a configuration contributing to $[m_\tau]$ uses at most kD underlying points. Relabelling those points proves

$$[m_\tau]\,\mathrm{SMG}_{J(v,k)} \text{ is constant for } v \ge kD. \tag{105}$$

The stable objects are isomorphism classes of τ-colored k-uniform multihypergraphs without isolated vertices. Ordered pairs are classified by intersection size, so

$$\boxed{[m_{(1,1)}]\,\mathrm{SMG}_{J(v,k)} = \min(k, v-k)+1.} \tag{106}$$

Thus no coefficientwise limit exists when $\min(k, v-k)\to\infty$.

37.4 Hamming and Doob product schemes

For $\Omega = [q]^d$ and $G = S_q \wr S_d$, Proposition 3 gives

$$F_r(g) = \prod_C \left|\mathrm{Fix}_{[q]}\left(a_C^{r/e_C(r)}\right)\right|^{e_C(r)}. \tag{107}$$

Consequently

$$\boxed{\mathrm{SMG}_{H(d,q)} = \frac{1}{(q!)^d d!}\sum_{\pi\in S_d}\sum_{h_1,\ldots,h_d\in S_q}\prod_{i,\ell\ge 1}(1-t_i^\ell)^{-\frac{1}{\ell}\sum_{r|\ell}\mu(\ell/r)F_r(g)}.} \tag{108}$$

For fixed d and $D = |\tau|$, once $q \ge D$ the orbit of the D symbols in each coordinate is exactly its equality partition in the partition lattice Π_D. If $K_\tau = \prod_j S_{\tau_j}$ permutes repeated objects within the colored multiset blocks, then

$$\boxed{[m_\tau]\,\mathrm{SMG}_{H(d,q)} = \#\big((K_\tau\times S_d)\backslash \Pi_D^d\big), \qquad q\ge D.} \tag{109}$$

On the other hand, ordered word pairs are classified by Hamming distance:

$$\boxed{[m_{(1,1)}]\,\mathrm{SMG}_{H(d,q)} = d+1.} \tag{110}$$

For the Doob graph $D(m,n) = \mathrm{Sh}^{\square m}\square K_4^{\square n}$, take

$$G_{m,n} = (\mathrm{Aut}(\mathrm{Sh})\wr S_m)\times(S_4\wr S_n).$$

Fixed points multiply between the two Cartesian blocks. The Shrikhande vertex action has four ordered-pair types and $S_4 \curvearrowright [4]$ has two. Permuting equal factors leaves multisets of types, so

$$\boxed{[m_{(1,1)}]\,\mathrm{SMG}_{D(m,n)} = \binom{m+3}{3}(n+1).} \tag{111}$$

In contrast, $H(2m+n,4)$ has pair coefficient $2m+n+1$ even though the two families have the same distance-regular parameters.

37.5 Grassmann, polar, polygon, and building actions

37.5.1 Grassmann schemes

Let $\Omega = \mathrm{Gr}_k(\mathbb{F}_q^n)$ and take the projective image of $\mathrm{GL}_n(q)$. Directly,

$$F_r(g) = \#\{U \le \mathbb{F}_q^n : \dim U = k,\ g^r U = U\}. \tag{112}$$

Equivalently, with $P = P_{k,n-k}$, Proposition 1 gives

$$F_r(g) = \frac{|C_G(g^r)|\,|(g^r)^G \cap P|}{|P|}.$$

Invariant-subspace counts may be computed from the rational canonical form of g^r. Ordered pairs are classified by $\dim(U \cap W)$:

$$\boxed{[m_{(1,1)}]\,\mathrm{SMG}_{\mathrm{Gr}_k(n,q)} = \min(k, n-k) + 1.} \tag{113}$$

The span of D participating k-spaces has dimension at most kD, and an isomorphism between supported configurations extends to the ambient space. Thus

$$[m_\tau]\,\mathrm{SMG}_{\mathrm{Gr}_k(n,q)} \text{ stabilizes for } n \ge kD \quad (q, k, \tau \text{ fixed}). \tag{114}$$

Fixed-rank field-size limits can fail: four points on a projective line carry a cross-ratio parameter.

37.5.2 Dual-polar spaces and generalized polygons

Let V be a finite formed space of Witt rank d and let $\Omega = G/P$ be the maximal isotropic or singular subspaces. Then

$$F_r(g) = \frac{|C_G(g^r)|\,|(g^r)^G \cap P|}{|P|}. \tag{115}$$

Pair orbits are indexed by intersection codimension, hence

$$\boxed{[m_{(1,1)}]\,\mathrm{SMG}_{\mathrm{DP}(d,q)} = d + 1.} \tag{116}$$

The same coset formula applies to a flag-transitive generalized polygon with point set G/P. If the point action is distance-transitive, its pair orbits are indexed by distance. Therefore the pair coefficients for generalized quadrangles, hexagons, and octagons are respectively $3, 4, 5$. Without flag/distance transitivity, the order parameters alone do not imply these values and do not determine the full series.

37.5.3 Buildings

Suppose G has a BN-pair with Weyl group W. For chambers G/B, Bruhat decomposition gives $B\backslash G/B \cong W$, so

$$\boxed{[m_{(1,1)}]\,\mathrm{SMG}_{G,G/B} = |W|.} \tag{117}$$

For residues of type J,

$$\boxed{[m_{(1,1)}]\,\mathrm{SMG}_{G,G/P_J} = |W_J\backslash W/W_J|.} \tag{118}$$

These conclusions require the natural strong or Weyl transitivity. Higher coefficients need not be controlled by W because triples of flags can carry field-valued moduli.

37.6 Translation schemes of forms

All families in this section use Proposition 2. The target representation is the complex permutation module on the additive form space; the finite-field matrices are used to construct its zero-one permutation matrices.

37.6.1 Bilinear forms

Let $A = M_{m\times n}(\mathbb{F}_q)$ and let (P,Q) act linearly by $L(X) = PXQ^{-1}$. For $g = (B,P,Q)$ define $B_r = (I+L+\cdots+L^{r-1})B$. Then

$$\boxed{F_r(g) = \begin{cases} q^{\dim\ker(I-L^r)}, & B_r \in \operatorname{im}(I-L^r), \\ 0, & \text{otherwise.} \end{cases}} \tag{119}$$

After translating one member of a pair to zero, differences are classified by rank:

$$\boxed{[m_{(1,1)}]\,\mathrm{SMG}_{\mathrm{Bil}(m,n,q)} = \min(m,n)+1.} \tag{120}$$

For fixed m, q, τ, translating one of $D = |\tau|$ matrices to zero leaves combined row support of dimension at most $m(D-1)$. Thus the coefficient is stable once $n \geq m(D-1)$.

37.6.2 Alternating, Hermitian, and quadratic forms

For $A = \mathrm{Alt}_n(\mathbb{F}_q)$ use $L(X) = PXP^T$; for Hermitian matrices over $\mathbb{F}_{q^2}$ use $L(X) = PXP^{\sigma T}$. Formula (119) applies verbatim with the appropriate linear operator. Rank classifies the one-form orbits in these two cases, while alternating ranks are even. Hence

$$[m_{(1,1)}]\,\mathrm{SMG}_{\mathrm{Alt}(n,q)} = \left\lfloor \frac{n}{2} \right\rfloor + 1, \tag{121}$$

$$[m_{(1,1)}]\,\mathrm{SMG}_{\mathrm{Herm}(n,q)} = n+1. \tag{122}$$

Let $A = \mathcal{Q}(n,q)$ be the additive space of quadratic forms, with $P\cdot Q = Q\circ P^{-1}$. Again (101) is exact. For odd q, each positive rank has two congruence types and rank zero has one, yielding

$$\boxed{[m_{(1,1)}]\,\mathrm{SMG}_{\mathcal{Q}(n,q)} = 2n+1 \qquad (q \text{ odd}).} \tag{123}$$

Even characteristic requires the usual rank/type refinement but no change to the affine fixed-point formula. The displayed pair coefficients already rule out rank-growing coefficientwise limits. Field-size limits can also fail in higher degree: after translating a triple of bilinear forms and making one difference invertible, the other retains similarity invariants such as eigenvalues.

37.7 Affine-polar, folded-cube, and halved-cube actions

For an affine-polar action take the additive formed space $A = V$ and the relevant isometry, similitude, or semisimilitude group H. Equation (101) is the complete exact answer. For $D = |\tau|$, translate one of the D points to zero. The other $D-1$ differences span a formed subspace of dimension at most $D-1$. Its restricted form and labelled vectors determine the orbit; Witt extension embeds equivalent supported configurations into all sufficiently large ambient spaces. Therefore

$$[m_\tau]\,\mathrm{SMG}_{\mathrm{AffPol}(N,q)} \text{ is eventually constant as } N\to\infty \quad (q,\tau \text{ fixed}). \tag{124}$$

For the folded cube, let

$$A_n = \mathbb{F}_2^n/\langle \mathbf{1}\rangle, \qquad G = A_n \rtimes S_n.$$

For $g = (b,\sigma)$, equation (101) becomes

$$F_r(g) = \begin{cases} 2^{\dim\ker(I-\sigma^r|A_n)}, & b_r \in \operatorname{im}(I-\sigma^r), \\ 0, & \text{otherwise.} \end{cases} \tag{125}$$

Weights w and $n-w$ represent the same difference orbit, so

$$\boxed{[m_{(1,1)}]\,\mathrm{SMG}_{\mathrm{FoldCube}_n} = \left\lfloor \frac{n}{2} \right\rfloor + 1.} \tag{126}$$

For the halved cube use $E_n = \{x \in \mathbb{F}_2^n : \sum_i x_i = 0\}$ and $G = E_n \rtimes S_n$. Formula (125) holds with A_n replaced by E_n. Difference weights are the even integers from zero to the largest even value at most n, giving

$$\boxed{[m_{(1,1)}]\,\mathrm{SMG}_{\frac{1}{2}Q_n} = \left\lfloor \frac{n}{2} \right\rfloor + 1.} \tag{127}$$

37.8 Distance-regular and strongly regular graphs

For any finite graph Γ, equations (97)–(98) apply with $G = \mathrm{Aut}(\Gamma)$ and $\Omega = V(\Gamma)$. If the action is distance-transitive of diameter D, then

$$[m_{(1,1)}]\,\mathrm{SMG}_\Gamma = D + 1. \tag{128}$$

More generally, the pair coefficient is the permutation rank of the vertex action.

The 4×4 rook graph and the Shrikhande graph both have strongly regular parameters $(16, 6, 2, 2)$, but their full vertex actions have ranks three and four. In the Cayley model on $\mathbb{Z}_4^2$, the Shrikhande vertex stabilizer has orbits

$$\{0\}, \quad \{\text{six neighbors}\}, \quad \{(2,0),(0,2),(2,2)\}, \quad \{\text{remaining six vertices}\}.$$

Therefore the σ-MGF is not determined by the intersection array, spectrum, or strongly regular parameters. At the other extreme, a graph with trivial automorphism group on N vertices has

$$\mathrm{SMG}_\Gamma = \prod_{i \ge 1} (1 - t_i)^{-N}.$$

37.9 Asymptotic classification

Let $D = |\tau|$. The proof mechanism behind the positive stability results is bounded support together with an extension theorem: a fixed-degree configuration lives in a bounded subobject and supported isomorphisms extend to ambient automorphisms. Most rank-growth failures below are already forced by an unbounded degree-two coefficient. The fixed-rank Grassmann $q \to \infty$ regime is different: its pair coefficient is constant, while cross-ratio moduli can first obstruct convergence in higher degree.

Family and regime	Coefficientwise behavior
$J(v,k)$, fixed k, $v \to \infty$	Stable for $v \ge kD$
$J(v,k)$, $\min(k, v-k) \to \infty$	No limit; pair coefficient diverges
$H(d,q)$, fixed d, $q \to \infty$	Stable for $q \ge D$
$H(d,q)$, fixed q, $d \to \infty$	No limit; pair coefficient $d+1$
$\mathrm{Gr}_k(n,q)$, fixed q, k, $n \to \infty$	Stable for $n \ge kD$
Grassmann, fixed rank, $q \to \infty$	Can fail through cross-ratio moduli
Bilinear, fixed m, q, $n \to \infty$	Stable for $n \ge m(D-1)$
Alternating/Hermitian/quadratic, rank $\to \infty$	No limit in degree two
Dual-polar, Witt rank $\to \infty$	No limit; pair coefficient $d+1$
Affine-polar, fixed q, rank $\to \infty$	Eventually stable for fixed τ
Building chambers, rank $\to \infty$	No limit when $|W| \to \infty$
Bounded residues in classical towers, fixed q	Stable by bounded support
Doob/folded/halved dimension $\to \infty$	No limit in degree two

37.10 Exact computational verification

37.10.1 Construction and comparison strategy

The companion module `verification.py` uses only exact finite arithmetic for the structural checks.

1. Enumerate the stated finite matrix group (or a faithful permutation image) and the finite set Ω.
2. Construct P_g by applying each group element to every point of Ω. The dimension is exactly $|\Omega|$.
3. Compute $F_r(g)$ directly from P_g^r and independently from the appropriate coset, affine, invariant-subspace, or wreath formula.
4. Recover $c_d(g)$ by Möbius inversion and compare with literal cycle decomposition.
5. Compute $[m_\tau]\,\mathrm{SMG}$ from Newton matrix traces, from cycle-factor coefficient extraction, and from direct orbits on $\prod_j \mathrm{Sym}^{\tau_j}\,\Omega$.

The marimo notebook presents these checks by group family. Existing Johnson, Hamming, Grassmann, and polar constructors are imported from the canonical association-scheme package in this workspace; the new notebook adds the affine-form, building, graph, Doob, folded, and halved constructors. No combined anthology PDF is modified.

37.10.2 Representative exact cases

Action	$\|\Omega\|$	$\|H\|$	$\|G\|$	$[m_{11}]$	Audit
S_v on k-subsets: $(4,2),(5,2),(5,1)$	$6,10,5$	–	$48,120,120$	$3,3,2$	exact
$S_q \wr S_d$: $(2,2),(2,3),(3,2)$	$4,8,9$	–	$8,48,72$	$3,4,3$	exact
$PGL_3(2)$ on $\mathrm{Gr}_1,\mathrm{Gr}_2$; $PGL_2(3)$ on Gr_1	$7,7,4$	–	$168,168,24$	$2,2,2$	exact
$Sp_4(2)$ on polar points / Lagrangians	15	–	720	3	exact
$GL_3(2)$ on complete flags	21	$\|B\|=8$	168	6	coset
$M_2(\mathbb{F}_2) \rtimes (GL_2(2)\times GL_2(2))$	16	36	576	3	affine
$\mathrm{Alt}_3(2) \rtimes GL_3(2)$	8	168	1344	2	affine
$\mathrm{Herm}_2(4/2) \rtimes GL_2(4)$ image	16	60	960	3	affine
$\mathcal{Q}(2,3) \rtimes GL_2(3)$ image	27	24	648	5	affine
$\mathbb{F}_2^4 \rtimes O_4^+(2)$	16	72	1152	3	affine polar
Folded and halved 4-cubes	8	24	192	3	affine
Shrikhande full vertex action	16	12	192	4	graph
4×4 rook graph	16	–	1152	3	graph
Doob $D(1,1)$ chosen product action	64	–	4608	8	product

For the supplemental affine and graph cases, the reported low-degree values are:

case	Bil	Alt	Herm	$\mathcal{Q}$	AffPol	Fold	$\frac{1}{2}Q$	Sh	Rook	$D(1,1)$
$[m_{11}]$	3	2	3	5	3	3	3	4	3	8

In every entry, the matrix-trace average, cycle formula, and direct orbit count agree. The quadratic example also illustrates why $[m_{(2)}]$ need not equal $[m_{(1,1)}]$: its values are 4 and 5, respectively, because forgetting the two colors identifies a difference with its negative.

Verification outcome. The suite completes 99 exact coefficient comparisons. Every displayed coefficient agrees across all implemented routes; every family-specific power-fixed-point formula reconstructs the literal cycle counts; the $GL_3(2)/B$ example verifies the centralizer/class-intersection formula element by element; and the Doob $D(1,1)$ example verifies the product fixed-point identity for powers $1 \le r \le 6$.

37.11 Scope and cautions

- The universal theorem is valid for every finite action. An association scheme is useful structure for calculating $F_r(g)$; it is not an additional hypothesis needed for the theorem.
- The automorphism group must be specified. Different subgroups of $\mathrm{Aut}(\Omega)$ generally give different averages and different orbit counts.
- Parameters such as an intersection array or strongly regular tuple do not determine the automorphism action, as the rook/Shrikhande comparison shows.
- Semilinear, complement, or exceptional automorphisms are incorporated by averaging over their extra cosets. Omitting such a coset computes the series for the smaller stated group, not for the full automorphism group.
- Exact finite formulas, stable formulas, and coefficientwise limits are distinct claims. This note states a stable range only where bounded support and extension give one directly.

38 The hyperoctahedral group on the binary cube

Abstract

Let $B_n = C_2^n \rtimes S_n$ act affinely on $X_n = \mathbb{F}_2^n$ and let $V_n = \mathbb{C}[X_n]$. We prove the proposed fixed-point distribution and moment formulas, derive the exact signed-cycle-type expression for the full σ-MGF, and describe an independent exhaustive verification. The program constructs every $2^n \times 2^n$ permutation matrix, computes the target directly, traverses the relevant monomial-basis orbits, and compares both results with the signed-cycle formula for $1 \le n \le 5$. All checks pass; the largest case has $|B_5| = 3840$ matrices of size 32×32.

38.1 The action and the imported target

Write elements of B_n as $g = (b, \pi)$, where $b \in \mathbb{F}_2^n$ and $\pi \in S_n$, and set

$$g \cdot x = b + \pi x, \qquad x \in X_n.$$

On the basis $\{e_x : x \in X_n\}$ of V_n, the corresponding matrix is

$$P_g e_x = e_{g \cdot x}. \tag{129}$$

Thus P_g is a $2^n \times 2^n$ permutation matrix. Tools 2 and 3 specialize to

$$\mathbb{E}[\mathrm{Exp}_\sigma(X_{B_n,V_n} h_1)] = \mathbb{E}_{g \in B_n} \prod_{d \ge 1} Q_d(\mathbf{t})^{c_d(g)}, \qquad Q_d(\mathbf{t}) = \prod_{a \ge 1} (1 - t_a^d)^{-1}, \tag{130}$$

$$[m_\tau]\, \mathbb{E}[\mathrm{Exp}_\sigma(X_{B_n,V_n} h_1)] = \#\left(B_n \backslash \prod_{j=1}^{s} \mathrm{Sym}^{\tau_j}(X_n) \right). \tag{131}$$

The new argument is the conditional fixed-point law of the affine cube action.

38.2 The fixed-point law

Define

$$F_n(g) = \mathrm{Tr}(P_g) = \#\,\mathrm{Fix}_{X_n}(g).$$

Theorem 1 (Exact conditional law)**.** *For uniform $b \in \mathbb{F}_2^n$ and fixed $\pi \in S_n$, let $C(\pi)$ be the number of coordinate cycles of π. Then*

$$F_n(b, \pi) = \begin{cases} 2^{C(\pi)}, & \text{with probability } 2^{-C(\pi)}, \\ 0, & \text{with probability } 1 - 2^{-C(\pi)}. \end{cases}$$

Proof. On a coordinate cycle $(i_1, \ldots, i_\ell)$, the fixed equation $x = b + \pi x$ becomes a cyclic system

$$x_{i_1} = b_{i_1} + x_{i_\ell}, \quad x_{i_2} = b_{i_2} + x_{i_1}, \quad \ldots \quad x_{i_\ell} = b_{i_\ell} + x_{i_{\ell-1}}.$$

Adding in $\mathbb{F}_2$ shows that it is solvable only if $\sum_{j=1}^{\ell} b_{i_j} = 0$. This condition is also sufficient: after one coordinate is chosen, the remaining coordinates are forced. Consequently each coordinate cycle imposes one independent parity condition on b and, when solvable, contributes two solutions. There are $C(\pi)$ independent conditions and $2^{C(\pi)}$ solutions. $\square$

Let $\left[{n \atop k}\right]$ be the unsigned Stirling number of the first kind. Since there are $\left[{n \atop k}\right]$ permutations with k cycles, Theorem 1 yields

$$\mathbb{P}(F_n = 2^k) = \frac{\left[{n \atop k}\right]}{n!\, 2^k}, \qquad 1 \le k \le n. \tag{132}$$

The remaining probability is at zero. Using $\sum_k \left[{n \atop k}\right] x^k = x(x+1)\cdots(x+n-1)$ gives

$$\mathbb{P}(F_n > 0) = \frac{(1/2)^{\overline{n}}}{n!} = \frac{\Gamma(n + \frac{1}{2})}{\sqrt{\pi}\,\Gamma(n+1)} = \frac{\binom{2n}{n}}{4^n} \sim \frac{1}{\sqrt{\pi n}}. \tag{133}$$

Hence $F_n \to 0$ in probability.

For every integer $m \geq 1$, conditioning on π instead gives

$$\mathbb{E}_b[F_n^m \mid \pi] = 2^{-C(\pi)} 2^{mC(\pi)} = 2^{(m-1)C(\pi)}.$$

Another application of the Stirling generating polynomial proves

$$\boxed{\mathbb{E}[F_n^m] = \frac{(2^{m-1})^{\overline{n}}}{n!} = \binom{n + 2^{m-1} - 1}{n}.} \tag{134}$$

In particular $\mathbb{E}F_n = 1$, $\mathbb{E}F_n^2 = n+1$, $\mathbb{E}F_n^3 = \binom{n+3}{3}$, and $\mathrm{Var}(F_n) = n$. Thus convergence in probability to zero coexists with divergent moments of every order $m \geq 2$.

There is also a direct Burnside proof of (134). Represent an ordered m-tuple of cube vertices by an $m \times n$ binary matrix. Coordinate permutations permute its columns, while a flip in one cube coordinate adds $\mathbf{1} = (1, \dots, 1)$ to the corresponding column. Each column is therefore remembered only as an element of $\mathbb{F}_2^m / \langle \mathbf{1} \rangle$, a set of size 2^{m-1}. The orbit is a multiset of n such column types, so

$$\#(B_n \backslash X_n^m) = \binom{n + 2^{m-1} - 1}{n} = \mathbb{E}[F_n^m].$$

Combining this with (131) gives

$$[m_{(1^m)}]\mathcal{H}_n = \binom{n + 2^{m-1} - 1}{n}. \tag{135}$$

38.3 Signed cycle types and the full cycle index

A signed coordinate-cycle type is a pair of sequences $(a_\ell, b_\ell)_{\ell \geq 1}$, where a_ℓ counts length-ℓ coordinate cycles with total flip parity zero and b_ℓ counts those with total flip parity one. They satisfy

$$\sum_{\ell \geq 1} \ell(a_\ell + b_\ell) = n.$$

For a uniform element of B_n, the probability of this type is

$$w(a,b) = \prod_{\ell \geq 1} \frac{1}{(2\ell)^{a_\ell + b_\ell} a_\ell! b_\ell!}. \tag{136}$$

This follows from the usual S_n cycle-type probability and the fact that the accumulated parity on every coordinate cycle is an independent fair bit.

Let

$$R_r(a,b) = \# \operatorname{Fix}_{X_n}(g^r)$$

for any element of signed type (a,b). A positive length-ℓ cycle contributes $2^{\gcd(\ell,r)}$ fixed choices. For a negative cycle, write the affine map as $T(x) = Rx + e_0$, with R a cyclic rotation. If $d = \gcd(\ell, r)$, the permutation R^r has d coordinate orbits. On each orbit, the translation in T^r has parity r/d. The fixed equation is therefore solvable exactly when r/d is even; when solvable it has 2^d solutions. Independence between coordinate cycles proves

$$R_r(a,b) = \begin{cases} 2^{\sum_{\ell \geq 1} (a_\ell + b_\ell) \gcd(\ell, r)}, & \dfrac{r}{\gcd(\ell, r)} \text{ is even for every } \ell \text{ with } b_\ell > 0, \\ 0, & \text{otherwise.} \end{cases} \tag{137}$$

Because $R_r = \sum_{d \mid r} d\, c_d$, Möbius inversion recovers the exact vertex-cycle counts:

$$c_d(a,b) = \frac{1}{d} \sum_{r \mid d} \mu(d/r) R_r(a,b). \tag{138}$$

Substituting (136)–(138) into (130) proves the full finite-n formula

$$\boxed{\mathcal{H}_n(\mathbf{t}) = \sum_{\substack{(a_\ell),(b_\ell) \\ \sum_\ell \ell(a_\ell+b_\ell)=n}} w(a,b) \prod_{d\geq 1} Q_d(\mathbf{t})^{c_d(a,b)}.} \tag{139}$$

This is exact, not an asymptotic or a truncation.

At total degree two, the orbit interpretation gives a transparent check. The group is transitive on vertices, and pairs of vertices are classified by their Hamming distance $0, 1, \ldots, n$. Therefore

$$\mathcal{H}_n = 1 + m_{(1)} + (n+1)\big(m_{(2)} + m_{(1,1)}\big) + \cdots . \tag{140}$$

In particular, the coefficients do not stabilize as $n \to \infty$.

38.4 Independent verification strategy

The companion program avoids using the statement being tested as its only computational route.

Literal matrix computation. For every one of the $2^n n!$ pairs (b, π), the program builds the vertex permutation and the matrix (129). It checks that the matrices are distinct permutation matrices. For a partition τ, it computes complete symmetric characters from actual matrix traces using Newton's recurrence

$$d\, h_d(P_g) = \sum_{r=1}^{d} \mathrm{Tr}(P_g^r) h_{d-r}(P_g), \qquad h_0 = 1, \tag{141}$$

and averages $\prod_j h_{\tau_j}(P_g)$ exactly.

Direct orbit computation. The monomial basis in (131) is constructed literally. A breadth-first traversal using the n coordinate flips and $n-1$ adjacent coordinate transpositions counts its orbits. This computation does not use traces, determinants, signed types, or the proposed formula.

Signed-type computation. All nonnegative solutions (a_ℓ, b_ℓ) of the size constraint are generated without enumerating group elements. Their rational weights (136) are checked to sum to one. Equations (137) and (138) then produce vertex cycles and the coefficient predicted by (139).

The tests also compare the entire convergent function, not only a finite Taylor truncation. At $(t_1, t_2) = (0.037, 0.061)$ they evaluate

$$\frac{1}{|B_n|} \sum_g \frac{1}{\det(I - t_1 P_g)\det(I - t_2 P_g)}$$

by dense determinants, by actual vertex cycles, and by the signed-type sum.

38.5 Exact results

The following coefficient vectors are ordered as

$$\tau = (1), (2), (1,1), (3), (2,1), (1,1,1).$$

Every entry agrees across literal matrices, direct orbit traversal, and the signed-type formula.

n	$\lvert B_n\rvert$	$\dim V_n$	$\mathbb{P}(F_n > 0)$	common coefficient vector
1	2	2	1/2	$(1, 2, 2, 2, 3, 4)$
2	8	4	3/8	$(1, 3, 3, 4, 7, 10)$
3	48	8	5/16	$(1, 4, 4, 7, 13, 20)$
4	384	16	35/128	$(1, 5, 5, 11, 22, 35)$
5	3840	32	63/256	$(1, 6, 6, 16, 34, 56)$

For $n = 1, \dots, 5$, the program additionally verifies, element by element, the conditional parity law and reconstructs every vertex-cycle count from (137)–(138). It checks moments $m = 1, 2, 3, 4$, giving 20 exact moment comparisons, and the six displayed coefficients, giving 30 three-way coefficient comparisons. In the largest numerical case, the matrix determinant average is

$$1.1620117926716296,$$

and its maximum discrepancy from the cycle and signed-type evaluations is 1.74×10^{-14}, consistent with floating-point roundoff. All integer and rational comparisons are exact.

The range $n \le 5$ is a practical exhaustive choice: it includes all 3840 elements at $n = 5$ and orbit spaces as large as $|X_5|^3 = 32768$, while the group order grows as $2^n n!$. The proof and signed-type algorithm remain valid for every n; only brute-force matrix enumeration is deliberately bounded.

38.6 Reproduction

From the repository root, run

```
marimo run marimo_notebooks/hyperoctahedral_cube.py
```

The notebook allows inspection of individual matrices and signed types and shows every comparison table. The noninteractive regression test is

```
pytest -q tests/test_hyperoctahedral_cube.py
```

38.7 Conclusion

The proposed formulas are correct. A random cube symmetry has no fixed vertex with probability tending to one, but its rare nonzero fixed sets are large enough that $\mathbb{E}F_n = 1$ and all higher moments diverge. The signed-cycle power formula determines every vertex cycle and hence the complete σ-MGF through (139). Exhaustive matrix, orbit, and signed-type computations agree throughout the representative range $1 \le n \le 5$.

39 Socle, product-type, and primitive-extension permutation actions

Canonical testing artifacts.
Exact backend: `lambda_distributions/dists/primitive_action_sigma_mgfs/verification.py`.
Regression: `lambda_distributions/dists/primitive_action_sigma_mgfs/test_verification.py`.
Marimo: `marimo_notebooks/primitive_actions.py`.

Scope and outcome

For a finite group G acting on a finite set Ω, we study the honest complex matrix representation $V = \mathbb{C}[\Omega]$ and its multigraded σ-MGF. A universal Reynolds–Molien theorem expresses the series in terms of the cycle counts of represented permutation matrices. We prove the specialized fixed-power formulas for almost-simple coset actions, the socle actions underlying simple and compound diagonal type, wreath products in product action, regular twisted-wreath actions, and holomorph-simple and holomorph-compound actions. A full primitive extension is obtained only after averaging the additional top-permutation and outer-automorphism cosets. We also prove the exact moment and rare-event statements attached to these formulas. An executable marimo notebook constructs representative matrix groups of dimensions $5, 9, 27, 36, 60$, and 3600 and checks fixed points, Möbius-recovered cycles, Molien coefficients, and direct configuration orbits. The audit comprises 43,642 exact checks, all passing.

39.1 The common representation and universal formula

Let G act on a finite set Ω, and let $V = \mathbb{C}[\Omega]$. For a composition or partition $\tau = (\tau_1, \dots, \tau_s)$ put

$$W_\tau(V) = \bigotimes_{j=1}^{s} \operatorname{Sym}^{\tau_j} V, \qquad a_\tau(G, \Omega) = \dim W_\tau(V)^G.$$

Our normalization is

$$\mathrm{SMG}_{G,\Omega}(\mathbf{t}) = \sum_\tau a_\tau(G, \Omega) m_\tau(\mathbf{t}).$$

For $g \in G$, define

$$F_r(g) = |\operatorname{Fix}_\Omega(g^r)|, \qquad c_d(g) = \#\{d\text{-cycles of } g \text{ on } \Omega\}.$$

Theorem 2 (universal permutation theorem)**.** *For every finite permutation action,*

$$F_r(g) = \sum_{d|r} d\, c_d(g), \qquad c_d(g) = \frac{1}{d} \sum_{r|d} \mu(d/r) F_r(g), \tag{142}$$

$$\mathrm{SMG}_{G,\Omega}(\mathbf{t}) = \frac{1}{|G|} \sum_{g \in G} \prod_{i \geq 1} \prod_{d \geq 1} (1 - t_i^d)^{-c_d(g)}, \tag{143}$$

$$[m_\tau]\, \mathrm{SMG}_{G,\Omega} = \# \left(G \backslash \prod_j \operatorname{Sym}^{\tau_j} \Omega \right). \tag{144}$$

In particular,

$$[m_{(1^s)}]\, \mathrm{SMG}_{G,\Omega} = \mathbb{E}_{g \in G} F_1(g)^s = \#(G \backslash \Omega^s). \tag{145}$$

Imported part and family-independent reduction. The sigma-MGF and orbit identities are Tools 2 and 3. A point in a d-cycle of g is fixed by g^r precisely when $d \mid r$, giving the first relation in (142); ordinary Möbius inversion gives the second. Substitution into the imported permutation formula proves the remaining displayed identities. □

Remark 1 (matrix construction)**.** *The verification code stores a permutation matrix P_g by the tuple $p_g(x)$ defined by $P_g e_x = e_{p_g(x)}$. This is a lossless sparse matrix representation with exactly one nonzero entry per column. Composition of these tuples is matrix multiplication. Thus the degree-3600 cases are fully constructed matrix representations; only wasteful dense zero storage is omitted.*

39.2 Almost-simple coset actions

Let $T \le G \le \mathrm{Aut}(T)$ and let G act transitively on $\Omega = G/M$, where $M < G$ is maximal. For $y \in G$, let y^G denote its conjugacy class.

Proposition 1 (class-power formula)**.** *For $g \in G$ and $r \ge 1$,*

$$\boxed{F_r(g) = \frac{|C_G(g^r)|\,|(g^r)^G \cap M|}{|M|}.} \tag{146}$$

Consequently,

$$\boxed{\mathrm{SMG}_{G,G/M} = \sum_{[g]\subseteq G} \frac{1}{|C_G(g)|} \prod_{i,d} (1 - t_i^d)^{-c_d(g)},} \tag{147}$$

where

$$c_d(g) = \frac{1}{d} \sum_{r \mid d} \mu(d/r) \frac{|C_G(g^r)|\,|(g^r)^G \cap M|}{|M|}.$$

Proof. The coset xM is fixed by g^r exactly when $x^{-1} g^r x \in M$. Every element of $(g^r)^G \cap M$ has $|C_G(g^r)|$ conjugators, while every fixed coset has $|M|$ representatives. Double counting proves (146). Theorem 2 and grouping the group average by conjugacy classes prove (147). □

There is no universal almost-simple limit. Natural symmetric and alternating actions exhibit the familiar Poisson behavior, while bounded-flag and projective regimes depend on which field or rank parameter grows. The exact finite formula above remains valid in every case; asymptotics require family-specific control of class powers and subgroup intersections.

39.3 Socle diagonal actions

Let T be nonabelian simple, $N = T^k$, $D = \mathrm{diag}(T)$, and $\Omega = N/D$. For a conjugacy class $C \subseteq T$, set $c_C = |C_T(x)|$ for $x \in C$.

Theorem 3 (socle diagonal fixed powers and moments)**.** *For $g = (g_1, \dots, g_k) \in T^k$,*

$$\boxed{F_r(g) = \begin{cases} c_C^{k-1}, & g_1^r, \dots, g_k^r \text{ all lie in the same class } C, \\ 0, & \text{otherwise.} \end{cases}} \tag{148}$$

Equations (142) and (143) therefore give the exact σ-MGF. If $X_k(g) = F_1(g)$, then for every $m \ge 1$,

$$\boxed{\mathbb{E} X_k^m = \sum_{C \in \mathrm{Cl}(T)} c_C^{(m-1)k-m}.} \tag{149}$$

In particular,

$$[m_{(1,1)}]\,\mathrm{SMG} = \sum_C c_C^{k-2}, \qquad \Pr(X_k > 0) = \sum_C c_C^{-k}. \tag{150}$$

Proof. A coset xD is fixed by g^r exactly when $x^{-1} g^r x = (y, \dots, y)$ for some $y \in T$. If the component powers are not conjugate, there is no solution. If all lie in C, each $y \in C$ has c_C possible conjugators in every coordinate. Dividing the $|C| c_C^k = |T| c_C^{k-1}$ solutions by $|D| = |T|$ proves (148). A random component lies in C with probability $|C|/|T| = 1/c_C$. Thus the event that all k components lie in C has probability c_C^{-k} and the fixed-point value there is c_C^{k-1}. Summing its mth power proves (149). □

The identity class alone contributes $|T|^{k-2}$ to the pair coefficient. Therefore the unnormalized series has no finite coefficientwise limit as $k \to \infty$. At the same time $\sum_C c_C^{-1} = 1$, so $\mathbb{E} X_k = 1$, while $\Pr(X_k > 0) \to 0$ and all higher moments grow. This is an exact rare-event law: $X_k \to 0$ in probability but not in moments.

Remark 2 (Full primitive diagonal-type groups)**.** *The theorem above is the exact series for the socle $N = T^k$ acting on $N/\operatorname{diag}(T)$. If a larger primitive group G contains N normally, the full G-series is the normalized average over every coset of N in G. Top permutations and outer automorphisms act on the common-class condition and may merge or split N-orbits. No average over N alone is labelled the full primitive-group series.*

39.4 Compound diagonal actions

Partition $b\ell$ factors into b blocks B_j of size ℓ and set

$$N = T^{b\ell}, \qquad D = \prod_{j=1}^{b} \operatorname{diag}_{B_j}(T), \qquad \Omega = N/D \cong \prod_{j=1}^{b} T^{\ell}/\operatorname{diag}(T).$$

Write $g = (g_{j,s})$. Applying Theorem 3 independently in each block proves

$$\boxed{F_r(g) = \prod_{j=1}^{b} \begin{cases} c_{C_j}^{\ell-1}, & g_{j,1}^r, \ldots, g_{j,\ell}^r \in C_j, \\ 0, & \text{otherwise.} \end{cases}} \tag{151}$$

Together with Theorem 2, this is the exact compound-diagonal σ-MGF.

Put $R_\ell(T) = \sum_C c_C^{\ell-2}$, the rank of one diagonal block. Without block permutations, pair orbits are independent, so

$$[m_{(1,1)}] = R_\ell(T)^b.$$

If the full S_b permutes the blocks, a pair orbit is instead a multiset of b single-block orbital types. Hence

$$\boxed{[m_{(1,1)}] = \binom{b + R_\ell(T) - 1}{R_\ell(T) - 1}.} \tag{152}$$

Thus block permutations reduce exponential divergence in b to polynomial divergence.

39.5 Wreath products in product action

Let H act transitively on a finite set Ω, and let $W_k = H^k \rtimes S_k$ act on Ω^k. Write $g = (h_1, \ldots, h_k; \pi)$. For a cycle C of π, let a_C be the product of its labels around the cycle; a different starting point conjugates a_C, which does not change any fixed-point number. Put $f_s(a) = |\operatorname{Fix}_\Omega(a^s)|$ and $d_C(r) = \gcd(|C|, r)$.

Theorem 4 (product-action fixed powers)**.** *For every $r \geq 1$,*

$$\boxed{F_r(g) = \prod_{C \in \operatorname{Cyc}(\pi)} f_{r/d_C(r)}(a_C)^{d_C(r)}.} \tag{153}$$

Consequently (142) and (143) give the exact product-action σ-MGF.

Proof. On a coordinate cycle C of length q, the permutation π^r splits into $d = \gcd(q, r)$ cycles. Following one new cycle to its starting coordinate applies an element conjugate to $a_C^{r/d}$. Each of the d cycles can be assigned independently a point fixed by that element, producing the factor $f_{r/d}(a_C)^d$. Different original cycles use disjoint coordinates and multiply. □

Theorem 5 (exact product-action moments)**.** *Let $R_m(H, \Omega) = \#(H \backslash \Omega^m)$. Then*

$$\boxed{\mathbb{E}_{g \in H \wr S_k} |\operatorname{Fix}_{\Omega^k}(g)|^m = \binom{k + R_m(H, \Omega) - 1}{R_m(H, \Omega) - 1}.} \tag{154}$$

Proof. By (145), the left side counts orbits on $(\Omega^k)^m$. Regard an m-tuple of words as k columns in Ω^m. The base group independently replaces each column by one of the R_m H-orbit types; the top S_k forgets column order. The remaining datum is a multiset of size k chosen from R_m types, counted by the displayed binomial coefficient. □

For $m = 2$, if H has permutation rank $R_2 = r_H$, then

$$[m_{(1,1)}]\,\mathrm{SMG} = \binom{k + r_H - 1}{r_H - 1}.$$

Every nontrivial transitive action has $r_H \geq 2$, so a finite coefficientwise limit is impossible as $k \to \infty$.

Let δ be the derangement proportion of H. Conditional on π, the cycle products a_C are independent uniform elements of H. A fixed word exists exactly when no cycle product is a derangement, hence

$$\Pr(F_1 > 0 \mid \pi) = (1-\delta)^{K(\pi)}.$$

Using the cycle-count generating identity for S_k gives

$$\boxed{\Pr(F_1 > 0) = \frac{(1-\delta)(2-\delta)\cdots(k-\delta)}{k!} \sim \frac{k^{-\delta}}{\Gamma(1-\delta)}.} \tag{155}$$

For finitely many pairs (q, C), the number of q-cycles whose cycle product lies in a conjugacy class $C \subseteq H$ converges jointly to independent Poisson variables of means $|C|/(q|H|)$. The input is therefore marked Poisson, but F_1 is a killed multiplicative functional of that input, not ordinarily a compound-Poisson sum.

39.6 Twisted-wreath regular actions

Let $N = T^k$ and $G = N \rtimes P$, where P acts on N by automorphisms. Use the affine action on the regular set $\Omega = N$,

$$x^{(n,p)} = n\,p(x).$$

For $g = (n, p)$ write

$$g^r = (n_r, p^r), \qquad n_r = n\,p(n)\cdots p^{r-1}(n),$$

and define the nonabelian Lang map $L_\alpha(x) = x\alpha(x)^{-1}$.

Theorem 6 (Lang-map formula and spike law)**.** *For every $r \geq 1$,*

$$\boxed{F_r(n,p) = \begin{cases} |\operatorname{Fix}_N(p^r)|, & n_r \in \operatorname{im} L_{p^r}, \\ 0, & \textit{otherwise.} \end{cases}} \tag{156}$$

Conditionally on p and with n uniform in N,

$$\boxed{F_1(n,p) = \begin{cases} |\operatorname{Fix}_N(p)|, & \textit{with probability } |\operatorname{Fix}_N(p)|^{-1}, \\ 0, & \textit{otherwise.} \end{cases}} \tag{157}$$

Consequently

$$\boxed{\mathbb{E}F_1^m = \frac{1}{|P|}\sum_{p\in P} |\operatorname{Fix}_N(p)|^{m-1} = \#(P\backslash N^{m-1}),} \tag{158}$$

and $[m_{(1,1)}]\,\mathrm{SMG} = \#(P\backslash N)$.

Proof. The fixed-point equation is $x = n_r p^r(x)$, equivalently $L_{p^r}(x) = n_r$. If $L_\alpha(x) = L_\alpha(y)$, then $z = y^{-1}x$ satisfies $z = \alpha(z)$; conversely, right multiplication by any $z \in \operatorname{Fix}_N(\alpha)$ preserves the Lang value. Thus every nonempty fiber of L_α is a right coset of $\operatorname{Fix}_N(\alpha)$. This proves (156). Its image has size $|N|/|\operatorname{Fix}_N(p)|$, so a uniform n belongs to it with probability $|\operatorname{Fix}_N(p)|^{-1}$, proving (157). Averaging the mth power gives the first equality in (158); Burnside's lemma for the action of P on N^{m-1} gives the second. □

If p permutes factors with automorphism labels, and β_C is the twist product around a coordinate cycle C, then direct propagation around each cycle gives

$$\boxed{|\operatorname{Fix}_N(p)| = \prod_C |\operatorname{Fix}_T(\beta_C)|.} \tag{159}$$

For the untwisted benchmark $P = S_k$, this becomes $|\operatorname{Fix}_N(p)| = |T|^{K(p)}$, so

$$\mathbb{E}F_1^m = \binom{k+|T|^{m-1}-1}{|T|^{m-1}-1}, \tag{160}$$

$$\Pr(F_1 > 0) = \frac{(1/|T|)(1+1/|T|)\cdots(k-1+1/|T|)}{k!} \sim \frac{k^{1/|T|-1}}{\Gamma(1/|T|)}. \tag{161}$$

This retains the same Bernoulli-spike mechanism as genuine primitive twisted-wreath groups.

39.7 Holomorph-simple and holomorph-compound actions

39.7.1 Holomorph-simple left-right action

Let $T \times T$ act on $\Omega = T$ by

$$x^{(a,b)} = a^{-1}xb.$$

Then x is fixed by $(a,b)^r$ precisely when $a^r = xb^rx^{-1}$.

Proposition 2 (holomorph-simple formulas)**.**

$$\boxed{F_r(a,b) = \begin{cases} |C_T(a^r)|, & a^r \sim b^r, \\ 0, & \textit{otherwise}. \end{cases}} \tag{162}$$

If $X(a,b) = F_1(a,b)$, then

$$\boxed{\mathbb{E}X^m = \sum_{C \in \operatorname{Cl}(T)} c_C^{m-2}, \qquad [m_{(1,1)}]\,\mathrm{SMG} = |\operatorname{Cl}(T)|.} \tag{163}$$

Proof. The fixed-point equation proves (162); if it is solvable, the set of conjugators is a centralizer coset. For a class C, the probability that independent uniform a, b both lie in C is $(|C|/|T|)^2 = c_C^{-2}$, and the fixed-point value is c_C. Summing its mth power proves (163). $\square$

Thus a family in which $|\operatorname{Cl}(T)|$ grows has no coefficientwise holomorph-simple limit. Adding outer automorphisms replaces ordinary conjugacy by the corresponding twisted-conjugacy equation.

39.7.2 Holomorph-compound product action

Let $(T^k \times T^k) \rtimes S_k$ act on T^k coordinatewise by left-right multiplication, with S_k permuting coordinates. This is exactly the product action of the base group $H = T \times T \curvearrowright T$. The base rank is $R_2(H,T) = |\operatorname{Cl}(T)|$, so

$$\boxed{[m_{(1,1)}]\,\mathrm{SMG} = \binom{k+|\operatorname{Cl}(T)|-1}{|\operatorname{Cl}(T)|-1}.} \tag{164}$$

All fixed-power and higher-moment formulas are those of Theorems 4 and 5 with this base action.

39.8 Exact verification strategy and results

39.8.1 Independent routes

The executable package uses four logically distinct routes.

1. **Matrix route.** It constructs every represented permutation matrix in the exhaustive cases and computes F_r from the cycles of its actual permutation.
2. **Structural route.** It computes the right side of the relevant class-intersection, common-conjugacy-class, cycle-product, Lang-map, or left-right power formula without consulting the represented matrix cycles.

3. **Molien route.** For cycle counts $c_d(g)$ it extracts $[t^q]\prod_d(1-t^d)^{-c_d(g)}$ exactly and averages products of these coefficients over the group.
4. **Orbit route.** It constructs products of multisets or ordered tuples and counts their orbits directly. This does not use determinant or character formulas.

The Möbius formula is also checked by reconstructing every cycle count from fixed points of powers.

39.8.2 Representative cases

Family	Constructed action	Matrix dimension	Exact comparison
Almost simple	$A_5 \curvearrowright A_5/A_4$	5	All 60 elements, powers 1–6, five τ values
Simple diagonal / HS	$A_5^2 \curvearrowright A_5$ by left-right multiplication	60	All 3600 elements, powers 1–6, direct pair orbits
Simple diagonal	$A_5^3 \curvearrowright A_5^2$	3600	Representative fixed powers; all simultaneous-conjugacy orbits
Compound diagonal	Two $A_5^2/\operatorname{diag}(A_5)$ blocks, with and without S_2	3600	Direct stabilizer orbits: 25 and 15
Product action	$S_3 \wr S_k \curvearrowright \{0,1,2\}^k$, $k=2,3$	$9, 27$	All 72 and 1296 elements, powers 1–6, moments $2,3$
Twisted wreath	$S_3^2 \rtimes S_2 \curvearrowright S_3^2$ affinely	36	All 72 elements, powers 1–6, moments $1,2,3$
Holomorph compound	$((S_3\times S_3)^2 \rtimes S_2) \curvearrowright S_3^2$	36	All 2592 elements, five powers, direct pair orbits

The smaller S_3 wreath benchmarks are intentional. The fixed-point identities are valid for arbitrary finite base groups, while the smaller base permits exhaustive enumeration of every matrix and target orbit. The A_5 cases separately test the hypotheses that require a nonabelian simple group.

39.8.3 Numerical results, all exact

For A_5, the centralizer sizes are $3,4,5,5,60$. Hence the direct diagonal pair coefficients are

$$k=2: \quad 5, \qquad k=3: \quad 3+4+5+5+60=77.$$

For two diagonal $k=2$ blocks, direct orbit traversal gives $5^2=25$ without block swaps and $\binom{2+5-1}{5-1}=15$ with the full S_2.

For $S_3 \wr S_k$ on $\{0,1,2\}^k$, the base has $R_2=2$ and $R_3=5$. Direct orbit counts, Burnside averages, and Theorem 5 all give

k	$[m_{(1,1)}]$	$[m_{(1,1,1)}]$
2	3	15
3	4	35

For the affine benchmark $S_3^2 \rtimes S_2$, the first three fixed-point moments are exactly $1, 21, 666$, and the direct orbit count $|S_2\backslash S_3^2|$ is 21. For the holomorph-compound benchmark, the direct pair-orbit count, Burnside average, and formula $\binom{2+3-1}{3-1}$ are all 6.

The final check count is

$$360+21600+16+432+7776+432+12960+66=43{,}642.$$

The summands are respectively: almost-simple class powers; degree-60 diagonal/HS powers; degree-3600 diagonal samples; product-action $k=2$; product-action $k=3$; twisted-wreath powers; holomorph-compound powers; and a separate universal Möbius audit on all A_5 and S_3 natural-action matrices. Every equality is integral; no floating-point tolerance is used.

39.9 Asymptotic classification and cautions

Type	Exact structural input	Typical asymptotic
Almost simple	$C_G(g^r)$ and $(g^r)^G \cap M$	Family-dependent
Simple diagonal	Common power-conjugacy class	Exponential moment explosion
Compound diagonal	Blockwise diagonal formula	Polynomial after block permutation
Product action	Marked cycle products in H	$F_1 \to 0$; polynomially divergent moments
Twisted wreath	Lang solvability and $\mathrm{Fix}_N(p^r)$	Bernoulli spikes
Holomorph simple	Conjugacy of left/right powers	Pair coefficient $\lvert \mathrm{Cl}(T)\rvert$
Holomorph compound	Product action over HS	Polynomial moment explosion

Limit class. These families typically combine marked-Poisson cycle inputs with multiplicative rare-event observables: $F_1 \to 0$ in probability, $\mathbb{E}F_1 = 1$, and higher moments diverge. Calling this an ordinary compound-Poisson Λ-limit would be misleading. The precise statement is convergence in probability of the scalar observable together with explicit divergence of selected moments; there is no raw moment/σ-MGF limit.

Three cautions delimit the statement. First, all representations here are the finite permutation modules $\mathbb{C}[\Omega]$; no canonical embedding of an abstract finite group into complex matrices is being assumed beyond the stated action. Second, coefficientwise nonconvergence follows as soon as one fixed coefficient diverges, but convergence in probability of F_1 is a different notion. Third, genuine primitive twisted-wreath groups impose additional structure on T and P; the exhaustive S_3 computation is a benchmark for the universal affine/Lang identity, not a claim that this small benchmark itself satisfies every primitivity hypothesis.

39.10 Reproducibility

The files accompanying this note are:

- `verification.py`: exact group constructors, sparse permutation matrices, fixed-power formulas, coefficient extraction, and orbit traversal;
- `test_verification.py`: automated regression tests for every family; and
- `marimo_notebooks/primitive_actions.py`: a marimo interface presenting the checks in the same group order as this document.

From the repository root, run

```
.venv/bin/python -m lambda_distributions.dists.primitive_action_sigma_mgfs.verification
.venv/bin/pytest -q lambda_distributions/dists/primitive_action_sigma_mgfs/test_verification.py
.venv/bin/marimo run marimo_notebooks/primitive_actions.py
```

References

[1] *Embedding permutation groups into wreath products in product action*, https://arxiv.org/abs/1108.3611.

[2] *Derangements in simple and primitive groups*, https://arxiv.org/abs/math/0208022.

[3] *Base sizes of primitive groups of diagonal type*, https://arxiv.org/abs/2303.14290.

[4] *Derangements in wreath products of permutation groups*, https://arxiv.org/abs/2207.01215.

[5] *Bases of twisted wreath products*, https://arxiv.org/abs/2102.02190.

Part IV

Finite affine, Lie-type, and exceptional finite families

40 Power characters, parabolic cycles, and bounded-support stability

Abstract

For an arbitrary finite group and honest complex representation we derive the sigma-MGF from character values on all powers. For permutation actions this becomes an exact fixed-point and cycle formula; for coset actions it becomes a centralizer–subgroup-intersection formula, and for conjugation it becomes a centralizer formula. We then prove coefficientwise fixed-field stability for bounded flags in the classical growing-rank towers. A marimo notebook checks the linear, parabolic, conjugation, and stability statements by explicit matrices or generator orbits. The proof is general; finite computations are reported only as independent exact verification.

Status. Equations (165)–(173) and the type-A threshold (175) are exact theorems. The formed-space statement is coefficientwise stability with a finite tower-dependent threshold, not a claimed universal numerical threshold. Complete flags are excluded: their degree-two coefficient already grows with the Weyl group.

40.1 Common target and universal character-power formula

Let G be finite and let $\rho : G \to GL(V)$ be an honest finite-dimensional complex representation with character χ_V. For a partition $\tau = (\tau_1, \dots, \tau_s)$ set

$$W_\tau(V) = \bigotimes_{j=1}^{s} \operatorname{Sym}^{\tau_j} V, \qquad |\tau| = \sum_j \tau_j,$$

and define

$$\mathrm{SMG}_{G,V} = \mathbb{E}_{g\in G}[\operatorname{Exp}_\sigma(X_V(g)h_1)] = \sum_\tau \dim W_\tau(V)^G\, m_\tau.$$

Theorem 1 (Universal power-character formula)**.** *One has the exact identities*

$$\mathrm{SMG}_{G,V} = \frac{1}{|G|}\sum_{g\in G}\prod_{i\geq 1}\det(I - t_i\rho(g))^{-1}, \tag{165}$$

$$= \frac{1}{|G|}\sum_{g\in G}\exp\left(\sum_{r\geq 1}\frac{\chi_V(g^r)}{r}p_r\right), \tag{166}$$

$$= \sum_{[g]\subseteq G}\frac{1}{|C_G(g)|}\exp\left(\sum_{r\geq 1}\frac{\chi_V(g^r)}{r}p_r\right). \tag{167}$$

Thus the series is determined by conjugacy classes, class power maps, and the character values. Its coefficient is

$$[m_\tau]\,\mathrm{SMG}_{G,V} = \frac{1}{|G|}\sum_{g\in G}\prod_{j=1}^{s}\left(\sum_{\lambda\vdash\tau_j}\frac{1}{z_\lambda}\prod_{r\geq 1}\chi_V(g^r)^{m_r(\lambda)}\right). \tag{168}$$

Reduction to the imported tools. Equation (165) is Tool 2. Applying the standard logarithmic determinant change of coordinates and using $\operatorname{Tr}(\rho(g)^r) = \chi_V(g^r)$ gives (166). Grouping the resulting average by conjugacy classes gives (167); expanding each symmetric-power character in power sums gives (168). No invariant-dimension or Molien identity is reproved here. □

Remark 1 (Character specializations)**.** *For a genuine Steinberg or a specified genuine Weil representation, insert its character into* (167)*. For a Deligne–Lusztig virtual character $R_T^G(\theta)$ the same expression is interpreted in the completed Grothendieck lambda-ring; its coefficients are virtual multiplicities and need not be nonnegative. These are specializations of the theorem, not claims that all such character values collapse to one uniform product.*

40.2 Permutation and parabolic quotient groups

Suppose G acts on a finite set X, and use the honest complex permutation module $V = \mathbb{C}[X]$. Let $c_r(g; X)$ count the r-cycles and put $F_d(g; X) = |\operatorname{Fix}_X(g^d)|$.

Proposition 1 (Permutation-cycle formula)**.** *For every finite action,*

$$c_r(g; X) = \frac{1}{r} \sum_{d|r} \mu(r/d) F_d(g; X), \tag{169}$$

$$\mathrm{SMG}_{G,X} = \frac{1}{|G|} \sum_{g \in G} \prod_{i,r \geq 1} (1 - t_i^r)^{-c_r(g;X)}, \tag{170}$$

$$[m_\tau]\,\mathrm{SMG}_{G,X} = \# \left(G \backslash \prod_j \mathrm{Sym}^{\tau_j} X \right). \tag{171}$$

Proof. An r-cycle has determinant factor $1 - t^r$ and contributes all of its r points to F_d precisely when $r \mid d$. Hence $F_d = \sum_{r|d} r c_r$, and M"obius inversion gives (169). The determinant factors give (170). The monomial basis of $W_\tau(\mathbb{C}[X])$ is the displayed product of multiset spaces, so its invariant dimension is the number of orbits, proving (171). ☐

Proposition 2 (Coset and parabolic cycles)**.** *For $X = G/H$,*

$$F_d(g; G/H) = \frac{|C_G(g^d)|}{|H|} \, |(g^d)^G \cap H|, \tag{172}$$

$$c_r(g; G/H) = \frac{1}{r} \sum_{d|r} \mu(r/d) \frac{|C_G(g^d)|}{|H|} \, |(g^d)^G \cap H|. \tag{173}$$

Substitution in (170) *gives the complete exact sigma-MGF. In particular this applies to $G = \mathbf{G}^F$ and every F-stable parabolic P, including projective points, Grassmannians, polar Grassmannians, and fixed-shape partial flags.*

Proof. The coset xH is fixed by g^d exactly when $x^{-1} g^d x \in H$. Each element of $(g^d)^G \cap H$ has $|C_G(g^d)|$ conjugators x. Divide by $|H|$ for the number of representatives of a coset and then apply (169). ☐

40.3 Conjugation groups

For conjugation on G itself,

$$F_d(g; G_{\mathrm{conj}}) = |C_G(g^d)|.$$

Therefore

$$c_r^{\mathrm{conj}}(g) = \frac{1}{r} \sum_{d|r} \mu(r/d) |C_G(g^d)|, \qquad \mathrm{SMG}_{G,\mathrm{conj}} = \frac{1}{|G|} \sum_g \prod_{i,r} (1 - t_i^r)^{-c_r^{\mathrm{conj}}(g)}. \tag{174}$$

On a fixed conjugacy class C, replace the fixed-point count by $|C \cap C_G(g^d)|$. Hence the conjugation series is determined by centralizers, class intersections, and power maps.

40.4 Fixed-field stability for bounded flags

Fix q and a flag shape $\mathbf{d} = (d_1 < \cdots < d_s)$, with $d = d_s$ independent of ambient rank. Let $X_n = \mathrm{Flag}_{\mathbf{d}}(\mathbb{F}_q^n)$.

Theorem 2 (Type-A bounded-support stability)**.** *For every partition τ of total degree D,*

$$[m_\tau]\,\mathrm{SMG}_{GL_n(q),X_n} \quad \textit{is independent of } n \textit{ for } n \geq dD. \tag{175}$$

The same holds for $PGL_n(q)$. For $SL_n(q)$ and $PSL_n(q)$ the safe range is $n > dD$. The stable coefficient is

$$a_{\tau,\mathbf{d}}(q) = \sum_{r=0}^{dD} \#\left(GL_r(q)\backslash \mathcal{C}^{\mathrm{span}}_{\tau,\mathbf{d}}(r)\right), \tag{176}$$

where the configuration space consists of τ-colored flag multisets whose top subspaces span $\mathbb{F}_q^r$.

Proof. A degree-D configuration uses D flags with multiplicity. The sum of their top subspaces, called the support, has dimension at most dD. Two configurations are $GL_n(q)$-conjugate exactly when their supported labelled configurations are linearly isomorphic, because every isomorphism between the supports extends across complements. Once $n \geq dD$, every possible support already embeds, so no new orbit type can appear. Sorting by the support dimension gives (176). Scalar matrices act trivially, so the PGL orbits agree. If $n > dD$, a nonzero complementary direction can correct the determinant of any extension while fixing the target support; this gives the stated SL and PSL range. □

Proposition 3 (Formed-space towers)**.** *Fix one symplectic, unitary, odd-orthogonal, or one of the two even-orthogonal towers, and let X_n be the totally isotropic or singular flags of fixed shape. For every τ there is a finite threshold $N_{\mathcal{T}}(\mathbf{d},\tau)$ after which $[m_\tau]\,\mathrm{SMG}_{G_n,X_n}$ is constant.*

Proof. The span of a degree-D configuration again has bounded dimension. Its orbit is determined by the labelled flags together with the restricted Hermitian, alternating, or quadratic form and the multiplicities. In characteristic two the quadratic form, not merely its polar form, is retained. There are only finitely many such bounded-dimensional supported types over a fixed field. Witt extension embeds and identifies each type once the chosen tower is large enough; taking the maximum of these finitely many embedding ranks gives the threshold. Determinant or spinor-norm correction may require enlarging the unused nondegenerate complement before passing to special, derived, or projective groups. □

Remark 2 (Why complete flags are excluded)**.** *For G/B, Bruhat decomposition gives $[m_{(1,1)}]\,\mathrm{SMG}_{G,G/B} = |B\backslash G/B| = |W|$. This grows with rank, so bounded shape is an essential hypothesis.*

40.5 Representation scope for finite Lie-type groups

Projective, subspace, flag, and conjugacy actions are honest complex permutation representations. Formula (167) also applies to every specified genuine complex character, including Steinberg and chosen Weil representations. The defining-characteristic action on $\mathfrak{g}(\mathbb{F}_q)$ requires care:

- as a finite set with conjugation action it is an honest permutation representation, with $F_d(g) = q^{\dim_{\mathbb{F}_q} C_{\mathfrak{g}}(g^d)}$;
- as a linear module over $\mathbb{F}_q$ it is not automatically a complex representation, so a cross-characteristic realization or lift must be specified before applying (166).

40.6 Group-separated exact verification

The executable companion is `marimo_notebooks/power_character_stability.py`. It calls a plain Python verification module, so the same checks run without a notebook server.

40.6.1 Linear group: $S_3 \cong GL_2(2)$ and its Steinberg module

Every one of the six integral 2×2 matrices of the standard S_3 representation is constructed. For each τ, the program builds the matrices of every $\mathrm{Sym}^{\tau_j} V$, takes their Kronecker product, averages the actual matrices, and computes the Reynolds rank. Independently it uses only $\chi(g^r)$ and Newton's recurrence to evaluate (168). In the order

$$(1), (2), (1,1), (3), (2,1), (1,1,1),$$

both routes give

$$(0, 1, 1, 1, 1, 1).$$

The largest projector-idempotence error is 5.56×10^{-17}.

40.6.2 Parabolic group: $GL_3(2)$ on G/P

The program enumerates all 168 invertible 3×3 matrices over $\mathbb{F}_2$ and their action on the seven projective points. It separately computes all six conjugacy classes, the point stabilizer P of order 24, centralizers, and class intersections with P. Formula (172) reproduces every needed fixed-point count and reconstructs the cycles of all 168 elements. Actual permutation-matrix averages, cycle factors, and direct multiset-orbit traversal agree on

$$(1, 2, 2, 4, 5, 6).$$

There are 715 direct fixed-point equalities and 168 full cycle reconstructions.

40.6.3 Conjugation group: S_3 on itself

The six conjugation permutations are built from the group law. Direct fixed points agree with $|C_G(g^d)|$ for every required power, and all six cycle decompositions are reconstructed. Matrix averages, (174), and direct orbits agree on

$$(3, 8, 11, 17, 32, 49).$$

40.6.4 Stable type-A groups

Transvections generate the actions without enumerating the large ambient groups. Direct configuration-orbit traversal gives:

τ for projective points	$n = 1$	$n = 2$	$n = 3$	$n = 4$
(1)	1	1	1	1
(2)	1	2	2	2
(1, 1)	1	2	2	2
(3)	1	3	4	4
(2, 1)	1	4	5	5
(1, 1, 1)	1	5	6	6

The onset is exactly consistent with $n \geq |\tau|$ when $d = 1$. For $GL_n(2)$ on $\mathrm{Gr}_2(n)$, both $\tau = (2)$ and $(1, 1)$ give three orbits at $n = 4$ and again at $n = 5$, matching the threshold $n \geq 2|\tau| = 4$ and the intersection-dimension classification.

Verification result. All 26 reported coefficient or stability rows pass. These computations test the formulas at small representative dimensions with distinct failure modes; the all-rank claims rest on the proofs above.

40.7 Reproduction

From the repository root, run

```
python3 -m for_this_guy.power_character_bounded_flag_stability.verification
.venv/bin/marimo run marimo_notebooks/power_character_stability.py
```

References

[1] S. Howe, *Random matrix statistics and zeroes of L-functions via probability in lambda-rings*, arXiv:2412.19295.

[2] J. Huang, *Some extensions of Witt's theorem*, arXiv:0705.3839.

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41 σ-MGF for finite affine groups

Abstract

Let $\mathrm{AGL}_n(q)$ act on $S = \mathbb{F}_q^n$ and hence on the permutation module $\mathbb{C}[S]$. This note proves the cycle-index form of its generalized Molien series, the tuple-of-multisets orbit interpretation, the stable range $n \geq |\tau| - 1$, and the Gaussian-binomial formula for fixed-point moments. The companion Marimo notebook constructs the affine permutations and their $q^n \times q^n$ permutation matrices. It then compares literal orbit enumeration with coefficient extraction, checks the linear fixed-point and Möbius formulas, and tests the moment formula. All 18 reported comparisons passed exactly, including $\mathrm{AGL}_3(2)$ of order 1344 and $\mathrm{AGL}_2(3)$ of order 432.

41.1 The statement being tested

For a finite group G acting on a finite set S, put

$$\mathrm{SMG}_G(\mathbf{t}) = \frac{1}{|G|} \sum_{g \in G} \prod_{a \geq 1} \det(I - t_a g \mid \mathbb{C}[S])^{-1}.$$

If $\tau = (\tau_1, \dots, \tau_\ell)$ is a partition, the coefficient of the monomial symmetric function m_τ is

$$[m_\tau]\, \mathrm{SMG}_G = \# \left(G \backslash \prod_{j=1}^{\ell} \mathrm{Mult}_{\tau_j}(S) \right). \tag{177}$$

Our affine group is

$$G_{n,q} = \mathrm{AGL}_n(q) = \mathbb{F}_q^n \rtimes \mathrm{GL}_n(q), \qquad (A, b)x = Ax + b.$$

The exact proposed formula is

$$\boxed{\mathcal{A}_{n,q}(\mathbf{t}) = \frac{1}{q^n |\,\mathrm{GL}_n(q)|} \sum_{A \in \mathrm{GL}_n(q)} \sum_{b \in \mathbb{F}_q^n} \prod_{r \geq 1} Q_r(\mathbf{t})^{c_r(A,b)},} \qquad Q_r(\mathbf{t}) = \prod_{a \geq 1} (1 - t_a^r)^{-1}. \tag{178}$$

Here $c_r(A, b)$ is the number of r-cycles of the affine permutation.

41.2 Imported orbit and cycle reductions

Equation (177) is Tool 3, and (178) is the permutation specialization of Tool 2. Order the q^n points of $\mathbb{F}_q^n$; the verifier constructs the corresponding matrices by

$$(P_{(A,b)})_{y,x} = \mathbf{1}_{\{y = Ax + b\}}.$$

The affine-specific work below is therefore the fixed-point calculation, Möbius recovery of cycles, stability, and moment enumeration.

There is also a useful coefficient extractor that avoids symbolic determinants. The number of size-d multisets fixed by g is

$$H_d(g) = [z^d] \prod_{r \geq 1} (1 - z^r)^{-c_r(g)}. \tag{179}$$

Therefore

$$[m_\tau] \mathcal{A}_{n,q} = \frac{1}{|G_{n,q}|} \sum_{g \in G_{n,q}} \prod_{j=1}^{\ell} H_{\tau_j}(g). \tag{180}$$

The implementation extracts (179) by integer dynamic programming. Separately, it explicitly moves every multiset tuple by every affine permutation and counts canonical orbit representatives. Thus the two numbers compared in the main test do not share an orbit-counting routine.

41.3 Linear fixed-point formula and Möbius recovery

For $g = (A, b)$,

$$g^d(x) = A^d x + b_d, \qquad b_d = (I + A + \cdots + A^{d-1})b.$$

Thus $(I - A^d)x = b_d$. Elementary linear algebra over $\mathbb{F}_q$ gives

$$F_d(A, b) := \#\,\mathrm{Fix}(g^d) = \begin{cases} q^{\dim\ker(I-A^d)}, & b_d \in \mathrm{im}(I - A^d), \\ 0, & b_d \notin \mathrm{im}(I - A^d). \end{cases} \tag{181}$$

Since every r-cycle contributes its r points to $\mathrm{Fix}(g^d)$ exactly when $r \mid d$,

$$F_d = \sum_{r|d} rc_r, \qquad \boxed{c_r = \frac{1}{r}\sum_{d|r} \mu(r/d)F_d.}$$

The checker computes ranks of both the coefficient matrix and its augmented matrix modulo q, compares (181) with fixed points of the explicit permutation power, and then reconstructs every cycle count by Möbius inversion. It does so for every element of $\mathrm{AGL}_2(2)$ and $\mathrm{AGL}_2(3)$ and for every required power through q^n.

41.4 Stable coefficients

Let $k = |\tau|$. A tuple of multisets has at most k support points; attach to each support point its nonzero multiplicity vector. Its affine span has dimension at most $k - 1$.

Suppose two such labeled configurations have affinely isomorphic spans. An affine isomorphism between their spans extends to an element of $\mathrm{AGL}_n(q)$: choose origins, extend the induced linear isomorphism of direction subspaces to an element of $\mathrm{GL}_n(q)$, and restore the translation. The converse follows by restriction. Hence the orbit is determined by the intrinsic labeled affine configuration, not the ambient space, once $n \geq k - 1$. Therefore

$$\boxed{[m_\tau]\mathcal{A}_{n,q} \text{ is independent of } n \text{ for } n \geq |\tau| - 1.}$$

The computed value for $\tau = (1, 1, 1)$ is 5 for both $\mathrm{AGL}_2(2)$ and $\mathrm{AGL}_3(2)$, directly exercising this threshold.

41.5 Fixed-point moments and the two limits

Let $F(g) = \#\,\mathrm{Fix}(g)$. Burnside gives

$$\mathbb{E}[F^k] = \#\big(\mathrm{AGL}_n(q)\backslash(\mathbb{F}_q^n)^k\big).$$

Translate the first point to zero and form the $n \times (k-1)$ matrix with columns $x_i - x_1$. Left multiplication by $\mathrm{GL}_n(q)$ preserves, and is classified by, its row space in $\mathbb{F}_q^{k-1}$. It follows that

$$\boxed{\mathbb{E}[F^k] = \sum_{r=0}^{\min(n,k-1)} \begin{bmatrix} k-1 \\ r \end{bmatrix}_q.} \tag{182}$$

In particular, $\mathbb{E}[F^3] = q+2$ for $n = 1$ and $q+3$ for $n \geq 2$. For fixed q, the moment stabilizes but is not the mean-one Poisson moment: when $q > 2$ the third moment already differs from 5; for $q = 2$, the fourth moment is 16, not 15. For fixed $n \geq 1$, the third moment diverges as $q \to \infty$, so the unrenormalized σ-MGF has no coefficientwise finite limit.

41.6 Exact verification evidence

All arithmetic in the finite fields, cycle extraction, Burnside averages, and Gaussian binomials is exact. No floating-point eigenvalues are used. The determinant identity is independently recovered from traces of the constructed permutation matrices via Newton identities.

Group	τ	$\lvert G\rvert$	configurations	literal orbits	formula
$\mathrm{AGL}_1(2)$	(3)	2	4	2	2
$\mathrm{AGL}_1(2)$	$(2,1)$	2	6	3	3
$\mathrm{AGL}_2(2)$	(3)	24	20	3	3
$\mathrm{AGL}_2(2)$	$(2,1)$	24	40	4	4
$\mathrm{AGL}_2(2)$	$(1,1,1)$	24	64	5	5
$\mathrm{AGL}_3(2)$	$(1,1,1)$	1344	512	5	5
$\mathrm{AGL}_1(3)$	(3)	6	10	3	3
$\mathrm{AGL}_2(3)$	$(2,1)$	432	405	5	5
$\mathrm{AGL}_1(5)$	$(1,1,1)$	20	125	7	7

Group	moment	direct average	formula (182)
$\mathrm{AGL}_1(2)$	$\mathbb{E}F^3$	4	4
$\mathrm{AGL}_2(2)$	$\mathbb{E}F^3$	5	5
$\mathrm{AGL}_3(2)$	$\mathbb{E}F^4$	16	16
$\mathrm{AGL}_1(3)$	$\mathbb{E}F^3$	5	5
$\mathrm{AGL}_2(3)$	$\mathbb{E}F^3$	6	6
$\mathrm{AGL}_1(5)$	$\mathbb{E}F^3$	7	7

Scope and reproducibility. The mathematical proof applies to every prime power q. The transparent enumerator supplied here implements prime fields $\mathbb{F}_q$ only; extension-field arithmetic is the sole missing software layer for nonprime prime powers. The Marimo notebook exposes the group, partition, and moment as controls and prints an actual permutation matrix. The adjacent standalone module runs the complete fixed suite and exits nonzero on any discrepancy.

Conclusion. The cycle-index formula, the linear fixed-point/Möbius formulas, the orbit interpretation, the stable range, and the Gaussian-binomial moments agree with explicit finite computations across the representative dimensions and field sizes tested.

42 Finite linear-group permutation actions

Abstract

For a finite group acting on a set S, we prove the determinant/cycle formula, its fixed-point reconstruction, and its interpretation as an orbit count on tuples of multisets. We then specialize to nonzero-vector, projective-point, and complete-flag actions of finite linear groups. Exact finite-field matrix enumerations verify all coefficients through degree three for $PSL_3(2)$, the degree-two vector claims for $GL_2(3)$, and the Bruhat obstruction in ranks two and three. The computation also exposes a sharp boundary phenomenon for $SL_2(3)$.

42.1 Imported finite permutation formula

For $V = \mathbb{C}[S]$, Tools 2 and 3 give

$$\mathbb{E}[\mathrm{Exp}_\sigma(X_{G,V}h_1)] = \frac{1}{|G|}\sum_{g\in G}\prod_{i\geq 1}\prod_{d\geq 1}(1-t_i^d)^{-c_d(g;S)}, \tag{183}$$

$$[m_\tau]\,\mathbb{E}[\mathrm{Exp}_\sigma(X_{G,V}h_1)] = \#\left(G\backslash\prod_b \mathrm{Sym}^{\tau_b} S\right). \tag{184}$$

The new calculations begin with the fixed-point counts for the vector, projective-point, and flag actions.

Fixed points determine the cycles

A point on a d-cycle is fixed by g^r exactly when $d \mid r$. Thus

$$\mathrm{Fix}_S(g^r) = \sum_{d|r} d\, c_d(g;S).$$

Möbius inversion gives

$$\boxed{c_d(g;S) = \frac{1}{d}\sum_{r|d}\mu(d/r)\,\mathrm{Fix}_S(g^r).} \tag{185}$$

For $S = G/P$, $\mathrm{Fix}_S(g^r)$ is the value at g^r of the permutation character $\mathrm{Ind}_P^G\,\mathbf{1}$. Grouping (2) by conjugacy class then yields the stated centralizer-weighted formula.

Direct orbit interpretation

A monomial basis vector of $\mathrm{Sym}^r\,\mathbb{C}[S]$ is a multiset of r points of S. Consequently a basis of W_τ is a tuple of such multisets, of respective sizes $\tau_1,\ldots,\tau_s$. Since G permutes this basis, the sum of the basis vectors in each orbit is one invariant basis vector. Hence

$$\boxed{[m_\tau]\,\mathrm{SMG}_{G,S} = |\{G\text{-orbits on tuples of multisets of sizes } \tau_1,\ldots,\tau_s\}|\,.} \tag{186}$$

Equation (4) supplies the independent direct route used in the computation.

42.2 Linear and projective specializations

For $S = \mathbb{F}_q^n \setminus \{0\}$ and $g \in GL_n(q)$,

$$\mathrm{Fix}_S(g^r) = q^{\dim\ker(g^r - I)} - 1. \tag{187}$$

For projective points, a fixed line is an eigenline whose eigenvalue lies in $\mathbb{F}_q^\times$. The eigenspaces for distinct eigenvalues contribute disjoint sets of lines, so

$$\mathrm{Fix}_{\mathbb{P}^{n-1}(\mathbb{F}_q)}(g^r) = \sum_{\lambda\in\mathbb{F}_q^\times}\frac{q^{\dim\ker(g^r-\lambda I)}-1}{q-1}. \tag{188}$$

Substitution of (5) or (6) into (3) and then (2) proves the proposed exact formulas.

Over $\mathbb{F}_2$, every projective line has one nonzero vector and $GL_n(2) = SL_n(2) = PGL_n(2) = PSL_n(2)$. Thus (5) directly gives the displayed $PSL_n(2)$ formula.

Stable range

A configuration of total degree $D = |\tau|$ involves at most D projective points and its span has dimension at most D. When $n \geq D$, two such configurations are $GL_n(q)$-equivalent exactly when the induced colored configurations on their spans are linearly isomorphic: any isomorphism of the spans extends after choosing complements. This proves coefficientwise stability for GL/PGL projective actions. The same argument for nonzero vectors is immediate. For SL/PSL, the determinant of an extension can be corrected on an unused one-dimensional summand, which requires the strict range $n > D$.

For projective points the stable expansion through degree three is

$$1 + m_{(1)} + 2m_{(2)} + 2m_{(1,1)} + 4m_{(3)} + 5m_{(2,1)} + 6m_{(1,1,1)} + O(\deg 4). \tag{189}$$

For three distinct points, collinear and noncollinear triples give two different orbits; this is the extra degree-three term relative to the symmetric-group/Poisson pattern.

For nonzero vectors, ordered pairs consist of one independent orbit and one dependent orbit $y = ax$ for each $a \in \mathbb{F}_q^\times$, giving

$$[m_{(1,1)}]\,\mathrm{SMG} = q.$$

For an unordered pair, the dependent ratios other than 1 are identified by $a \leftrightarrow a^{-1}$. Together with the repeated and independent types, this gives

$$[m_{(2)}]\,\mathrm{SMG} = 2 + \left\lfloor \frac{q-1}{2} \right\rfloor.$$

42.3 Complete flags and failure of rank stability

For $S = G/B$, orbitals are the double cosets $B\backslash G/B$. Bruhat decomposition identifies these with the Weyl group, whence

$$[m_{(1,1)}]\,\mathrm{SMG}_{G,G/B} = |W|. \tag{190}$$

In type A_{n-1} this is $n!$, so even the degree-two coefficient cannot stabilize as rank grows.

42.4 Exact verification method

The accompanying marimo notebook performs the following independent steps.

1. Enumerate every finite-field matrix and retain the required determinant or invertibility condition.
2. Induce its permutation on nonzero vectors, normalized projective representatives, or incident point–hyperplane flags.
3. Count orbits directly on tuples of multisets, without using character or cycle formulas.
4. Extract the same coefficient from the actual cycle counts in (2).
5. Reconstruct every cycle count from fixed points of powers using (3).
6. At $(t_1, t_2) = (0.07, 0.11)$, compare the direct matrix-determinant average with the cycle-product average.

The last two numerical averages both equal 1.247556190416 for the $PSL_3(2)$ projective action, with difference below displayed precision.

Action	τ	direct orbits	cycle formula	claimed
$PSL_3(2)$ on $\mathbb{P}^2(\mathbb{F}_2)$	(1)	1	1	1
	(2)	2	2	2
	$(1,1)$	2	2	2
	(3)	4	4	4
	$(2,1)$	5	5	5
	$(1,1,1)$	6	6	6
$PSL_2(2)$ on $\mathbb{P}^1(\mathbb{F}_2)$	(3)	3	3	3
$GL_2(3)$ on nonzero vectors	(2)	3	3	3
	$(1,1)$	3	3	3
$SL_2(3)$ on nonzero vectors	$(1,1)$	4	4	4
$PGL_2(3)$ on projective points	$(1,1)$	2	2	2
$PSL_2(3)$ on projective points	$(1,1)$	2	2	2
$GL_2(2)$ on complete flags	$(1,1)$	2	2	2
$GL_3(2)$ on complete flags	$(1,1)$	6	6	6

The constructed orders are $|GL_3(2)| = 168$, $|GL_2(3)| = 48$, and $|SL_2(3)| = 24$; their projective images in degree two have orders 24 and 12. All displayed tests pass exactly.

Sharpness found by verification. The $SL_2(3)$ value 4 at $\tau = (1,1)$ differs from the $GL_2(3)$ value 3. Here $n = |\tau| = 2$, so there is no unused complement on which to correct the determinant. Independent ordered pairs split into two $SL_2(3)$ orbits according to their determinant. This confirms that the proposed SL/GL agreement must retain the strict hypothesis $n > |\tau|$.

42.5 Scope of the conclusion

The matrix, orbit, fixed-point, and determinant routes agree for every representative case above. These computations verify the exact mechanism and the low-degree stable claims; the proofs of (1)–(8), rather than finite sampling, establish the formulas for all finite parameters in their stated ranges.

43 Finite polar-group permutation actions

Abstract

This note specializes the exact permutation σ-MGF to isotropic and singular point actions of finite unitary, symplectic, and orthogonal groups. Witt extension proves the stable three-orbit classification of pairs. The companion marimo notebook constructs $Sp_4(2)$, $O_4^+(2)$, and $U_4(2)$ as finite-field matrix groups, induces their point actions, and independently compares direct orbit counts, cycle products, fixed-point reconstruction, and matrix determinants. Every tested coefficient agrees exactly.

43.1 Exact formula for a polar action

Let a finite classical group G act on a finite set S of isotropic or singular one-spaces. For $g \in G$, let $c_d(g;S)$ denote the number of d-cycles of its action on S. The permutation-matrix eigenvalues on a d-cycle are the dth roots of unity, so

$$\boxed{\mathrm{SMG}_{G,S}(t_\bullet) = \frac{1}{|G|}\sum_{g\in G}\prod_{i,d\geq 1}(1-t_i^d)^{-c_d(g;S)}.} \tag{191}$$

Moreover,

$$\mathrm{Fix}_S(g^r) = \sum_{d|r} d\, c_d(g;S), \qquad c_d(g;S) = \frac{1}{d}\sum_{r|d}\mu(d/r)\,\mathrm{Fix}_S(g^r). \tag{192}$$

Thus one may compute (1) purely from the number of isotropic or singular eigenlines of every power g^r. Equivalently, for a parabolic point stabilizer P, this fixed-point count is $(\mathrm{Ind}_P^G \mathbf{1})(g^r)$.

For $\tau = (\tau_1, \ldots, \tau_s)$, expanding the determinants gives

$$[m_\tau]\,\mathrm{SMG}_{G,S} = \dim\left(\bigotimes_{b=1}^{s}\mathrm{Sym}^{\tau_b}\,\mathbb{C}[S]\right)^G. \tag{193}$$

The tensor product has a basis consisting of tuples of multisets of points. Because G permutes this basis, (3) is exactly the number of G-orbits on such tuples. This orbit count is the independent direct computation used below.

43.2 Why the degree-two coefficient is three

For two isotropic or singular points in a nondegenerate polar space of sufficient Witt rank, there are three possible geometric types:

1. the two points coincide;
2. they are distinct and perpendicular;
3. they are distinct and nonperpendicular.

Each relation is preserved by every isometry. Conversely, an isometry between either pair of generated subspaces extends to the ambient space by Witt's extension theorem. Hence each type is one orbit. This applies to both ordered pairs and unordered 2-multisets, proving

$$\boxed{[m_{(2)}]\,\mathrm{SMG} = [m_{(1,1)}]\,\mathrm{SMG} = 3.} \tag{194}$$

It follows that the stable expansion begins

$$\mathrm{SMG} = 1 + m_{(1)} + 3m_{(2)} + 3m_{(1,1)} + O(\deg 3).$$

The extra perpendicularity relation already distinguishes these limits from the Poisson/symmetric-group series in degree two.

Proposition (coefficientwise stability within a fixed polar tower). Fix the field, one classical type, and a total degree D. For each tuple of multisets of singular or isotropic points of total degree D, retain the labelled span together with the restricted form and the multiplicities of the labelled one-spaces; in characteristic-two orthogonal cases retain the restricted quadratic form rather than only its polar bilinear form. Two configurations in the same ambient nondegenerate polar space are conjugate precisely when these labelled formed spaces are isometric, because an isometry of their spans extends by Witt's lemma.

Only finitely many labelled formed spaces of dimension at most D exist over the fixed finite field. Let $m_0(D)$ be the maximum, over the types that occur somewhere in the chosen tower, of the least Witt rank in which the type embeds. Every coefficient of total degree D is therefore constant for Witt rank at least $m_0(D)$. Orthogonal plus, minus, and odd-dimensional towers are treated separately. This proves existence of a stable threshold but does not claim that the displayed bound is sharp. The degree-two value three was proved directly above and does not depend on this proposition.

43.3 The three exact matrix constructions

All arithmetic in the program is exact integer arithmetic.

$Sp_4(2)$**.** We enumerate $GL_4(2)$ and retain exactly the matrices preserving

$$B(x, y) = x_1 y_2 + x_2 y_1 + x_3 y_4 + x_4 y_3.$$

The resulting group has order 720. In characteristic two every one-space is isotropic, giving a permutation action on the 15 nonzero vectors (equally, the 15 projective points).

$O_4^+(2)$**.** We retain the matrices preserving the hyperbolic quadratic form

$$Q(x) = x_1 x_2 + x_3 x_4.$$

The resulting group has order 72 and acts on its 9 nonzero singular vectors.

$U_4(2)$**.** We implement $\mathbb{F}_4 = \mathbb{F}_2[\alpha]/(\alpha^2 + \alpha + 1)$ and the Hermitian form

$$h(x, y) = \sum_{j=1}^{4} \overline{x_j} y_j, \qquad \overline{z} = z^2.$$

Every ordered orthonormal basis is enumerated recursively, producing exactly

$$|U_4(2)| = 2^6 \prod_{i=1}^{4} (2^i - (-1)^i) = 77{,}760$$

matrices. Normalizing each nonzero vector by its first nonzero coordinate gives 45 isotropic projective points. The scalar center has order three, so the faithful induced permutation image has order 25,920.

43.4 Verification strategy and results

For each action, the notebook:

1. constructs the full finite-field matrix group and induced permutation;
2. directly enumerates orbits on ordered point pairs and 2-multisets;
3. extracts the same coefficients from the cycle product in (1);
4. reconstructs cycle counts from fixed points of powers using (2); and
5. numerically compares direct permutation-matrix determinants with the cycle products at $(t_1, t_2) = (0.07, 0.11)$.

The regression module also compares a degree-three labelled-formed-space orbit count in two consecutive ranks of each implemented tower. This tests the restricted-form encoding; it is not used as a proof of the proposition.

Action	τ	direct orbits	cycle formula	claimed
$Sp_4(2)$ on 15 isotropic points	(2)	3	3	3
	$(1,1)$	3	3	3
$O_4^+(2)$ on 9 singular points	(2)	3	3	3
	$(1,1)$	3	3	3
$U_4(2)$ on 45 isotropic points	(2)	3	3	3
	$(1,1)$	3	3	3

Every equality in the table is exact. All cycle counts in the tested $Sp_4(2)$ and projective $U_4(2)$ actions are also recovered exactly from fixed points of powers. For the symplectic action, the direct determinant average and cycle-product average both equal 1.304867127368, with zero difference at the displayed floating-point precision.

43.5 Representations requiring additional choices

The Weil-representation paragraph is governed by the general identity

$$\det(I - t\rho(g))^{-1} = \exp\left(\sum_{r\geq 1} \frac{\chi_\rho(g^r)}{r} t^r\right),$$

but it does not specify one numerical representation. A reproducible matrix test requires a choice of q and n, an additive character, a linearization, and whether ρ is the full Weil representation or a constituent. Even characteristic requires a different construction. Accordingly, no arbitrary choice was silently inserted into the verification.

Likewise, the exceptional/Suzuki/Ree formula is an exact template for a specified pair $(G(q), P(q))$, not a single representation. A numerical test requires the concrete group, field size, and parabolic action. Once supplied, the same fixed-point, cycle, determinant, and orbit pipeline applies without change.

Conclusion. The three classical polar families realize exactly the predicted three pair types in small but stable-rank examples. The agreement of full matrix-group enumeration, direct configuration orbits, cycle extraction, Möbius reconstruction, and determinants validates both the formula and the proposed verification method.

44 Finite local-ring permutation actions

Canonical testing artifacts.
Exact backend: `lambda_distributions/dists/local_ring_permutation_actions/verification.py`.
Regression: `lambda_distributions/dists/local_ring_permutation_actions/test_verification.py`.
Marimo: `marimo_notebooks/local_ring_actions.py`.

Scope and outcome

Let $R_a = \mathcal{O}/\pi^a\mathcal{O}$ be a finite chain ring with residue field $\mathbb{F}_q$. A subgroup of $M_n(R_a)$ is not canonically a complex matrix group; the representations studied here are instead the honest complex permutation modules $\mathbb{C}[X]$ on primitive vectors, free projective points, free summands, flags, formed-space submodules, and congruence adjoint quotients. We prove the fixed-point, cycle, and σ-MGF formulas and the low-degree orbit formulas. The fixed-level rank-stability statements are conditional on the explicit generating-tuple and formed-space extension properties stated below. An executable marimo notebook independently constructs representative groups over $\mathbb{Z}/4$ and $\mathbb{Z}/9$ and compares direct configuration orbits with Smith-kernel, homogeneous-space, Möbius-cycle, and adjoint-kernel calculations. All reported checks pass exactly.

44.1 Representation convention and universal permutation theorem

Fix a finite group H acting on a finite set X, and put $V = \mathbb{C}[X]$. For $h \in H$, write

$$F_d(h; X) = |\operatorname{Fix}_X(h^d)|, \qquad c_r(h; X) = \#\{r\text{-cycles of } h \text{ on } X\}.$$

A point on an r-cycle is fixed by h^d exactly when $r \mid d$, so

$$F_d(h; X) = \sum_{r \mid d} r c_r(h; X).$$

Möbius inversion proves

$$\boxed{c_r(h; X) = \frac{1}{r} \sum_{d \mid r} \mu(r/d) F_d(h; X).} \tag{195}$$

The permutation matrix of an r-cycle has all rth roots of unity as its eigenvalues. Its characteristic factor is therefore $1 - t^r$. Reynolds averaging and the symmetric-power identity give the exact series

$$\boxed{\mathrm{SMG}_{H,X}(t_1, t_2, \ldots) = \frac{1}{|H|} \sum_{h \in H} \prod_{i \geq 1} \prod_{r \geq 1} (1 - t_i^r)^{-c_r(h;X)}.} \tag{196}$$

Indeed, for a partition or composition $\tau = (\tau_1, \ldots, \tau_s)$,

$$[m_\tau]\,\mathrm{SMG}_{H,X} = \dim \left(\bigotimes_{j=1}^{s} \operatorname{Sym}^{\tau_j} \mathbb{C}[X] \right)^H.$$

A monomial basis for $\operatorname{Sym}^d \mathbb{C}[X]$ is indexed by d-element multisets in X. Orbit sums are a basis of the invariants, proving the independent interpretation

$$\boxed{[m_\tau]\,\mathrm{SMG}_{H,X} = \# \left(H \backslash \prod_j \operatorname{Sym}^{\tau_j} X \right).} \tag{197}$$

Equations (195)–(197) reduce every family in this note to a fixed-point problem on a finite set.

44.2 $\mathrm{GL}_n(R_a)$ on primitive vectors

Let

$$R_a = \mathcal{O}/\pi^a\mathcal{O}, \qquad R_a/\pi R_a \simeq \mathbb{F}_q, \qquad X^{\mathrm{prim}}_{n,a} = R_a^n \setminus \pi R_a^n.$$

For $A \in M_n(R_a)$, let $0 \le \nu_1(A), \dots, \nu_n(A) \le a$ be its truncated Smith exponents; a zero Smith entry receives exponent a. Define

$$K_b(A) = q^{\sum_{j=1}^n \min(\nu_j(A), b)}. \tag{198}$$

Smith reduction shows that $K_b(A)$ is the size of the kernel of A after reduction to R_b.

44.2.1 Fixed points and exact series

A primitive vector fixed by h^d lies in $\ker(h^d - I)$ but not in πR_a^n. Multiplication by π identifies R_{a-1}^n with πR_a^n, and under this identification the second kernel has size $K_{a-1}(h^d - I)$. Hence

$$\boxed{F_d^{\mathrm{prim}}(h) = K_a(h^d - I) - K_{a-1}(h^d - I).} \tag{199}$$

Substitution in (195) and (196) proves

$$\boxed{\mathrm{SMG}^{\mathrm{prim}}_{n,a} = \frac{1}{|\mathrm{GL}_n(R_a)|} \sum_h \prod_{i,r\ge 1} (1 - t_i^r)^{-\frac{1}{r}\sum_{d\mid r} \mu(r/d)[K_a(h^d-I) - K_{a-1}(h^d-I)]}.} \tag{200}$$

The identity element in (199) also gives $|X^{\mathrm{prim}}_{n,a}| = q^{an} - q^{(a-1)n}$.

44.2.2 Ordered-pair coefficient

For $n \ge 2$, transitivity allows the first primitive vector to be fixed as e_1. Write the second as $w = xe_1 + y$. Its component y in a chosen complement has valuation $r \in \{0, \dots, a\}$, with $r = a$ meaning $y = 0$. The stabilizer of e_1 moves y to $\pi^r e_2$ and changes x by an element of $\pi^r R_a$. For $r = 0$ there is one orbit. For $1 \le r \le a$, primitivity forces the residue of x modulo π^r to be a unit, giving $q^r - q^{r-1}$ orbits. Consequently

$$\boxed{[m_{(1,1)}]\,\mathrm{SMG}^{\mathrm{prim}}_{n,a} = 1 + \sum_{r=1}^{a} (q^r - q^{r-1}) = q^a.} \tag{201}$$

44.3 $\mathrm{GL}_n(R_a)$ on projective points

Let $X^{\mathrm{proj}}_{n,a} = \mathbf{P}^{n-1}(R_a)$ be the free rank-one direct summands. If $L = R_a v$ is fixed by h^d, then $h^d v = uv$ for a unique $u \in R_a^\times$. Conversely each primitive u-eigenvector generates a fixed line, and every line has $|R_a^\times|$ primitive generators. Thus

$$\boxed{F_d^{\mathrm{proj}}(h) = \frac{1}{|R_a^\times|} \sum_{u \in R_a^\times} \left[K_a(h^d - uI) - K_{a-1}(h^d - uI)\right].} \tag{202}$$

Together with (195) this yields

$$\boxed{\mathrm{SMG}^{\mathrm{proj}}_{n,a} = \frac{1}{|\mathrm{GL}_n(R_a)|} \sum_h \prod_{i,r\ge 1} (1 - t_i^r)^{-\frac{1}{r}\sum_{d\mid r} \mu(r/d) F_d^{\mathrm{proj}}(h)}.} \tag{203}$$

Counting primitive generators gives

$$|\mathbf{P}^{n-1}(R_a)| = q^{(a-1)(n-1)} \frac{q^n - 1}{q - 1}.$$

For $n \ge 2$, after fixing the first line, every second line has a unique relative-position parameter

$$R_a(e_1 + \pi^r e_2), \qquad 0 \le r \le a,$$

up to the stabilizer. The parameter is unchanged when the two lines are interchanged. It classifies both ordered pairs and two-element multisets, including equality at $r = a$. Therefore

$$\boxed{[m_{(1,1)}]\,\mathrm{SMG}^{\mathrm{proj}}_{n,a} = [m_{(2)}]\,\mathrm{SMG}^{\mathrm{proj}}_{n,a} = a + 1.} \tag{204}$$

For $n = 2$ this is also the complete-flag action. At $a = 2$ its three orbitals already exceed the field-level Weyl count $|S_2| = 2$, concretely exhibiting the extra valuation parameter in local-ring Bruhat theory.

44.4 Free summands, flags, and homogeneous spaces

Let $H = \mathrm{GL}_n(R_a)$ and let $P_{k,n-k}$ stabilize the standard free rank-k summand. For any finite homogeneous space H/P,

$$|\operatorname{Fix}_{H/P}(x)| = \frac{|C_H(x)|\,|x^H \cap P|}{|P|}. \tag{205}$$

To prove this, count $y \in H$ satisfying $y^{-1}xy \in P$. Every fixed coset has exactly $|P|$ such representatives. On the other hand, each element of $x^H \cap P$ has exactly $|C_H(x)|$ conjugators.

For $X^{(k)}_{n,a} = \mathrm{Gr}^{\mathrm{free}}_k(R^n_a) = H/P_{k,n-k}$, equations (205), (195), and (196) give

$$F^{(k)}_d(h) = \frac{|C_H(h^d)|\,|(h^d)^H \cap P_{k,n-k}|}{|P_{k,n-k}|}, \tag{206}$$

$$\mathrm{SMG}^{(k)}_{n,a} = \frac{1}{|H|}\sum_h \prod_{i,r\geq 1} (1 - t_i^r)^{-\frac{1}{r}\sum_{d|r}\mu(r/d)F^{(k)}_d(h)}. \tag{207}$$

Reduction modulo π and graph lifting over a fixed complement yield

$$|X^{(k)}_{n,a}| = q^{(a-1)k(n-k)} \begin{bmatrix} n \\ k \end{bmatrix}_q .$$

44.4.1 Pair coefficient for free summands

Assume $n \geq 2k$ and fix one summand U. For a second summand W, apply Smith reduction to the projection $W \to R^n_a/U$. Its uniquely determined exponents satisfy $0 \leq \lambda_1 \leq \cdots \leq \lambda_k \leq a$. Changes of bases in W, in U, and in a complementary k-summand put the pair in the form

$$U = \langle e_1, \ldots, e_k \rangle, \qquad W = \langle e_i + \pi^{\lambda_i} e_{k+i} : 1 \leq i \leq k \rangle.$$

Parabolic shears remove the remaining components, so the Smith exponents are also a complete invariant. There are as many nondecreasing k-tuples from $\{0, \ldots, a\}$ as multisets of size k from an $(a+1)$-element set:

$$\boxed{[m_{(1,1)}]\,\mathrm{SMG}^{(k)}_{n,a} = [m_{(2)}]\,\mathrm{SMG}^{(k)}_{n,a} = \binom{a+k}{k}.} \tag{208}$$

At $a = 1$ this reduces to the $k + 1$ field-level intersection types.

For a partial flag shape $\mathbf{d} = (d_1 < \cdots < d_s)$, put $X_{\mathbf{d},n,a} = H/P_{\mathbf{d},n}$. The same proof gives

$$F_d(h; X_{\mathbf{d},n,a}) = \frac{|C_H(h^d)|\,|(h^d)^H \cap P_{\mathbf{d},n}|}{|P_{\mathbf{d},n}|}, \tag{209}$$

followed by (195) and (196). Unlike the field case, $P_{\mathbf{d},n}\backslash H/P_{\mathbf{d},n}$ generally retains valuation data and is not indexed only by the Weyl group.

44.5 Conditional fixed-level stability as rank grows

Fix a, take $Q \leq \mathrm{GL}_d(R_a)$, and define

$$X_n(Q) = \mathrm{SplitInj}(R_a^d, R_a^n)/Q.$$

Primitive vectors, projective points, rank-k summands, and fixed-shape flags arise from $d = 1, Q = 1$; $d = 1, Q = R_a^\times$; $d = k, Q = \mathrm{GL}_k(R_a)$; and $d = d_s, Q = P_{\mathbf{d}}(R_a)$, respectively.

Let $D = |\tau|$. Choose a split frame for each of the D objects and concatenate the resulting blocks into an $n \times dD$ matrix A. Let $L(A)$ be its row submodule in R_a^{dD}. Every block projection $L(A) \to R_a^d$ is surjective. Write

$$\mathcal{L}_{\tau,d}(R_a) = \{L \leq R_a^{dD} : \text{every block projection is surjective}\}, \qquad \Gamma_{\tau,Q} = Q^D \rtimes \prod_j S_{\tau_j}.$$

Two matrices with the same row submodule are two generating n-tuples of the same finite R_a-module. We isolate the exact hypothesis used by the orbit argument:

Generating-tuple extension property $\mathrm{GT}(n, L)$**.** The group $\mathrm{GL}_n(R_a)$ is transitive on the surjections $R_a^n \twoheadrightarrow L$.

Every submodule of R_a^{dD} needs at most dD generators. Assuming $\mathrm{GT}(n, L)$ for every $L \leq R_a^{dD}$, the block changes and within-color reorderings are precisely $\Gamma_{\tau,Q}$. Hence

$$\boxed{[m_\tau]\,\mathrm{SMG}_{\mathrm{GL}_n(R_a),X_n(Q)} = \#\big(\Gamma_{\tau,Q}\backslash\mathcal{L}_{\tau,d}(R_a)\big), \qquad n \geq dD.} \tag{210}$$

The right side is independent of n. Under the same extension property, the $\mathrm{SL}_n(R_a)$ coefficient agrees in the uniform sufficient range $n > dD$: an unused coordinate corrects the determinant without moving the configuration. The conclusion then passes to projective quotients after factoring the action kernel. A proof attempt and the remaining module-theoretic lemma are recorded in the final provisional part.

44.6 Symplectic and orthogonal towers

Let G_n be a finite symplectic or orthogonal group over R_a and let X_n be a homogeneous orbit of split totally isotropic or singular free submodules or flags of fixed shape. The exact statement needs no new averaging:

$$\boxed{\mathrm{SMG}_{G_n,X_n} = \frac{1}{|G_n|}\sum_{g\in G_n}\prod_{i,r\geq 1}(1-t_i^r)^{-\frac{1}{r}\sum_{d\mid r}\mu(r/d)|\operatorname{Fix}_{X_n}(g^d)|}.} \tag{211}$$

When $X_n = G_n/P_n$, equation (205) computes its fixed points. For fixed a, shape, and coefficient degree, each configuration is supported on a formed submodule of bounded rank. Coefficient stability follows provided the relevant split unimodular Witt-extension theorem holds over R_a and every supported isometry extends in the stated tower. Under that hypothesis, Theorem 8 applies. In residue characteristic two, the quadratic form itself (not only its polar form) must be included in the support data. The unverified extension input is isolated in the final provisional part.

44.7 Reductive group schemes and adjoint congruence quotients

Let $\mathcal{G}$ be a reductive group scheme, $G_a = \mathcal{G}(R_a)$, and let ρ_a be any honest complex representation with character χ_a. Expanding each inverse determinant logarithm gives the class-compressed master formula

$$\boxed{\mathrm{SMG}_{G_a,\rho_a} = \sum_{[g]\subseteq G_a}\frac{1}{|C_{G_a}(g)|}\exp\left(\sum_{r\geq 1}\frac{\chi_a(g^r)}{r}p_r\right).} \tag{212}$$

If $X_a = G_a/P_a$ is a single parabolic orbit, combine (205) with (196). If $(\mathcal{G}/\mathcal{P})(R_a)$ has several G_a-orbits, the formula must be applied orbit by orbit; silently treating the set as one coset space would be incorrect.

Now let $c \leq a$ and $\mathfrak{g}_c = \mathrm{Lie}(\mathcal{G})(R_c)$. Let G_a act on this finite set by reduction modulo π^c and the adjoint action. Then

$$\boxed{F_d^{\mathrm{Ad},c}(g) = \left|\ker\left(\mathrm{Ad}(\bar{g}^d) - I : \mathfrak{g}_c \to \mathfrak{g}_c\right)\right|.} \tag{213}$$

If the Smith exponents of the displayed operator are $\eta_1,\ldots,\eta_N \in \{0,\ldots,c\}$, the kernel has size $q^{\eta_1+\cdots+\eta_N}$. Therefore

$$\boxed{\mathrm{SMG}_{G_a,\mathfrak{g}_c} = \frac{1}{|G_a|}\sum_{g}\prod_{i,r\geq 1}(1-t_i^r)^{-\frac{1}{r}\sum_{d\mid r}\mu(r/d)q^{\sum_j \eta_j(g^d)}}.} \tag{214}$$

44.8 Level growth and the correct asymptotic object

The rank-stable degree-two coefficients still grow with the level:

action	$[m_{(1,1)}]$ in stable rank
primitive vectors	q^a
projective points	$a+1$
rank-k free summands	$\binom{a+k}{k}$.

Thus no unnormalized coefficientwise limit exists as $a \to \infty$ for these actions. The natural replacement is the level-generating series. With the formal level-zero convention,

$$\sum_{a\geq 0} q^a u^a = \frac{1}{1-qu}, \qquad \sum_{a\geq 0}(a+1)u^a = \frac{1}{(1-u)^2}, \qquad \sum_{a\geq 0}\binom{a+k}{k}u^a = \frac{1}{(1-u)^{k+1}}. \tag{215}$$

For actual rings indexed from $a = 1$, subtract the constant term. The growth is exponential for primitive pairs and polynomial of degree 1 or k for projective points or rank-k summands.

44.9 Independent computational verification

The accompanying executable files are

`verification.py`, `marimo_notebooks/local_ring_actions.py`, `test_verification.py`.

The marimo notebook is separated into four group/action sections. It performs the following independent checks.

1. Enumerate matrices over $\mathbb{Z}/p^a$ and retain exactly those whose determinant is a unit; retain determinant one for $\mathrm{Sp}_2 = \mathrm{SL}_2$.
2. Construct primitive vectors and free rank-one summands as literal finite sets, then induce the matrix permutations.
3. Count orbits directly on tuples of multisets, without using characters, fixed-point formulas, or cycle indices.
4. Separately compute the cycle-index coefficient in (196); reconstruct cycles from fixed points of powers using (195).
5. Recover truncated Smith exponents from exact kernel sizes over every R_b and compare (199) and (202) with literal fixed points.
6. Check (205) from centralizers and conjugacy-class intersections with the projective parabolic.
7. Build all rank-two free summands of $(\mathbb{Z}/4)^4$ as unique graph lifts of planes in $\mathbb{F}_2^4$, then compute stabilizer orbits from parabolic generators.
8. Construct the adjoint action by explicit conjugation and compare every fixed-point count with (213).

44.9.1 Exact result ledger

Action	$\lvert G\rvert$	$\lvert X\rvert$	τ	direct/formula
$\mathrm{GL}_2(\mathbb{Z}/4)$, primitive	96	12	$(1,1)$	4/4
$\mathrm{GL}_2(\mathbb{Z}/4)$, projective	96	6	(2)	3/3
			$(1,1)$	3/3
$\mathrm{GL}_2(\mathbb{Z}/9)$, primitive	3888	72	$(1,1)$	9/9
$\mathrm{GL}_2(\mathbb{Z}/9)$, projective	3888	12	(2)	3/3
			$(1,1)$	3/3
$\mathrm{Sp}_2(\mathbb{Z}/4)$, isotropic lines	48	6	$(1,1)$	3/3
$\mathrm{Sp}_4(2)$, isotropic points	720	15	$(2),(1,1)$	3/3 each
$O_4^+(2)$, singular points	72	9	$(2),(1,1)$	3/3 each
$\mathrm{GL}_2(\mathbb{Z}/4)$, $M_2(\mathbb{F}_2)$ adjoint	96	16	$(1),(2),(1,1)$	$6/6, 34/34, 56/56$

The $\mathrm{GL}_2(\mathbb{Z}/4)$ Smith, homogeneous-space, and Möbius checks cover all 96 matrices and every relevant power. For $\mathrm{GL}_2(\mathbb{Z}/9)$ the orbit and cycle averages use all 3888 matrices, while the more expensive pointwise comparison uses 32 deterministic representatives. The rank-two summand construction contains exactly

$$2^{(2-1)2(4-2)} \begin{bmatrix} 4 \\ 2 \end{bmatrix}_q \Bigg|_{q=2} = 16 \cdot 35 = 560$$

summands. The stabilizer has exactly six orbits, with Smith types

$$(0,0),(0,1),(0,2),(1,1),(1,2),(2,2),$$

which is the complete list and agrees with $\binom{2+2}{2} = 6$.

Verification outcome. Ten newly tabulated coefficient comparisons pass exactly. All fixed-point/cycle reconstructions pass in the stated exhaustive or representative ranges. Four automated tests complete successfully, and the marimo notebook passes its static notebook check.

44.10 Scope and reproducibility

The proof applies to finite chain rings with residue cardinality q. The executable constructors specialize to $R_a = \mathbb{Z}/p^a\mathbb{Z}$, so their residue cardinality is $q = p$; this is sufficient to test non-field level behavior at two distinct primes. The computations prove no untested all-parameter statement: the universal Reynolds, Smith, coset-counting, and bounded-support arguments above provide those proofs.

From the workspace root, run

```
marimo run marimo_notebooks/local_ring_actions.py
pytest -q lambda_distributions/dists/local_ring_permutation_actions/test_verification.py
```

The standalone PDF generated from this source is designed to be included later, without modification, by the proof anthology's `\proofnote` mechanism.

References

[1] Y. Guo, J. Guo, and L. Huang, *Vector spaces and Grassmann graphs over residue class rings*, arXiv:1705.04610.

[2] U. Onn, A. Prasad, and L. Vaserstein, *A note on Bruhat decomposition of $GL(n)$ over local principal ideal rings*, arXiv:math/0506094.

[3] J.-P. Tignol, *Witt's extension theorem for quadratic spaces over semiperfect rings*, arXiv:1408.0522.

45 $PSL_2(q)$ on the projective line

Outcome. Formula (3.3) in the supplied statement is correct. Prime-field groups $q = 2, 3, 5, 7$ were constructed from all determinant-one 2×2 matrices and then quotiented by their projective action. All cycle-type multiplicities, 48 coefficients through degree four, and direct orbit counts agree. Determinant-level error is at most 6.67×10^{-16}.

45.1 Representation and coefficient meaning

Let $q = p^f$, $d = \gcd(2, q-1)$, and $G = PSL_2(q)$, so $|G| = q(q^2-1)/d$. Let $V = \mathbb{C}[\mathbf{P}^1(\mathbf{F}_q)]$, of dimension $q+1$. For a permutation with c_m cycles of length m,

$$\det(I - tg)^{-1} = \prod_m (1 - t^m)^{-c_m}, \qquad \prod_j \det(I - t_j g)^{-1} = \mathrm{Exp}_\sigma\left(\sum_m c_m p_m\right). \tag{1}$$

Tools 1 and 3 identify the coefficient at τ with $\dim(\bigotimes_s \mathrm{Sym}^{\tau_s} V)^G$ and with the number of G-orbits on tuples of multisets having sizes τ_s. The latter gives a genuinely independent combinatorial check of the class calculation.

45.2 Four element types

Put $L_- = (q-1)/d$ and $L_+ = (q+1)/d$.

type	cycle structure on $\mathbf{P}^1$	multiplicity
identity	1^{q+1}	1
nonidentity unipotent	$1^1 p^{q/p}$	$q^2 - 1$
split semisimple, order $m > 1$	$1^2 m^{(q-1)/m}$	$\frac{q(q+1)}{2}\varphi(m)$, $m \mid L_-$
nonsplit semisimple, order $m > 1$	$m^{(q+1)/m}$	$\frac{q(q-1)}{2}\varphi(m)$, $m \mid L_+$

The unipotent form $x \mapsto x+1$ fixes infinity and partitions the affine line into q/p cycles. A split element has two eigenlines; a nonsplit element has none. Each nontrivial semisimple element determines its eigenline pair, hence its unique torus. The split and nonsplit torus counts are $q(q+1)/2$ and $q(q-1)/2$ respectively, giving the displayed multiplicities.

45.3 Exact formula

Substituting the four rows into (1) proves

$$\begin{aligned}\mathrm{SMG}_{G,V} = \frac{1}{|G|}\Bigg[&\mathrm{Exp}_\sigma((q+1)p_1) + (q^2-1)\,\mathrm{Exp}_\sigma\left(p_1 + \frac{q}{p}p_p\right)\\ &+ \frac{q(q+1)}{2}\sum_{\substack{m \mid L_- \\ m>1}} \varphi(m)\,\mathrm{Exp}_\sigma\left(2p_1 + \frac{q-1}{m}p_m\right)\\ &+ \frac{q(q-1)}{2}\sum_{\substack{m \mid L_+ \\ m>1}} \varphi(m)\,\mathrm{Exp}_\sigma\left(\frac{q+1}{m}p_m\right)\Bigg].\end{aligned} \tag{2}$$

The multiplicities sum to $|G|$, an immediate global consistency check.

45.4 Poisson comparison and the exact boundary

Two-transitivity forces the same orbit counts as the stable symmetric-group action through total degree two:

$$[m_{(1)}] = 1, \qquad [m_{(2)}] = [m_{(1,1)}] = 2.$$

For ordered triples, equality patterns contribute $1 + 3$ orbits, while distinct triples contribute

$$\frac{(q+1)q(q-1)}{|G|} = d.$$

Thus

$$[m_{(1,1,1)}] = 4 + d = \begin{cases} 5 & q \text{ even}, \\ 6 & q \text{ odd}. \end{cases} \tag{3}$$

At degree four, equality patterns with $1, 2, 3, 4$ distinct values give

$$[m_{(1,1,1,1)}] = 1 + 7 + 6d + d(q-2) = 8 + d(q+4). \tag{4}$$

The last term is the projective cross-ratio parameter. Consequently the raw sigma–MGF has no finite coefficientwise limit as $q \to \infty$.

45.5 Verification strategy

For each prime q tested, the implementation enumerates every $(a, b, c, d) \in \mathbf{F}_q^4$ with $ad - bc = 1$, induces the fractional-linear permutation on $\mathbf{F}_q \cup \{\infty\}$, and removes scalar-kernel duplicates. It compares the actual cycle-type histogram with the four-type theorem, extracts every coefficient through degree four, independently counts orbits on tuples of multisets, and finally compares determinant averages of the permutation matrices at $(t_1, t_2) = (0.07, 0.11)$. The construction is intentionally transparent and is presently implemented for prime fields. The proof and formula cover all prime powers; testing $f > 1$ would require a small finite-field layer, not a change to the group-theoretic argument.

46 $\Delta(3n^2)$ and $\Delta(6n^2)$ monomial $SU(3)$ groups

Outcome. Both proposed formulas are correct. The standard 3-dimensional matrices were built for $n = 2, 3, 4, 5$ (orders 12–150). All 240 coefficients through total degree six agree with the exact Q_n/R_n formulas. Eight determinant-level comparisons have maximum error 8.89×10^{-16}.

46.1 Groups and imported sigma-MGF target

Put $\eta = e^{2\pi i/n}$ and

$$N_n = \{D(a, b) = \mathrm{diag}(\eta^a, \eta^b, \eta^{-a-b}) : a, b \in \mathbb{Z}/n\}.$$

The coefficient and determinant identities are Tools 1 and 2; the new content is the coset spectral calculation.

Let P be the permutation matrix of a 3-cycle. Then $\Delta(3n^2) = N_n \rtimes \langle P\rangle$. For $\pi \in S_3$ use the determinant-one lift $\widetilde{P}_\pi = \mathrm{sgn}(\pi)P_\pi$; then $\Delta(6n^2) = N_n \rtimes \widetilde{S}_3$.

46.2 Coset proof

Define

$$\mathcal{A}_n = \frac{1}{n^2}\sum_{a,b}\prod_j \frac{1}{(1-\eta^a t_j)(1-\eta^b t_j)(1-\eta^{-a-b}t_j)}. \tag{2}$$

For DP or DP^2, the three entries around the monomial 3-cycle multiply to $\det D = 1$, so $\det(I - tDP) = 1 - t^3$. Therefore

$$\boxed{\mathrm{SMG}_{\Delta(3n^2)} = \tfrac{1}{3}\mathcal{A}_n + \tfrac{2}{3}\,\mathrm{Exp}_\sigma(p_3).} \tag{3}$$

For a transposition coset in the determinant-one lift, let α be the product around the 2-cycle and β the fixed-coordinate entry. Since the underlying permutation has determinant -1, $-\alpha\beta = 1$ and $\beta = -\alpha^{-1}$. Thus

$$\det(I - tg) = (1 - \alpha t^2)(1 + \alpha^{-1}t).$$

As $D(a, b)$ varies, α is uniform in μ_n. Put

$$\mathcal{B}_n = \frac{1}{n}\sum_{a \bmod n}\prod_j \frac{1}{(1-\eta^a t_j^2)(1+\eta^{-a}t_j)}. \tag{4}$$

The identity, 3-cycle, and transposition proportions in S_3 are 1/6, 1/3, and 1/2. Hence

$$\boxed{\mathrm{SMG}_{\Delta(6n^2)} = \tfrac{1}{6}\mathcal{A}_n + \tfrac{1}{3}\,\mathrm{Exp}_\sigma(p_3) + \tfrac{1}{2}\mathcal{B}_n.} \tag{5}$$

46.3 Exact coefficient formulas

For $\tau = (\tau_1, \ldots, \tau_\ell)$, let $Q_n(\tau)$ count rows (x_s, y_s, z_s) with $x_s + y_s + z_s = \tau_s$ and $\sum_s x_s \equiv \sum_s y_s \equiv \sum_s z_s \pmod{n}$. Root-of-unity orthogonality gives $[m_\tau]\mathcal{A}_n = Q_n(\tau)$.

Let

$$R_n(\tau) = \sum_{\substack{x_s+2y_s=\tau_s \\ \sum y_s \equiv \sum x_s\ (n)}} (-1)^{\sum x_s}.$$

This follows by expanding $(1+\alpha^{-1}t)^{-1}$ and $(1-\alpha t^2)^{-1}$. Finally, $[m_\tau]\,\mathrm{Exp}_\sigma(p_3)$ is 1 exactly when every τ_s is divisible by 3. Therefore

$$[m_\tau]\,\mathrm{SMG}_{\Delta(3n^2)} = \tfrac{1}{3}Q_n(\tau) + \tfrac{2}{3}\mathbf{1}_{3|\tau_s\ \forall s}, \tag{6}$$

$$[m_\tau]\,\mathrm{SMG}_{\Delta(6n^2)} = \tfrac{1}{6}Q_n(\tau) + \tfrac{1}{3}\mathbf{1}_{3|\tau_s\ \forall s} + \tfrac{1}{2}R_n(\tau). \tag{7}$$

Their integrality is automatic from (1), even though (6)–(7) display rational weights.

46.4 Verification design and results

For each n, all matrices $D(a,b)P^k$ and $D(a,b)\widetilde{P}_\pi$ are constructed. The code checks group size and determinant one. It then compares: (i) eigenvalue-based symmetric-power character averages; (ii) exact enumeration of Q_n and R_n; and (iii) the direct determinant average at $(0.07, 0.11)$ against (3) or (5).

n	$\lvert\Delta(3n^2)\rvert$	$\lvert\Delta(6n^2)\rvert$	degree	checks/family	max. error	result
2	12	24	0–6	30	4.45×10^{-16}	**PASS**
3	27	54	0–6	30	6.67×10^{-16}	**PASS**
4	48	96	0–6	30	6.67×10^{-16}	**PASS**
5	75	150	0–6	30	8.89×10^{-16}	**PASS**

There are two families for each row, hence $2 \cdot 4 \cdot 30 = 240$ exact coefficient checks. As useful spot checks, for $n = 2$ the coefficients at $\tau = (3), (2,1), (1,1,1)$ are $(1,1,2)$ for $\Delta(12)$ and $(0,0,1)$ for $\Delta(24)$.

46.5 Novel diagnostic

The coset decomposition is also a debugging tool. A mismatch isolated to $\mathcal{B}_n$ almost always signals the missed minus sign in the determinant-one lift of a transposition. A mismatch isolated to $\mathrm{Exp}_\sigma(p_3)$ signals an incorrect monomial-cycle product. This three-channel comparison is stronger than a single ordinary Molien specialization.

Reproduction

```
.venv/bin/python -m \
  for_this_guy.generalized_molien_extensions.delta_su3.verification
```

47 Code–monomial matrix groups

Outcome. The proposed weighted-cycle formula and the dual-code orbit formula are correct under the stated prime-alphabet convention. Explicit matrices for three groups of orders 8, 24, and 18 were used to check all 57 coefficients through total degree five. All agree. Three independent determinant evaluations agree to machine precision (maximum error 0).

47.1 Imported coefficient identity

For the code–monomial module V, put $\mathrm{SMG}_{G_C,V} = \mathbb{E}[\mathrm{Exp}_\sigma(X_{G_C,V} h_1)]$. The determinant and coefficient identities are imported; the new theorem is the cycle-decorated dual-code filter.

47.2 Code–monomial theorem

Let r be prime, $C \le \mathbf{F}_r^n$, and let

$$H \le \mathrm{Aut}_{\mathrm{perm}}(C) := \{\pi \in S_n : \pi(C) = C\}$$

be a coordinate-permutation automorphism group. Put $\zeta = e^{2\pi i/r}$ and define

$$D_c = \mathrm{diag}(\zeta^{c_1}, \dots, \zeta^{c_n}), \qquad G_C = D_C \rtimes H.$$

Use $P_\pi e_i = e_{\pi(i)}$. On a coordinate cycle O of π, the weighted shift $D_c P_\pi$ returns after $|O|$ steps with scalar $\zeta^{c(O)}$, where $c(O) = \sum_{i \in O} c_i$. Therefore

$$\det(I - t D_c P_\pi) = \prod_{O \in \mathrm{Cyc}(\pi)} (1 - \zeta^{c(O)} t^{|O|}). \tag{2}$$

Substitution into the master average proves

$$\boxed{\mathrm{SMG}_{G_C,V}(\mathbf{t}) = \frac{1}{|C||H|} \sum_{\pi \in H} \sum_{c \in C} \prod_{j,O} (1 - \zeta^{c(O)} t_j^{|O|})^{-1}.} \tag{3}$$

47.2.1 Why the coefficient is an orbit count

A monomial basis of $\bigotimes_s \mathrm{Sym}^{\tau_s} V$ is indexed by arrays $A = (a_{si}) \ge 0$ with $\sum_i a_{si} = \tau_s$. Set

$$w(A)_i = \sum_s a_{si} \pmod r.$$

The diagonal matrix D_c multiplies the basis vector by $\zeta^{c \cdot w(A)}$. Character orthogonality says it survives all diagonal matrices precisely when $w(A) \in C^\perp$. The remaining group H permutes columns, so invariants in this permutation module are constant on H-orbits. Hence

$$\boxed{[m_\tau]\,\mathrm{SMG}_{G_C,V} = \#\big(H \backslash \{A : \sum_i a_{si} = \tau_s,\ w(A) \in C^\perp\}\big).} \tag{4}$$

This also proves integrality without relying on cancellation in (3).

The proof uses only coordinate permutations and therefore does not cover the full monomial or semilinear automorphism group of a code. If coordinate scalings or field automorphisms are included, their characters must be incorporated into the diagonal filter and the column-orbit argument must be replaced by the corresponding semilinear action. The verifier rejects an automorphism entry that is not a coordinate permutation.

47.3 Independent verification method

The implementation uses routes that share the group input but not the coefficient algorithm:

1. Enumerate every complex matrix $D_c P_\pi$. Obtain $\chi_{\mathrm{Sym}^d V}(g)$ from the eigenvalue series $\prod_i (1 - t\lambda_i)^{-1}$ and average products of characters.
2. Enumerate exponent arrays, impose $c \cdot w(A) = 0$ for every $c \in C$, and count column-permutation orbits. No eigenvalues occur in this route.
3. At $(t_1, t_2) = (0.07, 0.11)$ compare the direct matrix determinant average with (3), evaluated solely from weighted coordinate cycles.

47.4 Representative cases and results

Every partition of every total degree 0 through 5 was checked (19 coefficients per group).

Case	$\dim V$	$\lvert C\rvert$	$\lvert H\rvert$	checks	result
$W(B_2)$, full binary code	2	4	2	19	**PASS**
$W(D_3)$, even binary code	3	4	6	19	**PASS**
ternary repetition code $\rtimes S_3$	3	3	6	19	**PASS**

Case	(2)	(3)	$(2,1)$	$(1,1,1)$
$W(B_2)$	1	0	0	0
$W(D_3)$	1	1	1	1
ternary repetition	0	3	4	5

The numerical generalized-Molien values by both routes were

$$1.025494342880870, \qquad 1.028638382342067, \qquad 1.010621973731615.$$

47.5 Scope and reusable tool

For $\mathbf{F}_{p^f}$ with $f > 1$, the notation $c \mapsto \zeta^c$ is not a faithful additive embedding of the whole additive group into one cyclic root group. Formula (4) remains valid after replacing $C^\perp$ by the annihilator of the chosen diagonal-character group. The reusable computational idea is to test (3) numerically while testing (4) exactly: agreement localizes errors to either the weighted-cycle determinant or the character-orthogonality filter.

47.6 Extended equivariant code-monomial theory

Canonical testing artifacts.
Exact backend: `lambda_distributions/dists/code_monomial_sigma_mgfs/verification.py`.
Regression: `lambda_distributions/dists/code_monomial_sigma_mgfs/test_verification.py`.
Marimo: `marimo_notebooks/code_monomial.py`.

Scope and outcome

Let $C \le (\mathbb{Z}/r\mathbb{Z})^N$ be an additive code and let a coordinate-permutation group $P \le S_N$ preserve C. We construct the honest unitary monomial group $G(C,P) = D_C \rtimes P \le U(N)$ and prove two equivalent formulas for its complete generalized Molien, or σ-MGF, series: a weighted-cycle average over $C \times P$ and a root-filtered sum over dual codewords that are constant on permutation cycles. We also prove the exact coefficient-orbit interpretation, the complete-weight-enumerator specialization for $P = 1$, and the resulting coefficientwise convergence criterion and orbit-growth obstruction. An accompanying marimo notebook constructs five groups of dimensions 2 through 4 and orders $4, 8, 24, 18, 64$. All 95 coefficients through total degree five agree exactly; three independent evaluations of the full series agree to 4.45×10^{-16}.

47.6.1 Representation and target quantity

Fix an integer $r \ge 2$ and write $R = \mathbb{Z}/r\mathbb{Z}$ and $\zeta = e^{2\pi i/r}$. Let $C \le R^N$ be an additive subgroup. For $c = (c_1, \ldots, c_N) \in C$, define

$$D(c) = \operatorname{diag}(\zeta^{c_1}, \ldots, \zeta^{c_N}), \qquad D_C = \{D(c) : c \in C\}.$$

Let $P \le S_N$ preserve C under coordinate permutation, and use the convention $P_\pi e_j = e_{\pi(j)}$. Then conjugation by P_π permutes the diagonal entries of $D(c)$ and preserves D_C. Hence

$$G(C,P) = \{D(c)P_\pi : c \in C,\ \pi \in P\} = D_C \rtimes P \le U(N).$$

The parametrization by $C \times P$ is injective, so $|G(C,P)| = |C||P|$.

For any finite $G \le U(V)$ define

$$\mathrm{SMG}_{G,V}(\mathbf{t}) = \frac{1}{|G|} \sum_{g \in G} \prod_{i \ge 1} \det(I - t_i g)^{-1}. \tag{216}$$

For a partition $\tau = (\tau_1, \ldots, \tau_s)$, Tool 1 gives

$$[m_\tau]\, \mathbb{E}[\operatorname{Exp}_\sigma(X_{G,V} h_1)] = \dim \left(\bigotimes_{\alpha=1}^{s} \operatorname{Sym}^{\tau_\alpha} V \right)^G. \tag{217}$$

The new work is the dual-code and cycle-decorated evaluation of this imported coefficient.

47.6.2 Weighted-cycle and equivariant dual-code formulas

For $a \in R$, define the formal series

$$H_a(\mathbf{t}) = \sum_{\substack{\nu_1, \nu_2, \ldots \ge 0 \\ \sum_i \nu_i \equiv a \pmod r}} \prod_i t_i^{\nu_i}. \tag{218}$$

Character orthogonality on R gives the root filter

$$\boxed{H_a(\mathbf{t}) = \frac{1}{r} \sum_{s=0}^{r-1} \zeta^{-as} \prod_i (1 - \zeta^s t_i)^{-1}.} \tag{219}$$

For a positive integer ℓ, put $H_a^{[\ell]}(\mathbf{t}) = H_a(t_1^\ell, t_2^\ell, \ldots)$, and define

$$C^\perp = \{u \in R^N : \langle c, u \rangle = 0 \text{ in } R \text{ for every } c \in C\}.$$

Theorem 1 (Equivariant dual-code σ-MGF)**.** *For $G(C,P)$ on $V=\mathbb{C}^N$,*

$$\mathrm{SMG}_{C,P}(\mathbf{t}) = \frac{1}{|C||P|}\sum_{\pi\in P}\sum_{c\in C}\prod_{i\geq 1}\prod_{Q\in\mathrm{Cyc}(\pi)}\left(1-\zeta^{\sum_{j\in Q}c_j}t_i^{|Q|}\right)^{-1}, \tag{220}$$

$$= \frac{1}{|P|}\sum_{\pi\in P}\sum_{\substack{u\in C^\perp \\ u \text{ constant on every cycle of } \pi}}\prod_{Q\in\mathrm{Cyc}(\pi)} H_{u_Q}^{[|Q|]}(\mathbf{t}), \tag{221}$$

where u_Q is the common value of u on Q.

Proof. Consider a cycle $Q=(j_1\, j_2\, \cdots\, j_\ell)$ of π. On its coordinate subspace, $D(c)P_\pi$ is a weighted cyclic shift. Its ℓth power is scalar multiplication by $\zeta^{c_{j_1}+\cdots+c_{j_\ell}}$, and therefore

$$\det(I-tD(c)P_\pi \mid \mathbb{C}^Q) = 1-\zeta^{\sum_{j\in Q}c_j}t^\ell.$$

Multiplication over the disjoint cycles, over the variables t_i, and then averaging over $C\times P$ proves (220).

For a fixed cycle Q, expand its factor as

$$\prod_i\left(1-\zeta^{\sum_{j\in Q}c_j}t_i^{|Q|}\right)^{-1} = \sum_{\nu_1,\nu_2,\ldots\geq 0}\zeta^{(\sum_i\nu_i)(\sum_{j\in Q}c_j)}\prod_i t_i^{|Q|\nu_i}.$$

Assign to every $j\in Q$ the residue $u_j=\sum_i \nu_i \pmod r$. The resulting word u is constant on every cycle of π, and the total phase is $\zeta^{\langle c,u\rangle}$. Orthogonality on the finite additive group C gives

$$\frac{1}{|C|}\sum_{c\in C}\zeta^{\langle c,u\rangle} = \begin{cases}1, & u\in C^\perp,\\ 0, & u\notin C^\perp.\end{cases}$$

After this filter, the sum of the allowed exponents on Q is precisely $H_{u_Q}^{[|Q|]}$. Multiplying over cycles proves (221). This orthogonality argument is valid over $\mathbb{Z}/r\mathbb{Z}$ for composite as well as prime r. □

47.6.3 Pure diagonal groups and complete weight enumerators

When $P=1$, every coordinate is a one-cycle. Define the complete weight enumerator of a code $A\leq R^N$ by

$$\mathrm{cwe}_A(X_0,\ldots,X_{r-1}) = \sum_{u\in A}\prod_{j=1}^N X_{u_j}.$$

Corollary 1 (Pure diagonal specialization)**.** *For the diagonal group D_C,*

$$\boxed{\mathrm{SMG}_{D_C} = \mathrm{cwe}_{C^\perp}(H_0,\ldots,H_{r-1}).} \tag{222}$$

For $r=2$ this becomes

$$H_0=\tfrac{1}{2}\left[\prod_i(1-t_i)^{-1}+\prod_i(1+t_i)^{-1}\right],\quad H_1=\tfrac{1}{2}\left[\prod_i(1-t_i)^{-1}-\prod_i(1+t_i)^{-1}\right],$$

so $\mathrm{SMG}_{D_C}=W_{C^\perp}(H_0,H_1)$ for the ordinary binary weight enumerator W.

Proof. Set $P=1$ in (221). The contribution of a fixed $u\in C^\perp$ is $\prod_j H_{u_j}$, exactly the corresponding monomial in the complete weight enumerator. □

This is the first verification family below. It isolates the diagonal character filter before any coordinate-orbit quotient is imposed.

47.6.4 Exact coefficient formula

For $\tau = (\tau_1, \ldots, \tau_s)$, let $\mathcal{E}_\tau(C^\perp)$ be the set of matrices $A = (a_{\alpha j}) \in \mathbb{Z}_{\geq 0}^{s \times N}$ satisfying

$$\sum_{j=1}^{N} a_{\alpha j} = \tau_\alpha \quad (1 \leq \alpha \leq s), \qquad \left(\sum_\alpha a_{\alpha 1}, \ldots, \sum_\alpha a_{\alpha N} \right) \bmod r \in C^\perp.$$

The group P acts on these arrays by permuting columns.

Theorem 2 (Dual-code configuration orbits)**.** *For every partition τ,*

$$\boxed{[m_\tau]\,\mathrm{SMG}_{C,P} = \#\big(P \backslash \mathcal{E}_\tau(C^\perp)\big).} \tag{223}$$

Proof. A monomial basis of $\bigotimes_\alpha \mathrm{Sym}^{\tau_\alpha} \mathbb{C}^N$ is indexed by the exponent arrays with the stated row sums. The matrix $D(c)$ multiplies the basis vector indexed by A by

$$\zeta^{\sum_j c_j \sum_\alpha a_{\alpha j}}.$$

It is fixed by every $D(c)$ exactly when the vector of total coordinate exponents lies in $C^\perp$ modulo r. Thus the D_C-invariant monomial basis is indexed by $\mathcal{E}_\tau(C^\perp)$. The quotient group P permutes this basis by permuting columns, and its orbit sums form a basis of the invariant space. Equation (217) completes the proof. □

47.6.5 Coefficientwise asymptotics

Consider a sequence (C_N, P_N) with a fixed modulus r. The exact orbit formula immediately gives

$$\mathrm{SMG}_{C_N,P_N} \text{ converges coefficientwise} \quad \Longleftrightarrow \quad \#\big(P_N \backslash \mathcal{E}_\tau(C_N^\perp)\big) \text{ is eventually constant for every fixed } \tau. \tag{224}$$

Here ordinary convergence of a nonnegative integer sequence is equivalent to eventual constancy.

There is a code-independent obstruction. In each copy of $\mathrm{Sym}^r \mathbb{C}^N$, the vectors x_j^r are fixed by every $D(c)$ because $\zeta^{rc_j} = 1$. Consequently, for $\tau = (r, \ldots, r)$ with s parts,

$$\boxed{[m_{(r^s)}]\,\mathrm{SMG}_{C_N,P_N} \geq \#\big(P_N \backslash [N]^s\big).} \tag{225}$$

Thus bounded orbit counts on every fixed Cartesian power of the coordinate set are necessary for a raw coefficientwise limit. Two immediate examples are useful:

1. If $P_N = 1$, then $[m_{(r)}]\,\mathrm{SMG} \geq N$, so every length-growing pure diagonal family diverges in a fixed degree.
2. If P_N is the regular cyclic shift group, then it has N orbits on ordered coordinate pairs, so $[m_{(r,r)}]\,\mathrm{SMG} \geq N$.

Weight-enumerator convergence alone cannot control these coordinate-orbit counts. Conversely, (224) identifies the extra information needed: bounded-degree *equivariant* dual-code configurations.

47.6.6 Independent verification strategy

The executable verifier uses four logically separated calculations.

1. **Matrix construction.** It creates every dense complex matrix $D(c)P_\pi$, checks the expected group cardinality and distinctness, and checks $g^* g = I$.
2. **Direct full-series value.** At $(t_1, t_2) = (0.07, 0.11)$ it evaluates (216) using `numpy.linalg.det` on the constructed matrices.
3. **Two full-series comparators.** It evaluates (220) from codeword cycle sums and (221) by enumerating $C^\perp$, testing cycle constancy, and evaluating (219).

4. **Coefficient comparison.** For the matrix route it obtains $\mathrm{Tr}(\mathrm{Sym}^d g)$ from traces of actual matrix powers using Newton's recurrence
$$d\, h_d(g) = \sum_{k=1}^{d} \mathrm{Tr}(g^k) h_{d-k}(g), \qquad h_0 = 1,$$
then averages products over the matrix group. The independent exact route enumerates exponent arrays, imposes the dual-code condition, and counts P-orbits as in Theorem 2.

The proof is general; these computations are regression tests, not a finite substitute for the proof.

47.6.7 Verification organized by group

Every partition of every total degree from 0 through 5 was tested: 19 coefficients for each of five groups, or 95 exact comparisons in total.

Pure diagonal group For $C = (\mathbb{Z}/2\mathbb{Z})^2$ and $P = 1$, the group has order 4 on $\mathbb{C}^2$. This case separately tests Corollary 1; in particular the degree-two coefficient is 2 before the coordinate swap identifies the two squares.

Binary semidirect groups The full binary code of length two with $P = S_2$ produces the order-8 signed permutation group $W(B_2)$. The even binary code of length three with $P = S_3$ produces the order-24 group $W(D_3)$. These test nontrivial cycle types and dual codes of different dimensions.

Ternary semidirect group For the ternary repetition code $C = \{(a, a, a) : a \in \mathbb{Z}/3\mathbb{Z}\}$ and $P = S_3$, the constructed group has order 18 on $\mathbb{C}^3$. Its nonzero coefficients occur in a substantially different degree pattern from the binary groups.

Composite cyclic-alphabet group For
$$C = \{(a, b, a, b) : a, b \in \mathbb{Z}/4\mathbb{Z}\} \leq (\mathbb{Z}/4\mathbb{Z})^4$$
and the cyclic coordinate group generated by $(1\,2\,3\,4)$, the constructed group has order 64 on $\mathbb{C}^4$. This case directly tests the composite modulus asserted in Theorem 1.

Group	r	N	$\lvert C\rvert$	$\lvert P\rvert$	$\lvert G\rvert$	result
$D_{(\mathbb{Z}/2)^2}$, pure diagonal	2	2	4	1	4	**PASS**
$W(B_2)$	2	2	4	2	8	**PASS**
$W(D_3)$	2	3	4	6	24	**PASS**
ternary repetition $\rtimes S_3$	3	3	3	6	18	**PASS**
quaternary pair code $\rtimes C_4$	4	4	16	4	64	**PASS**

Selected coefficients are shown below; all remaining partitions through degree five are checked by the notebook and test suite.

Group	(1)	(2)	(3)	$(2,1)$	$(1,1,1)$	(4)
$D_{(\mathbb{Z}/2)^2}$	0	2	0	0	0	3
$W(B_2)$	0	1	0	0	0	2
$W(D_3)$	0	1	1	1	1	2
ternary repetition $\rtimes S_3$	0	0	3	4	5	0
quaternary pair $\rtimes C_4$	0	0	0	0	0	3

At the full-series test point the direct matrix values are respectively

1.0507589401246002, 1.0254943428808700, 1.0286383823420668, 1.0106219737316153, 1.0013332933694528.

Both formula routes reproduce every value. Across all groups, the maximum full-series discrepancy is 4.45×10^{-16}, the maximum coefficient rounding residual is 2.23×10^{-14}, and the maximum unitarity residual is 2.23×10^{-16}.

47.6.8 Scope and reproduction

The literal scalar model uses the cyclic additive alphabet $\mathbb{Z}/r\mathbb{Z}$. For a noncyclic alphabet such as the additive group of $\mathbf{F}_{p^s}$, there is no faithful map of the entire alphabet into a single cyclic root group by $a \mapsto \zeta^a$. One must instead choose the appropriate family of additive characters, character module, or Clifford–Weil representation. Likewise, a monomial automorphism involving coordinate scalings, or a semilinear automorphism involving a field automorphism, is not merely a coordinate permutation in P and needs a correspondingly generalized cycle calculation.

It is also useful to separate three levels of code information. For a pure diagonal binary group, the ordinary Hamming weight enumerator of $C^\perp$ determines the answer through Corollary 1. For $r > 2$, the ordinary weight enumerator forgets the symbols and must be replaced by the complete weight enumerator. Once P is nontrivial, even the complete weight enumerator forgets how the coordinate group acts. The exact required datum is the equivariant collection of orbit counts in (223), equivalently a cycle-indexed complete-weight enumerator. Thus ordinary weight-enumerator convergence does not by itself imply σ-MGF convergence.

The finite audit deliberately exercises the following boundaries:

1. $P = 1$ versus nontrivial P, so the complete-weight and equivariant formulas are tested separately;
2. prime moduli $2, 3$ versus the composite modulus 4;
3. full, even, repetition, and two-generator pair codes;
4. identity, symmetric, and cyclic coordinate groups.

It does not claim to test every asymptotic code family mentioned in broader surveys. Equations (224) and (225) are the proved reusable tests for such a family once its codes and coordinate groups are specified.

From the repository root, run

```
.venv/bin/python -m lambda_distributions.dists.code_monomial_sigma_mgfs.verification
.venv/bin/pytest -q lambda_distributions/dists/code_monomial_sigma_mgfs/test_verification.py
.venv/bin/marimo edit marimo_notebooks/code_monomial.py
```

The source and PDF in this directory are standalone artifacts intended for later inclusion. No combined PDF is edited or rebuilt.

Part V

Braid, Pauli, extraspecial, Clifford, and Barnes–Wall groups

48 Scalar cyclic finite braid images

Abstract

Using the imported sigma–MGF identities, we compute the scalar cyclic quotients of B_3 in arbitrary representation dimension. These examples isolate the scalar-center selection rule. Explicit Reynolds-projector ranks agree with both the elementwise power-trace formula and its conjugacy-class grouping in all tested dimensions $D = 1, 2, 3, 4$.

48.1 The representation and the proposed quantity

Fix $m \geq 2$, $D \geq 1$, and $\zeta = e^{2\pi i/m}$. Define

$$\rho(\sigma_1) = \rho(\sigma_2) = \zeta I_D.$$

The matrices are unitary and satisfy $\rho(\sigma_1)\rho(\sigma_2)\rho(\sigma_1) = \rho(\sigma_2)\rho(\sigma_1)\rho(\sigma_2)$, so this is a D-dimensional representation of B_3. Its image is C_m.

For a tuple $\tau = (\tau_1, \ldots, \tau_\ell)$ let

$$W_\tau = \bigotimes_{a=1}^{\ell} \mathrm{Sym}^{\tau_a}(\mathbb{C}^D), \qquad |\tau| = \sum_a \tau_a.$$

The coefficient under test is

$$[m_\tau]\,\mathrm{SMG}_{C_m}(\mathbf{t}), \qquad \mathrm{SMG}_{C_m}(\mathbf{t}) = \frac{1}{m}\sum_{j=0}^{m-1}\prod_{a\geq 1}\det(I - t_a\zeta^j I_D)^{-1}.$$

48.2 Scalar specialization of the imported formula

Tools 1 and 2 supply the common finite-group identities. In the present representation, $\zeta^j I_D$ acts on all of W_τ by $\zeta^{j|\tau|}$. Hence the Reynolds operator is either zero or identity:

$$\boxed{[m_\tau]\,\mathrm{SMG}_{C_m} = \mathbf{1}_{m||\tau|}\prod_a \binom{D+\tau_a-1}{\tau_a}.}$$

This proves the scalar selection rule and also determines the nonzero value, not merely its support.

48.3 Independent verification strategy

The program first closes the matrix group from the displayed generator and asserts order m, unitarity, and the braid relation. It then constructs an orthonormal occupation basis of every $\mathrm{Sym}^r\,\mathbb{C}^D$ inside $(\mathbb{C}^D)^{\otimes r}$. In that basis it forms

$$P_\tau = \frac{1}{m}\sum_{g\in C_m}\bigotimes_a \mathrm{Sym}^{\tau_a}(g)$$

and computes its rank. This is the direct target calculation.

The comparator never uses this projector. It obtains the complete homogeneous characters from Newton's recurrence

$$rh_r(g) = \sum_{k=1}^{r} \mathrm{Tr}(g^k) h_{r-k}(g), \qquad h_0 = 1,$$

then averages $\prod_a h_{\tau_a}(g)$ elementwise and, separately, by matrix conjugacy classes.

m	D	τ	$\dim W_\tau$	$\operatorname{rank} P_\tau$	formula
3	1	(1)	1	0	0
3	1	(3)	1	1	1
3	2	$(2,1)$	6	6	6
4	2	(4)	5	5	5
4	3	$(2,2)$	36	36	36
5	4	(5)	56	56	56

The maximum projector idempotence/Hermiticity error in this sweep is below 6×10^{-15}.

48.4 Reproduction and scope

From the repository root run

```
.venv/bin/python for_this_guy/finite_braid_images/
  cyclic_scalar/verification.py
.venv/bin/marimo edit marimo_notebooks/braid_cyclic_scalar.py
```

after joining each wrapped command. The proof is valid for every m, D, τ; the table is a small numerical regression suite, not evidence substituted for the proof.

49 The permutation braid image $B_3 \twoheadrightarrow S_3$

Abstract

We construct the adjacent-transposition quotient of B_3 in the three-dimensional permutation representation. We derive its sigma–MGF coefficient formula from the three spectral types of S_3, and independently recover the same integers as ranks of explicit Reynolds projectors on tensor products of symmetric powers.

49.1 Matrix group and braid relation

Let

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}.$$

These unitary matrices obey $ABA = BAB$. Sending $\sigma_1 \mapsto A$ and $\sigma_2 \mapsto B$ therefore defines a representation of B_3. Matrix closure gives the six permutation matrices, so the image is S_3 acting on $V = \mathbb{C}^3$.

49.2 Imported target and the three-class formula

For $W_\tau = \bigotimes_a \operatorname{Sym}^{\tau_a} V$, Tools 1 and 2 give

$$[m_\tau]\, \mathbb{E}[\operatorname{Exp}_\sigma(X_{S_3,V} h_1)] = \dim(W_\tau^{S_3}) = \frac{1}{6} \sum_{g \in S_3} \prod_a h_{\tau_a}(g).$$

The representation-specific input is that the three conjugacy classes have eigenvalues

$$(1,1,1), \qquad (1,1,-1), \qquad (1,\omega,\omega^2), \quad \omega^3 = 1.$$

Their generating series are respectively

$$\frac{1}{(1-t)^3}, \qquad \frac{1}{(1-t)(1-t^2)}, \qquad \frac{1}{1-t^3}.$$

Consequently, with

$$A_r = \binom{r+2}{2}, \qquad B_r = \left\lfloor \frac{r}{2} \right\rfloor + 1, \qquad C_r = \mathbf{1}_{3|r},$$

the proposed class formula becomes the completely explicit identity

$$\boxed{[m_\tau]\, \mathrm{SMG}_{S_3} = \frac{1}{6}\left(\prod_a A_{\tau_a} + 3 \prod_a B_{\tau_a} + 2 \prod_a C_{\tau_a} \right).}$$

This is also the character-and-power-map formula: Newton's recurrence $r h_r = \sum_{k=1}^r \operatorname{Tr}(g^k) h_{r-k}$ is obtained by taking the logarithm of the determinant generating series.

49.3 Direct versus formula verification

The direct route embeds the normalized occupation basis of $\operatorname{Sym}^r V$ into $V^{\otimes r}$, restricts $g^{\otimes r}$ to that basis, forms the tensor product over the parts of τ, and ranks the averaged matrix. The two formula routes compute the power-trace recurrence over all six matrices and over the three constructed conjugacy classes. Thus the direct route does not diagonalize a Reynolds projector by using its character in advance.

τ	$\dim W_\tau$	projector rank	six elements	three classes
(1)	3	1	1	1
(2)	6	2	2	2
$(1,1)$	9	2	2	2
(3)	10	3	3	3
$(2,1)$	18	4	4	4
$(2,2)$	36	8	8	8

All matrices close to exactly six elements, exactly three conjugacy classes are recovered, and the largest projector error is below 8×10^{-16}.

49.4 Reproduction

From the repository root run

```
.venv/bin/python for_this_guy/finite_braid_images/
  s3_permutation/verification.py
.venv/bin/marimo edit marimo_notebooks/braid_s3_permutation.py
```

after joining wrapped lines. This example verifies the universal finite-image claim in a non-scalar braid quotient; it is not asserted to be a particular fusion-space representation of a chosen modular category.

50 The finite Ising/Jones braid image

Abstract

We test the finite-image formulas on the two-dimensional Ising/Jones braid representation. The displayed braid generators close to a 192-element unitary lift with scalar center of order eight. We verify its one-sided sigma–MGF and the full-twist selection rule. We then pass to conjugation on $\mathrm{End}(\mathbb{C}^2)$, obtaining the canonical 24-element projective image in dimension four, and verify the projective formula there.

50.1 The finite braid image

Put

$$H = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \qquad S = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}, \qquad q = e^{-\pi i/8},$$

and define

$$\rho(\sigma_1) = qS, \qquad \rho(\sigma_2) = qHSH.$$

Direct multiplication gives the braid relation. Breadth-first matrix closure gives $|G| = 192$; quotienting its eight scalar matrices gives a projective image of order 24. The B_3 full twist is

$$\rho((\sigma_1\sigma_2)^3) = e^{-\pi i/4} I.$$

It therefore acts on $W_\tau = \bigotimes_a \mathrm{Sym}^{\tau_a} \mathbb{C}^2$ by $e^{-\pi i|\tau|/4}$, proving the necessary selection rule

$$\boxed{W_\tau^G = 0 \quad \text{unless} \quad 8 \mid |\tau|.}$$

The rule is necessary, not sufficient, as the degree-eight computations show.

50.2 Imported finite-image tools

Once the 192-element represented image is fixed, its one-sided σ-MGF and coefficients are Tools 2 and 1. The new issue is projective: scalar changes alter the one-sided series, whereas adjoint and balanced constructions are lift-independent.

50.3 Projective correction

Changing a lift g to zg changes its eigenvalues, so the one-sided series is not a projective invariant. Conjugation on $\mathrm{End}(V)$ is unchanged:

$$\mathrm{Ad}_{zg}(A) = (zg)A(zg)^{-1} = gAg^{-1}.$$

Thus the projectively intrinsic series is the ordinary finite Molien series of the adjoint representation,

$$\mathrm{SMG}^{\mathrm{proj}}(\mathbf{t}) = \frac{1}{|\overline{G}|} \sum_{\bar{g} \in \overline{G}} \prod_a \det(I - t_a \,\mathrm{Ad}_{\bar{g}})^{-1}.$$

If g is any lift, then

$$\mathrm{Tr}(\mathrm{Ad}_g^k) = \mathrm{Tr}(g^k)\overline{\mathrm{Tr}(g^k)} = |\chi(g^k)|^2,$$

which proves the proposed projective power-trace formula and its independence of lift.

50.4 Independent matrix checks

For each τ, the program explicitly restricts tensor powers to normalized symmetric occupation bases and ranks the averaged Reynolds matrix. Separately, it computes $h_r(g)$ from traces of powers using Newton's recurrence and averages $\prod_a h_{\tau_a}(g)$ first over elements and then over matrix conjugacy classes. The lift has 32 classes; the adjoint image has five.

representation	τ	$\dim W_\tau$	rank	elements	classes
lift on $\mathbb{C}^2$	(1)	2	0	0	0
	(4)	5	0	0	0
	(8)	9	0	0	0
	$(4,4)$	25	1	1	1
adjoint on $\mathrm{End}(\mathbb{C}^2)$	(1)	4	1	1	1
	(2)	10	2	2	2
	$(1,1)$	16	2	2	2
	(3)	20	2	2	2

All braid, unitarity, closure, integrality, and equality assertions pass. The largest projector error is below 4×10^{-15}.

50.5 Reproduction and interpretation

Run

```
.venv/bin/python for_this_guy/finite_braid_images/
  ising_clifford/verification.py
.venv/bin/marimo edit marimo_notebooks/braid_ising_clifford.py
```

after joining wrapped lines. The computation verifies the stated formulas for the displayed B_3 Ising representation. It does not claim that every root-of-unity Jones representation has finite image; finiteness remains an essential hypothesis.

51 Specified braid images and conditional quantum-image closures

Canonical testing artifacts.
Exact backend: `lambda_distributions/dists/braid_tqft_quantum_images_sigma_mgfs/verification.py`.
Regression: `lambda_distributions/dists/braid_tqft_quantum_images_sigma_mgfs/test_verification.py`.
Marimo: `marimo_notebooks/braid_tqft.py`.

Scope and outcome

A finite braid or TQFT image has a uniform measure; an infinite image does not, and its canonical unitary substitute is normalized Haar measure on the compact closure. We import the generalized Molien identity in both settings, record its class-power form, and test each mathematically distinct mechanism on a specified representation. The program constructs a 192-matrix Ising braid lift, the 27-matrix qutrit Heisenberg group, a 96-matrix rank-one Weil lift, the 24-element projective qubit Clifford group, and regular and affine phase-space permutation groups through degree 25. All 35 finite formula comparisons and all 1,440 affine fixed-power checks pass. Sixteen seeded Haar diagnostics corroborate the exact U, SU, O, and USp conclusions. The generic Jones, mapping-class, and Hadamard statements are recorded with their necessary image/closure hypotheses; no unspecified representation is treated as though it determined a matrix group.

Status: Mixed. The finite-image and fixed compact-closure formulas are exact. Generic TQFT image determination and closure classification are not proved here; those conditional research directions are moved to the final provisional part.

51.1 The probability distinction and the common target

Let V be a d-dimensional complex unitary module. For a finite represented image $I \leq U(V)$, define

$$\mathbb{E}[\mathrm{Exp}_\sigma(X_{I,V}h_1)]\,(\mathbf{t}) = \frac{1}{|I|}\sum_{x\in I}\prod_{a\geq 1}\det(I - t_a x)^{-1}. \tag{226}$$

If a unitary representation has infinite image, there is no uniform probability measure on that countable image. Put instead

$$K = \overline{\rho(\Gamma)} \leq U(V)$$

and use normalized Haar measure:

$$\mathbb{E}_{k\in K}[\mathrm{Exp}_\sigma(X_{K,V}h_1)]\,(\mathbf{t}) = \int_K \prod_{a\geq 1}\det(I - t_a k)^{-1}\,dk. \tag{227}$$

This agrees with (226) for finite K and is unchanged when a dense subgroup is replaced by its closure. Thus "braid representation" or "TQFT representation" is not enough data: one must specify the represented image, a finite central lift when the action is projective, or the compact closure.

For a partition or composition $\tau = (\tau_1, \ldots, \tau_s)$, set

$$W_\tau(V) = \bigotimes_{j=1}^{s}\mathrm{Sym}^{\tau_j} V, \qquad |\tau| = \sum_j \tau_j.$$

The coefficient tested throughout is

$$[m_\tau]\,\mathrm{SMG}_{G,V} = \dim W_\tau(V)^G.$$

51.2 Universal finite-image and compact-closure theorem

Theorem 1 (Generalized Molien/class-power identity)**.** *For a finite unitary image I with character χ_V,*

$$\mathbb{E}[\mathrm{Exp}_\sigma(X_{I,V}h_1)] = \frac{1}{|I|}\sum_{x\in I}\exp\left(\sum_{r\geq 1}\frac{p_r}{r}\chi_V(x^r)\right) \tag{228}$$

$$= \sum_{[x]\subseteq I}\frac{1}{|C_I(x)|}\exp\left(\sum_{r\geq 1}\frac{p_r}{r}\chi_V(x^r)\right), \tag{229}$$

where $p_r = \sum_a t_a^r$. Moreover

$$\boxed{[m_\tau]\,\mathbb{E}[\mathrm{Exp}_\sigma(X_{I,V}h_1)] = \dim W_\tau(V)^I.} \tag{230}$$

The same statements hold for compact K after replacing the finite average by normalized Haar integration.

Source. The coefficient and determinant statements are Tools 1 and 2. The power-character expression is the standard logarithmic determinant change of coordinates, and the class form merely groups the imported finite average by conjugacy class. The compact statement is the same imported identity with normalized Haar measure. □

The theorem shows that the exact finite answer is determined by conjugacy classes, power maps, and the represented character. It does *not* say that the abstract source group or a TQFT label determines those data.

51.2.1 Projective representations and scalar selection

Suppose a projective image becomes honest on a finite central extension

$$1 \longrightarrow Z \longrightarrow \widetilde{I} \longrightarrow I \longrightarrow 1.$$

Theorem 1 applies to $\widetilde{I}$. If $z \in Z$ acts as $\xi(z)I$, then z acts on $W_\tau(V)$ as $\xi(z)^{|\tau|}$. Therefore

$$[m_\tau]\,\mathrm{SMG}_{\widetilde{I},V} = 0 \quad \text{unless} \quad \xi(z)^{|\tau|} = 1 \text{ for every } z \in Z. \tag{231}$$

This is a necessary condition, not a promise of a nonzero invariant in every allowed degree. It also explains why the one-sided series depends on the chosen finite lift.

51.3 Finite Ising braid image

Let

$$H = \frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\\ 1 & -1\end{pmatrix}, \qquad R = e^{-\pi i/8}\begin{pmatrix}1 & 0\\ 0 & i\end{pmatrix},$$

and put

$$\rho(\sigma_1) = R, \qquad \rho(\sigma_2) = HRH. \tag{232}$$

Direct multiplication gives

$$\rho(\sigma_1)\rho(\sigma_2)\rho(\sigma_1) = \rho(\sigma_2)\rho(\sigma_1)\rho(\sigma_2),$$

so (232) is an honest two-dimensional representation of B_3. The verifier starts from these matrices and closes under multiplication. It obtains 192 matrices, 24 matrices modulo scalar phase, and exactly eight scalar matrices $\mu_8 I$. Thus (231) forces $8 \mid |\tau|$.

For each matrix A, the direct route diagonalizes A and expands $\det(I - tA)^{-1}$. The independent power route computes $\mathrm{Tr}(A^r)$ by matrix multiplication and uses Newton's recurrence

$$h_0(A) = 1, \qquad h_n(A) = \frac{1}{n}\sum_{r=1}^{n}\mathrm{Tr}(A^r)h_{n-r}(A). \tag{233}$$

The literal matrix averages agree:

τ	(1)	(2)	(4)	(8)	(7, 1)	(4, 4)	(1^8)
direct matrices	0	0	0	0	1	1	7
power formula	0	0	0	0	1	1	7

At $(t_1, t_2, t_3) = (0.023, -0.031, 0.041)$, direct determinants and the truncated power-character exponential agree to 2.2×10^{-18}; the braid-relation residual is 3.0×10^{-16}. The power series is truncated at $r = 60$ only for this nonzero numerical evaluation. All displayed coefficients use the exact finite recurrence (233) through the required degree.

Remark 1. *This example validates a particular finite Ising/Clifford braid image. It does not imply that generic Jones, Hecke, Temperley–Lieb, or BMW sectors have finite image. Density results for broad Jones–Wenzl families show why the finite/closure distinction is substantive [2].*

51.4 Monomial finite image: the qutrit Heisenberg group

Suppose a represented finite group is monomial in a chosen basis. Write $x = D(\alpha)P_\pi$. For a cycle $C = (j_1 \cdots j_\ell)$ of π, define $\alpha_C = \prod_{j\in C} \alpha_j$. Reordering the basis cycle by cycle gives

$$\det(I - tx) = \prod_{C\in\mathrm{Cyc}(\pi)} (1 - \alpha_C t^{|C|}), \tag{234}$$

and hence

$$\boxed{\mathrm{SMG}_{I,V} = \frac{1}{|I|} \sum_{x\in I} \prod_{a,C} (1 - \alpha_C(x) t_a^{|C|})^{-1}.} \tag{235}$$

The concrete test is the qutrit Heisenberg group

$$H_3 = \{\omega^c X^a Z^b : a, b, c \in \mathbb{F}_3\}, \qquad Xe_j = e_{j+1}, \quad Ze_j = \omega^j e_j,$$

where $\omega = e^{2\pi i/3}$. All 27 dense 3×3 matrices are built. The code separately reads the nonzero entry in each column, traverses the permutation cycles, multiplies their phases, and evaluates (235). The dense and cycle-phase multivariate averages agree to 1.8×10^{-20}.

Here the central phase rule and the two spectral sectors simplify to

$$[m_\tau]\,\mathrm{SMG}_{H_3,V} = \begin{cases} \dfrac{1}{9} \displaystyle\prod_j \binom{\tau_j + 2}{\tau_j} + \dfrac{8}{9} \prod_j \mathbf{1}_{3|\tau_j}, & 3 \mid |\tau|, \\ 0, & 3 \nmid |\tau|. \end{cases} \tag{236}$$

Indeed, the three central matrices contribute the first term after the phase filter, while every noncentral matrix has spectrum a scalar multiple of all three cube roots and contributes the second term.

τ	(1)	(2)	(3)	(2, 1)	(4)	(3, 3)	(6)
direct matrices	0	0	2	2	0	12	4
formula (236)	0	0	2	2	0	12	4

This is a representative finite Gaussian/Heisenberg core of the type that occurs in metaplectic and parafermionic constructions. Identifying a particular physical braid sector with this group requires its own generators; finiteness alone does not imply a universal asymptotic law.

51.5 Rank-one Weil lift and the phase issue

Over $\mathbb{F}_3$, define the Fourier and chirp matrices

$$F_{xy} = 3^{-1/2}\omega^{xy}, \qquad P_{xx} = \omega^{x^2/2}, \tag{237}$$

where $1/2 = 2$ in $\mathbb{F}_3$. Simultaneously track the generators

$$S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad T = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \quad \text{in } SL_2(\mathbb{F}_3).$$

Projective matrix closure gives all 24 elements of $SL_2(\mathbb{F}_3)$; retaining the generated scalar phases gives a 96-matrix lift.

The finite-field Weil character has the form

$$\chi_\omega(\widetilde{A}) = \gamma(\widetilde{A})|\ker(A-I)|^{1/2}, \qquad |\gamma(\widetilde{A})| = 1, \tag{238}$$

with a lift-dependent Weil-index phase [3]. Consequently

$$|\chi_\omega(\widetilde{A})|^2 = |\ker(A-I)|. \tag{239}$$

The verifier checks (239) for every $A \in SL_2(\mathbb{F}_3)$. The distribution is

$\lvert\ker(A-I)\rvert$	1	3	9
number of matrices	15	8	1
observed $\lvert\operatorname{Tr}\omega(A)\rvert^2$	1	3	9

For the full lift, the phase-sensitive class-power formula is

$$\mathrm{SMG}^{\mathrm{Weil}} = \sum_{[\widetilde{A}]} \frac{1}{|C(\widetilde{A})|} \exp\left(\sum_{r\ge1} \frac{p_r}{r} \chi_\omega(\widetilde{A}^r) \right). \tag{240}$$

The direct and power-character coefficient averages give

τ	(1)	(2)	(4)	$(3,1)$	$(4,4)$
direct lift	0	0	1	2	6
formula (240)	0	0	1	2	6

The full numerical MGF error is 2.3×10^{-16}.

Equation (239) controls balanced trace moments, but it does not determine the holomorphic series (240): the latter also needs the joint phases of $A, A^2, \ldots$. This is exactly why fixed-space magnitude stabilization cannot by itself prove convergence of the full Weil sigma–MGF.

The same rank-one construction may be viewed as a finite quotient of a torus mapping-class representation. Higher-genus or composite-level formulas remain exact class sums only after the relevant lift and Weil character have been specified; Chinese-remainder structure alone does not remove the local phase data.

51.6 Balanced series and unitary designs

Scalar phases often annihilate the one-sided series. The appropriate design-sensitive object is

$$\mathcal{B}_{G,V}(\mathbf{s},\mathbf{t}) = \frac{1}{|G|} \sum_{g\in G} \prod_i \det(I - s_i g)^{-1} \prod_j \det(I - t_j \overline{g})^{-1}. \tag{241}$$

Character expansion and Reynolds averaging prove

$$[m_\alpha(\mathbf{s}) m_\beta(\mathbf{t})]\mathcal{B}_{G,V} = \dim \left[\left(\bigotimes_i \operatorname{Sym}^{\alpha_i} V \right) \otimes \left(\bigotimes_j \operatorname{Sym}^{\beta_j} V^* \right) \right]^G. \tag{242}$$

Only $|\alpha| = |\beta|$ is intrinsically projective. For $\alpha = \beta = (1^k)$, the coefficient becomes the frame potential

$$\Phi_k(G) = \frac{1}{|G|} \sum_{g\in G} |\operatorname{Tr} g|^{2k} = \dim \operatorname{End}_G(V^{\otimes k}). \tag{243}$$

If G is a unitary t-design, every balanced coefficient with $|\alpha|, |\beta| \le t$ equals its Haar $U(d)$ value. Conversely, equality of (243) with Haar at degree k forces equality of the nested invariant spaces and hence equality of the kth moment projector. By Schur–Weyl duality,

$$\Phi_k(U(d)) = \sum_{\substack{\lambda \vdash k \\ \ell(\lambda) \le d}} (f^\lambda)^2.$$

Closing H and $S = \mathrm{diag}(1, i)$ modulo scalar phase gives the 24-element projective one-qubit Clifford group. Direct matrix averaging and the hook-length formula give

k	1	2	3	4
$\Phi_k(\mathrm{Cliff}_1)$	1	2	5	15
$\Phi_k(U(2))$	1	2	5	14

Thus the tested group agrees through degree three and first differs at degree four, consistent with the general multiqubit design theorem and its extra fourth-order stabilizer invariant [4].

51.7 Weyl–Heisenberg regular and affine phase-space actions

51.7.1 Regular action and the prime SIC formula

Let $A_d \cong (\mathbb{Z}/d\mathbb{Z})^2$ act regularly on itself, viewed as the d^2 projective outcomes in a Weyl–Heisenberg covariant SIC setting. An element a of order $o(a)$ has $d^2/o(a)$ cycles, so

$$\mathrm{SMG}_{A_d,\mathrm{Reg}} = \frac{1}{d^2} \sum_{a \in A_d} \prod_i (1 - t_i^{o(a)})^{-d^2/o(a)}. \tag{244}$$

For prime p, every nonidentity element has order p:

$$\boxed{\mathrm{SMG}_{A_p,\mathrm{Reg}} = \frac{1}{p^2} \left[\prod_i (1 - t_i)^{-p^2} + (p^2 - 1) \prod_i (1 - t_i^p)^{-p} \right].} \tag{245}$$

In particular

$$[m_{(1,1)}]\, \mathrm{SMG}_{A_p,\mathrm{Reg}} = p^2, \tag{246}$$

because only the identity fixes a point and the pair coefficient is the average squared fixed-point count.

The verifier constructs all permutation matrices for $p = 2, 3, 5$. Selected exact comparisons are

p	permutation degree	$[m_{11}]$	$[m_2]$	$[m_{p,p}]$
2	4	4	4	28
3	9	9	5	3,033
5	25	25	13	564,110,025

Every entry agrees with coefficient extraction from (245); full nonzero determinant evaluations agree to 2.3×10^{-16}. The growth in (246) proves that the raw regular-action series has no coefficientwise $p \to \infty$ limit. The contextual link between prime Weyl–Heisenberg SICs and Clifford normalizers is discussed in [5].

51.7.2 Affine symmetry groups

Let $\Gamma = A \rtimes K$ act on the additive space A. Write $g = (b, A_0)$ with action $x \mapsto A_0 x + b$ and put

$$b_r = (I + A_0 + \cdots + A_0^{r-1}) b.$$

Then

$$g^r(x) = A_0^r x + b_r,$$

so the fixed-point equation is $(I - A_0^r)x = b_r$. Therefore

$$\boxed{|\operatorname{Fix}(g^r)| = \begin{cases} |\ker(I - A_0^r)|, & b_r \in \operatorname{im}(I - A_0^r), \\ 0, & \text{otherwise.} \end{cases}} \tag{247}$$

Fixed points of powers recover cycle counts by Möbius inversion, hence (247) determines the full permutation sigma–MGF.

Ordered-pair orbits are classified by the difference of the two points, so

$$[m_{(1,1)}]\,\mathrm{SMG}_{A \rtimes K, A} = \#(K \backslash A) = 1 + \#(K \backslash (A \setminus \{0\})). \tag{248}$$

For $K = SL_2(\mathbb{F}_p)$, the nonzero vectors form one orbit and the answer is 2. The code constructs the full affine permutation groups for $p = 2, 3$:

p	degree	group order	fixed-power checks	pair coefficient
2	4	24	144	2
3	9	216	1,296	2

All $1,440$ comparisons of direct fixed points with (247), for powers $1 \le r \le 6$, pass exactly.

51.8 Compact closures and stable classical laws

If the represented image is infinite, Theorem 1 applies to its compact closure K. Four cases must be distinguished.

51.8.1 Unitary and special-unitary closure

If $K = U(d)$ in the defining representation, the scalar circle acts on every positive-degree $W_\tau(V)$ with weight $|\tau|$. Hence

$$\boxed{\mathrm{SMG}_{U(d),\mathbb{C}^d} = 1.} \tag{249}$$

For $SU(d)$, its center $\mu_d I$ implies that a coefficient vanishes unless $d \mid |\tau|$. Consequently, for every fixed positive τ, the coefficient is exactly zero once $d > |\tau|$, and

$$\mathrm{SMG}_{SU(d),\mathbb{C}^d} \longrightarrow 1 \quad \text{coefficientwise.} \tag{250}$$

The divisibility condition is only necessary. For example, $[m_{(1^d)}] = 1$ because the determinant tensor is invariant, whereas $[m_{(d)}] = 0$ for $d > 1$ in the defining module.

51.8.2 Orthogonal and compact symplectic closure

The first fundamental theorems for $O(d)$ and $USp(2n)$ say that stable tensor invariants are generated by pair contractions with, respectively, a symmetric or alternating bilinear form. Repackaging these contractions by symmetric powers yields the stable identities

$$\boxed{\mathrm{SMG}_{O(\infty)} = \mathrm{Exp}_\sigma(h_2), \qquad \mathrm{SMG}_{USp(\infty)} = \mathrm{Exp}_\sigma(e_2).} \tag{251}$$

This is coefficientwise stability, not equality in every degree at every finite rank; contraction diagrams can acquire relations outside the stable range.

The proof of (249)–(251) is algebraic. As an independent diagnostic, the notebook draws seeded Haar matrices by complex Gaussian QR for U, determinant-normalized QR for SU, real QR for O, and quaternionic Gram–Schmidt for USp. It averages the same symmetric-power characters used in the finite tests. Representative rows from $3,000$ samples each are:

group	τ	estimate	target	standard error
$U(4)$	$(1,1)$	$-0.0279-0.0160i$	0	0.0258
$SU(2)$	$(1,1)$	1.0064	1	0.0182
$SU(3)$	$(1,1,1)$	$0.9930-0.0013i$	1	0.0412
$O(6)$	(2)	0.9956	1	0.0182
$O(6)$	$(1,1,1,1)$	2.7695	3	0.1503
$O(6)$	$(2,2)$	2.1018	2	0.0731
$USp(4)$	(2)	-0.0209	0	0.0178
$USp(4)$	$(1,1,1,1)$	3.1935	3	0.1554
$USp(4)$	$(2,2)$	1.0291	1	0.0437

All sixteen diagnostics lie within the stated five-standard-error/absolute tolerance. Worst unitarity residuals are below 6.1×10^{-15}. These samples test the matrix construction and low-degree consequences; they are not substituted for the Reynolds and invariant-theory proofs.

51.9 How the remaining family labels fit

The universal theorem organizes the broader claims without assigning a formula to an unspecified representation.

- **Jones, Hecke, Temperley–Lieb, and BMW sectors.** For a specified finite image, use (229). For an infinite image, use (227). If the closure is eventually $U(d_n)$ or $SU(d_n)$, the one-sided coefficientwise limit is 1; orthogonal and symplectic closures give (251). Fusion-space dimension growth alone does not determine the closure.
- **Property (F) and finite mapping-class images.** Property (F) supplies finiteness, not a universal class table. One still needs the lift, power maps, and represented character. Twisted quantum-double and metaplectic examples often allow monomial or Weil compression; the Heisenberg and rank-one Weil sections demonstrate the two reductions.
- **Dense mapping-class images.** Projective density must first be translated to the honest compact closure relevant to the chosen lift. Once that closure and its central scalars are identified, the compact theorem applies.
- **Hessian and Heisenberg-normalizer groups.** The defining unitary action uses a Weil–Heisenberg class sum; the phase-space action uses (247). Their asymptotics depend on fixed-space phases or on the orbit structure of the complement.
- **Hadamard-generated groups.** Hadamard generation alone does not imply finiteness. In the finite qubit example, H and HS are both complex Hadamard and generate the same lift as H, S, since $H(HS) = S$. Infinite examples require the closure-Haar formula.

The general obstruction to a raw coefficientwise limit is already visible at balanced fourth tensor degree:

$$\dim \operatorname{End}_{I_n}(V_n^{\otimes 2}).$$

If this diverges, the corresponding coefficient diverges. Conversely, bounded low moments do not determine the full holomorphic series when power-character phases fail to stabilize.

51.10 Verification design and reproducibility

The program uses structurally different routes:

1. **Matrix closure.** Products of the displayed generators are stored until no new matrix appears. Projective closures normalize a single nonzero entry only when the coefficient being tested is phase neutral.
2. **Direct determinants.** Every finite represented matrix is inserted into (226).
3. **Power characters.** Matrix powers and (233) compute coefficients without diagonalizing; the numerical whole-function check uses (228) through power 60.
4. **Monomial cycles.** Nonzero column entries determine permutation cycles and accumulated phases, independently checking (234).

5. **Finite-field tracking.** Weil words are followed simultaneously in $SL_2(\mathbb{F}_3)$; affine fixed points are obtained from direct permutations and from (247).
6. **Orbit/design comparators.** Pair orbits and the hook-length Haar frame potential avoid the determinant formula.
7. **Haar diagnostics.** Seeded continuous samples include standard errors and group residuals and are explicitly reported as numerical evidence.

Run the regression test from the repository root:

```
.venv/bin/python -m pytest -q \
  lambda_distributions/dists/braid_tqft_quantum_images_sigma_mgfs/test_verification.py
```

Open the interactive laboratory with:

```
.venv/bin/marimo run \
  marimo_notebooks/braid_tqft.py
```

The suite reports **PASS**: 35 finite comparisons, 1,440 affine fixed-power checks, and 16 compact-closure diagnostics.

References

[1] S. Howe, *Random matrix statistics and zeroes of L-functions via probability in λ-rings*, https://arxiv.org/abs/2412.19295.

[2] M. H. Freedman, M. J. Larsen, and Z. Wang, *The two-eigenvalue problem and density of Jones representation of braid groups*, https://arxiv.org/abs/math/0103200.

[3] T. Thomas, *The Character of the Weil Representation*, https://arxiv.org/abs/math/0610644.

[4] H. Zhu, R. Kueng, M. Grassl, and D. Gross, *The Clifford group fails gracefully to be a unitary 4-design*, https://arxiv.org/abs/1609.08172.

[5] H. Zhu, *SIC–POVMs and Clifford groups in prime dimensions*, https://arxiv.org/abs/1003.3591.

52 Central extensions and projective representations

Canonical testing artifacts.
Exact backend: `lambda_distributions/dists/central_extensions_projective_representations/verification.py`.
Regression: `lambda_distributions/dists/central_extensions_projective_representations/test_verification.py`.
Marimo: `marimo_notebooks/central_extensions.py`.

Scope and outcome

For a finite or compact central extension acting with one scalar central character, we prove that the one-sided σ-MGF is obtained by a homogeneous root-of-unity projection. A continuous scalar center kills every positive degree. Since a projective matrix has no canonical one-sided eigenvalue distribution, we construct and prove lift independence of the charge-zero balanced series. We then specialize the theory to Schur and pin covers, finite Weil representations, extraspecial groups, Clifford groups, sporadic covers, and determinant-one kernels. The companion marimo notebook constructs nine representative matrix groups in dimensions 2, 3, 5, and 9. It performs 180 coefficient comparisons through total degree six, five full determinant-average checks, and nine balanced-moment checks. Every comparison passes; the maximum numerical discrepancy is 6.28×10^{-12}.

52.1 The generalized Molien target

Let K be a finite or compact group and let $\rho : K \to U(V)$ be a finite-dimensional unitary representation. For a partition $\tau = (\tau_1, \dots, \tau_\ell)$ set

$$W_\tau(V) = \bigotimes_{i=1}^{\ell} \mathrm{Sym}^{\tau_i} V, \qquad |\tau| = \sum_i \tau_i.$$

The one-sided σ-MGF is

$$\mathrm{SMG}_{K,V}(\mathbf{t}) = \int_K \prod_{i \ge 1} \det(I - t_i \rho(g))^{-1}\, dg. \tag{252}$$

Theorem 2 (Imported generalized Molien identity)**.** *With normalized counting measure in the finite case and normalized Haar measure in the compact case,*

$$\mathrm{SMG}_{K,V}(\mathbf{t}) = \sum_\tau \dim W_\tau(V)^K\, m_\tau(\mathbf{t}).$$

Equivalently,

$$\prod_i \det(I - t_i \rho(g))^{-1} = \exp\left(\sum_{k \ge 1} \frac{p_k(\mathbf{t})}{k} \operatorname{Tr}(\rho(g)^k) \right). \tag{253}$$

Source. This is Tools 1 and 2; it is retained with its equation labels because later central-character formulas refer to them. ☐

52.2 Central-character projection

Consider a central extension

$$1 \longrightarrow Z \longrightarrow \widetilde{G} \xrightarrow{\pi} G \longrightarrow 1 \tag{254}$$

and a representation $\rho : \widetilde{G} \to U(V)$ for which $\rho(z) = \chi(z) I_V$. Write $C = \chi(Z) \le U(1)$.

Theorem 3 (Finite central-character projector)**.** *Suppose $C = \mu_r$. Then*

$$[m_\tau]\, \mathrm{SMG}_{\widetilde{G},V} = 0 \qquad \textit{unless } r \mid |\tau|. \tag{255}$$

For any set-theoretic section $g \mapsto \widetilde{g}$,

$$\boxed{\mathrm{SMG}_{\widetilde{G},V}(\mathbf{t}) = \frac{1}{|G|}\sum_{g\in G}\frac{1}{r}\sum_{\zeta\in\mu_r}\prod_i \det(I-\zeta t_i\rho(\widetilde{g}))^{-1}.} \tag{256}$$

Thus adjoining scalar rth roots applies

$$\mathcal{P}_r F(\mathbf{t}) = \frac{1}{r}\sum_{\zeta\in\mu_r} F(\zeta\mathbf{t}), \tag{257}$$

which retains precisely the homogeneous degrees divisible by r.

Proof. The center acts on $W_\tau(V)$ by the scalar $\chi(z)^{|\tau|}$. A fixed vector can exist only when this scalar is one for all z, proving (255). For the integral proof, disintegrate normalized measure over the fibres of π. The pushforward of normalized measure on Z under χ is uniform measure on C. In total degree d, the fibre average is

$$\frac{1}{r}\sum_{\zeta\in\mu_r}\zeta^d = \begin{cases}1, & r \mid d,\\ 0, & r \nmid d.\end{cases}$$

This proves (256) and (257), independently of the chosen section and even when χ has a kernel. □

Corollary 1 (Continuous scalar saturation)**.** *If $C = U(1)$, then*

$$\boxed{\mathrm{SMG}_{\widetilde{G},V} = 1.}$$

Proof. The scalar average in total degree d is $\int_{U(1)} z^d\,dz$, equal to zero for every $d > 0$ and one for $d = 0$. □

This explains the trivial one-sided series of $U(d)$ in its defining module and of a full unitary normalizer containing every scalar matrix.

52.3 The canonical projective series

A projective matrix $[U_g] \in PU(V)$ has no canonical eigenvalue multiset: another lift $\zeta_g U_g$ multiplies every eigenvalue by ζ_g. Consequently a projective representation does not canonically define (252).

For partitions α, β, define the charge-zero balanced series

$$\boxed{\mathcal{B}^{\mathrm{proj}}_{G,[V]}(\mathbf{s},\mathbf{t}) = \sum_{|\alpha|=|\beta|}\dim(W_\alpha(V)\otimes W_\beta(V)^*)^{\widetilde{G}}\, m_\alpha(\mathbf{s})m_\beta(\mathbf{t}).} \tag{258}$$

Theorem 4 (Lift-independent projective Molien formula)**.** *For any choice of unitary lift U_g of each projective matrix,*

$$\boxed{\mathcal{B}^{\mathrm{proj}}_{G,[V]} = \mathrm{CT}_u\int_G \prod_i \det(I-us_iU_g)^{-1}\prod_j \det(I-u^{-1}t_jU_g^{-*})^{-1}\,dg.} \tag{259}$$

The answer is unchanged by arbitrary, element-dependent replacements $U_g \mapsto \zeta_g U_g$. It therefore depends only on the projective image, not on a Schur cover, cocycle representative, scalar phases, or a determinant-one lift.

Proof. The coefficient indexed by (α,β) in the integrand acquires $u^{|\alpha|-|\beta|}$. The constant term imposes equality of the two degrees. Under $U_g \mapsto \zeta_g U_g$, that coefficient is multiplied by $\zeta_g^{|\alpha|-|\beta|} = 1$. Haar averaging then gives the invariant dimension exactly as in Theorem 2. □

For a finite scalar center μ_r, the full two-sided series has sectors $|\alpha| - |\beta| \equiv 0 \pmod r$. Exact equality, rather than congruence, is the part intrinsic to the projective representation. The multilinear diagonal coefficients are the frame potentials

$$[m_{(1^k)}(\mathbf{s})m_{(1^k)}(\mathbf{t})]\mathcal{B}^{\mathrm{proj}} = \int_G |\operatorname{Tr} U_g|^{2k}\,dg. \tag{260}$$

52.4 Finite groups: class compression and power maps

If $\widetilde{G}$ is finite with character χ_V, grouping (253) by conjugacy classes gives

$$\boxed{\mathrm{SMG}_{\widetilde{G},V} = \frac{1}{|\widetilde{G}|}\sum_C |C| \exp\left(\sum_{k\geq 1}\frac{p_k(\mathbf{t})}{k}\chi_V(c_C^k)\right).} \tag{261}$$

Thus an exact calculation needs the represented character, class sizes, and class power maps. The projective balanced version is

$$\mathrm{CT}_u\, \frac{1}{|\widetilde{G}|}\sum_C |C| \exp\left[\sum_{k\geq 1}\frac{u^k p_k(\mathbf{s})\chi_V(c_C^k) + u^{-k}p_k(\mathbf{t})\overline{\chi_V(c_C^k)}}{k}\right]. \tag{262}$$

Equations (261) and (262) are the practical route for large Schur covers, spin Weyl covers, and sporadic covers whose dense matrices would be wasteful.

52.5 Schur and spin covers: the binary octahedral test

For a genuine spin module of a double cover $2.S_n$ or $2.A_n$, the central involution acts by $-I$. Theorem 3 immediately gives

$$[m_\tau]\,\mathrm{SMG}_{2.S_n,V_n} = [m_\tau]\,\mathrm{SMG}_{2.A_n,V_n} = 0 \qquad (|\tau| \text{ odd}). \tag{263}$$

For the standard basic-spin module, Clifford algebra identifies $\mathrm{End}(V_n)$ with all or one parity of the exterior algebra of the reflection module E_n. Its exterior powers are distinct hook modules. With the standard associate convention this gives

$$\frac{1}{|2.S_n|}\sum_g |\chi_{V_n}(g)|^4 = \begin{cases} n, & n \text{ odd},\\ n/2, & n \text{ even}.\end{cases} \tag{264}$$

The fourth balanced moment therefore grows linearly although $\dim V_n$ grows exponentially; it does not approach the Haar value 2.

The notebook uses the first nontrivial concrete model relevant here: the 48 unit quaternions of the binary octahedral group. Under

$$a + bi + cj + dk \longmapsto \begin{pmatrix} a+bi & c+di \\ -c+di & a-bi \end{pmatrix} \in SU(2),$$

they form a double cover $2.S_4 \to S_4$ and give the two-dimensional basic-spin representation. The list consists of

$$\{\pm 1, \pm i, \pm j, \pm k\}, \qquad \tfrac{1}{2}(\pm 1 \pm i \pm j \pm k), \qquad \tfrac{1}{\sqrt{2}}(\pm e_a \pm e_b)\ (a<b).$$

All signs are independent in the displayed families. Dense multiplication checks all 48^2 products. The fourth trace moment is 2, agreeing with (264) at $n = 4$.

k	direct $\lvert\mathrm{Tr}\rvert^{2k}$	phase-twisted lifts	Haar $U(2)$	conclusion
1	1	1	1	agree
2	2	2	2	agree
3	5	5	5	agree
4	15	15	14	first separation

The phase-twisted column multiplies the jth matrix by an unrelated 37th root of unity. This is a direct test that projective balanced quantities do not depend on a coherent linear lift. The non-multilinear test

$$(\alpha, \beta) = ((2,1),(1,1,1))$$

also remains 3 after phase twisting, checking charge-zero invariance beyond frame potentials.

52.6 Pin covers of Weyl groups

Let $W \le O(E)$ be a finite Weyl or Coxeter group, $\widetilde{W} \le \mathrm{Pin}(E)$ its pin cover, and S a genuine spinor. The central involution gives the parity rule

$$[m_\tau]\,\mathrm{SMG}_{\widetilde{W},S} = 0 \qquad (|\tau| \text{ odd}).$$

Moreover

$$\mathrm{End}(S) \cong \mathrm{Cl}(E) \text{ or } \mathrm{Cl}^0(E) \cong \bigwedge^{\bullet} E \text{ or } \bigwedge^{\text{even}} E \tag{265}$$

as W-modules. Hence

$$\boxed{\frac{1}{|\widetilde{W}|}\sum_{\widetilde{w}} |\operatorname{Tr} S(\widetilde{w})|^{2k} = \dim\left(\mathrm{End}(S)^{\otimes k}\right)^{W}.} \tag{266}$$

For $k = 2$, if $\mathrm{End}(S) = \bigoplus_\lambda m_\lambda U_\lambda$, the answer is $\sum_\lambda m_\lambda^2$. Distinct exterior powers in the classical towers therefore produce a rank-growing exterior-algebra law, not a Haar unitary law. The notebook's Q_8 and binary octahedral constructions are the rank-small pin/spin test cases; higher ranks are more efficiently tested by (265) or class power maps.

52.7 Finite Weil representations

Assume either that q is odd, or more generally that a Weyl operator basis $\{D_v : v \in W_n\}$ of $\mathrm{End}(\mathcal{H}_n)$ and unitary lifts U_g have been chosen with exact covariance. Let $W_n = \mathbb{F}_q^{2n}$ and let $\mathcal{H}_n$ be the full finite Schrödinger space, $\dim \mathcal{H}_n = q^n$. The required covariance is

$$U_g D_v U_g^{-1} = D_{gv}$$

after a compatible normalization. Since the D_v form a basis of $\mathrm{End}(\mathcal{H}_n)$,

$$\mathrm{End}(\mathcal{H}_n) \cong \mathbb{C}[W_n] \tag{267}$$

as an $\mathrm{Sp}(2n, q)$-module. Burnside's orbit lemma gives

$$\boxed{\frac{1}{|\operatorname{Sp}(2n,q)|}\sum_{g} |\operatorname{Tr} U_g|^{2k} = \#\big(\mathrm{Sp}(2n,q)\backslash W_n^k\big).} \tag{268}$$

Status: Exact under covariance. The orbit formula applies to the full Schrödinger module. It is not inferred from the name "Weil representation" alone, and it must be modified for an irreducible even or odd constituent.

There are two orbits on W_n: zero and nonzero, so the second trace moment is 2. For ordered pairs and $n \ge 2$, the orbit types are

- $(0, 0)$, $(0, w \ne 0)$, and $(v \ne 0, 0)$;
- $w = av$, one orbit for each $a \in \mathbb{F}_q^\times$;
- independent orthogonal pairs;
- independent pairs with $\langle v, w\rangle = c$, one orbit for each $c \in \mathbb{F}_q^\times$.

Thus the stable fourth moment is $3 + (q-1) + 1 + (q-1) = 2q + 2$. Witt extension proves that every fixed-k orbit count stabilizes once the ambient symplectic rank is large enough.

Remark 2 (Rank-one correction)**.** *When $n = 1$, an orthogonal pair in a two-dimensional symplectic space is dependent. The independent-orthogonal orbit is absent, so*

$$\mathbb{E}|\operatorname{Tr} U_g|^4 = 2q + 1 \qquad (n = 1). \tag{269}$$

This hypothesis is essential when testing a reasonable small dimension.

For odd prime q, the notebook independently constructs the projective Weil matrices from

$$F_{xy} = q^{-1/2} e^{2\pi i xy/q}, \qquad P_{xx} = e^{2\pi i (2^{-1}x^2)/q}.$$

Projective closure yields $|SL(2,3)| = 24$ matrices in dimension 3 and $|SL(2,5)| = 120$ matrices in dimension 5. Separately, the code enumerates the $SL(2,q)$-orbits on $(\mathbb{F}_q^2)^k$.

q	projective order	k	matrix average	orbit count
3	24	1	2	2
3	24	2	7	7
5	120	1	2	2
5	120	2	11	11

For an irreducible even or odd Weil constituent, one must insert the corresponding idempotent in (267); the full-module orbit count must not be copied unchanged. Likewise, over finite fields where a metaplectic extension splits, the actual scalar image, rather than the name of the cover, determines the selection rule.

52.8 Odd extraspecial groups and their central products

Let $E_n(p)$ be an extraspecial group of order p^{1+2n} and exponent p, where p is odd. Its Schrödinger module has dimension $N = p^n$. In the explicit model used by the notebook, for $a, b \in \mathbb{F}_p^n$ and $c \in \mathbb{F}_p$,

$$\rho(c,a,b)e_x = \omega^{c+b\cdot x} e_{x+a}, \qquad \omega = e^{2\pi i/p}. \tag{270}$$

These pN^2 matrices are unitary and faithful.

Theorem 5 (Odd extraspecial formula)**.** *For the module* (270)*,*

$$\boxed{\mathrm{SMG}_{E_n(p),V} = N^{-2}\mathcal{P}_p \prod_i (1-t_i)^{-N} + (1-N^{-2}) \prod_i (1-t_i^p)^{-N/p}.} \tag{271}$$

Consequently,

$$[m_\tau]\,\mathrm{SMG}_{E_n(p),V} = \mathbf{1}_{p||\tau|} \Bigg[N^{-2} \prod_i \binom{N+\tau_i-1}{\tau_i} + (1-N^{-2}) \mathbf{1}_{p|\tau_i\ \forall i} \prod_i \binom{N/p+\tau_i/p-1}{\tau_i/p} \Bigg]. \tag{272}$$

If scalar sth roots are adjoined with $p \mid s$, then

$$\mathrm{SMG}_{\mu_s E_n(p),V} = \mathcal{P}_s\,\mathrm{SMG}_{E_n(p),V}\,. \tag{273}$$

Proof. The center consists of p scalar matrices $\omega^c I$, a fraction N^{-2} of the group. Averaging their determinant factors gives the first term of (271). For a noncentral element, the weighted shift (270) has all pth roots of unity as eigenvalues, each with multiplicity N/p. Therefore $\det(I - tg) = (1-t^p)^{N/p}$, giving the second term. Extracting coefficients with $(1-t)^{-a} = \sum_{d\ge 0} \binom{a+d-1}{d} t^d$ proves (272). Equation (273) is Theorem 3. □

The notebook constructs $E_1(3)$, $E_2(3)$, and $E_1(5)$, of respective orders 27, 243, and 125 and dimensions 3, 9, and 5. Every coefficient through total degree six agrees with (272). At $(t_1,t_2) = (0.07, 0.11)$ the full determinant averages are:

group	dense matrix average	formula	absolute error
$E_1(3)$	1.006152412632117	1.006152412632117	7.8×10^{-18}
$E_2(3)$	1.016020200753344	1.016020200753347	2.7×10^{-15}
$E_1(5)$	1.000511444770786	1.000511444770785	1.6×10^{-15}

The trace is zero off the center and equals $N\omega^c$ on the central element $\omega^c I$. Hence

$$\mathbb{E}[T^a\overline{T}^b] = \mathbf{1}_{p|(a-b)} N^{a+b-2}, \tag{274}$$

so $\mathbb{E}|T|^2 = 1$ while $\mathbb{E}|T|^{2k} = N^{2k-2}$ for $k \geq 2$. Thus $T \to 0$ in probability as $n \to \infty$, but its higher balanced moments diverge: the limit is a rare central spike, not a Gaussian.

For fixed τ and $d = |\tau|$, the central term of (272) is asymptotic to $N^{d-2}/\prod_i \tau_i!$. When all τ_i are divisible by p, the noncentral term is asymptotic to $(N/p)^{d/p}/\prod_i(\tau_i/p)!$. Raw coefficients therefore generally have no finite coefficientwise limit.

52.9 Extraspecial 2-groups: separate plus and minus sectors

Let E_n^ε be extraspecial of order 2^{1+2n}, sign $\varepsilon \in \{+1, -1\}$, and let $N = 2^n$. Central elements again give the root-filtered sector. Noncentral elements whose squares are $+I$ have eigenvalues ± 1 equally often; those whose squares are $-I$ have eigenvalues $\pm i$ equally often. Counting the two quadratic-form types gives

$$\boxed{\begin{aligned} \mathrm{SMG}_{E_n^\varepsilon} = 2^{-2n}\mathcal{P}_2 \prod_i (1-t_i)^{-N} \\ + \left(\frac{1}{2} + \varepsilon 2^{-n-1} - 2^{-2n}\right) \prod_i (1-t_i^2)^{-N/2} \\ + \left(\frac{1}{2} - \varepsilon 2^{-n-1}\right) \prod_i (1+t_i^2)^{-N/2}. \end{aligned}} \tag{275}$$

The final two sectors can cancel in low degree; they may not be combined by discarding the sign.

The rank-one plus group is D_8, represented by $\pm I, \pm X, \pm Z, \pm XZ$. The rank-one minus group is Q_8, represented by $\pm I, \pm iX, \pm iZ, \pm(-XZ)$. Both 8-element sets pass exact closure. All coefficients through degree six agree with (275). At the same determinant test point, the errors are below 3.6×10^{-18}.

52.10 Clifford groups and determinant-one lifts

The full unitary Clifford normalizer contains $U(1)I$, so Corollary 1 gives

$$\mathrm{SMG}_{\mathrm{Cliff}_n} = 1$$

for the one-sided series. A finite phase lift with scalar image μ_s instead applies $\mathcal{P}_s$ and has lift-dependent coefficients in degrees divisible by s.

Let $K_n = \mathrm{Cliff}_n \cap SU(N)$ with $N = 2^n$. It contains $\{\zeta I : \zeta^N = 1\} \cong \mu_N$, and therefore

$$[m_\tau]\,\mathrm{SMG}_{K_n} = 0 \qquad \text{unless } N \mid |\tau|. \tag{276}$$

For every fixed positive degree, eventually $N > |\tau|$, whence

$$\mathrm{SMG}_{K_n} \longrightarrow 1 \qquad \text{coefficientwise.} \tag{277}$$

In the one-qubit case, K_1 is the binary octahedral group already tested. Adjoining μ_4 produces 96 distinct unitary matrices; all coefficients through total degree six equal the binary-octahedral coefficient multiplied by $\mathbf{1}_{4||\tau|}$. This directly tests the finite-lift rule.

Balanced quantities are the same for the full normalizer, any finite scalar lift, K_n, and the projective quotient. Multiqubit Clifford groups are unitary 3-designs, so their balanced series agrees with Haar unitary through balanced degree three and differs at degree four. The one-qubit row in the table above gives the concrete values $1, 2, 5, 15$ versus Haar $1, 2, 5, 14$.

More generally, suppose $\delta : \widetilde{G} \to \mu_q$ is a character and $K = \ker\delta$. Define

$$\mathrm{SMG}^{(j)}_{\widetilde{G}} = \frac{1}{|\widetilde{G}|} \sum_{g \in \widetilde{G}} \delta(g)^j \prod_i \det(I - t_i\rho(g))^{-1}.$$

Character orthogonality gives the exact determinant-one formula

$$\boxed{\mathrm{SMG}_{K,V} = \sum_{j=0}^{q-1} \mathrm{SMG}^{(j)}_{\widetilde{G}}\,.} \tag{278}$$

Indeed, $\sum_j \delta(g)^j$ equals q on K and zero off K, while $|\widetilde{G}| = q|K|$ when δ is onto. Passing to determinant one therefore adds determinant-twisted sectors; it does not merely "remove scalars."

52.11 Sporadic covers: exact data workflow

For a genuine irreducible module of a sporadic cover $2.G$, $3.G$, or $6.G$, let r be the order of its central scalar character. Then

$$[m_\tau]\,\mathrm{SMG}_{\widetilde{G},V} = 0 \qquad \text{unless } r \mid |\tau|,$$

and the complete exact answer is (261). An auditable calculation imports three items from the represented covering group:

1. conjugacy-class sizes;
2. the genuine character values on those classes;
3. every class power map needed to the chosen truncation degree.

The verification then compares a class-compressed sum against an element enumeration when matrices are available, or against an independent character table implementation otherwise. There is no intrinsic rank asymptotic for an isolated sporadic group. Useful comparisons are the first separating degree, balanced design strength, and equality of charge-zero series across different lifts.

No external sporadic character table was supplied with this workspace, so the notebook deliberately does not invent a sporadic numerical example. The binary octahedral and finite Weil cases exercise the same class/power and phase mechanisms with explicit matrices; inserting certified ATLAS data in (261) is the correct sporadic verification strategy.

52.12 Verification method and reproducibility

The companion files are

- `verification.py`: constructions and comparison routes;
- `marimo_notebooks/central_extensions.py`: the marimo interface, grouped by group family;
- `test_verification.py`: a headless regression test.

For each dense matrix g, the code computes $\mathrm{Tr}(\mathrm{Sym}^d\, g)$ from the recurrence

$$h_0(g) = 1, \qquad d\, h_d(g) = \sum_{k=1}^{d} \mathrm{Tr}(g^k) h_{d-k}(g). \tag{279}$$

Thus

$$[m_\tau]\,\mathrm{SMG} = \frac{1}{|G|} \sum_{g\in G} \prod_i h_{\tau_i}(g)$$

is obtained directly from actual matrix powers. It is compared with (272), coefficient extraction from (275), or the scalar projector. This is independent of the determinant-sector comparison, which evaluates the full rational functions at $(0.07, 0.11)$.

For the projective tests, frame potentials are computed from the generated matrices while the Weil comparator counts finite symplectic orbits. The binary octahedral matrices are also phase-twisted element by element. Matrix audits check expected order, distinctness, identity, unitarity, and exact closure where a quadratic scan is modest. The 243-element $E_2(3)$ case uses the exact Heisenberg multiplication law from (270) instead of a redundant 243^2 dense closure scan.

family	groups	dimensions	checks	independent comparator
odd extraspecial	3	$3, 9, 5$	$90 + 3$	spectral sectors and binomial formula
extraspecial 2	2	$2, 2$	$60 + 2$	signed $\pm 1/\pm i$ sectors
Schur/scalar lift	2	$2, 2$	30	homogeneous μ_4 projector
projective spin/Clifford	1 image	2	5	phase changes and Haar values
finite Weil	2	$3, 5$	4	symplectic orbit enumeration
total	9 constructions	2–9	194	all pass

Here "194" means 180 coefficient, five determinant, and nine balanced comparisons. The maximum residual, 6.28×10^{-12}, comes from floating-point roots of unity in an integer-valued coefficient; all stated integer answers are within the declared 2×10^{-9} tolerance.

52.13 Additional Schur-cover audit for genuine modules

Let

$$1 \longrightarrow \{1, z\} \longrightarrow \widetilde{S}_n \longrightarrow S_n \longrightarrow 1$$

be a Schur double cover, and let D be a genuine module, so z acts by $-I$. Theorem 1, grouped by conjugacy classes, gives

$$\boxed{\mathrm{SMG}_{\widetilde{S}_n, D} = \sum_{[\widetilde{g}]} \frac{1}{|C_{\widetilde{S}_n}(\widetilde{g})|} \exp\left(\sum_{r \geq 1} \frac{p_r}{r} \chi_D(\widetilde{g}^r) \right).} \tag{280}$$

Pairing the summands g and zg shows immediately that

$$\boxed{[m_\tau]\, \mathrm{SMG}_{\widetilde{S}_n, D} = 0 \quad \text{when } |\tau| \text{ is odd.}} \tag{281}$$

If $D \otimes D \cong \bigoplus_\eta m_\eta U_\eta$ and the required self-duality identifies fourth invariants with $\mathrm{End}_{\widetilde{S}_n}(D \otimes D)$, Schur's lemma gives

$$\dim(D^{\otimes 4})^{\widetilde{S}_n} = \sum_\eta m_\eta^2. \tag{282}$$

Without that self-duality, the always-valid statement pairs multiplicities of dual constituents instead. This qualification is essential when the chosen genuine constituent is not self-dual.

For the audit, Hermitian Clifford matrices γ_i satisfy $\gamma_i \gamma_j + \gamma_j \gamma_i = 2\delta_{ij} I$. The lifts

$$u_i = \frac{\gamma_i - \gamma_{i+1}}{\sqrt{2}}, \qquad 1 \leq i < 4,$$

generate the Pin preimage of S_4, an order-48 matrix group containing $-I$. The represented module has dimension four. Direct generator-fixed spaces and (280) give

τ	(1)	(2)	$(1,1)$	(3)	$(1,1,1,1)$
coefficient	0	1	2	0	16

All five comparisons pass, the odd rows vanish as predicted, and the maximum Clifford-relation residual is zero at machine precision. This test verifies the universal formula and central selection rule for a genuine matrix module; it does not purport to tabulate every strict-partition spin character.

52.14 Asymptotic taxonomy and conclusions

family	one-sided behavior	balanced/projective behavior
$2.S_n, 2.A_n$ spin	odd degrees vanish	basic-spin fourth moment is linear in n
spin Weyl covers	parity projection	exterior-algebra centralizer law
sporadic covers	degree modulo central character	exact finite class formula; no intrinsic tower
finite Weil modules	lift-dependent scalar rule	fixed-k symplectic orbit law stabilizes
extraspecial p-groups	coefficients generally diverge	rare central spikes; high moments diverge
full Clifford normalizer	exactly 1	Haar agreement through design degree
determinant-one Clifford	coefficientwise limit 1	same law as the projective quotient
scalar quotient	not canonically defined	charge zero is canonical

Central extensions therefore produce three distinct mechanisms:

finite scalar center:	congruence projection on total degree;
growing or continuous scalar center:	one-sided collapse to 1;
projective representation:	information survives canonically at charge zero.

The natural asymptotic objects are consequently projective or charge-graded Λ-distributions, not phase-dependent one-sided distributions on a scalar quotient.

53 Pauli–Heisenberg groups

Abstract

For odd characteristic we prove the two-spectrum formula for the defining Schrödinger representation and check it by constructing every matrix in $P_{3,1}$, $P_{5,1}$, $P_{9,1}$, and $H_3(\mathbb{F}_9)$. The tests evaluate the target invariant dimension directly, not from the spectral classification used in the proof. All 17 representative coefficient comparisons pass.

53.1 Statement

Let $q = p^f$ with p odd, $V = \mathbb{C}[\mathbb{F}_q^n]$, and $N = q^n$. Fix a nontrivial additive character $\psi : \mathbb{F}_q \to \mu_p$. With

$$X(a)e_x = e_{x+a}, \qquad Z(b)e_x = \psi(b \cdot x)e_x, \qquad W(a,b) = \psi\big(\tfrac{1}{2}a \cdot b\big)\, X(a)Z(b),$$

put $P_{q,n} = \{\zeta W(a,b) : \zeta \in \mu_p\}$. For a partition $\tau = (\tau_1, \ldots, \tau_\ell)$ define

$$A_\tau = \prod_i \binom{N + \tau_i - 1}{\tau_i}, \qquad B_\tau = \begin{cases} \prod_i \binom{N/p + \tau_i/p - 1}{\tau_i/p}, & p \mid \tau_i \ \forall i, \\ 0, & \text{otherwise.} \end{cases}$$

Then the claimed generalized Molien coefficient is

$$[m_\tau]\,\mathrm{SMG}_{P_{q,n}} = \begin{cases} 0, & p \nmid |\tau|, \\ q^{-2n} A_\tau + (1 - q^{-2n}) B_\tau, & p \mid |\tau|. \end{cases} \tag{283}$$

Equivalently,

$$\mathrm{SMG}_{P_{q,n}}(\mathbf{t}) = \frac{q^{-2n}}{p} \sum_{\zeta \in \mu_p} \prod_j (1 - \zeta t_j)^{-N} + (1 - q^{-2n}) \prod_j (1 - t_j^p)^{-N/p}. \tag{284}$$

53.2 Proof

There are p central matrices ζI_N, a fraction q^{-2n} of the group. Their average is the first term of (284). Now take $(a,b) \neq (0,0)$. For $1 \le k < p$, $W(a,b)^k$ has trace zero: if $ka \neq 0$ it is a translation with zero diagonal; otherwise its diagonal trace is a nontrivial additive-character sum. Also $W(a,b)^p = I$. Fourier inversion of the multiplicities of the pth roots of unity therefore gives multiplicity N/p for every root. Hence

$$\det(I - tW(a,b)) = (1 - t^p)^{N/p}.$$

A scalar in μ_p only permutes this spectrum, proving (284). Expanding the central factors gives A_τ and forces $p \mid |\tau|$; expanding the noncentral factors gives B_τ. This proves (283).

For the finite Heisenberg group $H_{2n+1}(\mathbb{F}_q)$, the Schrödinger matrix is $\rho_\psi(a,b,z) = \psi(z)W(a,b)$. When $f > 1$, $\ker\psi$ has size q/p, so each matrix in $P_{q,n}$ has exactly q/p preimages. Uniform averages over the abstract group and its image consequently agree.

53.3 Independent verification method

For every constructed matrix g, the code computes its eigenvalues $\lambda_1, \ldots, \lambda_N$ numerically and obtains $h_d(\lambda)$ from

$$\prod_{r=1}^{N} (1 - \lambda_r t)^{-1} = \sum_{d \ge 0} h_d(\lambda) t^d.$$

It then evaluates the target directly as

$$D_\tau = \frac{1}{|G|} \sum_{g \in G} \prod_i h_{\tau_i}(\operatorname{spec} g).$$

This is the character of $\bigotimes_i \operatorname{Sym}^{\tau_i} V$ averaged over G, hence the rank of its Reynolds projector. Only after computing D_τ does the program evaluate (283).

The $q = 9$ construction uses $\mathbb{F}_9 = \mathbb{F}_3[u]/(u^2 + 1)$ and $\psi(x) = \exp(2\pi i \operatorname{Tr}_{\mathbb{F}_9/\mathbb{F}_3}(x)/3)$. Thus the test includes a genuine prime-power field. The $H_3(\mathbb{F}_9)$ run enumerates all 729 abstract triples, retaining the threefold central kernel rather than silently replacing the group by its image.

53.4 Representative results

group	$\dim V$	matrices averaged	τ	direct = formula
$P_{3,1}$	3	27	(3)	2
$P_{3,1}$	3	27	$(3,3)$	12
$P_{5,1}$	5	125	$(4,1)$	14
$P_{5,1}$	5	125	$(5,5)$	636
$P_{9,1}$	9	243	(3)	5
$P_{9,1}$	9	243	$(3,3)$	345
$H_3(\mathbb{F}_9)$	9	729	$(2,1)$	5
$H_3(\mathbb{F}_9)$	9	729	$(3,3)$	345

All 17 built-in cases pass to tolerance 2×10^{-8} and round to nonnegative integers. Characteristic two is intentionally excluded here: the displayed half-phase normalization is an odd-characteristic statement; the phase-extended qubit lift is handled in its own verification.

53.5 Reproduction

Run

```
.venv/bin/python for_this_guy/quantum_matrix_groups/
  pauli_heisenberg/verification.py
.venv/bin/marimo edit marimo_notebooks/pauli_heisenberg.py
```

The line breaks after the directory slash are for display; join each command onto one shell line.

54 Scalar-extended Pauli groups

Abstract

Building on Theorem A(3) of the 14 July paper, we prove the coefficient form of the scalar-extended Pauli formula and test the complete matrix groups $\mu_4 P_{2,1}$, $\mu_4 P_{2,2}$, and $\mu_6 P_{3,1}$. All 13 representative checks pass. The calculation also locates a sign error in the supplied qubit-recovery prose: equation (13) specializes with a positive, projected noncentral sector, not a minus sign.

54.1 Formula and proof

Let $q = p^f$, $N = q^n$, and let s be divisible by p. For the finite lift $\widetilde{P}^{(s)}_{q,n} = \mu_s P_{q,n}$, its distinct matrices may be indexed by $\alpha \in \mu_s$ and $v \in \mathbb{F}_q^{2n}$, so its order is sq^{2n}. For odd p use the usual phase-normalized Weyl operators. For qubits we use $\{I, X, Z, iXZ\}$ on each tensor factor and adjoin μ_4.

If $v = 0$, the matrix is αI_N. If $v \neq 0$, the p spectral branches of the phase-normalized $W(v)$ occur with multiplicity N/p, and therefore

$$\det(I - t\alpha W(v)) = (1 - \alpha^p t^p)^{N/p}.$$

Averaging these two sectors proves

$$\mathrm{SMG}_{\widetilde{P}}(\mathbf{t}) = q^{-2n}\frac{1}{s}\sum_{\alpha\in\mu_s}\prod_j(1-\alpha t_j)^{-N} + (1-q^{-2n})\frac{1}{s}\sum_{\alpha\in\mu_s}\prod_j(1-\alpha^p t_j^p)^{-N/p}. \tag{285}$$

Root-of-unity orthogonality forces $s \mid |\tau|$ in both sectors. Thus, with

$$A_\tau = \prod_i \binom{N+\tau_i-1}{\tau_i}, \qquad B_\tau = \mathbf{1}_{p\mid\tau_i\ \forall i}\prod_i\binom{N/p+\tau_i/p-1}{\tau_i/p},$$

the coefficient formula is

$$[m_\tau]\,\mathrm{SMG}_{\widetilde{P}} = \begin{cases} 0, & s \nmid |\tau|, \\ q^{-2n}A_\tau + (1-q^{-2n})B_\tau, & s \mid |\tau|. \end{cases} \tag{286}$$

54.2 Qubit specialization and correction

For $p = q = 2$, $s = 4$, equation (285) gives

$$4^{-n}\Pi_4\prod_j(1-t_j)^{-2^n} + (1-4^{-n})\frac{1}{4}\sum_{\alpha\in\mu_4}\prod_j(1-\alpha^2 t_j^2)^{-2^{n-1}}.$$

The second summand is an average of determinant reciprocals and has positive coefficients in allowed multidegrees. In particular, for one qubit,

$$[t^4]\,\mathrm{SMG} = \tfrac{1}{4}\binom{5}{4} + \tfrac{3}{4}\binom{2}{2} = 2.$$

The supplied "qubit recovery" paragraph instead writes a minus before the noncentral term, which would give 1/2, not even an integer invariant dimension. Therefore that minus sign cannot be correct. Equation (13) itself, interpreted as the root-of-unity average above, is consistent with the direct matrices and with (286).

54.3 Verification strategy and results

For each matrix the verifier computes the coefficients h_d of $\det(I - tg)^{-1}$ from its eigenvalues, then independently averages $\prod_i h_{\tau_i}(g)$ over every group matrix. It compares this character average with (286). Representative results are:

group	τ	direct	formula
$\mu_4 P_{2,1}$	(2)	0	0
$\mu_4 P_{2,1}$	(4)	2	2
$\mu_4 P_{2,1}$	$(2,2)$	3	3
$\mu_4 P_{2,2}$	(4)	5	5
$\mu_4 P_{2,2}$	$(4,4)$	85	85
$\mu_6 P_{3,1}$	(3)	0	0
$\mu_6 P_{3,1}$	(6)	4	4
$\mu_6 P_{3,1}$	$(5,1)$	7	7

All 13 built-in comparisons pass to 2×10^{-8}. The cases (2) for $s = 4$ and (3) for $s = 6$ directly exercise the stronger scalar selection rule, while $(3, 1)$ and $(5, 1)$ test mixed symmetric powers.

54.4 Reproduction

```
.venv/bin/python for_this_guy/quantum_matrix_groups/
  scalar_extended_pauli/verification.py
.venv/bin/marimo edit marimo_notebooks/scalar_extended_pauli.py
```

Join each displayed command onto one shell line.

55 Extraspecial groups

Abstract

We verify the proposed spectral formulas on all matrices of the two extraspecial groups of order 3^3 and on the faithful representations of D_8 and Q_8. These examples are minimal but discriminating: they detect both the exponent-p versus exponent-p^2 split and the plus/minus Arf type. All 20 coefficient comparisons pass.

55.1 Odd p: what the character does not see

Let E be extraspecial of order p^{1+2n} and let ρ be a faithful irreducible representation of dimension $N = p^n$, with $\rho(z) = \omega I_N$. Central matrices contribute

$$p^{-2n}\Pi_p \prod_j (1-t_j)^{-N}.$$

In the exponent-p case every noncentral element has all pth roots once per block of size p, hence

$$\mathrm{SMG}_{E,\exp p} = p^{-2n}\Pi_p \prod_j (1-t_j)^{-N} + (1-p^{-2n}) \prod_j (1-t_j^p)^{-N/p}. \tag{287}$$

In the exponent-p^2 case, $g^p = z^{\ell(gZ)}$ for a nonzero linear functional $\ell : E/Z(E) \to \mathbb{F}_p$. Set $A_a(\mathbf{t}) = \prod_j (1-\omega^a t_j^p)^{-N/p}$. Counting fibers of ℓ gives

$$\mathrm{SMG}_{E,\exp p^2} = p^{-2n}\Pi_p \prod_j (1-t_j)^{-N} + (p^{-1} - p^{-2n})A_0 + p^{-1}\sum_{a=1}^{p-1} A_a. \tag{288}$$

The coefficients in (288) sum to one. The last sum is the spectral information missed by the ordinary character, which vanishes off the center.

For the direct exponent-9 test we use $E = C_9 \rtimes C_3 = \langle x, y : y^{-1}xy = x^4\rangle$ and the matrices

$$\rho(x) = \mathrm{diag}(\eta, \eta^4, \eta^7), \qquad \rho(y)e_j = e_{j-1}, \qquad \eta = e^{2\pi i/9}.$$

The exponent-3 comparison is the 3-dimensional Heisenberg representation.

55.2 Extraspecial 2-groups

For sign $\varepsilon = +1$ (plus type) or -1 (minus type), put $N = 2^n$ and

$$A_+ = \prod_j (1-t_j^2)^{-N/2}, \qquad A_- = \prod_j (1+t_j^2)^{-N/2}.$$

The quadratic form on $E/Z(E)$ has $2^{2n-1} + \varepsilon 2^{n-1}$ zeros and $2^{2n-1} - \varepsilon 2^{n-1}$ ones. Removing the identity coset from the first sector, restoring its two central lifts, and dividing by $|E|$ proves

$$\begin{aligned}\mathrm{SMG}_{E^\varepsilon} = {} & 2^{-2n}\Pi_2 \prod_j (1-t_j)^{-N} \\ & + (\tfrac{1}{2} + \varepsilon 2^{-n-1} - 2^{-2n})A_+ + (\tfrac{1}{2} - \varepsilon 2^{-n-1})A_-. \end{aligned}\tag{289}$$

For $n = 1$ the plus and minus representations are D_8 and Q_8.

55.3 Direct coefficient test

The verifier constructs every representation matrix, computes $h_d(\operatorname{spec} g)$ from $\det(I - tg)^{-1}$, and averages $\prod_i h_{\tau_i}$ over the group. It separately expands (287), (288), or (289). Selected outcomes are:

group	τ	direct	formula
exponent 3	(3)	2	2
exponent 9	(3)	1	1
exponent 3	$(3, 3)$	12	12
exponent 9	$(3, 3)$	11	11
D_8	(2)	1	1
Q_8	(2)	0	0
D_8	$(1, 1)$	1	1
Q_8	$(1, 1)$	1	1
D_8	$(2, 2)$	3	3
Q_8	$(2, 2)$	3	3

The unequal rows demonstrate that the tests would fail if the spectral sectors were collapsed. All 20 built-in cases pass to 2×10^{-8}.

55.4 Reproduction

```
.venv/bin/python for_this_guy/quantum_matrix_groups/
  extraspecial_groups/verification.py
.venv/bin/marimo edit marimo_notebooks/extraspecial_groups.py
```

Join each displayed command onto one shell line.

56 Clifford groups and balanced design invariants

Abstract

The ordinary one-sided sigma–MGF does not detect unitary-design order. We prove the correct two-sided coefficient statement, isolate a further phase subtlety, and verify the design-detecting coefficient by closing explicit projective Clifford matrix groups in dimensions 2 and 4. The computed frame potentials agree with Haar through degree three and differ at degree four: $(1,2,5,15)$ versus $(1,2,5,14)$ in dimension two, and $(1,2,6,29)$ versus $(1,2,6,24)$ in dimension four.

56.1 One-sided series and the scalar obstruction

For a finite lift $\widetilde{G} \le U(V)$, put

$$\mathrm{SMG}_{\widetilde{G},V}(\mathbf{t}) = \frac{1}{|\widetilde{G}|}\sum_{g\in\widetilde{G}}\prod_{i\ge 1}\det(I-t_i g)^{-1}.$$

For a compact group $K \le U(V)$, define instead

$$\mathrm{SMG}_{K,V}(\mathbf{t}) = \int_K \prod_{i\ge 1}\det(I-t_i k)^{-1}\,dk$$

with normalized Haar measure. Expanding each determinant gives

$$[m_\tau]\,\mathrm{SMG}_{\widetilde{G},V} = \dim\left(\bigotimes_j \mathrm{Sym}^{\tau_j} V\right)^{\widetilde{G}}.$$

If the compact group contains the full scalar circle, every positive-degree one-sided coefficient vanishes. Hence

$$\boxed{\mathrm{SMG}_{K,V} = 1.}$$

for $U(2^n)$ and for the full normalizer $N_{U(2^n)}(P_n)$. A finite scalar lift instead obeys only the congruence imposed by its finite scalar center and may have nonzero lift-dependent coefficients.

56.2 The two-sided series, correctly phase neutralized

For a specified finite lift $\widetilde{G} \le U(V)$ define

$$\mathcal{B}_{\widetilde{G}}(\mathbf{s},\mathbf{t}) = \frac{1}{|\widetilde{G}|}\sum_{g\in\widetilde{G}}\prod_i \det(I-s_i g)^{-1}\prod_j \det(I-t_j\bar{g})^{-1}.$$

Character expansion and averaging prove

$$[m_\alpha(\mathbf{s})m_\beta(\mathbf{t})]\mathcal{B}_{\widetilde{G}} = \dim\left(\mathrm{Sym}^\alpha V\otimes \mathrm{Sym}^\beta V^*\right)^{\widetilde{G}}. \tag{290}$$

There is a useful refinement. Replacing a representative g by zg multiplies the coefficient integrand by $z^{|\alpha|-|\beta|}$. Thus the entire two-sided series is not intrinsically projective. Its *phase-neutral projection*

$$\mathcal{B}^0_G = \sum_{|\alpha|=|\beta|}[m_\alpha m_\beta]\mathcal{B}_{\widetilde{G}}\, m_\alpha m_\beta \tag{291}$$

is independent of representatives. Equivalently it is the constant term in z after $(\mathbf{s},\mathbf{t})\mapsto(z\mathbf{s},z^{-1}\mathbf{t})$. For the full normalizer, continuous scalar integration performs exactly this projection. All design coefficients used below lie in (291).

56.3 The design-detecting coefficient

For $(1^k) = (1, \dots, 1)$, one has $\mathrm{Sym}^{(1^k)} V = V^{\otimes k}$. Hence (290) gives

$$[m_{(1^k)}(\mathbf{s})m_{(1^k)}(\mathbf{t})]\mathcal{B}_G^0 = \dim \mathrm{End}_G(V^{\otimes k}) \tag{292}$$

$$= \frac{1}{|G|}\sum_{g\in G} |\operatorname{Tr} g|^{2k}. \tag{293}$$

The last quantity is the kth unitary frame potential.

Let $P_G^{(k)}$ be the average of $g^{\otimes k} \otimes \bar{g}^{\otimes k}$. It is the orthogonal projection onto the corresponding G-fixed space. Since $G \subseteq U(d)$, the Haar fixed space is contained in the G-fixed space. Equality of the coefficient in (293) with the Haar coefficient is equality of the two ranks; the nested spaces, and therefore their orthogonal projections, are equal. This proves

$$\boxed{G \text{ is a unitary } k\text{-design} \iff \Phi_k(G) = \Phi_k(U(d)).}$$

Schur–Weyl duality supplies the exact Haar reference

$$\Phi_k(U(d)) = \sum_{\substack{\lambda\vdash k\\ \ell(\lambda)\le d}} (f^\lambda)^2,$$

which reduces to $k!$ when $d \ge k$.

56.4 Independent matrix verification

The program constructs

$$H = 2^{-1/2}\begin{pmatrix}1 & 1\\ 1 & -1\end{pmatrix}, \qquad S = \begin{pmatrix}1 & 0\\ 0 & i\end{pmatrix}$$

and, for two qubits, their tensor placements together with CNOT. Matrices are multiplied until closure, with equality taken modulo global phase. The resulting orders are asserted to be 24 and 11,520. No class table or design theorem is used in the numerical average. The Haar comparator is computed independently by the hook-length formula in the Schur–Weyl sum.

n	d	$\lvert\overline{\mathrm{Cliff}}_n\rvert$	k	direct Φ_k	Haar
1	2	24	1	1	1
			2	2	2
			3	5	5
			4	15	14
2	4	11,520	1	1	1
			2	2	2
			3	6	6
			4	29	24

Thus the first total degree at which the canonical phase-neutral coefficient can disagree is four, and the frame coefficient does disagree there. This is precisely the Clifford 3-design/4-design transition established in [1, 2]. The notebook establishes the displayed frame-potential values only for the one- and two-qubit projective groups through $k = 4$. The all-qubit 3-design and failure-of-4-design statements are external theorems supplied by those references.

As a separate regression test, closing the ordinary one-qubit lift generated by H, S gives 192 matrices and scalar center $\mu_8 I$. Its one-sided coefficients for $\tau = (1), (2), (4)$ vanish, while those for (8) and $(4, 4)$ are 1 and 2. This confirms both lift dependence and the scalar selection rule $8 \mid |\tau|$.

56.5 Reproduction and scope

Run

```
.venv/bin/python for_this_guy/quantum_matrix_groups/
  clifford_groups/verification.py
.venv/bin/marimo edit marimo_notebooks/clifford_groups.py
```

after joining each wrapped command onto one line. The sweep verifies the multiqubit statement for $n = 1, 2$ and $k \leq 4$; the cited theorems supply the all-n result. It makes no inference about odd-prime qudit Clifford groups.

References

[1] H. Zhu, "Multiqubit Clifford groups are unitary 3-designs," https://arxiv.org/abs/1510.02619.

[2] H. Zhu, R. Kueng, M. Grassl, and D. Gross, "The Clifford group fails gracefully to be a unitary 4-design," https://arxiv.org/abs/1609.08172.

57 The real extraspecial Pauli layer

Abstract

For the plus-type extraspecial real Pauli group $E_m \cong 2_+^{1+2m}$ on $V_m = \mathbb{C}^{2^m}$, we derive the complete four-sector determinant average. An independent program constructs every matrix $\pm X^a Z^b$ and checks 24 one-row and mixed coefficients in dimensions $2, 4, 8$. All comparisons are exact integers and all pass.

57.1 Representation and coefficient target

Put $N = 2^m$. On the basis $\{e_x : x \in \mathbb{F}_2^m\}$ define

$$X(a)e_x = e_{x+a}, \qquad Z(b)e_x = (-1)^{b\cdot x}e_x.$$

Then

$$E_m = \{\pm X(a)Z(b) : a, b \in \mathbb{F}_2^m\}, \qquad |E_m| = 2N^2.$$

For a partition $\tau = (\tau_1, \ldots, \tau_r)$, determinant expansion gives

$$[m_\tau]\,\mathrm{SMG}_{E_m,m} = \frac{1}{|E_m|}\sum_{g\in E_m}\prod_i h_{\tau_i}(\operatorname{spec} g), \tag{294}$$

where $\sum_{d\geq 0} h_d(\operatorname{spec} g)t^d = \det(I - tg)^{-1}$. This is exactly $\dim(\bigotimes_i \mathrm{Sym}^{\tau_i} V_m)^{E_m}$.

57.2 The four spectral sectors

The central elements $I, -I$ have determinants $(1-t)^N, (1+t)^N$. For $g = \pm X(a)Z(b)$ one has

$$g^2 = (-1)^{a\cdot b} I.$$

Off the center, the trace is zero. Hence an involution has $N/2$ eigenvalues of each sign, while an element squaring to $-I$ has $N/2$ eigenvalues each of i and $-i$. Their determinants are respectively $(1-t^2)^{N/2}$ and $(1+t^2)^{N/2}$.

The quadratic form $q(a, b) = a \cdot b$ is plus type. It has $(N^2 + N)/2$ zeros, including $(0, 0)$. Accounting for both central lifts therefore gives

$$\#\{g \notin Z(E_m) : g^2 = I\} = N^2 + N - 2, \qquad \#\{g : g^2 = -I\} = N^2 - N.$$

The counts sum with the two central elements to $2N^2$. Averaging proves

$$\boxed{\begin{aligned}\mathrm{SMG}_{E_m,m}(\mathbf{t}) = \frac{1}{2N^2}\Big[&\prod_j(1-t_j)^{-N} + \prod_j(1+t_j)^{-N}\\ &+ (N^2+N-2)\prod_j(1-t_j^2)^{-N/2}\\ &+ (N^2-N)\prod_j(1+t_j^2)^{-N/2}\Big].\end{aligned}} \tag{295}$$

Because $-I \in E_m$, the first two terms cancel in odd total degree and the other two contain only even powers. Thus every odd-total-degree coefficient vanishes.

57.3 Independent coefficient extraction

The verifier does not classify matrices after construction. For each explicit matrix it computes the eigenvalues, obtains h_d recursively from the product generating function, forms the integrand in (294), and averages. The formula side separately extracts coefficients from the four factors in (295) using binomial coefficients. Before coefficient extraction, every matrix is independently checked to have real entries and to satisfy $g^\top g = I$.

m	N	$\lvert E_m \rvert$	τ	direct	formula
1	2	8	$(3, 1)$	2	2
1	2	8	$(2, 2)$	3	3
2	4	32	(4)	5	5
2	4	32	$(4, 2)$	23	23
3	8	128	$(3, 1)$	15	15
3	8	128	(6)	29	29
3	8	128	$(4, 2)$	190	190

The full sweep uses eight partitions at each rank. It includes the odd case (1) and the mixed cases $(1, 1)$, $(3, 1)$, $(2, 2)$, and $(4, 2)$. All 24 pass.

57.4 Reproduction

```
.venv/bin/python for_this_guy/barnes_wall_lattice_automorphism_groups/
  extraspecial_real_pauli/verification.py
.venv/bin/marimo edit marimo_notebooks/
  extraspecial_real_pauli.py
```

Join each wrapped command onto one shell line. The computation is exhaustive at the stated ranks; no random sampling or precomputed character table is used.

58 The real Clifford layer

Abstract

We prove the exact reduction of the real Clifford sigma–MGF to coset averages over $O^+(2m,2)$ and state precisely what the self-dual-code theorem does and does not imply for mixed coefficients. Exhaustive matrix closures in dimensions 2 and 4 recover group orders 16 and 2304, quotient orders 2 and 72, lift independence, and 16 direct coefficient matches.

58.1 Exact quotient formula

Let $E_m \triangleleft \mathcal{C}_m = N_{O(V_m)}(E_m)$, with

$$\mathcal{C}_m/E_m \cong O^+(2m,2).$$

For A in the quotient and any lift $\widetilde{A}$, define

$$\Phi_m(A;\mathbf{t}) = \frac{1}{|E_m|}\sum_{e\in E_m}\prod_j \det(I - t_j e\widetilde{A})^{-1}.$$

Replacing $\widetilde{A}$ by $e_0\widetilde{A}$ permutes E_m, so Φ_m is well defined. Partitioning $\mathcal{C}_m$ into right cosets proves

$$\boxed{\mathrm{SMG}_{\mathcal{C}_m,m}(\mathbf{t}) = \frac{1}{|O^+(2m,2)|}\sum_{A\in O^+(2m,2)}\Phi_m(A;\mathbf{t}).} \tag{296}$$

Coefficient extraction gives

$$[m_\tau]\,\mathrm{SMG}_{\mathcal{C}_m,m} = \dim\left(\bigotimes_i \mathrm{Sym}^{\tau_i} V_m\right)^{\mathcal{C}_m}.$$

58.2 One-row code theorem and its boundary

For a binary self-dual code $C \le \mathbb{F}_2^{2k}$, its genus-m complete weight enumerator is

$$\mathrm{cwe}_m(C) = \sum_{(c^{(1)},\dots,c^{(m)})\in C^m}\prod_{j=1}^{2k} x_{(c_j^{(1)},\dots,c_j^{(m)})}.$$

The Clifford invariant theorem identifies the degree-$2k$ invariant space with the span of these enumerators. In the stable range $m \ge k-1$, they are independent and therefore

$$[m_{(2k)}]\,\mathrm{SMG}_{\mathcal{C}_m,m} = \#\{\text{binary self-dual codes of length } 2k\}/\cong\,. \tag{297}$$

Consequently the stable values in degrees $2,4,6,8$ are $1,1,1,2$. The identification with genus-m complete weight enumerators and their independence in this range are external inputs from the Nebe–Rains–Sloane invariant theorem cited below. This note proves the coset formula and verifies low-rank examples; it does not reprove that theorem. The degree-eight value is guaranteed by (297) only for $m \ge 3$; our smaller closures also happen to give two, but we report that as a direct low-rank computation rather than misuse the stable-range hypothesis.

For a general partition τ, block-symmetrizing the full code tensor produces explicit invariants. The ordinary invariant-ring theorem does not by itself show that these span every mixed fixed space. Thus (296) is exact for all τ, while a code-indexed equality for all mixed τ would require a separate multigraded theorem.

58.3 Explicit matrices and exhaustive checks

We generate the real Pauli matrices together with tensor placements of

$$H = 2^{-1/2}\begin{pmatrix}1 & 1\\ 1 & -1\end{pmatrix}$$

and, for $m = 2$, CNOT. Breadth-first closure yields

$$|\mathcal{C}_1| = 16 = 8 \cdot 2, \qquad |\mathcal{C}_2| = 2304 = 32 \cdot 72.$$

Every closed matrix is checked to be real orthogonal. Conjugation by every selected coset representative sends the Pauli group to itself. The program then partitions the closed group into Pauli cosets, averages each coset, and compares (296) with an independent direct average. Each lift is also multiplied by a varying Pauli matrix, and every corresponding pair of coset contributions is compared before the quotient average is taken.

m	$\lvert\mathcal{C}_m\rvert$	$\lvert O^+\rvert$	τ	direct	coset
1	16	2	(2)	1	1
1	16	2	(8)	2	2
1	16	2	$(2,2)$	2	2
2	2304	72	(4)	1	1
2	2304	72	(8)	2	2
2	2304	72	$(4,2)$	2	2

The full sweep tests eight partitions at each rank. Stable-code references are applied only when $m \geq k - 1$; all 16 rows pass.

58.4 Reproduction and references

```
.venv/bin/python for_this_guy/barnes_wall_lattice_automorphism_groups/
  real_clifford/verification.py
.venv/bin/marimo edit marimo_notebooks/
  real_clifford.py
```

Join wrapped commands onto one line. For the invariant theorem, see G. Nebe, E. Rains, and N. J. A. Sloane, *The invariants of the Clifford groups*, https://arxiv.org/abs/math/0001038.

59 Barnes–Wall lattice automorphism groups

Abstract

Away from the exceptional rank $m = 3$, the Barnes–Wall automorphism group is an index-two subgroup of the real Clifford group. We prove the resulting twisted sigma–MGF identity and verify it exhaustively in the concrete models $BW_2 = \mathbb{Z}^2$ and $BW_4 = D_4$. The constructed groups have orders 8 and 1152 and agree exactly with the Clifford matrices preserving the specified lattice bases. All 16 coefficient comparisons pass.

59.1 The index-two identity

Assume $m \neq 3$ and write

$$B_m = \mathrm{Aut}(BW_{2^m}) = \ker \varepsilon_m \quad \text{inside} \quad \mathcal{C}_m,$$

where $\varepsilon_m : \mathcal{C}_m \to \{\pm 1\}$ is the nontrivial quotient character. Define the twisted series

$$\mathrm{SMG}^{\varepsilon_m}_{\mathcal{C}_m,m}(\mathbf{t}) = \frac{1}{|\mathcal{C}_m|} \sum_{g \in \mathcal{C}_m} \varepsilon_m(g) \prod_j \det(I - t_j g)^{-1}.$$

Since $\mathbf{1}_{B_m}(g) = (1 + \varepsilon_m(g))/2$ and $|B_m| = |\mathcal{C}_m|/2$, direct substitution proves

$$\boxed{\mathrm{SMG}_{B_m,m} = \mathrm{SMG}_{\mathcal{C}_m,m} + \mathrm{SMG}^{\varepsilon_m}_{\mathcal{C}_m,m}\,.} \tag{298}$$

For $W_\tau = \bigotimes_i \mathrm{Sym}^{\tau_i} V_m$, coefficient extraction equivalently gives

$$\dim W_\tau^{B_m} = \dim W_\tau^{\mathcal{C}_m} + \dim \mathrm{Hom}_{\mathcal{C}_m}(\varepsilon_m, W_\tau).$$

The second nonnegative summand is precisely the lattice-sensitive sector.

The structural equality $\mathrm{Aut}(BW_{2^m}) = \ker \varepsilon_m \leq \mathcal{C}_m$ and the exceptional behavior at $m = 3$ are external inputs from the cited Barnes–Wall/Clifford-group literature. Equation (298) is proved conditionally from this index-two statement, and the verifier checks the concrete ranks $m = 1, 2$.

59.2 Concrete lattice models

For $m = 1$ we use $BW_2 = \mathbb{Z}^2$. Its automorphism group is the signed permutation group of order 8, equal to E_1, while the real Clifford group has order 16.

For $m = 2$ we use

$$BW_4 = D_4 = \{x \in \mathbb{Z}^4 : \textstyle\sum_i x_i \equiv 0 \pmod 2\}.$$

A basis is given by the columns of

$$L = \begin{pmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 1 \\ 0 & 0 & -1 & 1 \end{pmatrix}.$$

Starting with E_2, tensor Hadamard $H \otimes H$, CNOT, and controlled-Z, matrix closure gives a subgroup of order 1152. Independently, for each of the 2304 matrices $g \in \mathcal{C}_2$ the verifier tests whether $L^{-1} g L \in GL(4, \mathbb{Z})$. Exactly the same 1152 matrices pass. Thus the subgroup identification uses lattice preservation, not only an expected order. Every matrix in both closed groups is also checked to have real entries and satisfy $g^\mathsf{T} g = I$.

59.3 Direct versus twisted averages

For each partition τ, the direct side averages $\prod_i h_{\tau_i}(\mathrm{spec}\, g)$ over B_m. On the right side, the same integrand is averaged first with weight 1 and then with weight $\varepsilon_m(g)$ over all of $\mathcal{C}_m$. Membership in the independently identified lattice stabilizer determines the sign.

m	$\lvert B_m \rvert$	τ	direct	Clifford	twisted	sum
1	8	(4)	2	1	1	2
1	8	(8)	3	2	1	3
1	8	$(4, 2)$	4	2	2	4
2	1152	(4)	1	1	0	1
2	1152	(6)	2	1	1	2
2	1152	(8)	3	2	1	3
2	1152	$(4, 2)$	3	2	1	3

These rows exhibit a real distinction: the first tested one-row twisted contribution occurs in degree 4 for B_1 but only degree 6 for B_2. The full sweep tests eight partitions at each rank, including mixed powers; all 16 instances of (298) pass.

59.4 Scope and reproduction

No $m = 3$ conclusion is drawn: that rank is explicitly exceptional to the index-two statement. Ranks $m \geq 4$ are covered algebraically by (298), but exhaustive closure is not a reasonable verification strategy because the groups grow rapidly.

```
.venv/bin/python for_this_guy/barnes_wall_lattice_automorphism_groups/
  barnes_wall/verification.py
.venv/bin/marimo edit marimo_notebooks/barnes_wall.py
```

Join each wrapped command onto one line. Background on the group chain and the exceptional rank appears in G. Nebe, E. Rains, and N. J. A. Sloane, *The invariants of the Clifford groups*, https://arxiv.org/abs/math/0001038.

60 Lattice automorphism groups

Status: Conditional on certified class data. The universal class/power or frame-shape formula is exact. Concrete numerical tables in this section depend on external class sizes, power maps, character values, or frame shapes and are not independently derived by the Molien argument.

Canonical testing artifacts.
Exact backend: `lambda_distributions/dists/lattice_sigma_mgfs/verification.py`.
Regression: `lambda_distributions/dists/lattice_sigma_mgfs/test_verification.py`.
Marimo: `marimo_notebooks/lattices.py`.

Scope and outcome

For the canonical complex representation of the full automorphism group of a positive-definite integral lattice, we prove the exact frame-shape σ-MGF and its coefficient interpretation by Reynolds projection. We then prove the cycle formulas and coefficientwise limits for type-A root lattices, signed-permutation lattices, and repeated orthogonal sums; give the finite class reductions for Niemeier, Coxeter–Todd, extremal, and modular lattices; exhibit explicit rank-$2, 3, 4$ lattices with automorphism group exactly $\{\pm I\}$; and prove the contrasting exponential moment growth for tensor powers. A reproducible marimo notebook constructs all tractable matrix groups and compares direct matrix averages, recovered frame shapes, family cycle formulas, and independently computed common fixed spaces.

60.1 Canonical representation and the frame-shape theorem

Let L be a positive-definite integral lattice of rank d, put

$$G_L = \mathrm{Aut}(L), \qquad V_L = L \otimes_{\mathbb{Z}} \mathbb{C},$$

and let G_L act through its canonical lattice matrices. Positivity makes G_L finite. If $\tau = (\tau_1, \ldots, \tau_s)$ is a partition, define

$$W_\tau(V_L) = \bigotimes_{j=1}^{s} \mathrm{Sym}^{\tau_j} V_L, \qquad a_\tau(L) = \dim W_\tau(V_L)^{G_L}.$$

The symmetric series $\mathrm{SMG}_L = \sum_\tau a_\tau(L) m_\tau$ is the lattice σ-MGF.

Every $g \in G_L$ has finite order. Its characteristic polynomial can be written uniquely in frame-shape form

$$\det(xI - g) = \prod_{m \geq 1} (x^m - 1)^{a_m(g)}, \qquad \sum_m m a_m(g) = d. \tag{299}$$

The exponents are integers and may be negative; cyclotomic cancellation makes the product a polynomial. Equivalently,

$$\det(I - tg) = \prod_{m \geq 1} (1 - t^m)^{a_m(g)}. \tag{300}$$

Theorem 1 (frame-shape σ-MGF)**.** *For every positive-definite integral lattice,*

$$\begin{aligned} \mathrm{SMG}_L(\mathbf{t}) &= \frac{1}{|G_L|} \sum_{g \in G_L} \prod_{i \geq 1} \prod_{m \geq 1} (1 - t_i^m)^{-a_m(g)}, & (301) \\ &= \sum_{[g] \subseteq G_L} \frac{1}{|C_{G_L}(g)|} \prod_{i,m} (1 - t_i^m)^{-a_m(g)}, & (302) \\ &= \sum_{[g]} \frac{1}{|C_{G_L}(g)|} \exp\left(\sum_{r \geq 1} \frac{p_r}{r} \mathrm{Tr}(g^r \mid V_L) \right). & (303) \end{aligned}$$

Moreover,

$$\boxed{[m_\tau]\, \mathrm{SMG}_L = \dim W_\tau(V_L)^{G_L}.} \tag{304}$$

Proof. For any complex matrix A,

$$\sum_{q\geq 0} \operatorname{Tr}(\operatorname{Sym}^q A)t^q = \det(I - tA)^{-1} = \exp\left(\sum_{r\geq 1} \frac{t^r}{r} \operatorname{Tr}(A^r)\right). \tag{305}$$

Multiply over the variables t_i and average over G_L. The coefficient of $t_1^{\tau_1} \cdots t_s^{\tau_s}$ is the trace on $W_\tau(V_L)$ of the Reynolds idempotent

$$R_\tau = \frac{1}{|G_L|} \sum_{g\in G_L} \rho_{W_\tau}(g).$$

Its image is exactly $W_\tau(V_L)^{G_L}$, so its trace proves (304). Equation (300) gives (301); grouping elements into conjugacy classes gives (302); and the exponential identity in (305) gives (303). □

The frame exponents and power traces determine one another:

$$\operatorname{Tr}(g^r) = \sum_{m|r} m a_m(g), \qquad a_m(g) = \frac{1}{m} \sum_{e|m} \mu(m/e) \operatorname{Tr}(g^e). \tag{306}$$

Indeed, the sum of the rth powers of all mth roots of unity is m if $m \mid r$ and zero otherwise. Möbius inversion proves the second formula. This is also the exact recovery algorithm used by the verifier.

60.2 Universal parity and low-degree diagnostics

The central automorphism $-I$ belongs to every G_L. It acts on $W_\tau(V_L)$ by $(-1)^{|\tau|}$; therefore

$$\boxed{[m_\tau]\,\mathrm{SMG}_L = 0 \quad \text{if } |\tau| \text{ is odd.}} \tag{307}$$

Because the lattice form identifies V_L with its dual,

$$[m_{(1,1)}]\,\mathrm{SMG}_L = \dim \operatorname{End}_{G_L}(V_L), \qquad [m_{(2)}]\,\mathrm{SMG}_L \geq 1. \tag{308}$$

If V_L is complex irreducible, Schur's lemma and uniqueness of the invariant symmetric form give equality to one in both positions.

The fourth multilinear coefficient is already more sensitive. If $V_L \otimes V_L = \bigoplus_\rho b_\rho \rho$, then

$$[m_{(1,1,1,1)}]\,\mathrm{SMG}_L = \dim \operatorname{End}_{G_L}(V_L \otimes V_L) = \sum_\rho b_\rho^2. \tag{309}$$

Thus irreducibility in degree one does not control a growing family in degree four.

More generally, Newton's identity gives the exact moment formula

$$[m_\tau]\,\mathrm{SMG}_L = \mathbb{E}_{g\in G_L} \prod_j \left[\sum_{\lambda\vdash\tau_j} \frac{1}{z_\lambda} \prod_{r\geq 1} \operatorname{Tr}(g^r)^{m_r(\lambda)}\right]. \tag{310}$$

A coefficient of total degree D depends only on the joint moments of $\operatorname{Tr}(g), \ldots, \operatorname{Tr}(g^D)$. This observation is the correct bridge from exact finite formulas to coefficientwise asymptotics.

60.3 Type-A root lattices

For

$$A_{n-1} = \{(x_1, \ldots, x_n) \in \mathbb{Z}^n : \textstyle\sum_i x_i = 0\}, \qquad n \geq 3,$$

the full automorphism group is $\{\pm 1\} \times S_n$ on the standard hyperplane representation. If σ has $c_m(\sigma)$ cycles of length m, deleting the constant eigenline from the permutation representation gives

$$\det(I - t\varepsilon\sigma \mid V_{A_{n-1}}) = \frac{\prod_m (1 - (\varepsilon t)^m)^{c_m(\sigma)}}{1 - \varepsilon t}. \tag{311}$$

Let $m_m(\lambda)$ denote the multiplicity of m in a partition λ, and $z_\lambda = \prod_m m^{m_m(\lambda)} m_m(\lambda)!$. Averaging (311) by cycle type proves

$$\boxed{\mathrm{SMG}_{A_{n-1}}(\mathbf{t}) = \frac{1}{2} \sum_{\varepsilon=\pm1} \sum_{\lambda\vdash n} \frac{1}{z_\lambda} \prod_i \left[(1-\varepsilon t_i) \prod_{m\geq 1} \left(1-(\varepsilon t_i)^m\right)^{-m_m(\lambda)} \right].} \tag{312}$$

Large-rank limit

For a uniform permutation, the finite collection of cycle counts converges with all fixed moments to independent $Z_m \sim \mathrm{Poisson}(1/m)$. Applying the Poisson probability-generating function to (312) yields, coefficientwise,

$$\boxed{\mathrm{SMG}_{A_\infty} = \frac{1}{2} \sum_{\varepsilon=\pm1} \prod_i (1-\varepsilon t_i) \exp\left[\sum_{m\geq 1} \frac{1}{m} \left(\prod_i (1-(\varepsilon t_i)^m)^{-1} - 1 \right) \right].} \tag{313}$$

Only finitely many cycle moments enter a bounded-degree coefficient, so the usual finite-dimensional cycle-count convergence is sufficient.

60.4 Signed-permutation lattices: types B, C, and D

The cubic lattice $\mathbb{Z}^n$ has the signed-permutation group $C_2^n \rtimes S_n$. The same natural matrix group is the Weyl group in types B_n and C_n, and it is the full automorphism group of D_n for $n \geq 5$. The exceptional triality of D_4 must be handled by its own finite class sum.

For a permutation cycle of length ℓ, let $\delta \in \{\pm 1\}$ be the product of its coordinate signs. Its determinant is $1 - \delta t^\ell$. After averaging the independent sign product, put

$$B_\ell(\mathbf{t}) = \frac{1}{2} \left[\prod_i (1-t_i^\ell)^{-1} + \prod_i (1+t_i^\ell)^{-1} \right]. \tag{314}$$

The cycle index of S_n now gives

$$\mathrm{SMG}_{B_n}(\mathbf{t}) = \sum_{\lambda\vdash n} \frac{1}{z_\lambda} \prod_{\ell\geq 1} B_\ell(\mathbf{t})^{m_\ell(\lambda)}, \tag{315}$$

$$\sum_{n\geq 0} u^n\, \mathrm{SMG}_{B_n}(\mathbf{t}) = \exp\left(\sum_{\ell\geq 1} \frac{u^\ell}{\ell} B_\ell(\mathbf{t}) \right). \tag{316}$$

Since $B_\ell = 1 +$ terms of positive $\mathbf{t}$-degree, removing the factor $(1-u)^{-1}$ and taking coefficients proves the stable limit

$$\boxed{\mathrm{SMG}_{B_\infty} = \exp\left(\sum_{\ell\geq 1} \frac{B_\ell(\mathbf{t}) - 1}{\ell} \right).} \tag{317}$$

60.5 Repeated orthogonal sums and the stable law

Fix a lattice L with $G = \mathrm{Aut}(L)$ and let $G \wr S_k$ act on $V_L^{\oplus k}$ by independent lattice automorphisms and component permutations. Define

$$A_\ell(\mathbf{t}) = \mathrm{SMG}_L(t_1^\ell, t_2^\ell, \ldots).$$

Theorem 2 (marked compound-Poisson law)**.**

$$\sum_{k\geq 0} u^k \,\mathrm{SMG}_{G\wr S_k, V_L^{\oplus k}} = \exp\left(\sum_{\ell\geq 1} \frac{u^\ell}{\ell} A_\ell(\mathbf{t})\right), \tag{318}$$

$$\lim_{k\to\infty} \mathrm{SMG}_{G\wr S_k, V_L^{\oplus k}} = \exp\left[\sum_{\ell\geq 1} \frac{\mathrm{SMG}_L(t_1^\ell, t_2^\ell, \ldots) - 1}{\ell}\right] \tag{319}$$

coefficientwise in $\mathbf{t}$*.*

Proof. For a top permutation cycle C of length ℓ, multiply the G-labels around C to obtain its cycle product g_C. On the associated block,

$$\det(I - tx \mid C) = \det(I - t^\ell g_C \mid V_L).$$

For uniformly independent labels, g_C is uniform in G, independently for distinct cycles. The exponential cycle-index formula proves (318). Next write

$$\exp\left(\sum_{\ell\geq 1} \frac{u^\ell A_\ell}{\ell}\right) = \frac{1}{1-u} \exp\left(\sum_{\ell\geq 1} \frac{u^\ell (A_\ell - 1)}{\ell}\right).$$

In each fixed $\mathbf{t}$-degree, the second factor is a polynomial in u. Its coefficient sequence therefore stabilizes after multiplication by $(1-u)^{-1}$, and evaluation at $u = 1$ proves (319). ☐

This theorem applies to repeated copies of any fixed root lattice, the Coxeter–Todd lattice, or any other fixed component. It is a statement about the natural wreath subgroup; an orthogonal sum can have additional automorphisms when accidental isometries occur.

60.6 Isolated and finite lattice families

Rootful Niemeier lattices

Let N be rootful with root sublattice R, Weyl group $W(R) \triangleleft \mathrm{Aut}(N)$, and umbral quotient $G^N = \mathrm{Aut}(N)/W(R)$. Choose a lift $\widetilde{\gamma}$ of each $\gamma \in G^N$. The exact finite reduction is

$$\boxed{\mathrm{SMG}_N = \frac{1}{|G^N|} \sum_{\gamma\in G^N} \frac{1}{|W(R)|} \sum_{w\in W(R)} \prod_i \det(I - t_i w\widetilde{\gamma})^{-1}.} \tag{320}$$

Changing a lift multiplies it by an element of $W(R)$ and merely permutes the inner sum, so (320) is lift-independent. If a lift cycles isomorphic root components with length ℓ and total diagram twist δ_C, that component cycle reduces to

$$\frac{1}{|W(R_a)|} \sum_{x\in W(R_a)} \prod_i \det(I - t_i^\ell x\delta_C)^{-1}. \tag{321}$$

Thus all 23 rootful cases are finite Weyl/quotient class calculations. They are a rank-24 collection, not a rank-asymptotic tower.

Coxeter–Todd, extremal, modular, and neighbor constructions

For the rank-12 Coxeter–Todd lattice K_{12}, and for every specified extremal or modular lattice, the exact answer is simply

$$\mathrm{SMG}_L = \sum_{[g]\subseteq \mathrm{Aut}(L)} \frac{1}{|C(g)|} \prod_{i,m} (1 - t_i^m)^{-a_m(g)}. \tag{322}$$

This is complete once the matrix classes or weighted frame-shape table are supplied. Lattice extremality, modularity, strong perfection, a neighbor relation, or an inclusion $L_n \subset L_{n+1}$ does not by itself impose compatible automorphism representations. Consequently no universal coefficientwise limit follows from those geometric labels alone.

The notebook does not fabricate missing class data for these large isolated groups. It verifies the determinant and frame-shape mechanism on exhaustive small matrix groups; a numerical evaluation of (320) or (322) requires the corresponding complete external class table.

60.7 Explicit lattices with only scalar symmetry

The smallest possible full automorphism group is $\{\pm I\}$. If L has rank d and this full group, then

$$\boxed{\mathrm{SMG}_L = \frac{1}{2}\left[\prod_i (1-t_i)^{-d} + \prod_i (1+t_i)^{-d}\right].} \tag{323}$$

In particular,

$$[m_{(1,1)}] = d^2, \qquad [m_{(2)}] = \frac{d(d+1)}{2}. \tag{324}$$

The verifier uses the following concrete positive-definite integral Gram matrices:

$$Q_2 = \begin{pmatrix} 6 & -1 \\ -1 & 8 \end{pmatrix},$$
$$Q_3 = \begin{pmatrix} 6 & 1 & 2 \\ 1 & 7 & -1 \\ 2 & -1 & 10 \end{pmatrix},$$
$$Q_4 = \begin{pmatrix} 6 & 1 & 1 & 2 \\ 1 & 7 & 1 & 2 \\ 1 & 1 & 11 & 1 \\ 2 & 2 & 1 & 13 \end{pmatrix}.$$

Proposition 1. *For $d = 2, 3, 4$, the lattice with Gram matrix Q_d has full automorphism group $\{\pm I_d\}$.*

Proof. Each matrix is strictly diagonally dominant with positive diagonal, hence positive definite. Its Gershgorin lower bound δ_d satisfies $v^T Q_d v \geq \delta_d \|v\|_2^2$. If $A^T Q_d A = Q_d$ and v_j is column j of A, then $v_j^T Q_d v_j = (Q_d)_{jj}$. Therefore every coordinate of every column has absolute value at most

$$B_d = \left\lfloor \sqrt{\max_j (Q_d)_{jj}/\delta_d} \right\rfloor.$$

The values (δ_d, B_d) are $(5,1), (3,1), (2,2)$. Exhaustive exact enumeration of the finite boxes $[-B_d, B_d]^d$, retaining precisely the column tuples with pairings $v_i^T Q_d v_j = (Q_d)_{ij}$, returns two matrices in each rank: I_d and $-I_d$. This column-pairing condition is exactly $A^T Q_d A = Q_d$, so the enumeration is exhaustive. ☐

Equations (324) show immediate degree-two divergence in any rank-growing tower whose full automorphism groups stay scalar. This is the basic obstruction to a universal law for arbitrary lattice families.

60.8 Tensor products of lattice representations

Let $G = \mathrm{Aut}(L)$ act on $V = V_L$ with character χ. The local group G^k acts on $V^{\otimes k}$. Since

$$\mathrm{Tr}\big((g_1 \otimes \cdots \otimes g_k)^r\big) = \prod_{j=1}^{k} \chi(g_j^r),$$

Theorem 1 gives

$$\boxed{\mathrm{SMG}_{G^k,V^{\otimes k}} = \frac{1}{|G|^k}\sum_{g_1,\dots,g_k}\exp\left[\sum_{r\geq 1}\frac{p_r}{r}\prod_{j=1}^{k}\chi(g_j^r)\right].} \tag{325}$$

Reordering tensor factors in the multilinear coefficient gives

$$\boxed{[m_{(1^s)}]\,\mathrm{SMG}_{G^k,V^{\otimes k}} = \left(\dim(V^{\otimes s})^G\right)^k.} \tag{326}$$

If V is irreducible orthogonal of dimension greater than one, then $\dim(V^{\otimes 2})^G = 1$, while

$$\dim(V^{\otimes 4})^G = \dim \mathrm{End}_G(V\otimes V) \geq 2.$$

The last inequality holds because $V\otimes V$ contains the trivial module and at least one nontrivial constituent. Hence

$$[m_{(1,1,1,1)}]\,\mathrm{SMG}_{G^k,V^{\otimes k}} \geq 2^k. \tag{327}$$

Tensor towers therefore usually exhibit exponential moment divergence, unlike the repeated-sum law in Theorem 2.

If factor permutations are included, write $x = (g_1,\dots,g_k;\pi) \in G \wr S_k$. For a cycle C of π, let a_C be the ordered cycle product and $d_C(r) = \gcd(|C|, r)$. The rth power splits C into $d_C(r)$ cycles, each carrying a conjugate of $a_C^{r/d_C(r)}$. Contracting indices around those cycles proves

$$\boxed{\mathrm{Tr}(x^r \mid V^{\otimes k}) = \prod_C \chi\left(a_C^{r/d_C(r)}\right)^{d_C(r)}.} \tag{328}$$

Substitution into (303) is the exact tensor-wreath σ-MGF.

60.9 Exact computational verification

The companion marimo notebook calls a pure-Python exact verifier. No floating-point eigenvalue or numerical determinant enters a pass condition. The cases are grouped by the formulas they test.

Matrix constructions

- **Type A.** For $n = 3,4,5$, every $\pm\sigma$ is represented on the integral basis $e_1 - e_n, \dots, e_{n-1} - e_n$. The matrices preserve the Gram matrix with diagonal 2 and off-diagonal 1.
- **Signed permutations.** Every signed permutation matrix is enumerated in dimensions $2,3,4$ and checked against the identity Gram matrix. These are structural tests of (315); the dimension-4 case is not claimed to be the full D_4 automorphism group.
- **Scalar lattices.** All automorphisms of Q_2, Q_3, Q_4 are enumerated by the exhaustive bound in Proposition 1.
- **Repeated sums.** All 288 block-monomial matrices of $\mathrm{Aut}(A_2)\wr S_2$ on $A_2 \perp A_2$ are constructed.
- **Tensor products.** Kronecker matrices construct $\mathrm{Aut}(A_2)^2$ and $\mathrm{Aut}(A_2)\wr S_2$ on $V_{A_2}^{\otimes 2}$. Their abstract orders are 144 and 288, while the matrix-image orders are 72 and 144 because $(-I,-I)$ acts trivially.

Four independent checks

For each concrete matrix A, the direct route computes $T_r = \mathrm{Tr}(A^r)$ and then

$$h_0(A) = 1, \qquad q h_q(A) = \sum_{r=1}^{q} T_r h_{q-r}(A). \tag{329}$$

It averages $\prod_j h_{\tau_j}(A)$, which is the Reynolds trace on W_τ. Independently, the verifier recovers $a_m(A)$ by (306), checks the polynomial identity

$$\det(I - tA) = \prod_m (1-t^m)^{a_m(A)},$$

including all negative exponents by cross-multiplication, and expands the frame product. A third route uses only the family formulas: ordinary cycles, signed cycle products, marked wreath cycles, or the tensor trace (328).

Finally, for $\tau = (1), (2), (1,1), (3)$, the code explicitly constructs the integer matrices on each $\mathrm{Sym}^{\tau_j} V$, forms their tensor products, and solves all generator-fixed equations over $\mathbb{Q}$. This common-kernel dimension is compared with the Reynolds trace. Thus the fixed-space check is not merely another expansion of the averaged determinant.

Representative cases and results

family	matrix image	dimension	image order
type A	$\mathrm{Aut}(A_2), \mathrm{Aut}(A_3), \mathrm{Aut}(A_4)$	$2,3,4$	$12,48,240$
signed	$C_2 \wr S_2, C_2 \wr S_3, C_2 \wr S_4$	$2,3,4$	$8,48,384$
scalar	$\mathrm{Aut}(Q_2), \mathrm{Aut}(Q_3), \mathrm{Aut}(Q_4)$	$2,3,4$	$2,2,2$
orthogonal sum	$\mathrm{Aut}(A_2) \wr S_2$	4	288
local tensor	$\rho(\mathrm{Aut}(A_2)^2)$	4	72
tensor wreath	$\rho(\mathrm{Aut}(A_2) \wr S_2)$	4	144

Selected degree-four outputs illustrate the different asymptotics:

matrix image	$[m_2]$	$[m_{11}]$	$[m_{22}]$	$[m_{1111}]$
$\mathrm{Aut}(A_2)$	1	1	2	3
$\mathrm{Aut}(A_3)$	1	1	3	4
$\mathrm{Aut}(A_4)$	1	1	3	4
$C_2 \wr S_n$, $n = 2,3,4$	1	1	3	4
$\mathrm{Aut}(Q_2)$	3	4	9	16
$\mathrm{Aut}(Q_3)$	6	9	36	81
$\mathrm{Aut}(Q_4)$	10	16	100	256
$\mathrm{Aut}(A_2) \wr S_2$ on $V^{\oplus 2}$	1	1	4	6
$\rho(\mathrm{Aut}(A_2)^2)$ on $V^{\otimes 2}$	1	1	5	9
$\rho(\mathrm{Aut}(A_2) \wr S_2)$ on $V^{\otimes 2}$	1	1	4	6

All odd total-degree values vanish, as required by (307). For the local tensor case, the value $9 = 3^2$ at m_{1111} verifies (326); adjoining the factor swap takes invariants in the symmetric square of a three-dimensional space and gives $\binom{3+1}{2} = 6$.

The degree-four audit contains twelve partitions per case and three recorded coefficient routes, for $12 \cdot 12 \cdot 3 = 432$ values. The fixed-space audit has $12 \cdot 4 = 48$ comparisons. There are 1,250 distinct matrices; six frame checks per matrix (one determinant identity and five complete characters through degree four) give 7,500 elementwise checks.

60.10 Asymptotic classification and scope

family	exact formula	asymptotic conclusion
A_{n-1}	(312)	Poisson-cycle limit (313)
$\mathbb{Z}^n$, types B/C, D_n for $n \geq 5$	(315)	stable signed-cycle law (317)
$R^{\perp k}$ with component permutations	(318)	compound-Poisson limit (319)
rootful Niemeier	(320)	finite rank-24 collection
K_{12} and isolated extremal/modular lattices	(322)	sequence-dependent; repeated sums use (319)
$\mathrm{Aut}(L) = \{\pm I\}$	(323)	degree-two divergence
$V_L^{\otimes k}$ under G^k	(325)	degree-four exponential growth
neighbor or laminated towers	(302)	requires compatible group data

The structural distinction is sharp: orthogonal repeated sums with component permutations yield stable marked compound-Poisson frame-shape laws, whereas tensor powers and low-symmetry towers force divergent invariant moments. The universal object is therefore the weighted frame-shape distribution

$$\boxed{\mathrm{SMG}_L = \mathbb{E}_{g\in\mathrm{Aut}(L)} \prod_{i,m} (1-t_i^m)^{-a_m(g)},}$$

together with whatever compatibility a specified lattice tower supplies.

60.11 Reproducibility

From the repository root, run

```
python3 -m lambda_distributions.dists.lattice_sigma_mgfs.verification
python3 -m pytest -q lambda_distributions/dists/lattice_sigma_mgfs/test_verification.py
marimo edit marimo_notebooks/lattices.py
```

The LaTeX source is standalone and can be rebuilt in its folder with

```
latexmk -pdf -interaction=nonstopmode -halt-on-error \
  lattice_sigma_mgfs_proof_and_verification.tex
```

No command in this workflow modifies `proved_matrix_groups/all_new_matrix_groups_proofs.pdf`.

Part VI

Compact classical, exceptional, and Schur-functor families

61 Classical compact groups and weighted tori

Abstract

We prove the coefficient formulas proposed for the defining representations of SU(n), SO(n), O(n), and arbitrary compact tori. We then describe an independent matrix calculation: connected-group invariants are the common kernel of the derived representation, while torus integrals are replaced by provably alias-free finite diagonal grids. The accompanying marimo notebooks perform 100 exact integer comparisons. All pass. We also state two necessary corrections explicitly: orthogonal Schur shapes must have length at most n, and all formulas are interpreted coefficientwise (or after finite truncation).

61.1 Imported compact-group identity

For each compact group below, $\mathrm{SMG}_{G,V} = \mathbb{E}[\mathrm{Exp}_\sigma(X_{G,V}h_1)]$ and the coefficient formula are Tools 2 and 1. We begin with the branching rules distinguishing $SU(n)$, $SO(n)$, $O(n)$, and weighted tori.

61.2 The $\mathrm{SU}(n)$ coefficient formula

As a polynomial GL_n-module,

$$\operatorname{ch} W_\tau = h_\tau = \sum_{\substack{\lambda\vdash|\tau| \\ \ell(\lambda)\le n}} K_{\lambda,\tau} s_\lambda.$$

The restriction of $S_\lambda V$ to SU(n) is trivial precisely when its highest-weight differences vanish, namely when $\lambda = (q^n) = (q,\dots,q)$. In that case $S_{(q^n)}V \cong (\det V)^q$, which is trivial on SU(n). Hence

Theorem 1. *For the defining representation of* SU(n),

$$\boxed{\dim W_\tau(V)^{\mathrm{SU}(n)} = \begin{cases} K_{(q^n),\tau}, & |\tau| = nq, \\ 0, & n\nmid|\tau|. \end{cases}}$$

The first correction to the $U(n)$ answer occurs at degree n and includes $\tau = (1^n)$, represented by the determinant tensor.

61.3 The $\mathrm{SO}(n)$ and $\mathrm{O}(n)$ formulas

The orthogonal branching criterion says

$$\dim(S_\lambda V)^{\mathrm{O}(n)} = 1 \quad\Longleftrightarrow\quad \lambda = 2\delta,$$

provided $\ell(\lambda) \le n$; otherwise $S_\lambda(\mathbb{C}^n) = 0$. Because SO($n$) has index two in O($n$), an SO($n$)-fixed line is either trivial or determinant-isotypic under O(n). Tensoring by determinant adds one box to each of exactly n rows. Therefore

Theorem 2. *For $n \ge 2$ and the defining representation,*

$$\dim W_\tau(V)^{\mathrm{O}(n)} = \sum_{\substack{\delta \\ \ell(2\delta)\le n}} K_{2\delta,\tau},$$

$$\dim W_\tau(V)^{\mathrm{SO}(n)} = \sum_{\substack{\delta \\ \ell(2\delta)\le n}} K_{2\delta,\tau} + \sum_{\substack{\delta \\ \ell(\delta)\le n}} K_{2\delta+(1^n),\tau}.$$

The length restriction is essential in finite dimension. The orientation correction begins at degree n, occurs only in degrees congruent to n modulo two, and contains the Levi-Civita tensor at $\tau = (1^n)$.

61.4 Weighted compact tori

Let T^r act diagonally on $\mathbb{C}^d$ with integer weights $w_1, \dots, w_d \in \mathbb{Z}^r$. Orthogonality of torus characters gives

$$\boxed{\mathrm{SMG}_{T^r,V} = \mathrm{CT}_{z_1,\dots,z_r} \prod_{i\geq 1} \prod_{j=1}^{d} (1 - t_i z^{w_j})^{-1}.}$$

Consequently, $\dim W_\tau(V)^{T^r}$ is the number of arrays $a_{ij} \in \mathbb{Z}_{\geq 0}$ satisfying

$$\sum_j a_{ij} = \tau_i, \qquad \sum_{i,j} a_{ij} w_j = 0.$$

Lemma 1 (Alias-free finite grid)**.** *Suppose every tested monomial has each weight coordinate in $[-B, B]$. For any integer $M > 2B$, averaging over the explicit subgroup $(\mu_M)^r \subset T^r$ selects exactly the zero integer weight.*

Proof. The finite average selects weights congruent to zero modulo M. The only multiple of M in $[-B, B]$ is zero. Thus the finite matrix average equals the continuous Haar constant term throughout the tested truncation. ☐

61.5 Independent direct matrix computation

The notebook does not recover the left side from Schur decomposition. In the normalized occupancy basis $|a_1, \dots, a_n\rangle$ of $\mathrm{Sym}^d V$, a matrix unit E_{uv} acts infinitesimally by

$$d\,\mathrm{Sym}^d(E_{uv})|a\rangle = \begin{cases} \sqrt{a_v(a_u+1)}\,|a + e_u - e_v\rangle, & u \neq v, \\ a_u|a\rangle, & u = v. \end{cases}$$

On W_τ the derived action is the Kronecker sum over tensor factors. The common nullity for a basis of $\mathfrak{sl}_n(\mathbb{C})$ gives the $\mathrm{SU}(n)$ invariants; skew matrix units give $\mathrm{SO}(n)$ invariants. Adding the constraint $\rho(\mathrm{diag}(-1, 1, \dots, 1)) - I$ gives $\mathrm{O}(n)$ invariants. This is valid because invariants of a connected matrix group are exactly the vectors killed by its derived Lie algebra.

Kostka numbers on the formula side are counted separately by backtracking over semistandard tableaux. Thus the two columns share only the mathematical claim being tested, not an implementation route.

61.6 Test matrix and outcome

Family	Dimensions	Tested degrees	Direct mechanism	Checks
$\mathrm{SU}(n)$	$n = 2, 3$	0–4 representative τ	Lie-kernel nullity	15
$\mathrm{SO}(n)$	$n = 2, 3, 4$	through orientation degree	Lie-kernel nullity	22
$\mathrm{O}(n)$	$n = 2, 3, 4$	through orientation degree	kernel plus reflection	20
T^r	$r = 1, 2$; $d = 2, 3$	all τ through degree 4 or 5	finite grid average	43
Total				**100**

All 100 comparisons pass. The largest directly formed continuous-group target has dimension 256. Torus grids use the conservative bound $M = 2D \max_{j,k} |(w_j)_k| + 1$, so the no-aliasing lemma applies.

61.7 Reproduction

From the repository root, run the following modules with `python3 -m`:

- `for_this_guy.matrix_group_formula_verification.classical_compact.verification`
- `for_this_guy.matrix_group_formula_verification.tori.verification`

Open the corresponding `marimo_notebooks/classical_compact.py` and `marimo_notebooks/tori.py` files with `marimo edit` for the interactive tables. The verification ranges are deliberately small enough to inspect and large enough to include determinant/orientation corrections. These computations test the stated dimensions; they do not replace the proofs above or establish unlisted exceptional-group simplifications.

62 The compact G_2 sigma-MGF on V_7

Abstract

We prove the compact Molien–Weyl formula for the minimal seven-dimensional G_2-module, construct its maximal-torus matrices from the weights, and verify every Schur coefficient through degree three exactly. The result is $\mathrm{SMG}_{G_2,7} = 1 + h_2 + e_3 + O_{\geq 4}$, and the ordinary specialization is $(1-t^2)^{-1}$.

62.1 Imported compact identity

The compact Molien and coefficient formulas are imported. Weyl integration reduces $\mathbb{E}[\mathrm{Exp}_\sigma(X_{G_2,V_7}h_1)]$ to the maximal-torus constant term below.

62.2 Specialization to the minimal module

The weights of V_7 are zero together with the six short roots. Since $|W(G_2)| = 12$,

$$\boxed{\mathrm{SMG}_{G_2,7} = \frac{1}{12}\,\mathrm{CT}_z\left[D_{G_2}(z)\prod_a\left((1-t_a)^{-1}\prod_{\beta\in\Phi_{\text{short}}}(1-t_az^\beta)^{-1}\right)\right].}$$

The invariant metric lies in $\mathrm{Sym}^2 V_7$, while the octonionic associative form lies in $\bigwedge^3 V_7$. Their uniqueness and the absence of other invariants below degree four give

$$\boxed{\mathrm{SMG}_{G_2,7} = 1 + h_2 + e_3 + O_{\geq 4}.}$$

A standard invariant-theory theorem for the minimal G_2 module states that $\mathbb{C}[V_7]^{G_2} = \mathbb{C}[q]$, where q is the invariant quadratic norm. Using that external theorem, $\mathrm{SMG}_{G_2,7}(t,0,\ldots) = (1-t^2)^{-1}$. The exact Weyl computation in this note independently verifies the coefficients through degree three.

62.3 Exact matrix verification

The notebook generates the root system from the G_2 Cartan matrix and forms

$$M(\theta) = \mathrm{diag}\big(e^{i\langle\mu,\theta\rangle}\big)_{\mu\in\{0\}\cup\Phi_{\text{short}}} \in U(7).$$

It compares $\prod_a \det(I - t_aM)^{-1}$ directly with the product over the seven diagonal eigenvalues. For integration it evaluates Schur characters by Jacobi–Trudi and uses the one-sided Weyl denominator

$$\prod_{\alpha>0}(1-z^{-\alpha}) = \sum_{w\in W}\det(w)z^{w\rho-\rho}.$$

Thus every coefficient is an exact integer, with no quadrature tolerance. In the requested monomial basis the exact degree-at-most-three signature is

$$([m_{(2)}],[m_{(1,1)}],[m_{(3)}],[m_{(2,1)}],[m_{(1,1,1)}]) = (1,1,0,0,1).$$

λ	$\varnothing$	(1)	(2)	$(1,1)$	(3)	$(2,1)$	$(1,1,1)$
proposed	1	0	1	0	0	0	1
exact Weyl check	1	0	1	0	0	0	1

The sampled torus matrix has unitarity residual 3.2×10^{-16} and the two integrand evaluations differ by less than 9×10^{-16}. All seven coefficient checks pass.

Reproducibility. Run `marimo_notebooks/lie_group_g2.py` with marimo. The shared exact-arithmetic module is in the parent folder.

External invariant-theory source

See J. F. Adams, *Lectures on Exceptional Lie Groups*, Chicago Lectures in Mathematics (1996), for the minimal G_2 module and its invariant metric. The all-degree one-copy invariant-ring statement is an external input; the low-degree Weyl checks above are internal to this artifact.

63 The compact $\mathrm{Spin}(7)$ sigma-MGF on S_8

Abstract

For the complexification of the real eight-dimensional spin representation, we prove the compact Molien–Weyl formula, construct explicit maximal-torus matrices, and verify all coefficients through total degree four by exact constant terms. In the Schur basis the answer is $1 + h_2 + h_4 + s_{(2,2)} + e_4 + O_{\geq 6}$. The coefficient of $m_{(1,1,1,1)}$ is therefore 4, with the extra invariant supplied by the Cayley alternating four-form.

63.1 Imported coefficient formula

Tools 1 and 2 apply to the compact spin module. The new calculation begins with the B_3 weights and Weyl constant term.

63.2 Matrix model and Weyl constant term

Use type B_3 with orthonormal coordinates e_1, e_2, e_3. The weights of S_8 are

$$\mathrm{Wt}(S_8) = \left\{ \tfrac{1}{2}(\epsilon_1 e_1 + \epsilon_2 e_2 + \epsilon_3 e_3) : \epsilon_i \in \{\pm 1\} \right\}.$$

For $\theta \in \mathbb{R}^3$ this constructs the explicit unitary matrix

$$M(\theta) = \mathrm{diag}_{\epsilon \in \{\pm 1\}^3} \exp\left(\frac{i}{2} \sum_{j=1}^{3} \epsilon_j \theta_j \right) \in U(8).$$

These matrices form the image of a maximal torus of $\mathrm{Spin}(7)$. Together with the 18 roots and $|W(B_3)| = 48$, the Weyl integration formula turns the group integral into the exact constant term

$$\boxed{\mathrm{SMG}_{\mathrm{Spin}(7),8} = \frac{1}{48} \mathrm{CT}_z \left[\prod_{\alpha > 0} (1 - z^{\alpha})(1 - z^{-\alpha}) \prod_{a, \mu \in \mathrm{Wt}(S_8)} (1 - t_a z^{\mu})^{-1} \right].}$$

Thus it is unnecessary to approximate Haar measure by random matrices.

63.3 Predicted low-degree tensors

The central element $-1 \in \mathrm{Spin}(7)$ acts by $-I$ on S_8, so all odd-total-degree coefficients vanish. The tensor square is

$$S_8 \otimes S_8 = \mathbf{1} \oplus \mathbf{7} \oplus \mathbf{21} \oplus \mathbf{35}, \qquad \mathrm{Sym}^2 S_8 = \mathbf{1} \oplus \mathbf{35}, \quad \bigwedge^2 S_8 = \mathbf{7} \oplus \mathbf{21}.$$

It is multiplicity-free, hence

$$\dim (S_8^{\otimes 4})^{\mathrm{Spin}(7)} = \dim \mathrm{End}_{\mathrm{Spin}(7)}(S_8 \otimes S_8) = 4.$$

Three tensors are the metric pairings. The fourth is the Cayley form in $\bigwedge^4 S_8^*$. In Schur notation this yields

$$\boxed{\mathrm{SMG}_{\mathrm{Spin}(7),8} = 1 + h_2 + h_4 + s_{(2,2)} + e_4 + O_{\geq 6}.}$$

Changing to the monomial basis gives, through degree four,

τ	(2)	$(1,1)$	(4)	$(3,1)$	$(2,2)$	$(2,1,1)$	$(1,1,1,1)$
$[m_\tau]$	1	1	1	1	2	2	4

Every coefficient of total degree one or three is zero.

63.4 Independent exact verification

The marimo notebook generates the B_3 root system and the eight-weight orbit from the Cartan matrix. It then performs two distinct checks. First, for a concrete $M(\theta)$ it compares

$$\prod_a \det(I - t_a M(\theta))^{-1} \quad \text{with} \quad \prod_{a,\mu}(1 - t_a e^{i\langle \mu,\theta\rangle})^{-1}.$$

Second, it forms the characters $h_{\tau_1} \cdots h_{\tau_r}$ directly from the weights and applies the one-sided Weyl-denominator identity

$$\prod_{\alpha>0}(1 - z^{-\alpha}) = \sum_{w\in W} \det(w) z^{w\rho-\rho}.$$

The resulting constant terms are exact integers and do not use the quoted tensor-square decomposition as input.

The exact output is

τ	(2)	(1, 1)	(4)	(3, 1)	(2, 2)	(2, 1, 1)	(1, 1, 1, 1)
proposed	1	1	1	1	2	2	4
Weyl check	1	1	1	1	2	2	4

with every omitted partition of degree at most four returning zero. In the Schur basis the nonzero multiplicities are

$$[s_{(2)}] = [s_{(4)}] = [s_{(2,2)}] = [s_{(1,1,1,1)}] = 1.$$

Reproducibility. Run the marimo notebook `marimo_notebooks/lie_group_spin7.py`. Its shared exact-arithmetic core is in the parent folder. The test suite also executes this degree-four case noninteractively.

64 Compact Spin groups and a Pin component reduction

Canonical testing artifacts.
Exact backend: `lambda_distributions/dists/verification.py`.
Regression: `lambda_distributions/dists/test_verification.py`.
Marimo: `marimo_notebooks/spin_pin.py`.

Scope and outcome

We prove the exact compact Molien–Weyl formulas for the spin representation of $\mathrm{Spin}(2n+1)$ and the two half-spin representations of $\mathrm{Spin}(2n)$. Clifford algebra decompositions give the decisive fourth multilinear coefficients $n+1$ and, respectively, $n/2+1$ or $(n-1)/2$. They grow with rank, so the unnormalized spinor sigma-MGFs have no coefficientwise large-rank limit. We also state the exact two-component Pin formula, distinguish the Pin^+ and Pin^- lifts, and record the mixed spinor–vector and adjoint consequences. An accompanying marimo notebook independently constructs Weyl characters and Clifford matrices. All 76 exact and matrix-level checks pass through rank five where applicable.

64.1 Imported compact coefficient formula

Let G be compact and W a finite-dimensional unitary module. The following is Tools 2 and 1, followed by Weyl integration; it is stated with labels because later spin formulas use it.

Theorem 1 (Imported compact Molien–Weyl formula)**.**

$$\mathrm{SMG}_{G,W}(\mathbf{t}) = \int_G \prod_{i\geq 1} \det(I - t_i\rho(g))^{-1}\,dg, \tag{330}$$

$$= \int_G \exp\left(\sum_{r\geq 1} \frac{p_r(\mathbf{t})}{r}\chi_W(g^r)\right) dg. \tag{331}$$

If T is a maximal torus with positive roots Φ^+ and Weyl group W_G,

$$\boxed{\mathrm{SMG}_{G,W} = \frac{1}{|W_G|}\,\mathrm{CT}\left[\prod_{\alpha\in\Phi^+}(1-z^{\alpha})(1-z^{-\alpha})\exp\left(\sum_{r\geq 1}\frac{p_r}{r}\chi_W(z^r)\right)\right].} \tag{332}$$

For a W_G-invariant Laurent polynomial f, the Weyl denominator gives the exact finite lookup

$$\frac{1}{|W_G|}\,\mathrm{CT}\left[\prod_{\alpha>0}(1-z^{\alpha})(1-z^{-\alpha})f(z)\right] = \sum_{w\in W_G}\det(w)\,[z^{\rho_\Phi - w\rho_\Phi}]f(z). \tag{333}$$

No Reynolds or Cauchy proof is repeated here. The new content starts with the Clifford-theoretic weights, all-rank tensor-square decompositions, and the Pin component reduction.

64.2 Type B_n: $\mathrm{Spin}(2n+1)$ on its spin module

Let $V = V_{2n+1}$ be the vector representation and let Δ_n be the 2^n-dimensional complex spin representation. In orthonormal torus coordinates the weights are

$$\frac{1}{2}(\epsilon_1 e_1 + \cdots + \epsilon_n e_n), \qquad \epsilon_j \in \{\pm 1\},$$

each once. Hence

$$\chi_{\Delta_n}(z) = \prod_{j=1}^{n}(z_j^{1/2} + z_j^{-1/2}). \tag{334}$$

Since $|W(B_n)| = 2^n n!$, Theorem 1 gives

$$\boxed{\mathrm{SMG}_{\mathrm{Spin}(2n+1),\Delta_n} = \frac{1}{2^n n!}\,\mathrm{CT}\left[\mathcal{D}_{B_n}(z)\exp\left(\sum_{r\ge 1}\frac{p_r}{r}\prod_{j=1}^{n}(z_j^{r/2}+z_j^{-r/2})\right)\right].} \tag{335}$$

Equivalently, if

$$H_k^{\Delta_n}(z) = [u^k]\exp\left(\sum_{r\ge 1}\frac{u^r}{r}\chi_{\Delta_n}(z^r)\right),$$

then every requested coefficient is the finite calculation

$$[m_\tau]\,\mathrm{SMG} = \frac{1}{2^n n!}\,\mathrm{CT}\left[\mathcal{D}_{B_n}(z)\prod_j H_{\tau_j}^{\Delta_n}(z)\right]. \tag{336}$$

The nontrivial element in the kernel of $\mathrm{Spin}(2n+1) \to SO(2n+1)$ acts as $-I$ on Δ_n. It follows immediately that

$$[m_\tau]\,\mathrm{SMG} = 0 \qquad \text{if } |\tau| \text{ is odd.} \tag{337}$$

The spin module is self-dual. The Clifford-periodic invariant bilinear form is symmetric for $n \equiv 0,3 \pmod 4$ and alternating for $n \equiv 1,2 \pmod 4$. Therefore

$$[m_{(1,1)}]\,\mathrm{SMG} = 1, \qquad [m_{(2)}]\,\mathrm{SMG} = \begin{cases} 1, & n \equiv 0,3 \pmod 4, \\ 0, & n \equiv 1,2 \pmod 4. \end{cases} \tag{338}$$

One proof of the symmetry sign starts with the alternating form on the $SU(2) \cong \mathrm{Spin}(3)$ module and tensors the Pauli-matrix Clifford construction. At each two-dimensional Clifford extension the transpose sign changes in the periodic pattern $--++$.

Theorem 2 (odd-spin fourth moment)**.** *For every $n \ge 1$,*

$$\boxed{\dim(\Delta_n^{\otimes 4})^{\mathrm{Spin}(2n+1)} = n+1.} \tag{339}$$

Proof. The even Clifford algebra acts faithfully on Δ_n, and the symbol map from Clifford products to exterior forms is Spin-equivariant. Hodge duality then gives the multiplicity-free decomposition

$$\Delta_n \otimes \Delta_n \cong \mathrm{End}(\Delta_n) \cong \bigoplus_{k=0}^{n}\Lambda^k V_{2n+1}. \tag{340}$$

The summands are pairwise nonisomorphic and self-dual. Since Δ_n is self-dual,

$$\dim(\Delta_n^{\otimes 4})^G = \dim \mathrm{End}_G(\Delta_n \otimes \Delta_n) = n+1.$$

□

64.3 Type D_n: half-spin modules of $\mathrm{Spin}(2n)$

Let Δ_n^+ and Δ_n^- be the two half-spin modules, each of dimension 2^{n-1}. Set

$$A_n(z) = \prod_{j=1}^{n}(z_j^{1/2}+z_j^{-1/2}), \qquad B_n(z) = \prod_{j=1}^{n}(z_j^{1/2}-z_j^{-1/2}).$$

Separating spin weights by the parity of their minus signs gives

$$\chi_{\Delta_n^\pm}(z) = \frac{1}{2}(A_n(z) \pm B_n(z)). \tag{341}$$

Because $|W(D_n)| = 2^{n-1}n!$,

$$\boxed{\mathrm{SMG}_{\mathrm{Spin}(2n),\Delta_n^\pm} = \frac{1}{2^{n-1}n!}\,\mathrm{CT}\left[\mathcal{D}_{D_n}(z)\exp\left(\sum_{r\geq 1}\frac{p_r}{r}\chi_{\Delta_n^\pm}(z^r)\right)\right].} \tag{342}$$

The half-spin central character has order two for even n and order four for odd n. Thus

$$[m_\tau]\,\mathrm{SMG} = 0 \qquad \begin{cases} \text{if } |\tau| \text{ is odd}, & n \text{ even}, \\ \text{unless } 4 \mid |\tau|, & n \text{ odd}. \end{cases} \tag{343}$$

Moreover,

$$(\Delta_n^+)^* \cong \begin{cases} \Delta_n^+, & n \text{ even}, \\ \Delta_n^-, & n \text{ odd}, \end{cases}$$

and, in the self-dual case, the form is symmetric for $n \equiv 0 \pmod 4$ and alternating for $n \equiv 2 \pmod 4$. Consequently

$$[m_{(1,1)}]\,\mathrm{SMG} = \mathbf{1}_{2|n}, \qquad [m_{(2)}]\,\mathrm{SMG} = \mathbf{1}_{4|n}. \tag{344}$$

Theorem 3 (half-spin fourth moments)**.** *For $n \geq 2$,*

$$\boxed{\dim((\Delta_n^+)^{\otimes 4})^{\mathrm{Spin}(2n)} = \begin{cases} n/2+1, & n \text{ even}, \\ (n-1)/2, & n \text{ odd}. \end{cases}} \tag{345}$$

The balanced count is

$$\boxed{\dim\left((\Delta_n^+ \otimes \Delta_n^{+*})^{\otimes 2}\right)^{\mathrm{Spin}(2n)} = \left\lfloor \frac{n}{2} \right\rfloor + 1.} \tag{346}$$

Proof. The Clifford symbol map and the two Hodge eigenspaces in middle degree give

$$\Delta_n^+ \otimes \Delta_n^+ \cong \bigoplus_{\substack{0\leq k<n \\ k\equiv n \pmod 2}} \Lambda^k V_{2n} \;\oplus\; \Lambda^n_\varepsilon V_{2n}, \tag{347}$$

where Λ^n_ε is one of the two middle-degree irreducibles. The tensor square of Δ_n^- has the same lower-degree summands and the opposite middle constituent.

If n is even, Δ_n^+ is self-dual and (347) has $n/2+1$ pairwise nonisomorphic summands. The invariant fourth power is its endomorphism algebra, proving the first line of (345). If n is odd, invariants are

$$\mathrm{Hom}_G(\Delta_n^- \otimes \Delta_n^-, \Delta_n^+ \otimes \Delta_n^+).$$

Only the $(n-1)/2$ lower exterior powers occur on both sides; the middle constituents are opposite. This proves the second line.

Finally, $\Delta_n^+ \otimes \Delta_n^{+*} = \mathrm{End}(\Delta_n^+)$ is multiplicity-free. Its exterior-form constituents are the even degrees below n, together with one middle summand when n is even. Their number is $\lfloor n/2 \rfloor + 1$, proving (346). □

The special case $n = 3$ is a useful boundary check: $\mathrm{Spin}(6) \cong SU(4)$ and Δ_3^+ is the defining four-dimensional representation. Its first one-copy invariant is the degree-four determinant, so the fourth multilinear coefficient is one, while both quadratic coefficients vanish.

64.4 Asymptotic obstruction

Corollary 1. *Neither the unnormalized odd-spin sequence $\mathrm{SMG}_{\mathrm{Spin}(2n+1),\Delta_n}$ nor either half-spin sequence has a finite coefficientwise limit in the completed symmetric-function ring.*

Proof. The fixed coefficient $[m_{(1,1,1,1)}]$ equals $n+1$ in type B_n and is asymptotic to $n/2$ in type D_n. It diverges along every sequence with $n \to \infty$. Coefficientwise convergence would require this scalar coefficient to converge, which is impossible. $\square$

The conclusion is not contradicted by $\dim \Delta_n = 2^n$. Increasing rank opens a new exterior-power channel in (340) or every two ranks in (347). A normalized, logarithmic, or cumulant renormalization may still have a limit, but the raw sigma-MGF does not.

64.5 Pin groups: the disconnected component formula

Fix a concrete complex pinor representation Π_N^ε of $\mathrm{Pin}^\varepsilon(N)$, $\varepsilon \in \{+,-\}$, and choose u_ε in the odd component. Since

$$\mathrm{Pin}^\varepsilon(N) = \mathrm{Spin}(N) \sqcup u_\varepsilon \,\mathrm{Spin}(N),$$

normalized Haar measure assigns mass 1/2 to each component. Therefore

$$\begin{aligned}\mathrm{SMG}_{\mathrm{Pin}^\varepsilon(N),\Pi_N^\varepsilon} &= \frac{1}{2}\int_{\mathrm{Spin}(N)} \exp\left(\sum_{r\ge 1}\frac{p_r}{r}\chi_{\Pi_N^\varepsilon}(g^r)\right)dg \\ &\quad + \frac{1}{2}\int_{\mathrm{Spin}(N)} \exp\left(\sum_{r\ge 1}\frac{p_r}{r}\chi_{\Pi_N^\varepsilon}((u_\varepsilon g)^r)\right)dg.\end{aligned} \tag{348}$$

This is an exact identity, but the second integrand depends on the chosen Pin extension, not only on the restricted Spin character.

For $N = 2n$, the restriction is $\Delta_n^+ \oplus \Delta_n^-$. An odd Clifford generator anticommutes with chirality and is therefore block off-diagonal. Every odd power remains block off-diagonal, so

$$\chi_{\Pi_{2n}^\varepsilon}(x^{2r+1}) = 0 \qquad (x \notin \mathrm{Spin}(2n)). \tag{349}$$

Only the p_{2r} terms remain in the odd-component exponential. The two Pin groups must not be identified: in the concrete Clifford models used here,

$$u_+ = \gamma_1, \quad u_- = i\gamma_1, \qquad u_+^2 = I, \quad u_-^2 = -I. \tag{350}$$

Thus their even power characters can differ.

64.6 Mixed spinor–vector and adjoint modules

Character calculus immediately extends the exact constant term. For $W_{a,b} = \Delta^{\otimes a} \otimes V^{\otimes b}$,

$$\chi_{W_{a,b}}(z) = \chi_\Delta(z)^a \chi_V(z)^b, \tag{351}$$

where

$$\chi_{V_{2n+1}} = 1 + \sum_j (z_j + z_j^{-1}), \qquad \chi_{V_{2n}} = \sum_j (z_j + z_j^{-1}).$$

Substitution of (351) into (332) is the exact sigma-MGF at every finite rank. For a direct sum $\Delta \oplus V$, the character product is replaced by $\chi_\Delta + \chi_V$.

The adjoint representation factors through $SO(N)$ and is $\mathfrak{so}_N \cong \Lambda^2 V_N$. Its torus character is

$$\chi_{\mathrm{Ad}}(z) = \mathrm{rank}(G) + \sum_{\alpha \in \Phi} z^\alpha, \tag{352}$$

so (332) again gives an exact finite-rank formula. The exterior-square identity gives the directly testable trace formula

$$\boxed{\mathrm{Tr}(\mathrm{Ad}(g)^r) = \frac{1}{2}\left(\mathrm{Tr}(V(g)^r)^2 - \mathrm{Tr}(V(g)^{2r})\right).} \tag{353}$$

Unlike the genuine spinors, $\Lambda^2 V_N$ has a coefficientwise stable orthogonal regime. For fixed τ and the safe range $N > 2|\tau|$,

$$[m_\tau]\,\mathrm{SMG}_{\mathrm{Ad},\infty} = \langle \mathrm{Exp}_\sigma(h_2), h_\tau[e_2]\rangle\,. \tag{354}$$

The strict inequality is a sufficient stable range, not a claim of sharpness in every degree.

64.7 Finite restrictions and spin covers

If $H \le \mathrm{Spin}(N)$ is finite and W is a restricted spinor module, Tool 2, followed by the power-trace change of coordinates and class grouping, gives

$$\mathbb{E}[\mathrm{Exp}_\sigma(X_{H,W}h_1)] = \frac{1}{|H|}\sum_{h\in H}\exp\left(\sum_{r\ge1}\frac{p_r}{r}\chi_W(h^r)\right) = \sum_{[h]}\frac{1}{|C_H(h)|}\exp\left(\sum_{r\ge1}\frac{p_r}{r}\chi_W(h^r)\right). \tag{355}$$

The same formula applies to the inverse image $\widetilde{W}$ of a finite real reflection group in Pin and to a Schur double cover $\widetilde{S}_n$ with a chosen genuine spin character. The cover and power maps are essential: ordinary conjugacy classes can split upstairs. No representation-independent large-n law exists for $\widetilde{S}_n$ until a sequence of strict-partition labels is specified.

64.8 Independent computational verification

64.8.1 Exact Weyl route

For type B_n the verifier enumerates all signed permutations; for type D_n it retains those with an even number of sign changes. In doubled coordinates,

$$2\rho_{B_n} = (2n-1, 2n-3, \dots, 1), \qquad 2\rho_{D_n} = (2n-2, 2n-4, \dots, 0).$$

For each τ, dynamic programming expands $\prod_j h_{\tau_j}$ from the actual spin weights. The finite lookup (333) then returns the invariant multiplicity. The decompositions in Theorems 2 and 3 are not used by this route.

Group	$\dim W$	$[m_{(2)}]$	$[m_{(1,1)}]$	$[m_{(1^4)}]$	formula	result
Spin(3)	2	0	1	2	$n+1$	**PASS**
Spin(5)	4	0	1	3	$n+1$	**PASS**
Spin(7)	8	1	1	4	$n+1$	**PASS**
Spin(9)	16	1	1	5	$n+1$	**PASS**
Spin(11)	32	0	1	6	$n+1$	**PASS**

Group	$\dim \Delta^+$	$[m_{(2)}]$	$[m_{(1,1)}]$	$[m_{(1^4)}]$	balanced fourth	result
Spin(4)	2	0	1	2	2	**PASS**
Spin(6)	4	0	0	1	2	**PASS**
Spin(8)	8	1	1	3	3	**PASS**
Spin(10)	16	0	0	2	3	**PASS**

All odd-total-degree type-B checks vanish. The type-D suite separately checks the order-two/order-four selection rule. In total, 45 exact spin, half-spin, and balanced coefficient comparisons pass.

64.8.2 Direct Clifford-matrix route

Let X, Y, Z be the Pauli matrices. The implementation uses the Jordan–Wigner gamma matrices

$$\gamma_{2j+1} = Z^{\otimes j}\otimes X\otimes I^{\otimes(n-j-1)}, \qquad \gamma_{2j+2} = Z^{\otimes j}\otimes Y\otimes I^{\otimes(n-j-1)},$$

with $Z^{\otimes n}$ appended in odd dimension. They satisfy

$$\gamma_a\gamma_b + \gamma_b\gamma_a = 2\delta_{ab}I.$$

Concrete represented Spin elements are products

$$\rho(g) = \prod_{a<b} \exp\left(\frac{\theta_{ab}}{2}\gamma_a\gamma_b\right). \tag{356}$$

Because $(\gamma_a\gamma_b)^2 = -I$, each exponential in (356) is evaluated exactly as a sine–cosine matrix polynomial.

On a tensor square, the code constructs the positive quadratic Casimir

$$C = -\sum_{a<b} \left(J_{ab} \otimes I + I \otimes J_{ab}\right)^2, \qquad J_{ab} = \tfrac{1}{2}\gamma_a\gamma_b.$$

Its eigenspace dimensions provide an independent matrix signature of the irreducible summands.

Module	matrix dimension	Casimir blocks	observed block dimensions	result
Spin(3) spin	2	2	$1, 3$	**PASS**
Spin(5) spin	4	3	$1, 5, 10$	**PASS**
Spin(7) spin	8	4	$1, 7, 21, 35$	**PASS**
Spin(9) spin	16	5	$1, 9, 36, 84, 126$	**PASS**
Spin(4) half-spin	2	2	$1, 3$	**PASS**
Spin(8) half-spin	8	3	$1, 28, 35$	**PASS**

The type-B dimensions are exactly $\binom{2n+1}{0}, \ldots, \binom{2n+1}{n}$, as predicted by (340). Every Clifford anticommutator residual is zero in floating arithmetic for these Pauli matrices.

64.8.3 Pin, mixed, and adjoint checks

For $N = 2, 4, 6$, the verifier constructs $u_+ = \gamma_1$ and $u_- = i\gamma_1$, a nontrivial Spin torus element, and their product in the odd component. All six cases have the correct square, anticommute with chirality, and have zero traces in powers $1, 3, 5$. The largest unitarity residual is 5.7×10^{-16}.

Mixed spinor–vector Kronecker matrices in types B_3 and D_3 agree with the product-weight construction and (351). Explicit adjoint torus matrices verify (353) for $r = 1, 2, 3, 4$ with maximum residual 7.2×10^{-15}. Exact Weyl integration in types B_2, B_3, D_4 also recovers one invariant metric and one invariant multilinear structure tensor. These are 25 additional checks. Together with the six Casimir rows, the complete report contains 76 passing checks.

64.9 Reproduction and scope

From the repository root, run

```
.venv/bin/python lambda_distributions/dists/verification.py
.venv/bin/python -m pytest -q lambda_distributions/dists/test_verification.py
.venv/bin/marimo edit marimo_notebooks/spin_pin.py
```

The first command prints the six verification subtotals and fails with a nonzero exit status if any comparison fails. The notebook exposes the same suite with a rank selector and group-separated tables.

The exact coefficient computations establish the tested low-degree cases; the all-rank claims are proved by Reynolds/Weyl integration and the Clifford decompositions. The matrix computations are an independent finite-rank audit, not a substitute for those proofs. For Pin, the document proves the component reduction and the even-dimensional odd-trace simplification. A fully expanded Pin sigma-MGF still requires the power characters of the chosen Pin$^{\pm}$ extension.

References

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65 The compact F_4 sigma-MGF on V_{26}

Abstract

For the trace-zero Albert module we prove the compact Molien–Weyl expression, construct the 26-dimensional torus matrices, and exactly verify $\mathrm{SMG}_{F_4,26} = 1+h_2+h_3+O_{\geq 4}$. The one-copy Hilbert series is $((1-t^2)(1-t^3))^{-1}$.

65.1 Weyl specialization

By the imported compact Molien identity and Weyl integration, the weights of V_{26} give

$$\boxed{\mathrm{SMG}_{F_4,26} = \mathbb{E}[\mathrm{Exp}_\sigma(X_{F_4,V_{26}}h_1)] = \frac{1}{1152}\,\mathrm{CT}_z\left[D_{F_4}(z)\prod_a(1-t_a)^{-2}\prod_{\beta\in\Phi_{\mathrm{short}}}(1-t_az^\beta)^{-1}\right].}$$

Only this weight specialization is family-specific.

65.2 Invariant tensors

The trace-zero Albert algebra carries the restricted trace form and the restricted determinant. These are respectively a unique symmetric quadratic and a unique symmetric cubic invariant. No other Schur type occurs below degree four, so

$$\boxed{\mathrm{SMG}_{F_4,26} = 1+h_2+h_3+O_{\geq 4}.}$$

The standard invariant theory of the trace-zero Albert algebra gives $\mathbb{C}[V_{26}]^{F_4} = \mathbb{C}[q,c]$, with generators of degrees two and three. Therefore $\mathrm{SMG}(t,0,\ldots) = ((1-t^2)(1-t^3))^{-1}$. This all-degree invariant-ring description is an external input; the notebook verifies only the displayed low-degree terms.

65.3 Verification method and evidence

Starting with the F_4 Cartan matrix, the notebook reflects a short simple root to obtain 24 short roots and appends two zero weights. It constructs

$$M(\theta) = \mathrm{diag}(e^{i\langle\mu,\theta\rangle})_\mu \in U(26)$$

and checks the determinant integrand against its eigenvalue product. Schur characters are computed independently by Jacobi–Trudi. Their Haar integrals are exact: for a character f invariant under W,

$$\frac{1}{|W|}\,\mathrm{CT}(D_{F_4}f) = \mathrm{CT}\left(f\prod_{\alpha>0}(1-z^{-\alpha})\right),$$

and the Weyl-denominator identity supplies every coefficient on the right. The direct monomial-basis check returns

$$([m_{(2)}],[m_{(1,1)}],[m_{(3)}],[m_{(2,1)}],[m_{(1,1,1)}]) = (1,1,1,1,1).$$

λ	$\varnothing$	(1)	(2)	$(1,1)$	(3)	$(2,1)$	$(1,1,1)$
proposed	1	0	1	0	1	0	0
exact Weyl check	1	0	1	0	1	0	0

The generated root count is 48, the matrix dimension is 26, the unitarity residual is 2.8×10^{-16}, and the two integrand evaluations agree to 1.3×10^{-14}. Every degree-at-most-three coefficient passes.

Reproducibility. Run `marimo_notebooks/lie_group_f4.py` with marimo.

External invariant-theory source

See T. A. Springer and F. D. Veldkamp, *Octonions, Jordan Algebras and Exceptional Groups*, Springer Monographs in Mathematics (2000), for the Albert-algebra realization and its norm invariants.

66 The compact E_6 sigma-MGF on V_{27}

Abstract

For simply connected compact E_6 and its minuscule 27-dimensional module, we prove the constant-term formula, the mod-three selection rule, and the initial expansion $1 + h_3 + O_{\geq 6}$. An exact 27-weight computation verifies all Schur coefficients through degree three.

66.1 Weyl specialization

Because the 27 weights form $W(E_6)\omega_1$, the imported compact Molien identity and Weyl integration give

$$\boxed{\mathrm{SMG}_{E_6,27} = \mathbb{E}[\mathrm{Exp}_\sigma(X_{E_6,V_{27}} h_1)] = \frac{1}{51840}\,\mathrm{CT}_z\left[D_{E_6}(z)\prod_a \prod_{\mu\in W\omega_1}(1 - t_a z^\mu)^{-1}\right]}.$$

The new points are the central selection rule and exceptional cubic.

66.2 Selection rule and one-copy series

The generator of the center $Z(E_6) \cong \mathbb{Z}/3$ acts on V_{27} by a primitive cube root. It acts on every degree-d tensor construction by its dth power. Consequently

$$[m_\tau]\,\mathrm{SMG}_{E_6,27} = 0 \quad \text{unless} \quad 3 \mid |\tau|.$$

The exceptional determinant is the unique symmetric cubic invariant. Thus

$$\boxed{\mathrm{SMG}_{E_6,27} = 1 + h_3 + O_{\geq 6}}, \qquad \boxed{\mathrm{SMG}_{E_6,27}(t,0,\ldots) = \frac{1}{1-t^3}}.$$

The module belongs to simply connected E_6 and does not descend to the adjoint quotient E_6/Z_3.

The all-degree statement $\mathbb{C}[V_{27}]^{E_6} = \mathbb{C}[\det]$ is an external invariant-theory input for the minimal 27-dimensional module. The notebook verifies the cubic invariant and the displayed coefficients but does not independently prove generation in all degrees.

66.3 Exact verification

The code generates the 72 roots, obtains ω_1 by rationally solving $A^T c = e_1$, and closes its Weyl orbit under simple reflections. The resulting 27 weights define the explicit matrix $M(\theta) = \mathrm{diag}(e^{i\langle\mu,\theta\rangle}) \in U(27)$. The determinant formula and eigenvalue product are evaluated independently.

For coefficients, Jacobi–Trudi produces $\chi_{\mathbb{S}_\lambda V}$. The one-sided Weyl denominator

$$\prod_{\alpha>0}(1 - z^{-\alpha}) = \sum_{w\in W} \det(w) z^{w\rho-\rho}$$

makes its integral an exact signed weight sum. Orbit membership of $\rho - \mu$ is tested by reflecting it to the dominant chamber. The monomial-basis check gives zero in degrees one and two and

$$([m_{(3)}], [m_{(2,1)}], [m_{(1,1,1)}]) = (1, 1, 1).$$

λ	$\varnothing$	(1)	(2)	$(1,1)$	(3)	$(2,1)$	$(1,1,1)$
proposed	1	0	0	0	1	0	0
exact Weyl check	1	0	0	0	1	0	0

The orbit contains exactly 27 distinct weights. Unitarity holds to 4.3×10^{-16}, the integrand residual is below 1.8×10^{-15}, and all coefficient checks pass.

Reproducibility. Run `marimo_notebooks/lie_group_e6.py` with marimo.

External invariant-theory source

See T. A. Springer and F. D. Veldkamp, *Octonions, Jordan Algebras and Exceptional Groups*, Springer Monographs in Mathematics (2000), for the Albert determinant and the E_6 structure group.

67 The compact E_7 sigma-MGF on V_{56}

Abstract

For simply connected compact E_7 on its minuscule 56, we prove the Molien–Weyl expression and parity rule, exactly verify the full quadratic expansion with explicit torus matrices, and state the degree-four expansion using cited invariant theory. We also give a sparse quartic verification algorithm; the present artifact does not report that computation as executed.

67.1 Weyl specialization and parity

The imported compact Molien identity and Weyl integration yield

$$\boxed{\mathrm{SMG}_{E_7,56} = \mathbb{E}[\mathrm{Exp}_\sigma(X_{E_7,V_{56}}h_1)] = \frac{1}{2903040}\,\mathrm{CT}_z\left[D_{E_7}(z)\prod_a \prod_{\mu\in W\omega_7}(1-t_a z^\mu)^{-1}\right]}\,.$$

The nontrivial central element acts as $-I$, so odd total degree vanishes. This is the simply connected group; the 56 does not descend to E_7/Z_2.

67.2 Low-degree tensors

Let $\omega \in \bigwedge^2 V^*$ be the invariant symplectic form and let $q \in \mathrm{Sym}^4 V^*$ be the Cartan–Freudenthal quartic. At degree four, the three pairwise contractions made from two copies of ω decompose under S_4 as the Schur types $(2,2)$ and $(1,1,1,1)$. The quartic q supplies type (4). The one-copy statement $\mathbb{C}[V_{56}]^{E_7} = \mathbb{C}[q]$ and the exhaustion of degree-four tensor invariants by the symplectic pairings and q are external invariant-theory inputs; see Brown [1]. Using those inputs,

$$\boxed{\mathrm{SMG}_{E_7,56} = 1 + e_2 + h_4 + s_{(2,2)} + e_4 + O_{\geq 6}}, \qquad \boxed{\mathrm{SMG}(t,0,\ldots) = \frac{1}{1-t^4}}.$$

The one-variable specialization kills $e_2, s_{(2,2)}, e_4$ and retains the quartic generator. In particular $([m_{(2)}],[m_{(1,1)}]) = (0,1)$, while the predicted degree-four row is

$$([m_{(4)}],[m_{(3,1)}],[m_{(2,2)}],[m_{(2,1,1)}],[m_{(1,1,1,1)}]) = (1,1,2,2,4).$$

67.3 Executed quadratic verification and optional quartic confirmation

The notebook generates the 126 roots and the 56-weight orbit, constructs $M(\theta) = \mathrm{diag}(e^{i\langle\mu,\theta\rangle})$, and verifies the determinant integrand. Jacobi–Trudi characters are integrated exactly using

$$\frac{1}{|W|}\,\mathrm{CT}(Df) = \mathrm{CT}\left(f\prod_{\alpha>0}(1-z^{-\alpha})\right).$$

The complete quadratic computation is

λ	$\varnothing$	(1)	(2)	$(1,1)$
proposed	1	0	0	1
exact Weyl check	1	0	0	1

All entries pass. The matrix unitarity residual is 4.2×10^{-16} and the determinant/eigenvalue residual is 8.6×10^{-14}.

Status of the degree-four row. The row follows from the cited invariant-theory decomposition. The notebook output displayed here verifies only degrees at most two. A fully mechanical confirmation may be obtained as follows, but is not executed in this artifact. Do not form dense matrices on $V^{\otimes 4}$ (dimension 56^4). Instead: (i) generate the weight-sparse root-operator action on each of $\mathrm{Sym}^4 V$, $\mathbb{S}_{(2,2)}V$, and $\bigwedge^4 V$; (ii) retain only zero-weight basis vectors; (iii) solve the sparse raising-operator nullspace; and (iv) check dimensions $1, 1, 1$. Equivalently, stream the three Schur characters through the same exact Weyl functional. An executed result must report the zero-weight dimensions, exact raising-kernel ranks, arithmetic mode, and output hash.

Reproducibility. Run `marimo_notebooks/lie_group_e7.py` with marimo.

References

[1] R. B. Brown, "Groups of type E_7," *J. Reine Angew. Math.* **236** (1969), 79–102.

[2] J. F. Adams, *Lectures on Exceptional Lie Groups*, Chicago Lectures in Mathematics (1996).

68 The compact E_8 sigma-MGF on the adjoint module

Abstract

We prove the exact constant-term formula for the 248-dimensional adjoint module, verify its root and zero-weight data and its quadratic invariant exactly, and prove the alternating cubic row algebraically. We also record the correct one-copy Chevalley series and explain why it does not determine the multivariate sigma-MGF.

68.1 Exact formula

For every unitary compact-group representation,

$$\mathrm{SMG}_{K,V} = \int_K \prod_a \det(I - t_a\rho(g))^{-1} dg = \sum_\lambda \dim(\mathbb{S}_\lambda V)^K s_\lambda(\mathbf{t}).$$

Equivalently, $[m_\tau]\,\mathrm{SMG}_{K,V} = \dim(\bigotimes_i \mathrm{Sym}^{\tau_i} V)^K$. The adjoint E_8 weights are the 240 roots and zero with multiplicity eight. Weyl integration therefore proves

$$\boxed{\mathrm{SMG}_{E_8,248} = \frac{1}{696729600}\,\mathrm{CT}_z\left[D_{E_8}(z)\prod_a (1-t_a)^{-8}\prod_{\alpha\in\Phi(E_8)}(1-t_a z^\alpha)^{-1}\right].}$$

68.2 Universal adjoint tensors

For a compact simple Lie algebra, the Killing form κ is a unique symmetric quadratic invariant. In degree three,

$$c(x,y,z) = \kappa(x,[y,z])$$

is invariant, nonzero, and fully alternating. Moreover

$$(\bigwedge^3 \mathfrak{g})^G \cong \mathrm{Hom}_G(\bigwedge^2 \mathfrak{g}, \mathfrak{g}),$$

and for the simple Lie algebra $\mathfrak{e}_8$ the bracket spans this Hom-space. Chevalley restriction gives basic invariant degrees $2, 8, 12, 14, 18, 20, 24, 30$, so $(\mathrm{Sym}^3\,\mathfrak{g})^G = 0$. Since the only invariant in $\mathfrak{g}^{\otimes 3}$ is alternating, the Schur type $(2,1)$ also has multiplicity zero. Consequently

$$\boxed{\mathrm{SMG}_{E_8,248} = 1 + h_2 + e_3 + O_{\geq 4}.}$$

Chevalley restriction gives the one-copy series

$$\boxed{\mathrm{SMG}(t,0,\ldots) = \prod_{d\in\{2,8,12,14,18,20,24,30\}} (1-t^d)^{-1}.}$$

This specialization cannot see e_3 and therefore does not determine the full sigma-MGF. Thus the degree-at-most-three monomial signature is

$$([m_{(2)}], [m_{(1,1)}], [m_{(3)}], [m_{(2,1)}], [m_{(1,1,1)}]) = (1, 1, 0, 0, 1).$$

68.3 Executed quadratic check and optional sparse cubic confirmation

The notebook generates the E_8 root orbit (240 vectors), appends eight zero weights, and constructs a 248 by 248 diagonal unitary torus matrix. It checks the determinant integrand directly and integrates the quadratic Schur characters exactly with the Weyl-denominator identity.

λ	$\varnothing$	(1)	(2)	$(1,1)$
proposed	1	0	1	0
exact Weyl check	1	0	1	0

The computed dimension is 248, with 240 roots and eight zero weights; all quadratic entries pass. Floating-point work is confined to the illustrative torus matrix, while invariant multiplicities use rational weights and exact integer sums.

The cubic multiplicities $(\mathrm{Sym}^3, \mathbb{S}_{(2,1)}, \bigwedge^3) = (0,0,1)$ are therefore proved algebraically above. For an optional mechanical confirmation, construct only the zero-weight part of $\bigwedge^3 \mathfrak{e}_8$ and impose the eight simple-root raising operators. The expected nullspace is one-dimensional and represented by $c(x,y,z)$; the analogous expected nullities on Sym^3 and $\mathbb{S}_{(2,1)}$ are zero. This sparse route avoids materializing spaces of dimensions on the order of $\binom{248}{3}$. It is not reported as an executed computation in this artifact.

Reproducibility. Run `marimo_notebooks/lie_group_e8.py` with marimo.

69 The adjoint $SU(n)$ sigma-MGF

Abstract

We derive the compact Molien–Weyl formula for the adjoint representation of $SU(n)$ and test the manageable cases $n = 2, 3, 4$. Exact Schur-character integration confirms the quadratic Killing form, the alternating bracket tensor, and the symmetric cubic precisely when $n \geq 3$.

69.1 Formula

The adjoint weights of $SU(n)$ are zero with multiplicity $n-1$ and $e_i - e_j$ for $i \neq j$. The general Haar/Cauchy proof yields

$$\boxed{\mathrm{SMG}_{SU(n),\mathrm{Ad}} = \frac{1}{n!}\,\mathrm{CT}_z\left[D_{A_{n-1}}(z)\prod_a (1-t_a)^{-(n-1)}\prod_{i\neq j}(1-t_a z_i/z_j)^{-1}\right]}.$$

Indeed, expanding the determinant gives $\sum_\lambda s_\lambda(\mathbf{t})\chi_{\mathbb{S}_\lambda \mathfrak{sl}_n}$; Haar integration retains exactly the invariant multiplicity.

69.2 Low-degree structure

The Killing form supplies h_2, and $\kappa(X,[Y,Z])$ supplies e_3 for every $n \geq 2$. For $n \geq 3$ there is also the symmetric invariant

$$(X,Y,Z) \longmapsto \mathrm{Tr}(XYZ + XZY),$$

which supplies h_3. For $SU(2)$ it vanishes. Thus, through degree three,

$$\mathrm{SMG}_{SU(2),\mathrm{Ad}} = 1 + h_2 + e_3 + O_{\geq 4},$$

and, for $n \geq 3$,

$$\mathrm{SMG}_{SU(n),\mathrm{Ad}} = 1 + h_2 + h_3 + e_3 + O_{\geq 4}.$$

Chevalley restriction gives $\mathrm{SMG}(t, 0, \ldots) = \prod_{d=2}^{n}(1-t^d)^{-1}$.

69.3 Direct and exact verification

For each $n = 2, 3, 4$, the notebook generates the A_{n-1} roots, appends the correct zero weights, and constructs the diagonal adjoint torus matrix. It compares the determinant with the product over eigenvalues. It then computes every partition of degrees zero through three by Jacobi–Trudi and applies the exact Weyl functional

$$\frac{1}{|W|}\,\mathrm{CT}(Df) = \mathrm{CT}\left(f\prod_{\alpha>0}(1-z^{-\alpha})\right).$$

n	$\dim \mathfrak{sl}_n$	$[h_2]$	$[h_3]$	$[e_3]$	all other $d \leq 3$
2	3	1	0	1	0
3	8	1	1	1	0
4	15	1	1	1	0

All 21 partition-by-case comparisons pass. The tested representation dimensions $3, 8, 15$ are large enough to distinguish the exceptional $n = 2$ behavior from the stable $n \geq 3$ cubic while remaining directly inspectable.

Reproducibility. Run `marimo_notebooks/lie_group_su_adjoint.py` with marimo.

70 Schur-functor representations of $U(n)$

Outcome. The finite-dimensional plethystic trace formula is correct. For five pairs (n, λ), including $\lambda = (2, 1)$, 40 direct Schur-matrix comparisons pass with maximum error 2.98×10^{-15}. The important qualification is exact: for nonempty λ, the purely holomorphic $U(n)$ sigma-MGF is 1. The nontrivial statement is the joint law with the conjugate. Three seeded Haar diagnostics also pass.

70.1 Statement with the unitary correction

Let $W_n = \mathbb{C}^n$, let $U \in U(n)$, fix $\lambda \vdash d$, and put $V_{\lambda,n} = S_\lambda(W_n)$. If $X(U)$ is the eigenvalue alphabet of U, then the eigenvalue alphabet of $S_\lambda(U)$ is the plethysm $s_\lambda[X(U)]$. Hence, for every $r \geq 1$,

$$\boxed{\operatorname{Tr}(S_\lambda(U)^r) = p_r[s_\lambda][X(U)] = s_\lambda[X(U^r)].} \tag{357}$$

For a partition τ, the coefficient integrand of the sigma-MGF is

$$\prod_i h_{\tau_i}[s_\lambda][X(U)]. \tag{358}$$

Associativity of plethysm gives $h_\tau[s_\lambda[X]] = (h_\tau[s_\lambda])[X]$, so Haar integration yields

$$[m_\tau]\,\mathrm{SMG}^U_{\lambda,n} = \dim\left(\bigotimes_i \operatorname{Sym}^{\tau_i} S_\lambda(W_n)\right)^{U(n)}. \tag{359}$$

Proposition (central-character obstruction). If $d > 0$, then

$$\boxed{\mathrm{SMG}^U_{\lambda,n} = 1.} \tag{360}$$

Indeed, zI acts on $S_\lambda(W_n)$ by z^d and on the tensor product above by $z^{d|\tau|}$. For $|\tau| > 0$ this character is not identically one on $U(1)$, so its fixed space is zero. This is an exact finite-n proof, not an asymptotic statement.

The nontrivial object uses a conjugate alphabet. In the stable range its coefficient is

$$[m_\tau \bar{m}_\nu]\mathcal{J}^U_{\lambda,\infty} = \left\langle \operatorname{Exp}_\sigma(h_1\bar{h}_1),\, h_\tau[s_\lambda]\,\overline{h_\nu[s_\lambda]} \right\rangle. \tag{361}$$

This is the plethystic pushforward of the defining-representation joint law.

Stable-range qualification. Let $|\lambda| = d$ and consider a balanced coefficient indexed by outer partitions α, β, with $d|\alpha| = d|\beta| =: m$. Embed its coefficient space into $W_n^{\otimes m} \otimes (W_n^*)^{\otimes m}$. The first fundamental theorem for GL_n says that contraction tensors indexed by S_m span the invariants, and the second fundamental theorem says that they are linearly independent when $n \geq m$. Therefore the stable plethystic formula is valid whenever

$$n \geq m = d|\alpha| = d|\beta|.$$

If the two degrees are unequal, the coefficient is exactly zero for every n by the scalar-center argument.

70.2 Explicit polynomial-Gaussian law

The Frobenius formula gives

$$s_\lambda = \sum_{\alpha \vdash d} \frac{\chi^\lambda(\alpha)}{z_\alpha} p_\alpha, \qquad z_\alpha = \prod_j j^{m_j(\alpha)} m_j(\alpha)!. \tag{362}$$

Substituting $p_k[X(U^r)] = \operatorname{Tr}(U^{rk})$ in (357) gives the exact identity

$$\operatorname{Tr}(S_\lambda(U)^r) = \sum_{\alpha \vdash d} \frac{\chi^\lambda(\alpha)}{z_\alpha} \prod_{a \in \alpha} \operatorname{Tr}(U^{ra}). \tag{363}$$

For fixed powers, $\mathrm{Tr}(U^k)$ converges jointly in moments to independent circular complex Gaussians Z_k satisfying $\mathbb{E}Z_k = \mathbb{E}Z_k^2 = 0$ and $\mathbb{E}|Z_k|^2 = k$. Polynomial substitution in (363) therefore gives

$$\mathrm{Tr}(S_\lambda(U)^r) \longrightarrow Q^U_{\lambda,r} = \sum_{\alpha\vdash d} \frac{\chi^\lambda(\alpha)}{z_\alpha} \prod_{a\in\alpha} Z_{ra}. \tag{364}$$

For example, $Q^U_{(2),1} = \frac{1}{2}(Z_1^2 + Z_2)$ and $Q^U_{(1,1),1} = \frac{1}{2}(Z_1^2 - Z_2)$. Pure holomorphic moments vanish, while mixed moments generally do not.

70.3 Independent finite-dimensional verification

For each test the code draws a seeded Haar unitary by complex Gaussian QR factorization. It constructs a standard Young symmetrizer c_λ on $W_n^{\otimes d}$, obtains an orthonormal basis B of its image by singular-value decomposition, and forms the actual matrix

$$D_\lambda(U) = B^* U^{\otimes d} B. \tag{365}$$

This route does not use symmetric-function trace identities. The comparison route computes $p_k = \mathrm{Tr}(U^{rk})$, obtains h_j from Newton's recurrence

$$jh_j = \sum_{k=1}^{j} p_k h_{j-k}, \qquad h_0 = 1, \tag{366}$$

and evaluates $s_\lambda = \det(h_{\lambda_i - i + j})$ by Jacobi–Trudi. It compares both (357) for $r = 1, 2, 3$ and (358) for $\tau = (), (1), (2), (1, 1), (3)$.

n	tested λ	direct checks	largest purpose	status
2	$(1), (2)$	trace and sigma integrands	low-rank boundary	PASS
3	$(1,1), (3), (2,1)$	trace and sigma integrands	mixed Young symmetry	PASS
			40 comparisons; maximum absolute error	2.98×10^{-15}

The seeded $U(8)$ Haar run uses 1,200 matrices. It estimates $\mathbb{E}\chi_{(2)} = 0.0222 + 0.0343i$, $\mathbb{E}\chi_{(2)}^2 = 0.0322 - 0.0404i$, and $\mathbb{E}\chi_{(1,1)} = -0.0059 + 0.0148i$, all within 4.5 estimated standard errors of the exact target zero. These statistical rows check the implementation but do not replace the central-character proof.

70.4 Scope and reproducibility

The exact trace formula holds whenever $S_\lambda(\mathbb{C}^n)$ is defined ($\ell(\lambda) \leq n$). Coefficient stabilization is a separate assertion: the balanced unitary threshold is $n \geq d|\alpha| = d|\beta|$, while the orthogonal/symplectic Brauer bound is stated in their respective notes. Unitary holomorphic vanishing is exact for every allowed n. Run the verification module below from the repository root, or open `marimo_notebooks/schur_unitary.py` with marimo.

```
python -m for_this_guy.schur_functor_classical_groups.unitary.verification
```

References

S. Howe, *Random matrix statistics and zeroes of L-functions via probability in λ-rings*.
G. Lehrer and R. Zhang, *The Brauer Category and Invariant Theory*.

71 Schur-functor representations of $O(n)$

Outcome. The orthogonal formula is correct as a coefficientwise stable plethystic pushforward. The finite trace identity is exact before taking a limit. Five representative (n, λ) cases give 40 direct comparisons with maximum error 5.33×10^{-15}. Haar $O(9)$ tests recover the invariant quadratic form in $\mathrm{Sym}^2 W$, its absence from $\Lambda^2 W$, and the odd-degree parity obstruction.

71.1 Stable sigma-MGF and its proof

Let $W_n = \mathbb{C}^n$ with the defining action of the compact group $O(n)$, fix $\lambda \vdash d$, and put $V_{\lambda,n} = S_\lambda(W_n)$. For a partition τ,

$$[m_\tau]\,\mathrm{SMG}^{O}_{\lambda,n} = \dim\left(\bigotimes_i \mathrm{Sym}^{\tau_i} S_\lambda(W_n)\right)^{O(n)} = \mu_{O(n)}\big(h_\tau[s_\lambda]\big). \tag{367}$$

The second equality is simply associativity of plethysm: $h_\tau[s_\lambda[X]] = (h_\tau[s_\lambda])[X]$. The defining-representation stable sigma-MGF is $\mathrm{Exp}_\sigma(h_2)$. Applying its coefficient functional to the fixed symmetric function $h_\tau[s_\lambda]$ proves

$$\boxed{[m_\tau]\,\mathrm{SMG}^{O}_{\lambda,\infty} = \langle \mathrm{Exp}_\sigma(h_2), h_\tau[s_\lambda]\rangle\,.} \tag{368}$$

No new limiting invariant theory is needed after the defining case. A conservative sufficient stable range is $n \geq d|\tau|$: the relevant expression has tensor degree $d|\tau|$, where Brauer pairings are independent in that range.

Since $-I \in O(n)$ acts on $S_\lambda(W_n)$ by $(-1)^d$, it acts on the space in (367) by $(-1)^{d|\tau|}$. Therefore

$$\boxed{[m_\tau]\,\mathrm{SMG}^{O}_{\lambda,n} = 0 \quad \text{if } d|\tau| \text{ is odd.}} \tag{369}$$

This parity conclusion is exact at finite n.

71.2 Exact traces and the limiting law

For every $g \in O(n)$ and $r \geq 1$, functoriality gives $S_\lambda(g)^r = S_\lambda(g^r)$, whence

$$\boxed{\mathrm{Tr}(S_\lambda(g)^r) = s_\lambda[X(g^r)] = \sum_{\alpha \vdash d} \frac{\chi^\lambda(\alpha)}{z_\alpha} \prod_{a \in \alpha} \mathrm{Tr}(g^{ra}).} \tag{370}$$

This formula is exact for every allowed n; stability is needed only when replacing traces by Gaussian variables.

The joint moment convergence of finitely many orthogonal trace powers is an external classical-group trace theorem (Diaconis–Evans, with the normalization below). Let independent $A_k \sim N(0, 1)$ and set

$$Z^{O}_k = \sqrt{k}\, A_k + \mathbf{1}_{2|k}. \tag{371}$$

Joint moment convergence of fixed orthogonal trace powers and polynomial substitution in (370) give

$$\mathrm{Tr}(S_\lambda(g)^r) \longrightarrow Q^{O}_{\lambda,r} = \sum_{\alpha \vdash d} \frac{\chi^\lambda(\alpha)}{z_\alpha} \prod_{a \in \alpha} Z^{O}_{ra}. \tag{372}$$

For $\lambda = (2)$, $Q^{O}_{(2),1} = \frac{1}{2}((Z^{O}_1)^2 + Z^{O}_2)$ has mean 1, recording the invariant quadratic form. For $\lambda = (1, 1)$, the minus sign makes the mean zero.

71.3 Matrix construction and formula cross-check

The verification draws seeded Haar orthogonal matrices by real Gaussian QR with sign correction. A standard Young symmetrizer c_λ is built explicitly on $W_n^{\otimes d}$. If B is an orthonormal basis of $\operatorname{im} c_\lambda$, the direct representation matrix is

$$D_\lambda(g) = B^* g^{\otimes d} B. \tag{373}$$

The formula cross-check never uses this matrix: it computes the defining traces, uses Newton's recurrence $jh_j = \sum_{k=1}^{j} p_k h_{j-k}$, and evaluates the Jacobi–Trudi determinant $s_\lambda = \det(h_{\lambda_i - i + j})$. Tests compare $\mathrm{Tr}(D_\lambda(g)^r)$ for $r = 1, 2, 3$. They also compare

$$\prod_i h_{\tau_i}[D_\lambda(g)] \quad \text{against} \quad \prod_i h_{\tau_i}[s_\lambda][X(g)] \tag{374}$$

for $\tau = (), (1), (2), (1,1), (3)$, so the actual sigma-MGF integrands are covered.

n	tested λ	checks	purpose	status
3	$(1), (2), (3)$	trace and sigma integrands	defining/symmetric powers	PASS
4	$(1,1), (2,1)$	trace and sigma integrands	alternating/mixed symmetry	PASS
			40 comparisons; maximum absolute error	5.33×10^{-15}

The seeded Haar diagnostic uses 1,200 matrices in $O(9)$. It gives means 0.9979 for $\chi_{(2)}$ (target 1), -0.0289 for $\chi_{(1,1)}$ (target 0), and 0.0116 for $\chi_{(3)}$ (target 0). Each lies inside 4.5 estimated standard errors. Group residuals are at most 3.24×10^{-15}.

71.4 Reproducibility and interpretation

Run the module below from the repository root, or open `marimo_notebooks/schur_orthogonal.py` with marimo. Exact rows are the proof-driven regression test. Haar rows are seeded numerical diagnostics and are reported with standard errors; they are not presented as proofs of convergence.

```
python -m for_this_guy.schur_functor_classical_groups.orthogonal.verification
```

References

S. Howe, *Random matrix statistics and zeroes of L-functions via probability in λ-rings*.
G. Lehrer and R. Zhang, *The Brauer Category and Invariant Theory*.
P. Diaconis and S. N. Evans, *Linear functionals of eigenvalues of random matrices*, Trans. Amer. Math. Soc. 353 (2001), 2615–2633.

72 Schur-functor representations of $USp(2n)$

Outcome. Interpreting $Sp(2n)$ as the compact group $USp(2n)$, the stable formula is the plethystic pushforward of $\mathrm{Exp}_\sigma(e_2)$. Five representative cases in ambient dimensions 2 and 4 give 40 direct comparisons with maximum error 3.70×10^{-15}. A separate Haar $USp(8)$ run detects the invariant alternating form in $\Lambda^2 W$ and verifies odd-degree parity.

72.1 Convention and stable sigma-MGF

Here $Sp(2n)$ means the compact matrix group

$$USp(2n) = \{g \in U(2n) : g^T J g = J\}, \qquad J = \begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix}. \tag{375}$$

This convention matters: the complex algebraic group is not a Haar probability space, while its compact form is.

Let $W_n = \mathbb{C}^{2n}$, fix $\lambda \vdash d$, and put $V_{\lambda,n} = S_\lambda(W_n)$. For each partition τ,

$$[m_\tau]\,\mathrm{SMG}^{Sp}_{\lambda,n} = \dim \left(\bigotimes_i \mathrm{Sym}^{\tau_i} S_\lambda(W_n) \right)^{USp(2n)} = \mu_{Sp(2n)}\big(h_\tau[s_\lambda]\big). \tag{376}$$

Associativity of plethysm proves the last equality. Since the defining-representation stable sigma-MGF is $\mathrm{Exp}_\sigma(e_2)$, applying its coefficient functional to the fixed function $h_\tau[s_\lambda]$ proves

$$\boxed{[m_\tau]\,\mathrm{SMG}^{Sp}_{\lambda,\infty} = \langle \mathrm{Exp}_\sigma(e_2), h_\tau[s_\lambda] \rangle\,.} \tag{377}$$

A conservative sufficient stable range is $\dim W_n = 2n \geq d|\tau|$, obtained by embedding in tensor degree $d|\tau|$ and applying symplectic Brauer invariant theory.

The central element $-I$ acts by $(-1)^d$ on $S_\lambda(W_n)$. Thus the exact finite-dimensional parity rule is

$$\boxed{[m_\tau]\,\mathrm{SMG}^{Sp}_{\lambda,n} = 0 \quad \text{if } d|\tau| \text{ is odd.}} \tag{378}$$

72.2 Exact trace formula and Gaussian substitution

Functoriality and the Frobenius expansion of s_λ give, for every $g \in USp(2n)$,

$$\boxed{\mathrm{Tr}(S_\lambda(g)^r) = s_\lambda[X(g^r)] = \sum_{\alpha \vdash d} \frac{\chi^\lambda(\alpha)}{z_\alpha} \prod_{a \in \alpha} \mathrm{Tr}(g^{ra}).} \tag{379}$$

This is exact whenever $\ell(\lambda) \leq 2n$.

The joint moment convergence of finitely many symplectic trace powers is an external classical-group trace theorem (Diaconis–Evans, with the normalization below). Let independent $A_k \sim N(0,1)$ and set

$$Z^{Sp}_k = \sqrt{k}\,A_k - \mathbf{1}_{2|k}. \tag{380}$$

Joint moment convergence of fixed trace powers, followed by polynomial substitution in (379), gives

$$\mathrm{Tr}(S_\lambda(g)^r) \longrightarrow Q^{Sp}_{\lambda,r} = \sum_{\alpha \vdash d} \frac{\chi^\lambda(\alpha)}{z_\alpha} \prod_{a \in \alpha} Z^{Sp}_{ra}. \tag{381}$$

For $\lambda = (1,1)$, $Q^{Sp}_{(1,1),1} = \frac{1}{2}((Z^{Sp}_1)^2 - Z^{Sp}_2)$ has mean 1, representing the invariant symplectic form. For $\lambda = (2)$ the corresponding plus sign yields mean zero.

72.3 Finite matrix construction and formula cross-check

The sampler performs quaternionic Gram–Schmidt. If v is accepted, it also inserts $-J\bar{v}$; with columns ordered in pairs of blocks this gives both $g^*g = I$ and $g^T Jg = J$. The largest observed group residual in all reported sampling is 6.67×10^{-15}.

For the direct route, the code constructs a Young symmetrizer c_λ on $W_n^{\otimes d}$, computes an orthonormal basis B of its image, and forms

$$D_\lambda(g) = B^* g^{\otimes d} B. \tag{382}$$

For the formula route, only $\mathrm{Tr}(g^k)$ is retained. Newton's recurrence constructs the h_j, and Jacobi–Trudi constructs s_λ. The comparison covers $\mathrm{Tr}(D_\lambda(g)^r)$ for $r = 1, 2, 3$ and the sigma-MGF integrands $\prod_i h_{\tau_i}[D_\lambda(g)]$ for $\tau = (), (1), (2), (1, 1), (3)$.

ambient dimension	tested λ	checks	purpose	status
2	(1)	trace and sigma integrands	rank-one boundary	PASS
4	$(2), (1, 1), (3), (2, 1)$	trace and sigma integrands	all degree ≤ 3 types	PASS
			40 comparisons; maximum absolute error	3.70×10^{-15}

The seeded Haar diagnostic uses 1,200 matrices in $USp(8)$. It estimates means 0.0031 for $\chi_{(2)}$ (target 0), 0.9917 for $\chi_{(1,1)}$ (target 1), and 0.0818 for $\chi_{(3)}$ (target 0). All lie within 4.5 estimated standard errors. The last target also exercises (378).

72.4 Reproducibility and interpretation

Run the module below from the repository root, or open `marimo_notebooks/schur_symplectic.py` with marimo. The exact matrix/character identities certify the finite trace formula. The seeded Haar rows are diagnostics only; they neither prove Haar integration identities nor the $n \to \infty$ convergence theorem.

```
python -m for_this_guy.schur_functor_classical_groups.symplectic.verification
```

References

S. Howe, *Random matrix statistics and zeroes of L-functions via probability in λ-rings*.
G. Lehrer and R. Zhang, *The Brauer Category and Invariant Theory*.
P. Diaconis and S. N. Evans, *Linear functionals of eigenvalues of random matrices*, Trans. Amer. Math. Soc. 353 (2001), 2615–2633.

73 Balanced sigma-MGFs for local unitary tensor products

Outcome. For $G = \prod_{a=1}^{k} U(d_a)$ on $V = \bigotimes_{a=1}^{k} \mathbb{C}^{d_a}$, the ordinary one-sided sigma-MGF is exactly 1. The nontrivial object is the balanced two-sided series involving both V and V^*. Its stable power-sum coefficient is z_λ^{k-2}, its bipartite single-state coefficient is a restricted partition number, and its multilinear coefficient is an exact product of Schur–Weyl centralizer dimensions. A reproducible marimo notebook constructs the Kronecker-product matrices and compares direct Haar estimates with these formulas. All 46 numerical comparisons, 4 construction checks, and 5 scalar-center witnesses pass.

73.1 Representation and the correction at the outset

Put $V_a = \mathbb{C}^{d_a}$ and

$$G = U(d_1) \times \cdots \times U(d_k), \qquad V = V_1 \otimes \cdots \otimes V_k.$$

The representation is the concrete matrix homomorphism

$$\rho(g_1, \ldots, g_k) = g_1 \otimes \cdots \otimes g_k \in U(D), \qquad D = \prod_{a=1}^{k} d_a. \tag{383}$$

For a partition $\tau = (\tau_1, \ldots, \tau_\ell)$, write

$$\operatorname{Sym}^\tau V = \bigotimes_{j=1}^{\ell} \operatorname{Sym}^{\tau_j} V, \qquad |\tau| = \sum_j \tau_j.$$

The ordinary coefficient is

$$[m_\tau]\, \mathrm{SMG}_{G,V} = \dim(\operatorname{Sym}^\tau V)^G.$$

Proposition 1 (one-sided collapse).

$$\boxed{\mathrm{SMG}_{G,V} = 1.} \tag{384}$$

Proof. For $z \in U(1)$ the group contains $(zI_{d_1}, I_{d_2}, \ldots, I_{d_k})$, whose matrix in (1) is zI_V. It acts on $\operatorname{Sym}^\tau V$ as $z^{|\tau|}I$. If $|\tau| > 0$, one can choose z with $z^{|\tau|} \neq 1$, so a fixed vector must be zero. Only degree zero remains. □

The same center explains why local-unitary entanglement invariants use a state ψ and its conjugate $\bar{\psi}$: the weights z and z^{-1} cancel only when the two total degrees agree.

73.2 The balanced series and Molien formula

Introduce independent alphabets $\mathbf{s}$ and $\mathbf{t}$ and define

$$\mathcal{B}_{G,V}(\mathbf{s}, \mathbf{t}) = \int_G \operatorname{Exp}_\sigma(X_V(g) h_1(\mathbf{s}))\, \operatorname{Exp}_\sigma(X_{V^*}(g) h_1(\mathbf{t}))\, dg. \tag{385}$$

Coefficient extraction gives

$$\mathcal{B}_{G,V} = \sum_{\alpha,\beta} b_{\alpha,\beta} m_\alpha(\mathbf{s}) m_\beta(\mathbf{t}), \tag{386}$$

$$b_{\alpha,\beta} = \dim \left(\operatorname{Sym}^\alpha V \otimes \operatorname{Sym}^\beta V^*\right)^G = \dim \operatorname{Hom}_G(\operatorname{Sym}^\beta V, \operatorname{Sym}^\alpha V). \tag{387}$$

The center now gives the balanced selection rule

$$\boxed{b_{\alpha,\beta} = 0 \quad \text{unless} \quad |\alpha| = |\beta|.} \tag{388}$$

For finitely many variables, (3) is the balanced Molien integral

$$\mathcal{B}_{G,V} = \int_G \prod_i \det(I - s_i\rho(g))^{-1} \prod_j \det(I - t_j\rho(g)^{-1})^{-1}\, dg. \tag{389}$$

If $x_1^{(a)}, \ldots, x_{d_a}^{(a)}$ are the eigenvalues of g_a, then the eigenvalues of $\rho(g)$ are $x_{i_1}^{(1)} \cdots x_{i_k}^{(k)}$. Consequently (7), together with the Weyl integration formula for each local factor, is an exact constant-term prescription in every dimension.

73.3 Stable power-sum formula

For every positive integer r, Kronecker products satisfy the directly testable identity

$$\mathrm{Tr}(\rho(g)^r) = \prod_{a=1}^{k} \mathrm{Tr}(g_a^r). \tag{390}$$

Thus, for a partition λ, $p_\lambda(\rho(g)) = \prod_a p_\lambda(g_a)$, and product Haar measure gives

$$\mathbb{E}_G\left[p_\lambda(\rho(g))\overline{p_\mu(\rho(g))}\right] = \prod_{a=1}^{k} \mathbb{E}_{U(d_a)}\left[p_\lambda(g_a)\overline{p_\mu(g_a)}\right]. \tag{391}$$

Let $z_\lambda = \prod_i i^{m_i(\lambda)} m_i(\lambda)!$. Schur orthogonality, or equivalently the stable Hall inner product, gives

$$\mathbb{E}_{U(d)}[p_\lambda(U)\overline{p_\mu(U)}] = \delta_{\lambda\mu} z_\lambda \quad \text{when } d \geq |\lambda|, |\mu|. \tag{392}$$

Combining (9) and (10), if $N \leq \min_a d_a$ then

$$\mathbb{E}_G[p_\lambda(\rho)\overline{p_\mu(\rho)}] = \delta_{\lambda\mu} z_\lambda^k \quad (|\lambda|, |\mu| \leq N). \tag{393}$$

Each determinant exponential contributes a factor p_λ/z_λ. Therefore through bidegree (N, N),

$$\boxed{\mathcal{B}_{G,V} \equiv \sum_{|\lambda|\leq N} z_\lambda^{k-2} p_\lambda(\mathbf{s}) p_\lambda(\mathbf{t}).} \tag{394}$$

For $k = 1, 2, 3$, the coefficients are respectively $z_\lambda^{-1}, 1, z_\lambda$.

73.4 Exact single-state coefficients

For partitions $\lambda^{(1)}, \ldots, \lambda^{(k)} \vdash m$, define the generalized Kronecker coefficient

$$g(\lambda^{(1)}, \ldots, \lambda^{(k)}) = \dim\left([\lambda^{(1)}] \otimes \cdots \otimes [\lambda^{(k)}]\right)^{S_m}.$$

Applying Schur–Weyl duality to each $V_a^{\otimes m}$ and then taking diagonal S_m-invariants proves

$$\mathrm{Sym}^m(V_1 \otimes \cdots \otimes V_k) \cong \bigoplus_{\lambda^{(1)},\ldots,\lambda^{(k)}\vdash m} g(\lambda^{(1)}, \ldots, \lambda^{(k)}) \bigotimes_{a=1}^{k} S_{\lambda^{(a)}} V_a, \tag{395}$$

where terms with $\ell(\lambda^{(a)}) > d_a$ vanish. Schur orthogonality now gives the all-dimensional identity

$$\boxed{b_{(m),(m)} = \sum_{\substack{\lambda^{(a)}\vdash m \\ \ell(\lambda^{(a)})\leq d_a}} g(\lambda^{(1)}, \ldots, \lambda^{(k)})^2.} \tag{396}$$

For $k = 2$, $g(\lambda, \mu) = \delta_{\lambda\mu}$. Hence

$$\boxed{b_{(m),(m)} = \#\{\lambda \vdash m : \ell(\lambda) \leq \min(d_1, d_2)\}.} \tag{397}$$

This is $p(m)$ when both dimensions are at least m. In stable range, setting each alphabet in (12) to one variable yields the equivalent formula

$$\boxed{b_{(m),(m)} = \sum_{\lambda \vdash m} z_\lambda^{k-2}.} \tag{398}$$

For completeness, general $b_{\alpha,\beta}$ also has an exact finite formula. Decompose each $\mathrm{Sym}^{\alpha_j} V$ by (13), then combine local Schur functors with Littlewood–Richardson coefficients. If the resulting multiplicity of $\bigotimes_a S_{\mu^{(a)}} V_a$ is $M_\alpha(\mu^{(1)}, \ldots, \mu^{(k)})$, then

$$b_{\alpha,\beta} = \sum_{\substack{\mu^{(1)},\ldots,\mu^{(k)} \\ \ell(\mu^{(a)}) \le d_a}} M_\alpha(\mu^{(1)}, \ldots, \mu^{(k)}) M_\beta(\mu^{(1)}, \ldots, \mu^{(k)}). \tag{399}$$

This separates the two combinatorial ingredients: generalized Kronecker coefficients enter the symmetric powers of a tensor product, while Littlewood–Richardson coefficients enter products of symmetric powers.

73.5 Exact multilinear coefficients

For $\alpha = \beta = (1^r)$, $\mathrm{Sym}^{(1^r)} V = V^{\otimes r}$, so

$$b_{(1^r),(1^r)} = \dim \mathrm{End}_G(V^{\otimes r}). \tag{400}$$

Because the local factors act independently,

$$\mathrm{End}_G(V^{\otimes r}) \cong \bigotimes_{a=1}^{k} \mathrm{End}_{U(d_a)}(V_a^{\otimes r}).$$

If f^λ is the number of standard Young tableaux of shape λ, Schur–Weyl duality and the hook-length formula give

$$\boxed{b_{(1^r),(1^r)} = \prod_{a=1}^{k} \sum_{\substack{\lambda \vdash r \\ \ell(\lambda) \le d_a}} (f^\lambda)^2.} \tag{401}$$

When every $d_a \ge r$, the inner sum is $|S_r| = r!$, and therefore

$$\boxed{b_{(1^r),(1^r)} = (r!)^k.} \tag{402}$$

This is the dimension of the stable family of contraction diagrams indexed by k independent permutations in S_r.

73.6 Verification design

The accompanying implementation uses two independent routes.

Exact route. Integer partitions are generated recursively. The hook-length formula computes f^λ, which is substituted into (19). Equation (15) is evaluated by restricted partition counting, and (16) by direct computation of z_λ. No sampled matrices enter these tables.

Direct matrix route. For every local factor, a complex Gaussian matrix Z is QR-factorized and the diagonal phases of R are removed. The resulting Q is Haar distributed on $U(d_a)$. The code explicitly forms $\rho = Q_1 \otimes \cdots \otimes Q_k$, checks $\rho^* \rho = I$, and evaluates its power traces. Newton's recurrence

$$m h_m(\rho) = \sum_{j=1}^{m} \mathrm{Tr}(\rho^j) h_{m-j}(\rho) \tag{403}$$

then produces characters of $\mathrm{Sym}^m V$ directly. The sampled target quantities are

$$\begin{aligned}
&\mathbb{E}\prod_j h_{\tau_j}(\rho) && \text{(ordinary one-sided),}\\
&\mathbb{E}[p_\lambda(\rho)\overline{p_\mu(\rho)}] && \text{(balanced power-sum),}\\
&\mathbb{E}|h_m(\rho)|^2 && \text{(single-state),}\\
&\mathbb{E}|\operatorname{Tr}\rho|^{2r} && \text{(multilinear).}
\end{aligned}$$

The last two are character inner products, hence exactly the invariant multiplicities in (14) and (19). Numerical integration is corroboration, not a replacement for the proofs above.

73.7 Results

Each case used 6,000 deterministic-seed Haar samples. A numerical row passes when the absolute error is at most six empirical standard errors plus a small fixed rounding margin. Matrix unitarity and trace factorization were also checked at tolerance 10^{-11}.

Representation	Target	Direct estimate	Formula	Std. error	Result
$U(3)$	$\lvert h_3\rvert^2$	1.0367	1	0.0253	**PASS**
$U(3)$	$\lvert \operatorname{Tr}\rho\rvert^4$	2.0318	2	0.0593	**PASS**
$U(2)\times U(3)$	$\lvert h_2\rvert^2$	1.9533	2	0.0546	**PASS**
$U(2)\times U(3)$	$\lvert p_{(2)}\rvert^2$	3.8911	4	0.0638	**PASS**
$U(2)\times U(2)$	$\lvert h_3\rvert^2$	2.0797	2	0.1219	**PASS**
$U(2)^3$	$\lvert h_2\rvert^2$	4.0815	4	0.1641	**PASS**
$U(2)^3$	$\lvert \operatorname{Tr}\rho\rvert^4$	8.2364	8	0.6379	**PASS**
$U(2)^3$	$p_{(2)}\overline{p_{(1,1)}}$	0.0032	0	0.0284	**PASS**

The exact calculations make the stable-range boundary visible:

Dimensions	Sector	Degree	Exact	Stable expression
$(2,2)$	single-state	3	2	$p(3)=3$ (not applicable)
$(2,2)$	multilinear	3	25	$(3!)^2=36$ (not applicable)
$(2,3)$	multilinear	3	30	$(3!)^2=36$ (not applicable)
$(2,2,2)$	multilinear	3	125	$(3!)^3=216$ (not applicable)

Thus the dimension restrictions are not cosmetic: applying a stable formula outside $r,m\le \min_a d_a$ produces wrong answers.

The scalar-center test separately constructs, for each degree $1\le n\le 5$, $z=e^{2\pi i/(n+1)}$ and the group matrix $\rho(zI_2,I_3)=zI_6$. Its action on degree n is $z^n\neq 1$ in every case, giving five direct matrix witnesses for Proposition 1.

73.8 Reproduction and scope

The folder `for_this_guy/local_unitary_tensor_products` contains:

- `marimo_notebooks/local_unitary_tensor_products.py`, the interactive marimo report;
- `verification.py`, the direct and exact implementation;
- `test_verification.py`, deterministic regression tests;
- this LaTeX source and its compiled PDF.

The notebook exposes the sample count and reports every tested coefficient, not only the representative rows printed here. Its global $U(3)$ case is explicitly labeled as a comparator; it is not conflated with the local tensor-product groups. All local-unitary cases remain grouped in this single family because they are specializations of the same representation (1) and the same proof.

74 Unitary conjugation on endomorphisms

Abstract

For the conjugation representation of $U(n)$ on $W = \mathrm{End}(\mathbb{C}^n)$, this note proves the trace form of the generalized Molien integral, an exact finite-rank Littlewood–Richardson coefficient formula, and the stable description by colored necklaces. The formulas are checked by an independent matrix computation. The accompanying marimo notebook constructs unitary generators and their n^2-dimensional conjugation matrices, builds the full derived action on tensor products of symmetric powers, and computes its common kernel. All 18 representative comparisons pass for $1 \le n \le 4$, total degrees through 3, and full target dimensions through 405. The checks include the nonstable value 5 at $n = 2$, $\tau = (1, 1, 1)$, versus the stable value 6.

74.1 Imported target

For a compact group G acting on W, Tools 2 and 1 give

$$\mathrm{SMG}_{G,W}(\mathbf{t}) = \mathbb{E}[\mathrm{Exp}_\sigma(X_{G,W} h_1)] = \int_G \prod_{a \ge 1} \det(I - t_a \rho_W(g))^{-1}\, dg, \tag{404}$$

$$[m_\tau]\, \mathrm{SMG}_{G,W} = \dim \left(\bigotimes_a \mathrm{Sym}^{\tau_a} W \right)^G. \tag{405}$$

Writing $P_k = \sum_a t_a^k$, the logarithmic determinant expansion gives

$$\boxed{\mathrm{SMG}_{G,W}(\mathbf{t}) = \int_G \exp\left(\sum_{k \ge 1} \frac{P_k}{k} \mathrm{Tr}(\rho_W(g)^k) \right) dg.} \tag{406}$$

These identities are quoted; the new calculation begins with the conjugation trace identity and Littlewood–Richardson decomposition.

74.2 The conjugation representation

Take $G = U(n)$, $V = \mathbb{C}^n$, and $W = \mathrm{End}(V) = V \otimes V^*$. The action is $A \mapsto gAg^{-1}$. If g has eigenvalues $z_1, \dots, z_n$, its eigenvalues on W are $z_i z_j^{-1}$. Hence

$$\mathrm{Tr}(\mathrm{Ad}(g)^k) = \sum_{i,j} z_i^k z_j^{-k} = \mathrm{Tr}(g^k)\,\mathrm{Tr}(g^{-k}) = |\,\mathrm{Tr}(g^k)|^2. \tag{407}$$

Substitution into (406) proves the exact finite-rank formula

$$\boxed{\mathrm{SMG}_{U(n),\mathrm{End}(V)}(\mathbf{t}) = \int_{U(n)} \exp\left(\sum_{k \ge 1} \frac{P_k}{k} |\,\mathrm{Tr}(g^k)|^2 \right) dg.} \tag{408}$$

74.3 Exact finite-rank coefficients

Theorem 1 (Finite-rank formula)**.** *For every partition $\tau = (\tau_1, \dots, \tau_s)$,*

$$\boxed{[m_\tau]\, \mathrm{SMG}_{U(n),\mathrm{End}(V)} = \sum_{\substack{\lambda^{(a)} \vdash \tau_a \\ \ell(\lambda^{(a)}) \le n}} \sum_{\substack{\nu \vdash |\tau| \\ \ell(\nu) \le n}} \left(c^{\nu}_{\lambda^{(1)}, \dots, \lambda^{(s)}} \right)^2.} \tag{409}$$

Proof. Cauchy decomposition gives, as a $GL(V) \times GL(V)$ module,

$$\operatorname{Sym}^r(V \otimes V^*) \cong \bigoplus_{\substack{\lambda \vdash r \\ \ell(\lambda) \le n}} \mathbb{S}_\lambda V \otimes \mathbb{S}_\lambda V^*.$$

Tensor this decomposition over $a = 1, \ldots, s$. For a fixed tuple $(\lambda^{(1)}, \ldots, \lambda^{(s)})$, the Littlewood–Richardson rule gives

$$\bigotimes_a \mathbb{S}_{\lambda^{(a)}} V \cong \bigoplus_{\substack{\nu \\ \ell(\nu) \le n}} (\mathbb{S}_\nu V)^{\oplus c^\nu_{\lambda^{(1)}, \ldots, \lambda^{(s)}}}.$$

The associated dual factor has the same multiplicities. Schur's lemma says that the invariant dimension of this summand is the sum of the squares of those multiplicities. Summing over the partition tuple proves (409). □

74.4 Stable contractions and colored necklaces

Let $d = |\tau|$ and $H_\tau = S_{\tau_1} \times \cdots \times S_{\tau_s} \le S_d$. Since

$$\bigotimes_a \operatorname{Sym}^{\tau_a} W = (W^{\otimes d})^{H_\tau}, \qquad W^{\otimes d} = V^{\otimes d} \otimes (V^*)^{\otimes d},$$

the first fundamental theorem for $GL(V)$ supplies invariant contraction tensors T_σ, indexed by $\sigma \in S_d$. When $n \ge d$ these tensors are linearly independent, hence form a basis. A block permutation $h \in H_\tau$ acts by

$$hT_\sigma = T_{h\sigma h^{-1}}.$$

Therefore the fixed dimension is the number of orbits of this permutation basis.

Corollary 1 (Stable coefficient)**.** *If $n \ge d = |\tau|$, then*

$$\boxed{[m_\tau] \operatorname{SMG}_{U(n), \operatorname{End}(V)} = \#(S_d/H_\tau\text{-}conjugacy).} \tag{410}$$

Consequently all coefficients of total degree at most n have stabilized.

Give the d letters colors so that exactly τ_a letters have color a. Conjugating by H_τ relabels letters within each color and preserves only the cyclic color word attached to each cycle of a permutation. Thus an orbit is precisely a multiset of nonempty colored necklaces. If $\mathcal{N}$ is the set of all nonempty necklaces and

$$\operatorname{wt}(N) = \prod_a t_a^{\#\{\text{letters of color } a \text{ in } N\}},$$

the multiset construction gives the stable Euler product

$$\boxed{\operatorname{SMG}_{U(\infty), \operatorname{End}}(\mathbf{t}) = \prod_{N \in \mathcal{N}} \frac{1}{1 - \operatorname{wt}(N)}.} \tag{411}$$

This proves the proposed colored-cycle interpretation and fixes its meaning coefficientwise.

74.5 Independent matrix verification

The notebook does not use (409) to obtain its direct target. It constructs the group representation and then solves the invariant problem in the full target space.

Group and conjugation matrices

For column-major vectorization,

$$\operatorname{vec}(gAg^*) = (\overline{g} \otimes g) \operatorname{vec}(A). \tag{412}$$

The code constructs n^2 explicit unitary generators by exponentiating the standard skew-Hermitian basis

$$iE_{jj}, \qquad E_{ij} - E_{ji}, \qquad i(E_{ij} + E_{ji}) \quad (i < j).$$

It checks unitarity, the homomorphism identity $\operatorname{Ad}(gh) = \operatorname{Ad}(g) \operatorname{Ad}(h)$, and (407) for powers $1 \le k \le 4$.

Direct invariant dimension

Differentiating (412) gives the $n^2 \times n^2$ matrix

$$d\,\mathrm{Ad}(X) = I \otimes X - X^{\mathsf{T}} \otimes I, \qquad d\,\mathrm{Ad}(X)(A) = XA - AX. \tag{413}$$

If $B = (b_{ij})$ acts on a q-dimensional space and $x^\alpha = x_1^{\alpha_1} \cdots x_q^{\alpha_q}$ is a monomial basis element of Sym^r, its derived action is implemented directly as

$$d\,\mathrm{Sym}^r(B)x^\alpha = \sum_{i,j} \alpha_j b_{ij} x^{\alpha - e_j + e_i}. \tag{414}$$

On a tensor product, the code sums (414) over the tensor factors. It forms the positive semidefinite Gram matrix

$$Q = \sum_{X \text{ in the basis of } \mathfrak{u}(n)} d\rho_\tau(X)^* d\rho_\tau(X).$$

Its nullity is the common-kernel dimension. Because $U(n)$ is connected, this common kernel is exactly the $U(n)$-fixed space. This is a deterministic floating-point calculation, not a Haar-sampling estimate. Each reported row includes the scale-dependent rank threshold, largest null eigenvalue, smallest positive eigenvalue, spectral-gap ratio, and invariance residual. A row without a gap ratio of at least 10^4 is labelled inconclusive rather than rounded to a nullity.

Three comparison routes

The finite formula is computed by enumerating LR tableaux with all length cutoffs retained. In stable cases, the notebook separately (i) enumerates H_τ-conjugacy orbits in S_d and (ii) expands the necklace Euler product through multidegree τ. Agreement of all applicable columns therefore tests the representation construction, the finite formula, the stability threshold, and the necklace bijection.

n	τ	full target dim.	direct nullity	finite LR	stable orbit/Euler
1	(1)	1	1	1	1
1	(2)	1	1	1	–
1	$(1,1)$	1	1	1	–
1	(3)	1	1	1	–
1	$(2,1)$	1	1	1	–
2	(1)	4	1	1	1
2	(2)	10	2	2	2
2	$(1,1)$	16	2	2	2
2	(3)	20	2	2	–
2	$(2,1)$	40	3	3	–
2	$(1,1,1)$	64	5	5	–
3	(1)	9	1	1	1
3	(2)	45	2	2	2
3	$(1,1)$	81	2	2	2
3	(3)	165	3	3	3
3	$(2,1)$	405	4	4	4
4	(1)	16	1	1	1
4	(2)	136	2	2	2

Table 2: All 18 representative comparisons pass. A dash means $n < |\tau|$, so the stable formula is not asserted.

For $n = 1, 2, 3, 4$, the maximum errors in unitarity, the homomorphism identity, and the trace identity are respectively below 3.2×10^{-16}, 1.5×10^{-16}, and 1.8×10^{-15}. The test suite also checks the stable values

$$\tau = (3), (2,1), (1,1,1), (2,2) \quad \longmapsto \quad 3, 4, 6, 10$$

by both orbit enumeration and necklace coefficient extraction.

74.6 Reproduction and scope

The noninteractive companion files are grouped under `proved_matrix_groups/packages/unitary_conjugation_endomorphisms/`. From the repository root, run

```
python3 proved_matrix_groups/packages/unitary_conjugation_endomorphisms/
  unitary_conjugation_core.py
python3 -m pytest proved_matrix_groups/packages/
  unitary_conjugation_endomorphisms/test_verification.py
marimo edit marimo_notebooks/
  unitary_conjugation.py
```

Join each wrapped command onto one shell line. The notebook's interactive direct calculation uses a safety cap of 800 on the full target dimension; the combinatorial formulas can be explored farther. The proof itself has no dimension restriction beyond $n \geq 1$ for the finite formula and the explicitly stated stable condition $n \geq |\tau|$.

75 Proctor's odd symplectic group and compact replacements

Outcome. The proposed normalized Haar expectation for the full complex odd symplectic group is not defined: the group contains both a vector group and $\mathbb{C}^\times$. Three valid replacements were proved and tested. Exact constant-term calculations verify the two compact formulas through total degree six. The full algebraic invariant series is exactly 1. For finite fields, four matrix groups of orders 24, 432, 12,000, and 11,520 were enumerated in dimensions three and five; all 38 tested coefficients agree between direct permutation-matrix averages and the proposed cycle formula. Selected coefficients also agree with explicit orbit counts.

75.1 The obstruction comes before computation

Let $V = W \oplus L$ over $\mathbb{C}$, where $\dim W = 2n$, $L = \mathbb{C}e_0$, and ω is a nondegenerate alternating form on W. Extend it by

$$\Omega((w,s),(w',s')) = \omega(w,w'), \qquad \operatorname{rad}\Omega = L.$$

Consider the full stabilizer

$$G = \{g \in GL(V) : \Omega(gv, gw) = \Omega(v,w)\}.$$

Because the radical is intrinsic, $gL = L$. In the decomposition $W \oplus L$, every element therefore has the form

$$g = \begin{pmatrix} A & 0 \\ \ell & a \end{pmatrix}, \qquad A \in \mathrm{Sp}(W), \quad \ell \in W^*, \quad a \in \mathbb{C}^\times. \tag{415}$$

Conversely, every matrix in (1) preserves Ω. Thus

$$G \cong W^* \rtimes \big(\mathrm{Sp}_{2n}(\mathbb{C}) \times \mathbb{C}^\times\big). \tag{416}$$

The vector group W^* is the unipotent radical, so G is nonreductive; both W^* and $\mathbb{C}^\times$ are noncompact. A locally compact group has a normalized Haar probability exactly when it is compact. Consequently

$$\frac{1}{\operatorname{vol}(G)} \int_G f(g)\, dg$$

is not a defined normalization. This is a mathematical obstruction, not a numerical convergence problem. A finite box cutoff would depend on coordinates and cutoff shape and is not a group-invariant substitute.

There is a useful convention warning. Some authors impose $g(e_0) = e_0$. We write this subgroup as

$$G^1 = W^* \rtimes \mathrm{Sp}_{2n}(\mathbb{C}), \tag{417}$$

which is obtained by setting $a = 1$. The full stabilizer G and G^1 give different answers below and should not be conflated.

75.2 A structural identity and what it sees

Block triangularity gives, for every scalar t,

$$\det(I - tg) = (1 - at)\det(I - tA), \qquad \mathrm{Spec}(g) = \mathrm{Spec}(A) \cup \{a\}. \tag{418}$$

The shear coordinate ℓ can change Jordan structure but is invisible to the characteristic polynomial. Therefore any probability measure ν for which the coefficientwise moments exist satisfies

$$\int_G \prod_i \det(I - t_i g)^{-1}\, d\nu(g) = \int_{\mathrm{Sp}_{2n}(\mathbb{C})\times\mathbb{C}^\times} \prod_i \frac{\det(I - t_i A)^{-1}}{1 - at_i}\, d\bar{\nu}(A,a), \tag{419}$$

where $\bar{\nu}$ is the pushforward to the Levi quotient. This gives a useful diagnostic for heat-kernel or user-chosen noncompact probabilities: the answer depends on the chosen measure, but never on its conditional distribution in the unipotent coordinate.

The code tests (4) modulo q on deterministic samples from every constructed finite group. It is important that this structural test is separate from the permutation-representation test in Section 75.5.

75.3 Two compact replacements

A maximal compact subgroup of the full stabilizer is

$$K = \mathrm{Sp}(n) \times U(1), \qquad V|_K = W \oplus \chi,$$

where $\chi(z) = z$ on the radical line. Write

$$\mathrm{SMG}_{H,X}(\mathbf{t}) = \int_H \prod_i \det(I - t_i h|_X)^{-1}\, dh$$

for a compact group with normalized Haar probability. Direct-sum multiplicativity and product Haar measure give

$$\mathrm{SMG}_{K,V} = \mathrm{SMG}_{\mathrm{Sp}(n),W}\, \mathrm{SMG}_{U(1),\chi}\,.$$

Since

$$\prod_i (1 - z t_i)^{-1} = \sum_{d \geq 0} z^d h_d(\mathbf{t}), \qquad \int_{U(1)} z^d\, dz = \delta_{d0},$$

we obtain the exact identity

$$\boxed{\mathrm{SMG}_{K,V} = \mathrm{SMG}_{\mathrm{Sp}(n),W}\,.} \tag{420}$$

The radical line disappears because it has a nontrivial $U(1)$ weight.

For the fixed-radical group G^1, a maximal compact subgroup is $\mathrm{Sp}(n)$ and $V|_{\mathrm{Sp}(n)} = W \oplus \mathbf{1}$. The extra eigenvalue is now 1, so

$$\boxed{\mathrm{SMG}_{\mathrm{Sp}(n),W\oplus\mathbf{1}} = \mathrm{SMG}_{\mathrm{Sp}(n),W} \prod_i (1 - t_i)^{-1}.} \tag{421}$$

Exact rank-one check

For $n = 1$, restrict to the maximal torus of $SU(2) = \mathrm{Sp}(1)$ with weights z, z^{-1}. Weyl integration is the exact constant-term operation

$$\int_{SU(2)} f = \mathrm{CT}_z\big((1 - z^2) f(z)\big).$$

The verification represents every complete homogeneous character as a Laurent polynomial. It extracts both the z and $U(1)$ constant terms and tests all 30 partitions through total degree six. Representative results are:

τ	K direct	$\mathrm{Sp}(1), W$	G^1 compact direct	formula (7)
$()$	1	1	1	1
(1)	0	0	1	1
(2)	0	0	1	1
$(1,1)$	1	1	2	2
$(2,1)$	0	0	2	2
$(2,2)$	1	1	3	3

All 30 comparisons pass exactly, with integer arithmetic.

75.4 The full algebraic invariant series

Even though Haar averaging fails, the covariant invariant series is meaningful:

$$\mathrm{SMG}^{\mathrm{inv}}_{G,V} = \sum_\tau \dim \left(\bigotimes_j \mathrm{Sym}^{\tau_j} V \right)^G m_\tau. \tag{422}$$

Theorem. For the full stabilizer G in (1),

$$\boxed{\mathrm{SMG}^{\mathrm{inv}}_{G,V} = 1.} \tag{423}$$

Proof. Embed $\bigotimes_j \mathrm{Sym}^{\tau_j} V$ into $V^{\otimes d}$, where $d = |\tau|$. The subgroup $\mathrm{diag}(I_W, a)$ acts with weight a^r on tensors having r radical-line factors. Its invariant subspace is thus contained in $W^{\otimes d}$.

For $\ell \in W^*$, differentiate the action of

$$u_\ell = \begin{pmatrix} I_W & 0 \\ \ell & 1 \end{pmatrix}.$$

On $x \in W^{\otimes d}$ the derivative is a sum of d contractions, one for each tensor position, and the output of the jth contraction has its unique e_0 in position j. These outputs lie in distinct direct summands. Hence invariance forces every positional contraction against every ℓ to vanish. Already at the first position, the contractions by a basis of W^* recover all coordinates of x, so $x = 0$. Thus there are no positive-degree invariants. □

This proof gives a compact computational certificate: after the torus weight filter, stack the first-position contractions against a basis of W^*. The result is a coordinate permutation of rank $(2n)^d$. The implementation records full rank for $n = 1, 2$ and $1 \le d \le 4$ (eight checks), so every recorded kernel has dimension zero. This certificate is stronger and cheaper than building large Reynolds matrices for a nonreductive group.

75.5 Finite odd symplectic groups and a genuine complex representation

Let $E = W \oplus L$ over a prime field $\mathbb{F}_q$. The same block proof gives

$$G^{\mathrm{odd}}_{2n+1}(q) \cong W^* \rtimes \big(\mathrm{Sp}_{2n}(q) \times \mathbb{F}_q^\times\big), \tag{424}$$

and hence

$$\begin{aligned} |G^{\mathrm{odd}}_{2n+1}(q)| &= q^{2n}(q-1)|\,\mathrm{Sp}_{2n}(q)|, \\ |\,\mathrm{Sp}_{2n}(q)| &= q^{n^2}\prod_{j=1}^{n}(q^{2j}-1). \end{aligned} \tag{425}$$

Matrices with entries in $\mathbb{F}_q$ are not automatically complex matrices. We therefore use the honest complex permutation representation on $S = \mathbb{P}(E)$. If P_g is its permutation matrix and $c_r(g)$ is the number of r-cycles, then

$$\det(I - tP_g)^{-1} = \prod_{r\ge 1}(1-t^r)^{-c_r(g)}. \tag{426}$$

Consequently the proposed finite formula is

$$\boxed{\mathrm{SMG}_{G(q),\mathbb{C}[S]}(\mathbf{t}) = \frac{1}{|G(q)|}\sum_{g\in G(q)}\prod_i\prod_{r\ge 1}(1-t_i^r)^{-c_r(g)}.} \tag{427}$$

For a partition τ, Burnside's lemma also gives

$$[m_\tau]\,\mathrm{SMG}_{G(q),\mathbb{C}[S]} = \#\left(G(q)\backslash \prod_j \mathrm{MSet}_{\tau_j}(S)\right). \tag{428}$$

There are exactly two orbits on $\mathbb{P}(E)$: the radical point and all nonradical points. The latter assertion follows because $\mathrm{Sp}(W)$ is transitive on nonzero vectors of W, while $\ell(w)$ adjusts the radical coordinate. Therefore

$$\boxed{[m_{(1)}]\,\mathrm{SMG}_{G(q),\mathbb{C}[\mathbb{P}(E)]} = 2.} \tag{429}$$

75.6 Independent verification algorithms

For each (n, q), the program performs the following steps.

1. Enumerate every $2n \times 2n$ matrix A satisfying $A^{\mathsf{T}} J A = J$, then every triple (A, ℓ, a) in (10). Check the observed orders against (11), preservation of the degenerate form, and deterministic closure samples.
2. Enumerate normalized representatives of every projective point and construct the permutation P_g for every group matrix.
3. **Direct matrix route:** compute $p_k = \mathrm{tr}(P_g^k)$ from explicit permutation powers and use Newton's recurrence
$$dh_d = \sum_{k=1}^{d} p_k h_{d-k}.$$
Average $\prod_j h_{\tau_j}$ over all matrices.
4. **Cycle route:** independently find the cycles and expand the right side of (12) by dynamic programming. Average the resulting products and compare every coefficient with the direct route.
5. **Orbit route:** in the two smaller cases, enumerate the orbits on multisets for $\tau = (1), (2), (1, 1)$ and compare with (14).

The direct route does not call the cycle-product routine. This separation matters: otherwise (13) would only be tested against a rearrangement of itself.

75.7 Exact results

n	q	$\dim E$	$\lvert \mathrm{Sp}_{2n}(q)\rvert$	$\lvert G(q)\rvert$	$\lvert \mathbb{P}(E)\rvert$	tested degrees
1	2	3	6	24	7	all partitions through 4
1	3	3	24	432	13	all partitions through 4
1	5	3	120	12,000	31	all partitions through 3
2	2	5	720	11,520	31	all partitions through 3

All four form, order, radical-point, determinant, closure, and coefficient checks pass. In total, 38 matrix-average coefficients match 38 cycle-formula coefficients exactly. Selected values illustrate both stability and genuine dimension dependence:

(n, q)	$m_{(1)}$	$m_{(2)}$	$m_{(1,1)}$	$m_{(3)}$	$m_{(2,1)}$	$m_{(1,1,1)}$
$(1, 2)$	2	5	6	11	18	25
$(1, 3)$	2	5	6	12	19	28
$(1, 5)$	2	5	6	14	22	30
$(2, 2)$	2	6	7	19	31	43

For $(n, q) = (1, 2)$ and $(1, 3)$, explicit orbit enumeration gives respectively 2, 5, and 6 orbits for $\tau = (1), (2), (1, 1)$, exactly matching both averaging routes.

75.8 Interpretation, limitations, and new tools

The results separate four notions that are easy to blur:

- The full complex-group Haar expression is undefined, and no computation can validate it without first choosing a new measure.
- Maximal-compact averaging is canonical up to conjugacy, but it forgets the unipotent radical. For the full stabilizer it also removes the radical line by $U(1)$ weight; for G^1 it retains that line as a trivial summand.

- The full covariant invariant series is well-defined but collapses to 1 in the defining representation. Interesting odd symplectic representation theory therefore requires other modules, mixed V, V^* constructions, or polynomial-function invariants.
- Finite-field permutation actions give exact, nontrivial, reproducible Λ-distributions, but they are different representations of different finite groups and should not be presented as complex matrix realizations of the defining modular matrices.

Two reusable tools emerge. First, the *Levi visibility test* (4)–(5) immediately decides which nonreductive coordinates an eigenvalue statistic can detect. Second, the *torus-filter plus derivative certificate* proves invariant vanishing using small full-rank contraction maps instead of a nonexistent Reynolds average.

75.9 Reproducibility

From the repository root, run

```
.venv/bin/python \
  for_this_guy/proctor_odd_symplectic_group/verification.py
```

and open the interactive report with

```
.venv/bin/marimo edit \
  marimo_notebooks/proctor_odd_symplectic.py
```

The verification uses only exact integer and rational arithmetic. Exhaustive group enumeration is intentionally limited to prime fields and to the four displayed cases; extending to larger ranks should replace all-matrix search by standard symplectic generators.

References

[1] S. Okada, *A bialternant formula for odd symplectic characters and its application*, 2019, https://arxiv.org/abs/1905.12964.

[2] W. Burnside, *Theory of Groups of Finite Order*, second edition, 1911.

[3] A. Haar, *Der Massbegriff in der Theorie der kontinuierlichen Gruppen*, Annals of Mathematics 34 (1933), 147–169.

Part VII

Polyhedral, sporadic, and special 24-dimensional groups

76 Finite rotation groups and baseline matrix-group formulas

Abstract

We verify finite-group instances of the generalized Molien formula by full matrix enumeration. The examples cover the tetrahedral, octahedral, and icosahedral rotation representations; the full H_3 reflection group; restricted monomial groups including $G(r,p,n)$; an exterior-power representation; and a regular representation. In total, 492 coefficient comparisons pass, together with independent Reynolds-projector checks. A dual-code cycle counter for monomial groups is included as a reusable tool.

76.1 Imported finite-group tools

The universal determinant and coefficient identities are imported Tools 2 and 1. This section is a spectral compression and verification study, beginning with rotation-angle classes.

76.2 Polyhedral rotation groups

A rotation through angle θ in three dimensions has spectrum $1, e^{i\theta}, e^{-i\theta}$ and hence

$$\det(I - tR_\theta)^{-1} = \frac{1}{(1-t)(1-2\cos(\theta)t+t^2)}.$$

Notice the multiplication by t in the middle term; this corrects a typographical ambiguity in the supplied expression. Writing Φ_θ for the product of this factor over all variables t_i, class enumeration gives

$$\begin{aligned}
\mathrm{SMG}^{\mathrm{rot}}_{A_4} &= \frac{1}{12}(\Phi_0 + 8\Phi_{2\pi/3} + 3\Phi_\pi),\\
\mathrm{SMG}^{\mathrm{rot}}_{S_4} &= \frac{1}{24}(\Phi_0 + 8\Phi_{2\pi/3} + 6\Phi_{\pi/2} + 9\Phi_\pi),\\
\mathrm{SMG}^{\mathrm{rot}}_{A_5} &= \frac{1}{60}(\Phi_0 + 20\Phi_{2\pi/3} + 15\Phi_\pi + 12\Phi_{2\pi/5} + 12\Phi_{4\pi/5}).
\end{aligned}$$

The notebook constructs the first two groups as determinant-one signed permutation matrices (restricting to a tetrahedron for A_4). It constructs A_5 by enumerating all orthogonal maps preserving the twelve icosahedron vertices. No class labels are used on the direct side.

Adjoining the negatives of the 60 icosahedral rotations gives the order-120 H_3 reflection group. Its ten spectral classes are checked against complete matrix averaging. Independently, the one-variable invariant series

$$\frac{1}{(1-t^2)(1-t^6)(1-t^{10})}$$

matches the directly averaged dimensions through degree ten.

76.3 A dual-code cycle formula for monomial groups

Let $A \le (\mathbb{Z}/r\mathbb{Z})^n$ be a linear code stable under a permutation group $H \le S_n$. Put

$$G = D(A) \rtimes H, \qquad D(x) = \mathrm{diag}(\zeta^{x_1}, \dots, \zeta^{x_n}), \quad \zeta = e^{2\pi i/r}.$$

For a permutation σ, expand the determinant factor cycle by cycle. For every cycle C and every variable index a, choose a multiplicity $m_{C,a} \geq 0$ satisfying

$$\sum_C |C| m_{C,a} = \tau_a.$$

The diagonal phase exponent is constant on C and equals $e_j = \sum_a m_{C,a} \pmod r$. Character orthogonality over A gives

$$\frac{1}{|A|} \sum_{x \in A} \zeta^{\langle x,e \rangle} = \mathbf{1}_{\{e \in A^\perp\}}.$$

Thus the coefficient is obtained by counting these cycle assignments whose exponent vector lies in the dual code, then averaging over H. This removes the diagonal loop entirely and is useful well beyond the test cases.

For the complex reflection group $G(r,p,n)$, the code is $A = \{x : \sum_j x_j \equiv 0 \pmod p\}$. The notebook also tests a specialized counter: for each $q = 0, \dots, p-1$, take label vectors

$$u \in \prod_a \{0, \dots, \tau_a\}, \qquad \sum_a u_a \equiv q\, r/p \pmod r,$$

and count multisets of exactly n labels with componentwise sum τ. Summing these counts over q yields the coefficient. Dynamic programming implements this finite multiset count independently of matrix averaging.

76.4 Alternate representations

If V has eigenvalues $\lambda_1, \dots, \lambda_d$, then $\bigwedge^k V$ has eigenvalues $\prod_{j \in J} \lambda_j$ over k-subsets J. The notebook constructs $\bigwedge^2$ of the three-dimensional standard reflection representation of S_4 using matrix minors, while the formula side uses the spectral products. All group elements are enumerated.

For the regular representation, an element g of order $o(g)$ acts as $|G|/o(g)$ cycles of length $o(g)$. Hence

$$\boxed{\mathrm{SMG}_{G,\mathrm{Reg}} = \frac{1}{|G|} \sum_{g \in G} \prod_i (1 - t_i^{o(g)})^{-|G|/o(g)}.}$$

For S_3, the formula uses the order distribution $N_1 = 1, N_2 = 3, N_3 = 2$ and is compared with six explicit 6×6 left-regular permutation matrices.

76.5 Test matrix and outcome

Cases	Independent comparison	Checks
A_4, S_4, A_5 rotations	all matrices versus angle classes	135
Full H_3 reflection group	120 matrices versus ten spectra; degrees $2, 6, 10$	139
$D(A) \rtimes C_3$ and five $G(r,p,n)$ cases	matrix average versus dual-code or multiset counter	180
$\bigwedge^2$ standard S_4 and regular S_3	explicit matrices versus transformed spectra/order counts	38
Total		**492**

Every comparison passes. Additional Reynolds projectors satisfy $\|R^2 - R\|_F < 9 \times 10^{-16}$. Representative ranks are: rank one on $\mathrm{Sym}^2(\mathbb{C}^3)$ for H_3; rank one on $\mathrm{Sym}^2(\mathbb{C}^3) \otimes \mathbb{C}^3$ for $G(2,2,3)$; and rank five on Sym^2 of the six-dimensional regular S_3 representation.

76.6 Reproduction and scope

From the repository root, run these modules with `python3 -m`:

- `for_this_guy.matrix_group_formula_verification.h3_reflection.verification`
- `for_this_guy.matrix_group_formula_verification.restricted_monomial.verification`
- `for_this_guy.matrix_group_formula_verification.representation_variants.verification`

Their canonical marimo apps are cataloged in `marimo_notebooks/README.md`. The verification establishes the formulas for the displayed finite cases and truncations. The common finite-group identity is imported as Tools 1–2; these tests do not claim elementary simplifications for every finite group of Lie type, spin representation, or exceptional representation mentioned in the broader proposal.

77 Three representations associated with A_5

Outcome. All 45 multivariate coefficients indexed by partitions through total degree seven agree in each of the five-dimensional permutation, four-dimensional deleted permutation, and three-dimensional rotation representations. The explicit groups all have order 60. Separate Reynolds projectors confirm the low-degree invariant dimensions.

77.1 Imported target

Tools 1 and 2 identify the desired series as $\mathbb{E}[\mathrm{Exp}_\sigma(X_{G,V}h_1)]$ and its m_τ coefficient as $\dim(\bigotimes_a \mathrm{Sym}^{\tau_a} V)^G$. This section therefore starts with the three representation-specific class spectra; it does not rederive the common identity.

77.2 Three class-spectrum formulas

Write $F_{(\lambda_1,\dots,\lambda_n)}(\mathbf{t}) = \prod_{a,i}(1-\lambda_i t_a)^{-1}$, $\omega = e^{2\pi i/3}$, and $\xi = e^{2\pi i/5}$. The formulas tested are

$$\begin{aligned}\mathrm{SMG}_{\mathrm{perm}} &= \frac{1}{60}\big[F_{(1,1,1,1,1)} + 15F_{(1,1,1,-1,-1)} + 20F_{(1,1,1,\omega,\omega^{-1})}\\ &\qquad + 24F_{(1,\xi,\xi^2,\xi^3,\xi^4)}\big], \end{aligned} \tag{3}$$

$$\begin{aligned}\mathrm{SMG}_{\mathrm{del}} &= \frac{1}{60}\big[F_{(1,1,1,1)} + 15F_{(1,1,-1,-1)} + 20F_{(1,1,\omega,\omega^{-1})}\\ &\qquad + 24F_{(\xi,\xi^2,\xi^3,\xi^4)}\big], \end{aligned} \tag{4}$$

$$\begin{aligned}\mathrm{SMG}_{\mathrm{rot}} &= \frac{1}{60}\big[F_{(1,1,1)} + 15F_{(1,-1,-1)} + 20F_{(1,\omega,\omega^{-1})}\\ &\qquad + 12F_{(1,\xi,\xi^{-1})} + 12F_{(1,\xi^2,\xi^{-2})}\big]. \end{aligned} \tag{5}$$

For (3), a permutation of cycle type (1^5), $(2^2,1)$, $(3,1^2)$, or (5) has precisely the displayed roots of unity as eigenvalues. Removing the always-fixed all-ones line deletes one eigenvalue 1 and proves (4). For (5), a rotation through angle θ has spectrum $(1, e^{i\theta}, e^{-i\theta})$; the rotation counts are $1, 15, 20, 12, 12$ for angles $0, \pi, 2\pi/3, 2\pi/5, 4\pi/5$.

The split of the fifth turns is essential in (5): the two classes have different 3D spectra. They may be combined in (3)–(4) because a 5-cycle and its nontrivial powers have the same full permutation spectrum.

77.3 Independent matrix construction

For the first representation, all 60 even permutations of five symbols are converted directly to 5×5 permutation matrices. For the deleted representation, let B be an orthonormal basis for

$$\{x \in \mathbb{C}^5 : x_1 + \cdots + x_5 = 0\};$$

the matrices are $B^* P_\pi B$. Neither route uses the spectral tables above.

For the rotation representation, start from the twelve icosahedron vertices

$$(0,\pm1,\pm\varphi), \quad (\pm1,\pm\varphi,0), \quad (\pm\varphi,0,\pm1), \qquad \varphi = (1+\sqrt{5})/2.$$

Mapping a fixed independent vertex frame to every compatible target frame and retaining the orientation-preserving isometries reconstructs 60 distinct 3×3 matrices.

77.4 Verification protocol

For every partition τ through total degree seven, the direct route computes h_{τ_a} recursively from the eigenvalues of every explicit matrix and averages $\prod_a h_{\tau_a}$. The formula route uses only the weighted spectra in (3), (4), or (5). It never receives the matrices.

As a third route, the code explicitly constructs each $\mathrm{Sym}^2(V)$ inside $V^{\otimes 2}$ using normalized occupancy states, averages the resulting matrices, and checks projector rank, trace, and idempotence. The same check is also performed on $V \otimes V$.

77.5 Results

Representation	$\|G\|$	$\dim V$	degree	checks	result
A_5 permutation	60	5	0–7	45	**PASS**
A_5 deleted permutation	60	4	0–7	45	**PASS**
A_5 rotation	60	3	0–7	45	**PASS**

Representation	$[m_{(1)}]$	$[m_{(2)}]$	$[m_{(1,1)}]$	$[m_{(1,1,1)}]$
permutation	1	2	2	5
deleted permutation	0	1	1	1
rotation	0	1	1	1

The symmetric-square projector ranks are respectively $2, 1, 1$. Their traces agree with those integers to 3×10^{-16}, and the largest idempotence error is 3.27×10^{-16}. These checks recover the claims that the permutation representation has a fixed line while the deleted and rotation representations do not.

The value $[m_{(1,1,1)}] = 1$ in the rotation case is the invariant orientation tensor in $V^{\otimes 3}$. This is not contradicted by the absence of an invariant cubic in $\mathrm{Sym}^3(V)$: $(1, 1, 1)$ denotes three separate degree-one factors, not the partition (3).

77.6 Reproducibility and scope

Run the command-line suite from the repository root:

```
.venv/bin/python -m for_this_guy.matrix_group_formula_verification.a5_representations.verification
```

Open `marimo_notebooks/a5_representations.py` with marimo to change the degree bound and inspect every row.

The finite computation is a transcription and convention test, not the source of the all-degree theorem. All-degree validity follows from (1)–(2) and the complete class spectra; the numerical suite checks the matrix constructions, class sizes, fifth-root choices, and coefficient extraction in representative small dimensions.

78 Binary polyhedral groups

Outcome. Explicit 2×2 complex matrix groups of orders 24, 48, and 120 were reconstructed for $2T$, $2O$, and $2I$. For each group, all 67 multivariate coefficients through total degree eight agree with the proposed paired-lift formula. Closure, unitarity, determinant one, the central element $-I$, and the odd-degree filter all pass.

78.1 Imported target

For each binary polyhedral matrix group, the σ-MGF and invariant coefficients are imported Tools 2 and 1. The new calculation pairs the two lifts of each rotation class.

78.2 Proof of the binary-lift formula

Let $\widetilde{\Gamma} \le SU(2)$ be the inverse image of a finite rotation group $\Gamma \le SO(3)$. If $R \in \Gamma$ rotates through angle θ, its two lifts q and $-q$ have spectra

$$(e^{i\theta/2}, e^{-i\theta/2}), \qquad (-e^{i\theta/2}, -e^{-i\theta/2}).$$

Define

$$\begin{aligned}
\Psi_\theta^+(\mathbf{t}) &= \prod_a (1 - 2\cos(\theta/2)t_a + t_a^2)^{-1}, \\
\Psi_\theta^-(\mathbf{t}) &= \prod_a (1 + 2\cos(\theta/2)t_a + t_a^2)^{-1}, \\
B_\theta &= \tfrac{1}{2}(\Psi_\theta^+ + \Psi_\theta^-).
\end{aligned}$$

Pair the two terms in (1) over each $R \in \Gamma$. Because $|\widetilde{\Gamma}| = 2|\Gamma|$, this gives

$$\mathrm{SMG}_{\widetilde{\Gamma},\mathbb{C}^2} = \frac{1}{|\Gamma|} \sum_\theta N_\Gamma(\theta) B_\theta. \tag{3}$$

This proves the proposed formula without requiring a separate conjugacy table for the binary group.

Substituting the rotation counts yields

$$\mathrm{SMG}_{2T} = \frac{1}{12}(B_0 + 8B_{2\pi/3} + 3B_\pi), \tag{4}$$

$$\mathrm{SMG}_{2O} = \frac{1}{24}(B_0 + 6B_{\pi/2} + 8B_{2\pi/3} + 9B_\pi), \tag{5}$$

$$\mathrm{SMG}_{2I} = \frac{1}{60}(B_0 + 15B_\pi + 20B_{2\pi/3} + 12B_{2\pi/5} + 12B_{4\pi/5}). \tag{6}$$

78.3 The central parity filter

The matrix $-I$ belongs to each binary group. On $W_\tau = \bigotimes_a \mathrm{Sym}^{\tau_a}(\mathbb{C}^2)$ it acts as $(-1)^{|\tau|}$. If $|\tau|$ is odd and v is invariant, then $v = (-I)v = -v$, so $v = 0$. Therefore

$$[m_\tau]\,\mathrm{SMG}_{2T} = [m_\tau]\,\mathrm{SMG}_{2O} = [m_\tau]\,\mathrm{SMG}_{2I} = 0 \quad \text{for odd } |\tau|. \tag{7}$$

The same cancellation is visible termwise in the average defining B_θ.

78.4 Independent matrix construction

The parent rotation matrices are constructed geometrically: signed permutation matrices give the octahedral group; the subgroup preserving the four vertices $(1,1,1), (1,-1,-1), (-1,1,-1), (-1,-1,1)$ gives the tetrahedral group; and compatible vertex-frame maps of the icosahedron give the icosahedral group.

For each rotation matrix, a standard trace-and-diagonal algorithm recovers one unit quaternion $q = w + xi + yj + zk$. It is mapped to

$$\rho(q) = \begin{pmatrix} w + iz & y + ix \\ -y + ix & w - iz \end{pmatrix} \in SU(2). \tag{8}$$

Both $\rho(q)$ and $-\rho(q)$ are retained. Since the sign choice for an individual lift is arbitrary but both signs are present, the resulting set is the full preimage and is independent of those choices.

This construction does not use (4)–(6). The implementation verifies all pairwise products against the reconstructed set, as well as unitarity, determinant one, and the presence of $-I$.

78.5 Verification protocol and results

For every partition τ through total degree eight, the direct route recursively computes complete homogeneous characters from all explicit matrices and averages their products. The independent route uses only the parent angle counts and paired half-angle spectra in (3). All odd rows are also checked separately against zero.

Finally, explicit matrices on $V \otimes V$ and $\mathrm{Sym}^2(V)$ are averaged into Reynolds projectors. The alternating determinant tensor gives a one-dimensional invariant in $V \otimes V$, while $\mathrm{Sym}^2(V)$ has no invariant vector.

Group	order	dimension	degree	checks	result
$2T$	24	2	0–8	67	**PASS**
$2O$	48	2	0–8	67	**PASS**
$2I$	120	2	0–8	67	**PASS**

All three sets are closed under multiplication. The largest observed unitarity error is 3.17×10^{-16} and the largest determinant error is 2.27×10^{-16}. Every odd-degree coefficient is exactly zero after integer validation.

For each group, the 4×4 Reynolds projector on $V \otimes V$ has rank and trace 1; its largest idempotence error is 1.04×10^{-16}. The symmetric-square projector has rank 0. Consequently the computation recovers the proposed low-degree distinction

$$[m_{(2)}] = 0, \qquad [m_{(1,1)}] = 1.$$

78.6 Reproducibility and scope

Run from the repository root:

```
.venv/bin/python -m for_this_guy.matrix_group_formula_verification.binary_polyhedral.verification
```

Open `marimo_notebooks/binary_polyhedral.py` with marimo for interactive degree selection and complete row tables.

The finite checks validate the matrix lift, angle convention, class counts, central signs, and coefficient code in reasonable low dimensions. The all-degree conclusion is the formal proof (1)–(3); it does not depend on the chosen cutoff or floating-point experiment.

79 Solvable and finite-subgroup families

Canonical testing artifacts.
Exact backend: `lambda_distributions/dists/solvable_finite_subgroup_sigma_mgfs/verification.py`.
Regression: `lambda_distributions/dists/solvable_finite_subgroup_sigma_mgfs/test_verification.py`.
Marimo: `marimo_notebooks/solvable_finite.py`.

Scope and outcome

A group name alone does not determine a σ-MGF: the series depends on a specified complex representation. We prove the formulas for two distinct finite-lamplighter representations, split metacyclic induced modules, finite affine and Frobenius permutation actions, unitriangular, Borel, and parabolic actions, finite defining subgroups of $SU(3)$, $SU(4)$, $Sp(4)$, and $SO(4)$, and symplectic-reflection wreath products. The accompanying marimo notebook constructs every representative matrix group and compares direct Reynolds averages with weighted-cycle, fixed-power, class-spectrum, or wreath-recursion formulas. Dimensions range from 2 to 27; the audit performs 7,580 fixed-power checks and 70 coefficient comparisons, all passing. The finite theorems, computational checks, and asymptotic statements are kept logically separate.

79.1 The representation-dependent target

Let $\rho : G \to GL(W)$ be a finite complex representation. For a partition or composition $\tau = (\tau_1, \dots, \tau_s)$, put

$$W_\tau = \bigotimes_{j=1}^{s} \operatorname{Sym}^{\tau_j} W, \qquad a_\tau(G, W) = \dim W_\tau^G.$$

The normalization used here is

$$\mathrm{SMG}_{G,W}(\mathbf{t}) = \sum_\tau a_\tau(G, W) m_\tau(\mathbf{t}).$$

Theorem 1 (Imported Reynolds–Molien identity)**.** *For every finite complex representation,*

$$\mathrm{SMG}_{G,W}(\mathbf{t}) = \frac{1}{|G|} \sum_{g \in G} \prod_{i \geq 1} \det(I - t_i \rho(g))^{-1}, \tag{430}$$

$$= \frac{1}{|G|} \sum_{g \in G} \exp\left(\sum_{r \geq 1} \frac{p_r(\mathbf{t})}{r} \chi_W(g^r) \right), \tag{431}$$

$$[m_\tau] \, \mathrm{SMG}_{G,W} = a_\tau(G, W). \tag{432}$$

Source. Equations (430) and (432) are Tools 2 and 1; the power-trace form is the standard logarithmic determinant expansion. □

If G acts on a finite set X, we always mean the honest complex permutation module $W = \mathbb{C}[X]$. Write

$$F_r(g) = |\operatorname{Fix}_X(g^r)|, \qquad c_d(g) = \#\{d\text{-cycles of } g \text{ on } X\}.$$

Corollary 1 (permutation form)**.** *For every finite permutation action,*

$$F_r(g) = \sum_{d|r} d c_d(g), \qquad c_d(g) = \frac{1}{d}\sum_{r|d} \mu(d/r) F_r(g), \tag{433}$$

$$\mathrm{SMG}_{G,X} = \frac{1}{|G|}\sum_{g\in G}\prod_{i,d\geq 1}(1-t_i^d)^{-c_d(g)}, \tag{434}$$

$$[m_\tau]\,\mathrm{SMG}_{G,X} = \#\left(G\backslash \prod_j \mathrm{Sym}^{\tau_j} X\right). \tag{435}$$

In particular, $[m_{(1^s)}]\,\mathrm{SMG} = \mathbb{E}F_1(g)^s = \#(G\backslash X^s)$.

Proof. A point in a d-cycle is fixed by g^r exactly when $d \mid r$; Möbius inversion gives (433). A d-cycle has determinant factor $1-t^d$, which proves (434). Monomials in $\mathrm{Sym}^q\,\mathbb{C}[X]$ are indexed by q-element multisets, and orbit sums form a basis for the invariant space, proving (435). □

79.2 Finite lamplighter groups

79.2.1 Natural monomial representation

Put $L_{r,n} = \mu_r^n \rtimes C_n$, where the generator P cyclically shifts the coordinates, and use the n-dimensional representation

$$(z_1,\ldots,z_n;P^s) \longmapsto \mathrm{diag}(z_1,\ldots,z_n)P^s.$$

Define

$$A_{r,\ell}(\mathbf{t}) = \frac{1}{r}\sum_{\zeta^r=1}\prod_{i\geq 1}(1-\zeta t_i^\ell)^{-1}.$$

Root-of-unity filtering also gives

$$A_{r,\ell} = \sum_{q\geq 0} h_{qr}(t_1^\ell, t_2^\ell,\ldots).$$

Theorem 2 (lamplighter monomial formula)**.**

$$\boxed{\mathrm{SMG}^{\mathrm{mon}}_{L_{r,n}} = \frac{1}{n}\sum_{\ell|n}\varphi(\ell)A_{r,\ell}^{n/\ell}.}$$

Proof. The shift P^s has $d=\gcd(n,s)$ cycles, each of length $\ell = n/d$. On one coordinate cycle C, weighted permutation elimination gives

$$\det(I - t\,\mathrm{diag}(z)P^s\mid_C) = 1-\left(\prod_{j\in C} z_j\right)t^\ell.$$

When the labels are averaged independently, every cycle product is an independent uniform element of μ_r. Thus a shift of cycle length ℓ contributes $A_{r,\ell}^{n/\ell}$. Exactly $\varphi(\ell)$ shifts have this cycle length. □

There is no raw coefficientwise limit with r fixed and $n\to\infty$. The identity-shift sector has weight $1/n$ and contributes $A_{r,1}^n/n$. Choosing the h_r term from two factors gives

$$[m_{(r,r)}]\,\mathrm{SMG}^{\mathrm{mon}}_{L_{r,n}} \geq \frac{1}{n}\binom{n}{2}[m_{(r,r)}]h_r^2 = \frac{(r+1)(n-1)}{2}.$$

79.2.2 Affine action on lamp configurations

The same abstract group acts on $X = C_r^n$ by $x \mapsto b + P^s x$. For $g = (b, P^s)$ set $b_k = (I + P^s + \cdots + P^{(k-1)s})b$.

Proposition 1 (lamp fixed powers and pair coefficient)**.**

$$F_k(g) = \begin{cases} r^{\gcd(n,ks)}, & b_k \in \operatorname{im}(I - P^{ks}), \\ 0, & \textit{otherwise}, \end{cases} \tag{436}$$

$$[m_{(1,1)}]\, \mathrm{SMG}^{\mathrm{aff}}_{L_{r,n}} = \boxed{\frac{1}{n}\sum_{d\mid n}\varphi(d) r^{n/d}}. \tag{437}$$

Proof. Iteration gives $g^k(x) = b_k + P^{ks}x$. Hence fixed points solve $(I - P^{ks})x = b_k$. A soluble fiber has the size of the kernel, namely $r^{\gcd(n,ks)}$, proving (436). For ordered pairs, translate the first entry to 0. The remaining difference is acted on only by cyclic shifts. Burnside's lemma gives the necklace count (437). ☐

For fixed $r \geq 2$, the necklace count is asymptotic to r^n/n; thus this representation has exponential degree-two growth. This does not contradict Theorem 2: the two series use different representations.

79.3 Split metacyclic induced representations

Let

$$M(m, n, u) = \langle a, b : a^m = b^n = 1,\ bab^{-1} = a^u\rangle, \qquad u^n \equiv 1 \pmod m,$$

and suppose the chosen character has a full orbit of length n. With $\zeta = e^{2\pi i/m}$, define

$$\rho(a)e_j = \zeta^{u^{-j}} e_j, \qquad \rho(b)e_j = e_{j+1}.$$

For $g = a^x b^y$, let $\mathcal{C}_y$ be the cycles of addition by y on $\mathbb{Z}/n\mathbb{Z}$, and put

$$\alpha_C(x, y) = \zeta^{x \sum_{j\in C} u^{-j}}.$$

Theorem 3 (metacyclic weighted-cycle formula)**.**

$$\boxed{\mathrm{SMG}_{M(m,n,u),\rho} = \frac{1}{mn}\sum_{x=0}^{m-1}\sum_{y=0}^{n-1}\prod_{i\geq 1}\prod_{C\in\mathcal{C}_y}(1 - \alpha_C(x, y)t_i^{|C|})^{-1}.}$$

Proof. The relations hold because shifting the diagonal weights sends u^{-j} to $u^{-(j-1)} = u\,u^{-j}$. On a cycle C of the permutation $j \mapsto j + y$, the product of the diagonal labels is $\alpha_C(x, y)$. Weighted permutation elimination gives the characteristic factor $1 - \alpha_C(x, y)t^{|C|}$. Multiplication over cycles and Reynolds averaging prove the result. ☐

If the character orbit has length $d < n$, the same proof applies to its d-dimensional induced module. A nonsplit relation $b^n = a^c$ inserts the corresponding scalar in cycles crossing that relation; it must not be silently omitted. The arithmetic quantities $\gcd(n, y)$ and $\sum_{j\in C} u^{-j} \pmod m$ prevent a group-independent asymptotic law.

79.4 General affine and Frobenius permutation actions

79.4.1 Affine groups

Let $V = \mathbb{F}_q^d$, $H \leq GL(V)$, and let $V \rtimes H$ act naturally on V. For $g = (b, h)$ set $b_r = (I + h + \cdots + h^{r-1})b$.

Theorem 4 (affine fixed powers and moments)**.**

$$F_r(b,h) = \begin{cases} q^{\dim\ker(I-h^r)}, & b_r \in \operatorname{im}(I-h^r), \\ 0, & \textit{otherwise}, \end{cases} \tag{438}$$

$$[m_{(1^s)}]\,\mathrm{SMG}_{V\rtimes H,V} = \#(H\backslash V^{s-1}). \tag{439}$$

For $H = GL_d(q)$ this specializes to

$$\boxed{[m_{(1^s)}]\,\mathrm{SMG} = \sum_{j=0}^{\min(d,s-1)} \begin{bmatrix} s-1 \\ j \end{bmatrix}_q .} \tag{440}$$

Proof. Iteration gives $g^r(x) = b_r + h^r x$, and solving the affine linear equation gives (438). Translation sends the first point of an s-tuple to zero and leaves the diagonal H-action on its $s-1$ differences, proving (439). Regard an element of V^{s-1} as a d-by-$(s-1)$ matrix. Two matrices lie in the same $GL_d(q)$ orbit under left multiplication exactly when they have the same row space. The possible row spaces are the j-subspaces of $\mathbb{F}_q^{s-1}$ with $j \le d$, proving (440). □

For fixed q and fixed s, the right side stabilizes once $d \ge s-1$. For general H_d, coefficientwise convergence is equivalent to stabilization of the H_d-orbit counts on every fixed number of vectors.

79.4.2 Finite Frobenius groups

Let $G = K \rtimes H$ be a Frobenius group in its natural action on $G/H \simeq K$, and put $N = |K|$. Define

$$E_d = \prod_i (1-t_i^d)^{-N/d}, \qquad R_d = \prod_i (1-t_i)^{-1}(1-t_i^d)^{-(N-1)/d}.$$

Theorem 5 (Frobenius cycle inventory)**.**

$$\mathrm{SMG}_{G,K} = \boxed{\frac{1}{N|H|}\left[E_1 + \sum_{1\ne k\in K} E_{\mathrm{ord}(k)} + N \sum_{1\ne h\in H} R_{\mathrm{ord}(h)}\right]}, \tag{441}$$

$$[m_{(1,1)}]\,\mathrm{SMG}_{G,K} = \boxed{1 + \frac{N-1}{|H|}}. \tag{442}$$

Proof. A nonidentity $k \in K$ is a fixed-point-free translation; if its order is d, its cycles have type $d^{N/d}$. Every element outside K lies in a unique conjugate of a complement and fixes one point. Its nontrivial powers fix the same point, so an element of order d has cycle type $1^1 d^{(N-1)/d}$. The N complements have disjoint nonidentity parts, which proves (441). After translating the first member of an ordered pair to $1 \in K$, the complement acts freely on $K \setminus \{1\}$. Its orbit count proves (442). □

With $|H|$ fixed and $N \to \infty$, the pair coefficient grows linearly. Sharper transitivity can keep degree two bounded, but higher orbit parameters still require separate control.

79.5 Unitriangular, Borel, and parabolic actions

Finite-field matrices are not complex matrices until a complex representation is chosen. Here the choices are permutation modules on $V_n = \mathbb{F}_q^n$, $V_n^\times$, projective points, and flag varieties.

Proposition 2 (vector and projective fixed powers)**.** *For $g \in GL_n(q)$,*

$$F_r(g;V_n) = q^{\dim\ker(g^r-I)}, \tag{443}$$

$$F_r(g;V_n^\times) = q^{\dim\ker(g^r-I)} - 1, \tag{444}$$

$$F_r(g;\mathbb{P}^{n-1}) = \sum_{\lambda\in\mathbb{F}_q^\times} \frac{q^{\dim\ker(g^r-\lambda I)}-1}{q-1}. \tag{445}$$

Proof. The first two statements count fixed vectors, with or without zero. A projective point is fixed exactly when its vectors form an eigenline. The nonzero vectors in distinct eigenspaces are disjoint, and every line has $q-1$ representatives, giving (445). ☐

For the upper-unitriangular and upper-triangular Borel groups, elementary row operations give

$$\#(UT_n(q)\backslash V_n^\times) = n(q-1), \qquad \#(UT_n(q)\backslash \mathbb{P}^{n-1}) = n, \tag{446}$$
$$\#(B_n(q)\backslash V_n^\times) = n, \qquad \#(B_n(q)\backslash \mathbb{P}^{n-1}) = n. \tag{447}$$

For $UT_n(q)$ a nonzero vector is classified by the position and value of its last nonzero coordinate; projectivization removes the value. The diagonal torus in $B_n(q)$ removes it before projectivization. On all of V_n, add one orbit for zero. Consequently these rank-growing actions already diverge in degree one.

Let G be a finite reductive group and let P, Q be parabolics with Weyl subgroups W_P, W_Q. Bruhat decomposition and the coset fixed-point count give

$$[m_{(1)}]\,\mathrm{SMG}_{P,G/Q} = |P\backslash G/Q| = |W_P\backslash W/W_Q|, \tag{448}$$
$$|\operatorname{Fix}_{G/Q}(p^r)| = \frac{|C_G(p^r)|\,|(p^r)^G \cap Q|}{|Q|}. \tag{449}$$

For $P = Q = B_n(q)$, (448) is $|W| = n!$. Equation (449) follows by counting the $x \in G$ for which $x^{-1}p^r x \in Q$, then dividing by $|Q|$.

79.6 Finite defining subgroups of unitary and symplectic groups

79.6.1 Class spectra and scalar selection

For $G \le U(d)$ let $\lambda_1(g), \dots, \lambda_d(g)$ be the eigenvalues of the defining matrix.

Proposition 3 (finite unitary class formula)**.**

$$\mathrm{SMG}_{G,\mathbb{C}^d} = \boxed{\sum_{[g]} \frac{1}{|C_G(g)|} \prod_{i\ge 1} \prod_{j=1}^{d} (1-\lambda_j(g)t_i)^{-1}}, \tag{450}$$
$$= \sum_{[g]} \frac{1}{|C_G(g)|} \exp\left(\sum_{r\ge 1} \frac{p_r}{r} \chi(g^r) \right). \tag{451}$$

Thus a character table determines the complete series only together with its class power maps. If G contains a scalar ζI of order s, then

$$[m_\tau]\,\mathrm{SMG}_G = 0 \quad \textit{unless} \quad s \mid |\tau|.$$

Proof. Group the average in Theorem 1 by conjugacy classes to get (450); the logarithmic determinant identity gives (451). The scalar acts on W_τ as $\zeta^{|\tau|}$, so a fixed vector requires that scalar to be one. ☐

79.6.2 General monomial theorem and the SU(3) type-C and type-D series

Let $G = D \rtimes P \le U(d)$ with D diagonal and $P \le S_d$. For a cycle C of $\pi \in P$, write $z_C = \prod_{j\in C} z_j$.

Theorem 6 (monomial class formula)**.**

$$\boxed{\mathrm{SMG}_{D\rtimes P} = \frac{1}{|D||P|} \sum_{\pi\in P} \sum_{z\in D} \prod_{i\ge 1} \prod_{C\in \mathrm{Cyc}(\pi)} (1 - z_C t_i^{|C|})^{-1}.}$$

Proof. Weighted permutation elimination gives $\det(I - tD(z)P_\pi) = \prod_C (1 - z_C t^{|C|})$. Theorem 1 then applies. This includes the determinant-one monomial type-C and type-D families in $SU(3)$. ☐

If D_m is the m-torsion in a fixed compact diagonal torus T normalized by P, then in total degree at most D only characters of bounded exponent occur. For all sufficiently large m, such a character is trivial on D_m exactly when it is trivial on T. Hence

$$\mathrm{SMG}_{D_m \rtimes P} \longrightarrow \mathrm{SMG}_{T \rtimes P} \quad \text{coefficientwise.}$$

For the full coordinate torus and the determinant-one torus, $m > D$ suffices.

79.6.3 Two exceptional SU(3) class formulas

Write

$$\mathcal{D}(\alpha, \beta, \gamma) = \prod_{i \geq 1} \frac{1}{(1 - \alpha t_i)(1 - \beta t_i)(1 - \gamma t_i)}.$$

For the 3-dimensional rotation representation of $\Sigma(60) \cong A_5$,

$$\begin{aligned} \mathrm{SMG}_{\Sigma(60)} = \frac{1}{60}\big[&\mathcal{D}(1,1,1) + 15\mathcal{D}(1,-1,-1) + 20\mathcal{D}(1,\omega,\omega^2) \\ &+ 12\mathcal{D}(1,\zeta_5,\zeta_5^4) + 12\mathcal{D}(1,\zeta_5^2,\zeta_5^3)\big]. \end{aligned} \tag{452}$$

For either conjugate 3-dimensional representation of $\Sigma(168) \cong PSL_2(7)$,

$$\begin{aligned} \mathrm{SMG}_{\Sigma(168)} = \frac{1}{168}\big[&\mathcal{D}(1,1,1) + 21\mathcal{D}(1,-1,-1) + 56\mathcal{D}(1,\omega,\omega^2) + 42\mathcal{D}(1,i,-i) \\ &+ 24\mathcal{D}(\zeta_7,\zeta_7^2,\zeta_7^4) + 24\mathcal{D}(\zeta_7^3,\zeta_7^5,\zeta_7^6)\big]. \end{aligned} \tag{453}$$

Both identities are immediate from Proposition 3 and the displayed class spectra. Isolated exceptional groups have no intrinsic rank asymptotic. A sequence whose scalar-center order tends to infinity has series tending coefficientwise to 1 by (3).

79.6.4 SU(4), Sp(4), and SO(4)

For $G \leq SU(4)$, Proposition 3 with its defining character χ_4 is the complete answer. For $G \leq Sp(4)$, the spectrum is $\{\lambda, \lambda^{-1}, \mu, \mu^{-1}\}$ and hence

$$\mathrm{SMG}_G = \sum_{[g]} \frac{1}{|C_G(g)|} \prod_i \frac{1}{(1 - \lambda_g t_i)(1 - \lambda_g^{-1} t_i)(1 - \mu_g t_i)(1 - \mu_g^{-1} t_i)}.$$

If $-I \in G$, all odd-total-degree coefficients vanish.

For $SO(4)$ use $\mathrm{Spin}(4) \simeq SU(2)_L \times SU(2)_R$. If $\widetilde{G}$ is the inverse image of G, the complexified vector representation is the tensor product of the two defining planes, so

$$\boxed{\mathrm{SMG}_{G,\mathbb{C}^4} = \frac{1}{|\widetilde{G}|} \sum_{(a,b) \in \widetilde{G}} \exp\left(\sum_{r \geq 1} \frac{p_r}{r} \chi_2(a^r) \chi_2(b^r) \right)}.$$

The kernel element $(-I, -I)$ acts trivially, so averaging over the spin cover equals averaging over the represented image.

79.7 Symplectic-reflection wreath products

Let $K \leq SL_2(\mathbb{C})$ act on its defining symplectic plane E, and let $G_n = K \wr S_n$ act on $E^{\oplus n}$. Define

$$A_\ell(\mathbf{t}) = \frac{1}{|K|} \sum_{k \in K} \prod_{i \geq 1} \det(I - t_i^\ell k)^{-1}.$$

Theorem 7 (wreath exponential and stable law)**.**

$$\sum_{n\geq 0} u^n \operatorname{SMG}_{K\wr S_n, E^{\oplus n}} = \boxed{\exp\left(\sum_{\ell\geq 1} \frac{u^\ell}{\ell} A_\ell\right)}, \tag{454}$$

$$\lim_{n\to\infty} \operatorname{SMG}_{K\wr S_n, E^{\oplus n}} = \boxed{\exp\left(\sum_{\ell\geq 1} \frac{A_\ell - 1}{\ell}\right) = \operatorname{Exp}_\sigma(\operatorname{SMG}_K - 1)}. \tag{455}$$

The limit is coefficientwise.

Proof. For a permutation cycle C of length ℓ, multiply its K-labels around the cycle to obtain k_C. Block elimination gives $\det(I - tg_C) = \det(I - t^\ell k_C)$. The cycle product is uniform in K, and the exponential formula for labelled permutation cycles gives (454). Split $A_\ell = 1 + (A_\ell - 1)$ to write the right side as

$$\frac{1}{1-u} \exp\left(\sum_{\ell\geq 1} \frac{u^\ell (A_\ell - 1)}{\ell}\right).$$

In a fixed positive $\mathbf{t}$-degree only finitely many ℓ occur, and the second factor has bounded u-degree after that coefficient is extracted. Multiplication by $(1-u)^{-1}$ makes the coefficients eventually constant, proving (455). ☐

This is a genuine marked compound-Poisson law, unlike the rare-sector divergence of the lamplighter and fixed-complement Frobenius examples.

79.8 Verification strategy and results

79.8.1 Independent computational routes

The marimo notebook imports one pure verification module. Every check uses at least two of the following routes.

1. **Direct represented matrices.** Dense complex matrices are used for the monomial and defining representations. A permutation matrix is stored losslessly by its unique nonzero row in each column.
2. **Direct target quantity.** For a matrix A, complete homogeneous characters are computed from its eigenvalues, and
$$\frac{1}{|G|} \sum_g \prod_j h_{\tau_j}(\operatorname{Spec} \rho(g))$$
is rounded only after its imaginary part and distance to the nearest integer are checked. Permutation coefficients are computed exactly from cycle factors. Ordered-pair and difference-tuple orbits are also enumerated literally where applicable.
3. **Structural formula.** Lamplighter and metacyclic cases use root-filtered or label-product cycles; affine cases use image and kernel tests; Frobenius cases use the predicted full cycle inventory; flag actions use centralizers and class intersections; unitary cases use class spectra; and wreath products use the coefficient recurrence obtained by differentiating (454).
4. **Analytic spot check.** The full determinant products are evaluated at $(t_1, t_2) = (0.031, -0.047)$. This checks all degrees at once and is separate from the coefficient table.

79.8.2 Constructed groups and reasonable dimensions

Representation	dim.	group/cover	fixed checks	coeff. checks	result
$L_{2,3}, L_{3,2}, L_{2,4}$, monomial	2–4	18–64	–	15	**PASS**

Representation	dim.	group/cover	fixed checks	coeff. checks	result
$L_{2,3}, L_{2,4}, L_{3,3}$, affine lamps	8–27	24–81	823	–	**PASS**
$M(5,2,4), M(7,3,2)$, induced	2–3	$10, 21$	relations	10	**PASS**
$AGL(2,2), AGL(3,2)$, natural	$4, 8$	$24, 1344$	6634	5	**PASS**
$\mathbb{F}_5 \rtimes \mathbb{F}_5^\times$, D_{14}	$5, 7$	$20, 14$	cycles	10	**PASS**
$UT_3(2), B_2(3)$, vectors/projective	7–8	$8, 12$	123	–	**PASS**
$B_3(2)$ on complete flags	21	8	class powers	–	**PASS**
$A_5, PSL_2(7), \Delta(12)$ in $SU(3)$	3	$60, 168, 12$	relations	15	**PASS**
A_5 in $SU(4)$	4	60	classes	5	**PASS**
$Q_8 \wr S_2$ in $Sp(4)$	4	128	center	5	**PASS**
$(Q_8 \times Q_8)/\{\pm(I, I)\}$ in $SO(4)$	4	64/32	power traces	5	**PASS**
Total			7,580	70	**PASS**

For the $SU(3)$ exceptional cases the notebook constructs actual generators. For $PSL_2(7)$, with $\eta = e^{2\pi i/7}$, it uses

$$A = \operatorname{diag}(\eta, \eta^2, \eta^4), \qquad B = \frac{i}{\sqrt{7}} \begin{pmatrix} \eta^2 - \eta^5 & \eta - \eta^6 & \eta^4 - \eta^3 \\ \eta - \eta^6 & \eta^4 - \eta^3 & \eta^2 - \eta^5 \\ \eta^4 - \eta^3 & \eta^2 - \eta^5 & \eta - \eta^6 \end{pmatrix}.$$

Closure produces 168 distinct unitary determinant-one matrices with order inventory $1, 21, 56, 42, 48$ for element orders $1, 2, 3, 4, 7$. The maximum unitarity and determinant errors are below 5.4×10^{-14}.

The $SO(4)$ test constructs all 64 matrices $a \otimes b$ with $a, b \in Q_8 \leq SU(2)$. Deduplication leaves 32 represented matrices, exactly detecting the kernel $\{(I, I), (-I, -I)\}$. For powers $1 \leq r \leq 4$ it verifies

$$\operatorname{Tr}((a \otimes b)^r) = \operatorname{Tr}(a^r) \operatorname{Tr}(b^r)$$

to machine precision, and the cover and image Molien values agree within 6.7×10^{-16}.

79.8.3 Representative coefficient values

Representation	m_1	m_2	m_{11}	m_3	m_{21}
$L_{2,3}$ monomial	0	1	1	0	0
$L_{3,2}$ monomial	0	0	0	1	1
$M(5,2,4)$ induced	0	1	1	0	0
$M(7,3,2)$ induced	0	0	0	1	1
$\mathbb{F}_5 \rtimes \mathbb{F}_5^\times$ natural	1	2	2	3	5
D_{14} natural	1	4	4	8	16
$A_5 \leq SU(3)$	0	1	1	0	0
$PSL_2(7) \leq SU(3)$	0	0	0	0	0
$\Delta(12) \leq SU(3)$	0	1	1	1	1
$A_5 \leq SU(4)$	0	1	1	1	1
$Q_8 \wr S_2 \leq Sp(4)$	0	0	1	0	0
$(Q_8 \times Q_8)/\pm \leq SO(4)$	0	1	1	0	0

The affine-lamp pair coefficients are $4, 6, 11$ for $(r, n) = (2, 3), (2, 4), (3, 3)$, matching literal pair orbits and the necklace formula. For $AGL(2,2)$, the moments $[m_{(1^s)}]$ for $s = 2, 3, 4$ are $2, 5, 15$; for $AGL(3,2)$ the tested values for $s = 2, 3$ are $2, 5$. In each case direct difference-tuple orbits, affine Burnside averages, and Gaussian-binomial sums agree.

79.9 Asymptotic summary and scope

Family and representation	large-parameter behavior
$C_r^n \rtimes C_n$, natural monomial	$[m_{(r,r)}] \gg n$; no raw coefficientwise limit.
$C_r^n \rtimes C_n$, affine lamps	$[m_{11}] \sim r^n/n$; no raw coefficientwise limit.
Split metacyclic induced modules	arithmetic and subsequence dependent.
$V_d \rtimes GL_d(q)$, q fixed	every fixed (1^s) moment stabilizes once $d \geq s-1$.
Frobenius groups with fixed complement	$[m_{11}] = 1 + (\lvert K\rvert - 1)/\lvert H\rvert$ grows linearly.
$UT_n(q), B_n(q)$ on nonzero/projective points	$[m_1]$ grows linearly in n.
$B_n(q)$ on complete flags	$[m_1] = n!$.
$SU(3)$ diagonal torsion series	coefficientwise torus-normalizer limit as torsion becomes dense.
Exceptional fixed-dimensional unitary groups	isolated exact class sums; no intrinsic rank limit.
$K \wr S_n \leq Sp(2n)$	$\mathrm{Exp}_\sigma(\mathrm{SMG}_K - 1)$, a genuine coefficientwise limit.

A full crystallographic group is generally infinite and noncompact; it has no uniform probability measure and therefore no σ-MGF of the present kind. One must first choose a finite quotient, a compact quotient action, or a nonuniform probability law. A finite linear point group preserving a complex lattice is covered by Proposition 3, or by Theorem 6 when it is monomial.

Reproducibility. Run `python -m new_dists.solvable_finite_subgroup_sigma_mgfs.verification` for the noninteractive audit, or open `marimo_notebooks/solvable_finite.py` with marimo for the group-organized laboratory. The source keeps the direct matrix route and each family-specific formula in separate functions so that agreement cannot result from comparing an expression with itself.

80 The H_3 reflection group

Status: Formula exact; reconstruction diagnostic. The Molien formula is exact once the represented group is specified. Floating-point closure and rank tests do not certify the algebraic reflection group; an exact $\mathbb{Q}(\sqrt{5})$ construction or certified character table is required for a fully exact computational certificate.

Outcome. Explicit rotation matrices verify the proposed A_4, S_4, and A_5 angle formulas in 45 multivariate coefficients each. The 120 matrices of $W(H_3)$ were reconstructed from the icosahedron; all 139 coefficients through total degree ten agree with the signed class-spectrum formula, and the one-variable series agrees with $((1-t^2)(1-t^6)(1-t^{10}))^{-1}$.

80.1 Imported target

The target is $\mathbb{E}[\mathrm{Exp}_\sigma(X_{G,V}h_1)]$ from Tools 1 and 2. The calculation below does not reprove those identities: it constructs the H_3 matrices independently and compares their invariant coefficients with a weighted spectral computation.

80.2 Floating-point reconstruction diagnostic for $W(H_3)$

Let $\varphi = (1+\sqrt{5})/2$. Use the twelve vertices

$$\Omega = \{(0, \pm 1, \pm\varphi), (\pm 1, \pm\varphi, 0), (\pm\varphi, 0, \pm 1)\}. \tag{3}$$

Choose an ordered, linearly independent base triple B of vertices. For every ordered target triple T having the same Gram matrix, form the unique linear map $R = TB^{-1}$. Retain R when

$$R^{\mathsf{T}}R = I, \qquad \det R = 1, \qquad R\Omega = \Omega.$$

This produces 60 distinct rotations. Adjoining their negatives produces 120 orthogonal transformations. This is the full symmetry group of the icosahedron, which is the real reflection group $W(H_3)$.

This "vertex-frame reconstruction" is useful when no computer algebra class table is available, but the present implementation uses floating-point coordinates and therefore supplies a diagnostic rather than an exact group certificate. The implementation reports its equality tolerance, the minimum separation between reconstructed matrices, the maximum closure-matching distance, and the condition number of the chosen base triple; it then tests every one of the 120^2 products against the reconstructed set. No closure claim here is described as exact.

The regenerated diagnostic reports equality tolerance 5×10^{-10}, minimum separation 1.6625 between distinct reconstructed matrices, maximum closure-matching distance 3.77×10^{-16}, base-triple condition number 1.852, and maximum orthogonality residual 4.16×10^{-16}.

80.3 Polyhedral rotation groups

For a three-dimensional rotation through angle θ, define

$$\Phi_\theta(\mathbf{t}) = \prod_{a \ge 1} \left((1-t_a)(1-2\cos\theta\, t_a + t_a^2)\right)^{-1}.$$

The spectrum is $(1, e^{i\theta}, e^{-i\theta})$, so direct substitution of the rotation counts gives

$$\begin{aligned}
\mathrm{SMG}_{A_4,\mathrm{rot}} &= \frac{1}{12}(\Phi_0 + 3\Phi_\pi + 8\Phi_{2\pi/3}),\\
\mathrm{SMG}_{S_4,\mathrm{rot}} &= \frac{1}{24}(\Phi_0 + 6\Phi_{\pi/2} + 8\Phi_{2\pi/3} + 9\Phi_\pi),\\
\mathrm{SMG}_{A_5,\mathrm{rot}} &= \frac{1}{60}(\Phi_0 + 15\Phi_\pi + 20\Phi_{2\pi/3} + 12\Phi_{2\pi/5} + 12\Phi_{4\pi/5}).
\end{aligned}$$

The direct route independently constructs the 12 tetrahedral matrices as the determinant-one signed permutation matrices preserving a fixed tetrahedron, the 24 octahedral matrices as all determinant-one signed permutation matrices, and the 60 icosahedral matrices by the vertex-frame construction above. For each group, all 45 coefficients indexed by partitions through total degree seven agree with the displayed angle formula.

group	order	coefficient checks	result
A_4 tetrahedral rotations	12	45	**PASS**
S_4 octahedral rotations	24	45	**PASS**
A_5 icosahedral rotations	60	45	**PASS**

80.4 The explicit H_3 spectral formula

Write

$$A_z(\mathbf{t}) = \prod_{a\geq 1}(1-zt_a)^{-1}, \qquad \omega = e^{2\pi i/3}, \qquad \xi = e^{2\pi i/5}.$$

The orientation-preserving subgroup has the following spectra and class sizes:

size	eigenvalues	geometric type
1	$(1,1,1)$	identity
15	$(1,-1,-1)$	half turns
20	$(1,\omega,\omega^2)$	third turns
12	$(1,\xi,\xi^{-1})$	first fifth-turn class
12	$(1,\xi^2,\xi^{-2})$	second fifth-turn class

Multiplication by central inversion negates all three eigenvalues and supplies the other five classes. Grouping (1) by these ten spectra gives

$$\begin{aligned}\mathrm{SMG}_{H_3} = \frac{1}{120}\Big[&A_1^3 + A_{-1}^3 + 15(A_1A_{-1}^2 + A_{-1}A_1^2)\\ &+ 20(A_1A_\omega A_{\omega^2} + A_{-1}A_{-\omega}A_{-\omega^2})\\ &+ 12\sum_{j=1}^{2}(A_1A_{\xi^j}A_{\xi^{-j}} + A_{-1}A_{-\xi^j}A_{-\xi^{-j}})\Big].\end{aligned} \tag{4}$$

Equation (4) is exact in all multidegrees. It is more informative than the ordinary invariant degrees because mixed coefficients depend on the entire class spectrum.

80.4.1 Ordinary Molien specialization

Setting only $t_1 = t$ nonzero gives

$$\mathrm{SMG}_{H_3}(t,0,\ldots) = \frac{1}{(1-t^2)(1-t^6)(1-t^{10})}. \tag{5}$$

The right side has degree-d coefficient equal to the number of nonnegative solutions of $2a+6b+10c=d$. This elementary count is used as a third route, independent of both explicit matrix spectra and the multivariate class calculation.

80.5 Verification protocol

1. Reconstruct the 60 rotations from (3), adjoin negatives, and test order, orthogonality, signed determinants, and closure.
2. For every partition τ through total degree ten, compute the direct matrix average in (2) over all 120 matrices.

3. Compute the same coefficient using only the ten weighted spectra in (4). The explicit matrices are not passed to this route.
4. For every $0 \leq d \leq 10$, compare the direct one-variable average with the solution count for $2a + 6b + 10c = d$.
5. Construct the six-dimensional $\mathrm{Sym}^2(V)$ matrices and average them to a Reynolds projector.

80.6 Results

Check	observed	status
number of reconstructed matrices	120	**PASS**
closure products tested	14,400	**PASS**
maximum $\|g^{\mathsf{T}}g - I\|_F$	4.16×10^{-16}	**PASS**
signed determinant condition	all matrices	**PASS**
partitions through degree 10	139 coefficients	**PASS**
ordinary Molien degrees 0 through 10	11 coefficients	**PASS**

Some matched multivariate coefficients are

τ	(1)	(2)	(3)	(4)	$(2,1)$	$(2,2)$	$(3,2)$
dimension	0	1	0	1	0	2	0

The coefficients of (5) for $d = 0, \ldots, 10$ are

$$1, 0, 1, 0, 1, 0, 2, 0, 2, 0, 3,$$

and agree termwise with direct matrix averaging.

For $\tau = (2)$, the target $\mathrm{Sym}^2(V)$ has dimension six. The explicit Reynolds projector has

$$\operatorname{rank} P = 1, \qquad \operatorname{tr} P = 0.9999999999999998, \qquad \|P^2 - P\|_F = 1.26 \times 10^{-16}.$$

The invariant is the expected quadratic form.

80.7 Reproducibility and limitations

Run from the repository root:

```
.venv/bin/python -m for_this_guy.matrix_group_formula_verification.h3_reflection.verification
```

The companion marimo notebook exposes the maximum tested degree and displays all coefficient rows.

The group reconstruction uses double-precision values of φ, not an exact quadratic-number field. This does not affect the theoretical proof of (4), but it makes the group diagnostics numerical. To control this, vertices and matrices are canonicalized to nine decimal places, every product is tested, and all averaged characters must lie within 5×10^{-7} of an integer. The actual orthogonality and projector residuals are around 10^{-16}.

The finite checks are diagnostic evidence; the all-degree validity of (4) comes from the master identity and the complete class spectra. The computation specifically verifies that the class weights, signs, fifth-root classes, and matrix convention have been transcribed consistently.

81 The M_{24} permutation representation

Status: Conditional on certified class data. The universal class/power or frame-shape formula is exact. Concrete numerical tables in this section depend on external class sizes, power maps, character values, or frame shapes and are not independently derived by the Molien argument.

Abstract

We prove the cycle-shape formula for the sigma–MGF of the natural 24-dimensional permutation representation of M_{24}. Two published ATLAS permutations are converted into 24×24 matrices. Independent generator-orbit enumeration is compared with a full conjugacy-class sum. Agreement holds in every tested case. The computation also sharpens the 5-transitivity statement: equality with S_{24} holds through total degree five, while $[m_{(6)}]$ is 12 rather than 11 and $[m_{(1^6)}]$ is 204 rather than 203.

81.1 Exact formula

Let g have cycle shape $\prod_{r\geq 1} r^{c_r(g)}$. On an r-cycle the eigenvalues are all rth roots of unity, so

$$\det(I - tP_r) = \prod_{j=0}^{r-1}(1 - t\zeta_r^j) = 1 - t^r.$$

Multiplying over cycles and averaging the generalized Molien integrand gives

$$\boxed{\mathrm{SMG}_{M_{24},\mathrm{perm}}(\mathbf{t}) = \sum_C \frac{|C|}{|M_{24}|} \prod_{i\geq 1} \prod_{r\geq 1} (1 - t_i^r)^{-c_r(C)}.} \tag{456}$$

This is an all-degree identity, not an asymptotic expansion.

The coefficient of m_τ is

$$\dim \left(\bigotimes_j \mathrm{Sym}^{\tau_j} \mathbb{C}^{24} \right)^{M_{24}}.$$

Each symmetric power has a basis indexed by multisets. A permutation-group fixed vector is constant on each orbit of the resulting basis. Therefore

$$[m_\tau]\, \mathrm{SMG}_{M_{24},\mathrm{perm}} = \# \left(M_{24} \backslash \prod_j \mathrm{MSet}_{\tau_j}(\{1, \ldots, 24\}) \right). \tag{457}$$

Equations (456) and (457) are mathematically equivalent, but they produce genuinely different algorithms: weighted class expansion versus graph traversal under generators. Their agreement is the central cross-check.

81.2 Why equality with S_{24} stops after degree five

An S_{24} orbit of a product of multisets is determined by the multiplicity profile carried by its support points. If $|\tau| \leq 5$, that support has at most five points. The 5-transitivity of M_{24} maps any ordered listing of one support to the corresponding listing of another, so every S_{24} orbit remains a single M_{24} orbit. Consequently

$$\mathrm{SMG}_{M_{24},\mathrm{perm}} \equiv \mathrm{SMG}_{S_{24},\mathrm{perm}} \pmod{\deg > 5}.$$

For one symmetric power, the S_{24} coefficient is the partition number $p(k)$; for (1^k) it is the Bell number B_k (provided $k \leq 24$). The claim is not extended above degree five. Indeed, generator traversal finds two orbits on six-element subsets. This splits the all-distinct equality pattern and creates the first extra orbit.

81.3 Constructed matrices and checks

The verifier uses the ATLAS standard generators

$$\begin{aligned} a &= (1,4)(2,7)(3,17)(5,13)(6,9)(8,15)(10,19)(11,18) \\ &\qquad (12,21)(14,16)(20,24)(22,23), \\ b &= (1,4,6)(2,21,14)(3,9,15)(5,18,10)(13,17,16)(19,24,23). \end{aligned}$$

They are converted to permutation matrices by $P_g e_j = e_{g(j)}$. The code asserts the standard-generator fingerprints

$$o(a) = 2, \quad o(b) = 3, \quad o(ab) = 23, \quad o(ababababbababbabb) = 4,$$

and checks that the aggregated class sizes sum to $|M_{24}| = 244{,}823{,}040$.

For a class shape, the coefficient of t^k in $\prod_r (1-t^r)^{-c_r}$ is obtained with exact integer convolution. In parallel, breadth-first traversal acts with $a^{\pm 1}, b^{\pm 1}$ on every multiset through degree six and every ordered tuple through degree four. At degree six the tensor calculation uses equality patterns together with the directly enumerated two orbits of six-element subsets.

coefficient	direct orbits	class formula	S_{24}
$m_{(1)}$	1	1	1
$m_{(2)}$	2	2	2
$m_{(3)}$	3	3	3
$m_{(4)}$	5	5	5
$m_{(5)}$	7	7	7
$m_{(6)}$	12	12	11
$m_{(1^2)}$	2	2	2
$m_{(1^3)}$	5	5	5
$m_{(1^4)}$	15	15	15
$m_{(1^6)}$	204	204	203

The degree-one invariant is the all-ones line. The degree-six discrepancies are a useful regression test: a program that silently substitutes 5-transitivity beyond its valid range returns the wrong values 11 and 203.

81.4 Reproduction and provenance

Run

```
.venv/bin/python for_this_guy/sporadic_matrix_groups/
  m24_permutation/verification.py
.venv/bin/marimo edit marimo_notebooks/m24_permutation.py
```

after joining wrapped lines. The full cycle-shape/class-size table is stored as exact integers in `sporadic_matrix_groups/common.py`; classes with the same permutation cycle shape are aggregated because (456) cannot distinguish them.

class	aggregated size	permutation cycle shape
$1A$	1	1^{24}
$2A$	11,385	$1^8 2^8$
$2B$	31,878	2^{12}
$3A$	226,688	$1^6 3^6$
$3B$	485,760	3^8
$4A$	637,560	$2^4 4^4$
$4B$	1,912,680	$1^4 2^2 4^4$
$4C$	2,550,240	4^6
$5A$	4,080,384	$1^4 5^4$
$6A$	10,200,960	$1^2 2^2 3^2 6^2$
$6B$	10,200,960	6^4
$7AB$	11,658,240	$1^3 7^3$
$8A$	15,301,440	$1^2 248^2$
$10A$	12,241,152	$2^2 10^2$
$11A$	22,256,640	$1^2 11^2$
$12A$	20,401,920	24612
$12B$	20,401,920	12^2
$14AB$	34,974,720	12714
$15AB$	32,643,072	13515
$21AB$	23,316,480	321
$23AB$	21,288,960	123

The sizes sum to 244,823,040. Source: Online ATLAS of Finite Group Representations, retrieved 18 July 2026. The SHA-256 of the canonical serialized table is

`103ba576555f01bea74df73f8e2f788b02bee4e8b20df3ed198eb005d16de044`.

The SHA-256 of the software data file is

`3fbc8e5cde4ac5670696ecc1c5f692a2afb2065469a83ce486aa8b28d046d579`.

Generator-orbit enumeration is independent of the class table. The class-sum calculation is an exact cross-check conditional on the externally supplied ATLAS data; the complete table and checksums make that input auditable from this PDF.

Generator data and the group order are from the online ATLAS of Finite Group Representations, https://brauer.maths.qmul.ac.uk/Atlas/spor/M24/. The calculation verifies the stated representation and coefficients without enumerating all 244 million group elements.

82 The Co_0 Leech-space representation

Status: Conditional on certified class data. The universal class/power or frame-shape formula is exact. Concrete numerical tables in this section depend on external class sizes, power maps, character values, or frame shapes and are not independently derived by the Molien argument.

Abstract

We prove the frame-shape class formula for the natural 24-dimensional complexified Leech-lattice representation of Co_0 and the vanishing of all odd total-degree coefficients. Since direct enumeration of Co_0 is not a reasonable notebook computation, we test the determinant mechanism at the correct dimension on the explicit signed-coordinate subgroup $\{\pm P_g : g \in M_{24}\}$. Direct invariant counts and exact class expansion agree through degree six. The distinction between this stress test and a full Co_0 class-table evaluation is kept explicit.

82.1 Frame-shape formula

Let $V = \Lambda_{\mathrm{Leech}} \otimes_{\mathbb{Z}} \mathbb{C}$. If a conjugacy-class representative $g \in Co_0$ has frame shape $\pi_g = \prod_{r\geq 1} r^{k_r(g)}$, by definition

$$\det(xI - g) = \prod_{r\geq 1} (x^r - 1)^{k_r(g)}, \qquad \sum_r r k_r(g) = 24.$$

Substitute $x = t^{-1}$ and multiply by t^{24} to obtain

$$\det(I - tg) = \prod_{r\geq 1} (1 - t^r)^{k_r(g)}.$$

The generalized Molien formula, grouped by conjugacy class, is therefore

$$\boxed{\mathrm{SMG}_{Co_0,V}(\mathbf{t}) = \sum_{C\subseteq Co_0} \frac{|C|}{|Co_0|} \prod_{i,r} (1 - t_i^r)^{-k_r(C)}.} \tag{458}$$

Negative frame exponents cause no difficulty: each class contribution is a rational function whose product is the characteristic determinant.

82.2 Central parity theorem

The central element $-I$ belongs to Co_0. On $\mathrm{Sym}^\tau V = \bigotimes_j \mathrm{Sym}^{\tau_j} V$ it acts by $(-1)^{|\tau|}$. A fixed vector in odd total degree would satisfy $v = -v$, hence vanish. Thus

$$\boxed{[m_\tau]\,\mathrm{SMG}_{Co_0,V} = 0 \quad \text{whenever } |\tau| \text{ is odd.}} \tag{459}$$

In particular $[m_{(1)}] = 0$, unlike the M_{24} permutation representation, whose all-ones vector is fixed. The two 24-dimensional series therefore differ in their first positive degree.

This proof is representation theoretic and does not require any frame-shape table. As a data-integrity check on a full table, the class contributions in every odd degree must cancel exactly.

82.3 A tractable subgroup at the true dimension

In the standard coordinate construction of the Leech lattice, its monomial automorphism group contains M_{24} coordinate permutations and the central sign. We use

$$H = \{\pm P_g : g \in M_{24}\} \leq Co_0.$$

The notebook explicitly constructs three 24×24 integer orthogonal generators $P_a, P_b, -I$, using the ATLAS permutations a, b documented in the companion M_{24} note. This is large enough to exercise the actual ambient dimension while retaining an exact aggregated class computation.

If g has cycle counts $c_r(g)$, then

$$\det(I - tP_g) = \prod_r (1 - t^r)^{c_r(g)}, \qquad \det(I + tP_g) = \prod_r (1 - (-t)^r)^{c_r(g)}.$$

Averaging the two signs projects onto even total degree. Hence

$$\mathrm{SMG}_H(t) = \frac{1}{2|M_{24}|} \sum_{g \in M_{24}} \left\{ \prod_r (1 - t^r)^{-c_r(g)} + \prod_r (1 - (-t)^r)^{-c_r(g)} \right\}. \tag{460}$$

82.4 Independent verification

For even k, $-I$ acts trivially on $\mathrm{Sym}^k V$, so the H-fixed space has a basis indexed by M_{24} orbits of degree-k multisets. For odd k, (459) makes the dimension zero. This gives a direct generator-orbit algorithm independent of the class expansion (460). Exact results are:

k	direct invariant dimension	signed class formula	parity
1	0	0	odd
2	2	2	even
3	0	0	odd
4	5	5	even
5	0	0	odd
6	12	12	even

The degree-six value also exercises the first point at which the underlying M_{24} multiset orbits differ from those of S_{24}.

82.5 What this does and does not verify

The proof of (458) is exact for full Co_0, and the proof of (459) establishes every odd-degree zero for full Co_0. The matrix sweep validates determinant expansion, sign cancellation, and direct invariant counting in dimension 24 on a genuine subgroup. It does not claim that the even coefficients $2, 5, 12$ are full-Co_0 coefficients; fixed spaces can shrink when the acting group grows. A full numerical evaluation of (458) requires the complete Co_0 frame-shape and class-size table, which is deliberately not reconstructed from partial data.

Run

```
.venv/bin/python for_this_guy/sporadic_matrix_groups/
  co0_leech/verification.py
.venv/bin/marimo edit marimo_notebooks/co0_leech.py
```

after joining wrapped lines. For background on the Leech representation and Conway fixed lattices, see https://arxiv.org/abs/1505.06420.

83 Sporadic groups and represented covers

Status: Conditional on certified class data. The universal class/power or frame-shape formula is exact. Concrete numerical tables in this section depend on external class sizes, power maps, character values, or frame shapes and are not independently derived by the Molien argument.

Canonical testing artifacts.
Exact backend: `lambda_distributions/dists/sporadic_group_sigma_mgfs/verification.py`.
Regression: `lambda_distributions/dists/sporadic_group_sigma_mgfs/test_verification.py`.
Marimo: `marimo_notebooks/sporadic.py`.

83.1 A representation is part of the data

A sporadic abstract group does not by itself determine a sigma–MGF. Fix a finite-dimensional complex representation

$$\rho : G \longrightarrow GL(V), \qquad \chi(g) = \operatorname{Tr} \rho(g).$$

For formal variables $\mathbf{t} = (t_1, t_2, \ldots)$ and power sums $p_r = \sum_i t_i^r$, define

$$\mathrm{SMG}_{G,V}(\mathbf{t}) = \frac{1}{|G|} \sum_{g \in G} \prod_{i \geq 1} \det(I - t_i \rho(g))^{-1}. \tag{461}$$

Different representations of the same group generally give different functions. This note groups together two representations arising from the same M_{11} action: the full permutation module and its deleted constituent.

83.2 Universal class-power theorem

Let $x_1, \ldots, x_s$ represent the conjugacy classes of G, and put $z_a = |C_G(x_a)|$. Then

$$\boxed{\mathrm{SMG}_{G,V}(\mathbf{t}) = \sum_{a=1}^{s} \frac{1}{z_a} \exp\left(\sum_{r \geq 1} \frac{p_r}{r} \chi(x_a^r) \right).} \tag{462}$$

For a partition $\tau = (\tau_1, \ldots, \tau_\ell)$, with $m_\tau = t_1^{\tau_1} \cdots t_\ell^{\tau_\ell}$ before symmetrization,

$$\boxed{[m_\tau]\, \mathrm{SMG}_{G,V} = \dim \left(\bigotimes_{j=1}^{\ell} \operatorname{Sym}^{\tau_j} V \right)^G.} \tag{463}$$

Thus class centralizers, class power maps, and one ordinary character suffice to determine the complete sigma–MGF.

Source of the universal identities

Equation (463) is Tool 1, and the determinant form underlying (462) is Tool 2. The displayed class-power form is obtained only by the standard logarithmic determinant change of coordinates and grouping the imported average by conjugacy class. Thus the sporadic input is precisely the certified class sizes, power maps, and character values.

83.3 The M_{11} matrix group

The online ATLAS gives the following standard generators in the natural 11-point action:

$$a = (2, 10)(4, 11)(5, 7)(8, 9),$$
$$b = (1, 4, 3, 8)(2, 5, 6, 9).$$

The verifier closes these permutations under multiplication and inversion; it obtains exactly 7,920 elements. With $P_g e_j = e_{g(j)}$, this constructs all 11×11 zero-one matrices P_g rather than merely storing abstract class data. The standard-generator fingerprint is

$$o(a) = 2, \qquad o(b) = 4, \qquad o(ab) = 11, \qquad o(ababbabbb) = 5.$$

For the deleted representation, let

$$W = \{x \in \mathbb{C}^{11} : x_1 + \cdots + x_{11} = 0\}, \qquad f_j = e_j - e_{11} \quad (1 \le j \le 10).$$

The code constructs the restriction $D_g = P_g|_W$ in this basis. Explicitly,

$$(D_g)_{ij} = \delta_{i,g(j)} - \delta_{i,g(11)} \quad (1 \le i, j \le 10), \tag{464}$$

where a delta is zero if its second index is 11. Every entry is integral. All 7,920 matrices are distinct in each representation, and sampled products satisfy $P_g P_h = P_{gh}$ and $D_g D_h = D_{gh}$ exactly.

83.4 Exact formulas for the two M_{11} representations

If $\lambda = 1^{c_1} 2^{c_2} \cdots$ is a permutation cycle shape, set

$$Z_\lambda(\mathbf{t}) = \prod_{i \ge 1} \prod_{r \ge 1} (1 - t_i^r)^{-c_r}, \qquad \widetilde{Z}_\lambda(\mathbf{t}) = \prod_{i \ge 1} (1 - t_i) Z_\lambda(\mathbf{t}).$$

The second identity follows from $\mathbb{C}^{11} \cong \mathbf{1} \oplus W$. Direct enumeration gives the eight aggregated spectra below; the two order-eight classes and two order-eleven classes may be combined because their spectra in these representations agree.

class row	size	centralizer	cycle shape
$1A$	1	7,920	1^{11}
$2A$	165	48	$1^3 2^4$
$3A$	440	18	$1^2 3^3$
$4A$	990	8	$1^3 4^2$
$5A$	1,584	5	$1\,5^2$
$6A$	1,320	6	$2\,3\,6$
$8AB$	1,980	8 in each class	$1\,2\,8$
$11AB$	1,440	11 in each class	11

Consequently the natural permutation formula is

$$\boxed{\begin{aligned} \mathrm{SMG}_{M_{11},\mathbb{C}^{11}} = \frac{1}{7920}\big(&Z_{1^{11}} + 165 Z_{1^3 2^4} + 440 Z_{1^2 3^3} + 990 Z_{1^3 4^2} \\ &+ 1584 Z_{1\,5^2} + 1320 Z_{2\,3\,6} + 1980 Z_{1\,2\,8} + 1440 Z_{11}\big). \end{aligned}} \tag{465}$$

The deleted formula is the same weighted sum with every Z replaced by $\widetilde{Z}$:

$$\boxed{\mathrm{SMG}_{M_{11},W} = \left(\prod_{i \ge 1} (1 - t_i)\right) \mathrm{SMG}_{M_{11},\mathbb{C}^{11}}\,.} \tag{466}$$

Equation (466) is a relation between generalized Molien integrands and hence between their averages; it does not assert a tensor factorization of invariant spaces.

The same cycle data recover every character of a power:

$$\chi_{\mathrm{perm}}(g^r) = \sum_{d \mid r} d\, c_d(g), \qquad \chi_W(g^r) = \chi_{\mathrm{perm}}(g^r) - 1. \tag{467}$$

Substitution in (462) gives a third, power-character evaluation of (465)–(466).

83.5 Verification strategy

The program uses four routes with deliberately different inputs.

1. **Direct matrices.** For every generated matrix A, traces of A^r are computed by integer matrix multiplication. Complete symmetric characters are recovered from Newton's exact recurrence

$$h_0(A) = 1, \qquad h_d(A) = \frac{1}{d}\sum_{r=1}^{d} \mathrm{Tr}(A^r) h_{d-r}(A).$$

   The target coefficient is then the literal average $|G|^{-1} \sum_g \prod_j h_{\tau_j}(\rho(g))$.

2. **Class determinants.** Only the eight table rows are used to expand $\prod_r (1-t^r)^{-c_r}$; the deleted factor $1-t$ is then applied.

3. **Power characters.** Equation (467) and the same Newton recurrence expand the exponential formula, without using the determinant factorization in the implementation.

4. **Configuration orbits.** For selected permutation coefficients, the two generators traverse the finite set

$$\prod_j \mathrm{MSet}_{\tau_j}(\{1, \ldots, 11\}).$$

   The number of orbits must equal (463). This route uses neither the class table nor matrix traces.

Finally, at three nonzero variables the code evaluates every determinant numerically and compares the full average with the closed class formula. This catches mistakes that could be hidden by a short coefficient truncation.

83.6 Results

All 29 partitions of positive total degree at most six were tested in each representation, giving 58 exact three-way comparisons. Representative rows are shown here.

representation	τ	matrix	determinant	power character	result
permutation	(1)	1	1	1	**PASS**
permutation	(2)	2	2	2	**PASS**
permutation	$(2,1)$	4	4	4	**PASS**
permutation	$(2,2)$	9	9	9	**PASS**
permutation	$(1,1,1,1)$	15	15	15	**PASS**
permutation	(6)	13	13	13	**PASS**
deleted	(1)	0	0	0	**PASS**
deleted	(2)	1	1	1	**PASS**
deleted	$(2,1)$	1	1	1	**PASS**
deleted	$(2,2)$	3	3	3	**PASS**
deleted	$(1,1,1,1)$	4	4	4	**PASS**
deleted	(6)	5	5	5	**PASS**

The separate generator-orbit counts for $\tau = (1), (2), (1,1), (3), (2,1), (2,2), (1,1,1,1)$ are respectively

$$1,\ 2,\ 2,\ 3,\ 4,\ 9,\ 15,$$

and agree with the class coefficients. In particular, the last value is the Bell number $B_4 = 15$, as expected from the 4-transitivity of this action.

At $(t_1, t_2, t_3) = (0.037, -0.051, 0.083)$, the full determinant averages are

representation	direct matrix average	class formula	absolute error
permutation	1.0866701548337285	1.0866701548337498	2.13×10^{-14}
deleted	1.0085468522143926	1.0085468522144532	6.06×10^{-14}

The discrepancies are at floating-point roundoff scale; all coefficient and group-law checks are exact integers.

83.7 Other sporadic constructions and scope

83.7.1 Other ordinary representations

For any of the 26 sporadic simple groups, and for any chosen ordinary complex character, equation (462) applies verbatim. For a nontrivial irreducible representation of a sporadic simple group, the kernel is normal and therefore trivial. This observation does not select a representation; one must still state which character is intended.

83.7.2 Central covers

If $\widetilde{G}$ is a central cover and a central element z acts on V as the scalar $\omega(z)$, it acts on $\bigotimes_j \operatorname{Sym}^{\tau_j} V$ as $\omega(z)^{|\tau|}$. Hence

$$[m_\tau]\,\mathrm{SMG}_{\widetilde{G},V} = 0 \quad \text{unless} \quad \omega(z)^{|\tau|} = 1 \quad \text{for every } z \in Z(\widetilde{G}).$$

Projective representations of the simple quotient must therefore be treated as honest representations of the appropriate cover.

83.7.3 Lattice and conjugation actions

For a lattice action, class eigenvalues or frame shapes may be substituted directly into (461). For conjugation on the basis $\mathbb{C}[G]$, the fixed-point character is $\chi(g^r) = |C_G(g^r)|$; Möbius inversion then recovers the conjugation cycle counts. These are separate representation choices and are not silently combined with the M_{11} action tested here.

83.7.4 Asymptotics

There is no intrinsic limiting sequence over the sporadic simple groups: there are only 26. For the one-variable coefficients of either fixed faithful M_{11} representation, ordinary rational-function asymptotics do apply. The identity matrix supplies the unique pole of order $\dim V$ at $t = 1$, up to root-of-unity oscillations of lower polynomial order, so

$$\begin{aligned}
[t^d]\,\mathrm{SMG}_{M_{11},\mathbb{C}^{11}}(t) &= \frac{1}{7920}\binom{d+10}{10} + O(d^9),\\
[t^d]\,\mathrm{SMG}_{M_{11},W}(t) &= \frac{1}{7920}\binom{d+9}{9} + O(d^8).
\end{aligned}$$

The exact sequences are quasipolynomial because every eigenvalue is a root of unity.

83.8 Reproduction and provenance

From the repository root, run

```
.venv/bin/python -m lambda_distributions.dists.sporadic_group_sigma_mgfs.verification
.venv/bin/python -m pytest -q \
  lambda_distributions/dists/sporadic_group_sigma_mgfs/test_verification.py
.venv/bin/marimo edit marimo_notebooks/sporadic.py
```

The generator data, group order, standard-generator fingerprint, representation degree, and class centralizers are recorded by the online ATLAS page for M_{11}; the natural action and its cycle types are recorded on the ATLAS 11-point representation page. The program independently enumerates the group and reproduces every aggregated class multiplicity used in (465).

This is a standalone inclusion-ready proof. It does not modify or rebuild the combined proof PDF.

Part VIII

Nonuniform measures, heat flow, and limiting laws

84 Nonuniform laws on cyclic groups

Outcome. The averaging-operator and convolution formulas were checked on explicit diagonal models of C_3, C_4, C_5 in dimensions $1, 2, 3$. Arbitrary nonuniform measures and four walk times were tested for every partition through total degree five. All **288** comparisons pass; the maximum absolute error is 2.91×10^{-14}. Three additional unexpanded determinant evaluations also pass.

84.1 The general coefficient theorem

Let G be finite, $\rho : G \to U(V)$, and let ν be a probability measure. For finitely many active variables define

$$\mathrm{SMG}_{\nu,V}(\mathbf{t}) = \sum_{g \in G} \nu(g) \prod_a \det(I - t_a \rho(g))^{-1}. \tag{1}$$

The standard identity $\det(I - tA)^{-1} = \sum_{r \geq 0} \chi_{\mathrm{Sym}^r V}(A) t^r$ implies, by multiplying the series variable by variable, that

$$[\mathbf{t}^\tau]\, \mathrm{SMG}_{\nu,V} = \sum_g \nu(g) \chi_{W_\tau}(g) = \mathrm{Tr}\left(\sum_g \nu(g) \rho_{W_\tau}(g) \right), \qquad W_\tau = \bigotimes_a \mathrm{Sym}^{\tau_a} V. \tag{2}$$

This proves the proposed replacement of the Haar invariant dimension by the trace of a nonuniform averaging operator. Positivity of the measure does not force the trace to be real or integral; the C_5 test below deliberately exhibits a complex coefficient.

If $\nu = \kappa^{*r}$ and the convention is that a step product is $X_1 \cdots X_r$, then representation multiplicativity gives

$$\widehat{\nu}(W) = \widehat{\kappa}(W)^r. \tag{3}$$

Equations (2)–(3) prove the convolution formula without a centrality assumption.

84.2 Specialization to C_m

Write $C_m = \{0, \ldots, m-1\}$ additively and $\zeta_m = e^{2\pi i/m}$. A diagonal representation is determined by weights $w_1, \ldots, w_d \pmod m$:

$$\rho(k) = \mathrm{diag}(\zeta_m^{k w_1}, \ldots, \zeta_m^{k w_d}). \tag{4}$$

Let $M_\tau(q)$ denote the multiplicity of the character $k \mapsto \zeta_m^{kq}$ in W_τ. Extracting weights from

$$\prod_{j=1}^d (1 - x z^{w_j})^{-1} = \sum_{r \geq 0} \sum_q M_{(r)}(q) x^r z^q \tag{5}$$

and convolving the multiplicities for the factors in W_τ gives the finite Fourier formula

$$[\mathbf{t}^\tau]\, \mathrm{SMG}_{\nu,V} = \sum_{q \bmod m} M_\tau(q) \widehat{\nu}(q), \qquad \widehat{\nu}(q) = \sum_{k=0}^{m-1} \nu(k) \zeta_m^{kq}. \tag{6}$$

For a walk this becomes

$$[\mathbf{t}^\tau]\, \mathrm{SMG}_{\kappa^{*r},V} = \sum_q M_\tau(q) \widehat{\kappa}(q)^r. \tag{7}$$

Thus the abstract matrix Fourier transform reduces to a transparent scalar filter on each weight.

84.3 Independent verification routes

The direct route builds every matrix in (4), computes the power traces $p_j = \operatorname{Tr}(\rho(k)^j)$, and obtains complete characters from Newton's recurrence

$$h_0 = 1, \qquad h_r = \frac{1}{r}\sum_{j=1}^{r} p_j h_{r-j}. \tag{8}$$

It then averages $\prod_a h_{\tau_a}$ using the fully enumerated probability distribution. The formula route never uses matrix traces: it constructs the integer weight multiplicities in (5) and applies (6) or (7). For walks, the direct distribution is obtained by repeated cyclic convolution, whereas the formula route raises Fourier scalars to the rth power.

As a full target check, the code also evaluates (1) at $(t_1, t_2) = (0.07, -0.04)$ using matrix determinants and compares it with the product over the weights in (4). These values stay safely inside every pole.

84.4 Cases and results

The representations and the nonuniform arbitrary measure are

Group	weights	dimension	$\nu(k)$ before normalization
C_3	(1)	1	$1, 2, 3$
C_4	$(1, -1)$	2	$1, 2, 3, 4$
C_5	$(0, 1, 2)$	3	$1, 2, 3, 4, 5$

The walk step is $0.35\delta_0 + 0.40\delta_1 + 0.25\delta_{-1}$ and is tested at $r = 0, 1, 2, 5$. There are 19 partitions through total degree five, hence $3 \cdot (1 + 4) \cdot 19 = 285$ coefficient comparisons plus three determinant comparisons.

Case	coefficient	direct value	formula value	result
C_3, arbitrary	$\tau = (3)$	1.0000000000	1.0000000000	**PASS**
C_4, walk $r = 5$	$\tau = (2, 1)$	-0.0210525000	-0.0210525000	**PASS**
C_5, arbitrary	$\tau = (2, 2)$	$2.1666667 + 0.2293970i$	$2.1666667 + 0.2293970i$	**PASS**

The complex C_5 row is a useful diagnostic: the implementation is genuinely testing non-Haar averaging, not silently rounding to invariant dimensions.

84.5 Reproducibility and scope

Run `cyclic_groups/verification.py`, or open `marimo_notebooks/nonuniform_cyclic.py` in marimo. The notebook exposes the degree cutoff and every comparison. The finite computations verify all coefficients in the stated range; equations (1)–(7) are all-degree proofs. No sampling is used.

85 Nonuniform laws on symmetric groups

Outcome. Explicit permutation matrices for S_3, S_4, S_5 were compared with the cycle-count formula under six families of nonuniform measures. A separate S_3 calculation decomposes every tested tensor-symmetric character into all three irreducibles and verifies both central and noncentral Fourier operators. All **267** checks pass; the maximum absolute error is 4.45×10^{-15}.

85.1 Permutation matrices and the cycle transform

For the defining permutation representation $V = \mathbb{C}^n$, a d-cycle contributes all dth roots of unity. Consequently

$$\det(I - tP_\pi) = \prod_{d\geq 1} (1 - t^d)^{C_d(\pi)}. \tag{1}$$

With

$$Q_d(\mathbf{t}) = \prod_a (1 - t_a^d)^{-1}, \tag{2}$$

direct substitution in the determinant definition proves

$$\mathrm{SMG}_{\nu,n}(\mathbf{t}) = \mathbb{E}_\nu \left[\prod_{d\geq 1} Q_d(\mathbf{t})^{C_d(\pi)} \right]. \tag{3}$$

This identity is valid for every probability measure, whether or not it is a class measure. It is the joint cycle-count probability-generating function evaluated at the transformed variables Q_d.

For coefficient extraction, each factor is expanded using

$$(1 - t^d)^{-c} = \sum_{j\geq 0} \binom{c + j - 1}{j} t^{dj}. \tag{4}$$

The verification uses integer dynamic programming for (4), separately for every permutation. This route does not diagonalize a matrix.

85.2 Measures tested

All probabilities are constructed on the fully enumerated group:

1. the lazy random-transposition walk with $\kappa = 0.35\delta_e + 0.65\,\mathrm{Unif}(C_2)$, after three steps;
2. the lazy random 3-cycle walk with laziness 0.4, after two steps;
3. the lazy adjacent-transposition walk with laziness 0.25, after four steps;
4. the inverse 2-riffle law
$$\mathbb{P}(\pi) = 2^{-n} \binom{n + 1 - d(\pi)}{n}, \tag{5}$$
where $d(\pi)$ is the number of descents;
5. Ewens measure with $\theta = 1.7$, $\mathbb{P}(\pi) \propto \theta^{K(\pi)}$;
6. the general cycle weight $\mathbb{P}(\pi) \propto 1.2^{C_1} 0.7^{C_2} 1.6^{C_3}$.

For S_3, S_4, S_5 and every partition through total degree four, the direct route averages characters computed from the explicit matrices using Newton's identities. The cycle route applies (4) to the cycle structure. Thus the two routes share the probability law but not the spectral computation.

The normalizing generating-function formulas follow immediately from the exponential formula for labeled cycles. For example,

$$\mathrm{SMG}^{\mathrm{Ewens}}_{n,\theta} = \frac{[z^n]\exp\Big(\theta\sum_{d\geq 1} Q_d z^d/d\Big)}{[z^n]\exp\Big(\theta\sum_{d\geq 1} z^d/d\Big)}. \tag{6}$$

Equation (6) records the corrected typography: the normalization is $(\theta)_n$, and the summation index is $d \geq 1$.

85.3 Fourier proof and an independent S_3 test

Decompose

$$W_\tau = \bigotimes_a \mathrm{Sym}^{\tau_a} V \cong \bigoplus_{\lambda\vdash n} M_{\tau,\lambda}\otimes V_\lambda, \qquad m_{\tau,\lambda} = \dim M_{\tau,\lambda}. \tag{7}$$

Linearity of trace gives, for an arbitrary measure,

$$[\mathbf{t}^\tau]\,\mathrm{SMG}_{\nu,n} = \sum_{\lambda\vdash n} m_{\tau,\lambda}\,\mathrm{Tr}\,\widehat{\nu}(\lambda), \qquad \widehat{\nu}(\lambda) = \sum_\pi \nu(\pi)\rho_\lambda(\pi). \tag{8}$$

For a convolution walk, $\widehat{\kappa^{*r}}(\lambda) = \widehat{\kappa}(\lambda)^r$. If κ is central, Schur's lemma makes this operator scalar. In the lazy transposition case,

$$\widehat{\kappa}(\lambda) = \beta_\lambda I, \qquad \beta_\lambda = a + (1-a)\frac{\chi^\lambda((12))}{d_\lambda}. \tag{9}$$

For adjacent transpositions the full matrix power must be retained.

The S_3 test constructs the trivial, sign, and two-dimensional standard matrices explicitly. Multiplicities in (7) are obtained independently by the uniform character inner product, then (8) is evaluated for an arbitrary law, lazy central and adjacent walks, and the riffle law. For laziness $a = 0.35$, the central scalars on $(\mathbf{1}, \mathrm{sgn}, \mathrm{std})$ are

$$(1, -0.3, 0.35). \tag{10}$$

The adjacent operator on the standard representation instead has two distinct eigenvalues 0.025 and 0.675, directly exposing the claimed matrix-valued behavior.

85.4 Results

There are 12 partitions through total degree four. The cycle calculation therefore contributes $3 \cdot 6 \cdot 12 = 216$ comparisons. The four S_3 Fourier laws contribute 48 more, and the three irreducible operator checks bring the total to 267.

Case	coefficient	matrix/direct	proposed formula	result
S_3, inverse 2-riffle	(3)	5.750000	5.750000	**PASS**
S_4, Ewens $\theta = 1.7$	$(2,1)$	7.429557	7.429557	**PASS**
S_5, transposition walk	$(2,2)$	39.683180	39.683180	**PASS**
S_3, arbitrary Fourier law	$(2,1)$	1.904762	1.904762	**PASS**
S_3, adjacent walk Fourier law	$(2,2)$	9.671793	9.671793	**PASS**

85.5 Warnings, corrections, and scope

Without laziness, a transposition walk alternates parity and does not converge to uniform measure on S_n. A walk on odd-length cycles stays in A_n. These are support obstructions, not failures of (3) or (8).

The supplied continuous-time class-walk expressions contained a stray comma in the exponential; the intended factor is $\exp(t(r_\lambda - 1))$. Continuous-time graph interchange processes were not numerically tested here, but their operator formula follows from replacing matrix powers in (8) by the matrix exponential of the generator.

Run

```
.venv/bin/python -m \
  for_this_guy.nonuniform_lambda_distributions.symmetric_groups.verification
```

or open `marimo_notebooks/nonuniform_symmetric.py` in marimo.

86 $U(1)$ heat kernels and weight attenuation

Outcome. Three diagonal $U(1)$ representations of dimensions $1, 2, 3$ were checked at five heat times for all partitions through total degree five. Direct quadrature against the heat kernel agrees with the weight/Casimir formula in all **285** cases. The maximum absolute error is 5.73×10^{-14}.

86.1 Heat measure and normalization

Identify $U(1)$ with $e^{i\theta}$, $0 \le \theta < 2\pi$, and choose the Laplacian normalization for which the character $e^{iq\theta}$ has Casimir q^2. Brownian motion with generator $\frac{1}{2}\Delta$ has density relative to normalized Haar measure

$$k_t(\theta) = \sum_{q\in\mathbb{Z}} e^{-q^2 t/2} e^{-iq\theta}. \tag{1}$$

Orthogonality immediately yields

$$\int_{U(1)} e^{iw\theta} k_t(\theta)\, d\theta/(2\pi) = e^{-w^2 t/2}. \tag{2}$$

This fixes the otherwise convention-dependent factor in the Casimir exponent.

Let V be the diagonal representation with integer weights $a_1, \ldots, a_d$,

$$\rho(e^{i\theta}) = \operatorname{diag}(e^{ia_1\theta}, \ldots, e^{ia_d\theta}). \tag{3}$$

For $W_\tau = \bigotimes_j \operatorname{Sym}^{\tau_j} V$, write

$$W_\tau = \bigoplus_{w\in\mathbb{Z}} M_\tau(w)\, \mathbb{C}_w. \tag{4}$$

Combining (2)–(4) proves the coefficient formula

$$\boxed{[\mathbf{t}^\tau]\, \mathrm{SMG}_t = \sum_{w\in\mathbb{Z}} M_\tau(w) e^{-w^2 t/2}.} \tag{5}$$

This is the compact-group heat formula in its simplest exact form: each irreducible weight is attenuated by its Casimir eigenvalue.

86.2 Endpoints and interpretation

At $t = 0$, every exponential in (5) is one, hence

$$[\mathbf{t}^\tau]\, \mathrm{SMG}_0 = \dim W_\tau, \qquad \mathrm{SMG}_0(\mathbf{t}) = \prod_j (1 - t_j)^{-d}. \tag{6}$$

This is evaluation at the identity matrix. As $t \to \infty$, only weight zero survives:

$$\lim_{t\to\infty} [\mathbf{t}^\tau]\, \mathrm{SMG}_t = M_\tau(0) = \dim W_\tau^{U(1)}. \tag{7}$$

Thus (5) gives a literal spectral interpolation between dimension and Haar invariants. For a purely positive defining weight there are no positive-degree Haar invariants. Mixed positive and negative weights, however, can cancel and leave nontrivial weight-zero terms.

86.3 Representations and verification method

The tested weight lists are

Name	weights	dimension	purpose
defining	(1)	1	isolates the scalar Casimir r^2
weight pair	$(1,-1)$	2	creates many weight-zero invariants
mixed	$(0,1,-2)$	3	combines a fixed line with asymmetric weights

For each representation, $t = 0, 0.25, 0.8, 3, 20$ and all 19 partitions through total degree five are tested.

The direct route uses 512 equally spaced angles. At each angle it constructs the matrix (3), computes complete characters from the matrix power traces, and integrates their product against the Fourier heat kernel (1), truncated at $|q| \le 32$. The formula route never evaluates an angle or matrix: it builds the integer multiplicities $M_\tau(w)$ and applies (5).

Why the quadrature is especially strong

This is not a generic Monte Carlo comparison. Through total degree five the largest character frequency is at most $5 \max_j |a_j| = 10$. Multiplication by the truncated kernel produces frequencies at most 42. Since 512 exceeds twice this bandwidth, the periodic trapezoidal rule integrates the entire trigonometric polynomial exactly in exact arithmetic. At the smallest positive time, the first omitted heat coefficient is $e^{-33^2/8} < 10^{-59}$. The observed errors are therefore floating-point roundoff, not sampling noise or appreciable heat-kernel truncation.

At $t = 0$ the heat density is a delta distribution, so the code evaluates the identity matrix directly as in (6). This avoids approximating a singular endpoint with a finite Fourier sum.

86.4 Results

Representation and time	coefficient	quadrature	Casimir formula	result
defining, $t = 0.8$	(3)	0.0273237224	0.0273237224	**PASS**
weight pair, $t = 3$	$(2,1)$	0.8925233825	0.8925233825	**PASS**
mixed, $t = 20$	$(2,2)$	5.0003631994	5.0003631994	**PASS**

The last value is already close to the integer Haar multiplicity five, while the small positive correction records the surviving nonzero weights at finite time.

86.5 Scope and reproducibility

The computation verifies the compact heat formula for $U(1)$ with several representation dimensions and both one-variable and mixed coefficients. It does not by itself verify the branching rules proposed for $U(n)$, $SU(n)$, $O(n)$, or $USp(2n)$. Those require separate highest-weight and Casimir normalizations. The disconnected-group warning is also essential: Brownian motion from the identity in $O(n)$ remains in $SO(n)$ unless component-changing jumps are added.

Run `compact_u1_heat/verification.py`, or open `marimo_notebooks/u1_heat.py` in marimo.

87 Non-Haar Fourier-weighted sigma-MGFs

Canonical testing artifacts.
Exact backend: `lambda_distributions/dists/non_haar_fourier_sigma_mgfs/verification.py`.
Regression: `lambda_distributions/dists/non_haar_fourier_sigma_mgfs/test_verification.py`.
Marimo: `marimo_notebooks/non_haar_fourier.py`.

Scope and outcome

For a probability law on a finite or compact matrix group, Haar averaging is replaced by a Fourier-weighted operator rather than a Reynolds projection. We prove the exact Schur and monomial formulas, the full matrix Fourier formula for arbitrary random walks, its scalar character-ratio specialization for central walks, and the Casimir-weighted heat formula. Verification is split by group family. For S_3, every natural and deleted-representation matrix is constructed and arbitrary convolution is compared with all irreducible Fourier blocks. For $U(1)$, direct heat-kernel quadrature of 2×2 matrices is compared with weight decomposition. For Ewens laws on $S_3, \dots, S_6$, exact matrix enumeration is compared with the cycle-index formula. The accompanying marimo notebook exposes every comparison.

87.1 The non-Haar master theorem

Let G be finite or compact, let dg denote normalized Haar measure, and let $\rho : G \to U(V)$ be a finite-dimensional unitary representation. A probability measure μ on G need not be invariant. For variables $\mathbf{t} = (t_1, t_2, \dots)$ define

$$\mathrm{SMG}_{\mu,V}(\mathbf{t}) = \int_G \prod_{i \geq 1} \det(I - t_i \rho(g))^{-1} \, d\mu(g). \tag{468}$$

The statement is formal coefficientwise. It is also analytic when only finitely many variables are nonzero and $|t_i| < 1$. Write $p_r = \sum_i t_i^r$. For a partition $\tau = (\tau_1, \dots, \tau_s)$ put

$$W_\tau(V) = \bigotimes_{j=1}^{s} \mathrm{Sym}^{\tau_j} V. \tag{469}$$

Theorem 1 (non-Haar σ-MGF)**.** *The following identities hold coefficientwise:*

$$\mathrm{SMG}_{\mu,V}(\mathbf{t}) = \int_G \exp\left(\sum_{r \geq 1} \frac{p_r}{r} \chi_V(g^r) \right) d\mu(g), \tag{470}$$

$$= \sum_\lambda s_\lambda(\mathbf{t}) \int_G \chi_{S_\lambda V}(g) \, d\mu(g), \tag{471}$$

$$[m_\tau] \, \mathrm{SMG}_{\mu,V} = \int_G \chi_{W_\tau(V)}(g) \, d\mu(g). \tag{472}$$

If $S_\lambda V \cong \bigoplus_{\pi \in \widehat{G}} m_\pi(S_\lambda V)\pi$, then

$$\boxed{\mathrm{SMG}_{\mu,V} = \sum_\lambda s_\lambda \sum_{\pi \in \widehat{G}} m_\pi(S_\lambda V) \, \mathrm{Tr} \, \widehat{\mu}(\pi), \qquad \widehat{\mu}(\pi) = \int_G \pi(g) \, d\mu(g).} \tag{473}$$

The analogous monomial formula replaces $S_\lambda V$ by $W_\tau(V)$.

Proof. If A has eigenvalues $\alpha_1, \dots, \alpha_d$, then

$$\det(I - tA)^{-1} = \prod_j (1 - t\alpha_j)^{-1} = \exp\left(\sum_{r \geq 1} \frac{t^r}{r} \, \mathrm{Tr}(A^r) \right). \tag{474}$$

Apply this identity to $A = \rho(g)$ and multiply over i to obtain (470). The Cauchy identity gives

$$\prod_i \det(I - t_i\rho(g))^{-1} = \sum_\lambda s_\lambda(\mathbf{t})\chi_{S_\lambda V}(g), \tag{475}$$

which proves (471). Alternatively, the coefficient of t^q in one determinant inverse is $\chi_{\mathrm{Sym}^q V}(g)$. Multiplying the selected coefficients proves (472).

Finally, characters add over direct sums. On each irreducible summand π, the integral of its character is $\operatorname{Tr}\int \pi(g)d\mu(g)$. Summing with multiplicity proves (473). No invariance of μ was used. □

Remark 1 (what replaces Reynolds projection)**.** *For Haar measure, $\int_G \rho_W(g)dg$ is the idempotent projection onto W^G, so $\int \chi_W = \dim W^G$. For general μ the operator $T_{\mu,W} = \int \rho_W(g)d\mu(g)$ is usually neither idempotent nor central. The coefficient is its trace. Thus a non-Haar law produces a Fourier-weighted averaging operator, not an invariant dimension.*

87.2 Random walks and the cutoff-visible spectrum

Let ν be a step distribution and $\mu_k = \nu^{*k}$. With the convention in (473), representation multiplicativity and Fubini give

$$\widehat{\nu^{*k}}(\pi) = \widehat{\nu}(\pi)^k. \tag{476}$$

Corollary 1 (arbitrary walk)**.** *For every partition τ,*

$$\boxed{[m_\tau]\,\mathrm{SMG}_{\nu^{*k},V} = \sum_{\pi\in\widehat{G}} m_\pi(W_\tau(V))\operatorname{Tr}\big(\widehat{\nu}(\pi)^k\big).} \tag{477}$$

If ν is central, Schur's lemma gives

$$\widehat{\nu}(\pi) = \theta_\pi I_{d_\pi}, \qquad \theta_\pi = \frac{1}{d_\pi}\int_G \chi_\pi(g)d\nu(g), \tag{478}$$

and hence

$$\boxed{[m_\tau]\,\mathrm{SMG}_{\nu^{*k},V} = \sum_{\pi} m_\pi(W_\tau(V))d_\pi\theta_\pi^k.} \tag{479}$$

The trivial term in (479) is the Haar coefficient. Therefore

$$[m_\tau](\mathrm{SMG}_{\nu^{*k},V} - \mathrm{SMG}_{\mathrm{Haar},V}) = \sum_{\pi\neq\mathbf{1}} m_\pi(W_\tau(V))d_\pi\theta_\pi^k. \tag{480}$$

For a fixed finite group this gives the exact leading exponential decay after combining all terms with maximal $|\theta_\pi|$. For a sequence (G_n, V_n, ν_n) it also identifies the correct bounded-degree cutoff criterion: only irreducibles occurring in the fixed $W_{n,\tau}$ are visible. This is weaker than total-variation mixing of the full group-valued walk.

Remark 2 (centrality is not optional)**.** *Equation (479) must not be used for a noncentral generator law. The trace of $\widehat{\nu}(\pi)$ does not determine $\operatorname{Tr}(\widehat{\nu}(\pi)^k)$. Section 87.4 includes a concrete 2×2 counterexample to that shortcut.*

87.3 Compact heat kernels

Let the Brownian generator be $\frac{1}{2}\Delta_G$, and let $\kappa_\pi \geq 0$ be the Casimir eigenvalue on π for this normalization. The heat semigroup diagonalizes on matrix coefficients:

$$\widehat{\mu}_t(\pi) = e^{-t\kappa_\pi/2} I_{d_\pi}. \tag{481}$$

Substitution in Theorem 1 proves the following.

Theorem 2 (heat formula)**.**

$$\mathrm{SMG}^{\mathrm{heat}}_{G,V}(t) = \sum_{\lambda} s_\lambda \sum_{\pi} m_\pi(S_\lambda V) d_\pi e^{-t\kappa_\pi/2}, \tag{482}$$

$$[m_\tau]\,\mathrm{SMG}^{\mathrm{heat}}_{G,V}(t) = \sum_{\pi} m_\pi(W_\tau(V)) d_\pi e^{-t\kappa_\pi/2}. \tag{483}$$

At $t = 0$ the coefficient is $\dim W_\tau(V)$, and as $t \to \infty$ it tends to $\dim W_\tau(V)^G$. If a positive-Casimir constituent occurs, let γ_τ be the smallest positive Casimir appearing in $W_\tau(V)$. Then

$$[m_\tau]\,\mathrm{SMG}_t = \dim W_\tau(V)^G + e^{-t\gamma_\tau/2} \sum_{\kappa_\pi=\gamma_\tau} m_\pi(W_\tau) d_\pi + o(e^{-t\gamma_\tau/2}). \tag{484}$$

This theorem applies without conceptual change If no positive-Casimir constituent occurs, the coefficient is already constant and the remainder in the preceding asymptotic formula is zero. to classical and exceptional compact groups once the representation and metric normalization are fixed. For a growing-rank statement, that normalization is essential: rescaling the metric rescales every κ_π and therefore the asserted cutoff time.

87.4 Finite matrix group: two walks on S_3

87.4.1 Constructed representations

For $\sigma \in S_3$, let P_σ be its 3×3 permutation matrix, $P_\sigma e_j = e_{\sigma(j)}$. The natural module decomposes as

$$\mathbb{C}^3 \cong \mathbf{1} \oplus V_{\mathrm{std}}. \tag{485}$$

The verifier also constructs the deleted matrices on $V_{\mathrm{std}} = \{x_1 + x_2 + x_3 = 0\}$ in the basis $f_1 = e_1 - e_3, f_2 = e_2 - e_3$. Their entries are

$$(D_\sigma)_{ij} = \delta_{i,\sigma(j)} - \delta_{i,\sigma(3)}, \qquad 1 \le i, j \le 2, \tag{486}$$

where a delta is zero if its second index is 3. All six matrices in each image are generated explicitly and multiplication is exact over $\mathbb{Z}$.

The three irreducible character rows, ordered by identity, transposition, and three-cycle, are

π	d_π	$\chi_\pi(1)$	$\chi_\pi((12))$	$\chi_\pi((123))$
trivial	1	1	1	1
sign	1	1	-1	1
standard	2	2	0	-1

87.4.2 Central lazy-transposition law

Set

$$\nu_c = \frac{1}{4}\delta_e + \frac{1}{4} \sum_{r \text{ a transposition}} \delta_r. \tag{487}$$

It is central, and (478) gives

$$\theta_{\mathrm{triv}} = 1, \qquad \theta_{\mathrm{sign}} = -\frac{1}{2}, \qquad \theta_{\mathrm{std}} = \frac{1}{4}. \tag{488}$$

Thus every tested coefficient has the completely explicit form

$$m_{\mathrm{triv}}(W_\tau) + m_{\mathrm{sign}}(W_\tau)(-1/2)^k + 2m_{\mathrm{std}}(W_\tau)(1/4)^k. \tag{489}$$

87.4.3 Noncentral two-generator law

Set

$$\nu_{nc} = \frac{1}{2}\delta_{(23)} + \frac{1}{2}\delta_{(123)}. \tag{490}$$

Using (486), its standard Fourier block is

$$\widehat{\nu}_{nc}(\mathrm{std}) = \frac{1}{2}D_{(23)} + \frac{1}{2}D_{(123)} = \begin{pmatrix} 0 & -1/2 \\ 0 & -1/2 \end{pmatrix}. \tag{491}$$

This is not scalar. The correct term in (477) is $m_{\mathrm{std}}(W_\tau)\,\mathrm{Tr}(\widehat{\nu}_{nc}(\mathrm{std})^k)$; a character-ratio substitution would discard matrix information.

87.4.4 Exact verification strategy

For each representation, partition τ through total degree four, and $k \in \{0, 1, 2, 3, 5\}$, the program compares:

1. **Direct convolution and matrices.** It forms ν^{*k} on all six elements. For each matrix A, it computes

$$h_0(A) = 1, \qquad h_d(A) = \frac{1}{d}\sum_{r=1}^{d} \mathrm{Tr}(A^r) h_{d-r}(A), \tag{492}$$

   then averages $\prod_j h_{\tau_j}(A)$ literally.
2. **Full Fourier route.** Multiplicities are computed independently from

$$m_\pi(W_\tau) = \frac{1}{6}\sum_{g \in S_3} \chi_{W_\tau}(g)\chi_\pi(g^{-1}), \tag{493}$$

   and inserted into (477) using exact rational matrix powers.
3. **Central scalar route.** For ν_c only, the code separately checks that every Fourier block is scalar and evaluates (489).

walk	representation	(k, τ)	direct	Fourier	result
central	natural	$(3, (1))$	33/32	33/32	**PASS**
central	natural	$(3, (2, 2))$	63/8	63/8	**PASS**
central	deleted	$(5, (1))$	1/512	1/512	**PASS**
noncentral	natural	$(3, (1))$	7/8	7/8	**PASS**
noncentral	deleted	$(3, (2, 1))$	3/4	3/4	**PASS**
noncentral	deleted	$(5, (2, 2))$	61/32	61/32	**PASS**

The displayed noncentral values are generated by the same suite as the full table. All 220 exact comparisons pass.

87.5 Compact matrix group: $U(1)$ heat flow

Take

$$\rho(e^{i\phi}) = \begin{pmatrix} e^{i\phi} & 0 \\ 0 & e^{-i\phi} \end{pmatrix} \tag{494}$$

on a two-dimensional representation with weights $+1, -1$. For the standard circle Laplacian, the weight-q character has heat multiplier $e^{-tq^2/2}$. Moreover

$$\chi_{\mathrm{Sym}^d V}(e^{i\phi}) = \sum_{a=0}^{d} e^{i(d-2a)\phi}. \tag{495}$$

If $c_\tau(q)$ is the coefficient of z^q in

$$\prod_{j=1}^{\ell(\tau)} (z^{\tau_j} + z^{\tau_j - 2} + \cdots + z^{-\tau_j}), \tag{496}$$

then Theorem 2 becomes the finite exact sum

$$\boxed{[m_\tau]\,\mathrm{SMG}_t = \sum_{q\in\mathbb{Z}} c_\tau(q)e^{-tq^2/2}.} \tag{497}$$

At $t=0$ this is $\prod_j(\tau_j+1)$; at Haar time it tends to $c_\tau(0)$. The first nonzero $|q|$ in (496) gives the leading decay rate, an explicit instance of (484).

87.5.1 Independent numerical route

The verifier does not use (496) in its direct route. On a 256-node periodic grid it constructs (494), recovers every $h_d(\rho(e^{i\phi}))$ from matrix traces using (492), and multiplies by

$$K_t(\phi) = 1 + 2\sum_{q=1}^{48} e^{-tq^2/2}\cos(q\phi). \tag{498}$$

The trapezoidal average is compared with (497). The grid resolves all retained Fourier modes; the omitted tail is far below the reported tolerance at the smallest tested time $t = 0.2$.

t	τ	direct quadrature	weight formula	error
0.2	(1)	1.809674836071920	1.809674836071919	6.7×10^{-16}
0.2	$(2,2)$	6.085073220131871	6.085073220131868	3.6×10^{-15}
0.7	$(1,1)$	2.493193927883212	2.493193927883213	1.3×10^{-15}
2.0	(5)	0.736005701978834	0.736005701978834	4.4×10^{-16}

All 54 comparisons through degree five pass. The largest absolute error over the complete suite is 1.5988×10^{-14}. For all 18 partitions, the $t=0$ dimension and $t\to\infty$ Haar-weight identities also pass.

87.6 Ewens laws on symmetric matrix groups

For $\sigma\in S_n$, let $K(\sigma)$ be its number of cycles and take

$$\mathbb{P}_{n,\theta}(\sigma) = \frac{\theta^{K(\sigma)}}{\theta^{(n)}}, \qquad \theta^{(n)} = \theta(\theta+1)\cdots(\theta+n-1). \tag{499}$$

The normalization follows from $\sum_{\sigma\in S_n}\theta^{K(\sigma)} = \theta^{(n)}$.

If $m_j(\lambda)$ is the number of parts of size j in $\lambda\vdash n$ and $z_\lambda = \prod_j j^{m_j(\lambda)}m_j(\lambda)!$, a permutation of cycle shape λ satisfies

$$\det(I - tP_\sigma)^{-1} = \prod_{j\ge1}(1-t^j)^{-m_j(\lambda)}. \tag{500}$$

Proposition 1 (exact Ewens formula)**.** *For the natural permutation representation,*

$$\boxed{\mathrm{SMG}_{n,\theta}(\mathbf{t}) = \frac{n!}{\theta^{(n)}}\sum_{\lambda\vdash n}\frac{\theta^{\ell(\lambda)}}{z_\lambda}\prod_{i,j}(1-t_i^j)^{-m_j(\lambda)}.} \tag{501}$$

Proof. There are $n!/z_\lambda$ permutations of shape λ. They all have $\ell(\lambda)$ cycles, hence the same Ewens weight, and (500) gives their common matrix integrand. Grouping the direct element sum by shape proves (501). □

For fixed j, the Ewens cycle counts converge jointly to independent Poisson variables of means θ/j. Consequently, coefficientwise in the positive-degree variables,

$$\mathrm{SMG}_{\theta,\infty} = \exp\left[\sum_{j\ge1}\frac{\theta}{j}\left(\prod_i(1-t_i^j)^{-1} - 1\right)\right]. \tag{502}$$

For a coefficient of bounded degree, only finitely many j contribute, so the formal interpretation is unambiguous.

87.6.1 Exact verification strategy and results

The direct route enumerates all $n!$ permutation matrices, computes their symmetric characters from matrix powers, and applies (499). The independent route stores only cycle lengths, expands (500), and aggregates class sizes. Both use rational arithmetic for

$$n = 3, 4, 5, 6, \qquad \theta \in \{1/2, 1, 2\}, \qquad |\tau| \le 4.$$

This gives 132 coefficient comparisons. At $(t_1, t_2) = (1/10, -1/20)$, 12 further comparisons evaluate the full rational determinant average without truncating in degree. Every comparison passes.

n	θ	τ	matrix enumeration	cycle formula	result
3	1/2	(1)	3/5	3/5	**PASS**
3	1/2	(2, 2)	4	4	**PASS**
3	2	(1)	3/2	3/2	**PASS**
6	1/2	(2, 1)	326/231	326/231	**PASS**
6	1	(2, 2)	9	9	**PASS**
6	2	(1, 1)	32/7	32/7	**PASS**

87.7 Related measures and representation choices

87.7.1 Conjugacy-class walks

If ν_K is uniform on a conjugacy class K, then $\theta_\pi = \chi_\pi(K)/d_\pi$. Therefore

$$[m_\tau]\,\mathrm{SMG}_{\nu_K^{*k},V} = \sum_\pi m_\pi(W_\tau) d_\pi \left(\frac{\chi_\pi(K)}{d_\pi}\right)^k. \tag{503}$$

This includes random transpositions and central reflection walks. If a generator set is a union of classes, its weighted average character ratio is used. Noncentral elementary-generator walks require (477) instead.

87.7.2 Character-weighted laws

For an irreducible η, character orthogonality makes $|\chi_\eta(g)|^2 dg$ a probability measure. Formula (472) becomes

$$[m_\tau]\,\mathrm{SMG}_{|\chi_\eta|^2,V} = \langle \chi_{W_\tau}, \chi_\eta \chi_{\eta^*} \rangle_G = \dim \mathrm{Hom}_G(W_\tau, \eta \otimes \eta^*). \tag{504}$$

This is a tensor-product multiplicity, not generally an invariant dimension. Plancherel measure itself lives on $\widehat{G}$, so an additional construction is needed before it defines a law on group elements.

87.7.3 Finite-field and Clifford cautions

A finite-field matrix action is not automatically a complex representation. One must specify a complex permutation, Weil, or other cross-characteristic representation before applying the theorem. For scalar-rich unitary groups, one-sided moments may vanish in many degrees; a balanced series involving both g and g^* can be more informative. These are changes of representation or observable, not exceptions to Theorem 1.

87.8 Reproducibility and audit summary

The package contains one pure-Python verifier and one marimo interface. No external character table, symbolic algebra service, or random seed is used. Finite computations use `fractions.Fraction`; only the compact quadrature uses floating point.

family	independent routes	cases	comparisons	result
S_3 walks	convolution/matrix traces; full Fourier blocks; central scalars	2 laws, 2 reps	220	**PASS**
$U(1)$ heat	matrix-integrand quadrature; weight/Casimir sum	3 times, 18 partitions	54	**PASS**
Ewens S_n	element matrices; cycle-shape aggregation	4 ranks, 3 parameters	132	**PASS**
Ewens full series	element determinants; class determinants	12 laws	12	**PASS**

Run the tests from the repository root:

```
.venv/bin/python -m pytest -q \
  lambda_distributions/dists/non_haar_fourier_sigma_mgfs/test_verification.py
```

Open the grouped interactive notebook with:

```
.venv/bin/marimo edit \
  marimo_notebooks/non_haar_fourier.py
```

Conclusion. The tested cases verify the intended replacement precisely: Haar measure extracts the trivial representation, whereas a non-Haar law weights every constituent by a Fourier trace. Central walks reduce to character ratios, noncentral walks retain full Fourier matrices, heat flow supplies Casimir exponentials, and Ewens weighting is captured exactly by cycle-shape aggregation.

88 Limiting laws for symmetric-group k-subset actions

Abstract

For fixed k, we prove the joint limit of all power traces of the permutation representation of S_n on k-subsets. The limit is an explicit functional of independent $Z_j \sim \mathrm{Poisson}(1/j)$ variables and is non-Poisson for $k \geq 2$. An independent program constructs 1,584 representation matrices in dimensions $6, 10, 15, 20$ and performs 6,336 exact power-trace comparisons.

88.1 The representation and exact formula

Let $\Omega_{n,k} = \binom{[n]}{k}$ and let $\rho_{n,k}$ permute its standard basis. Since the trace of a permutation matrix counts fixed basis vectors,

$$p_r(X_\rho(\sigma)) = \mathrm{Tr}(\rho(\sigma)^r) = |\,\mathrm{Fix}_{\Omega_{n,k}}(\sigma^r)|.$$

Write $A_j(\sigma)$ for the number of j-cycles of σ. A j-cycle splits under the rth power into $d = \gcd(j,r)$ cycles of length j/d. An invariant subset independently includes or excludes each resulting cycle, so

$$\boxed{F_{k,r}(\sigma) = [u^k] \prod_{j=1}^{kr} \left(1 + u^{j/\gcd(j,r)}\right)^{\gcd(j,r)A_j(\sigma)}.} \tag{505}$$

The cutoff is valid: if $j > kr$, then $j/\gcd(j,r) > k$.

For a partition λ, direct averaging therefore gives

$$\mathrm{SMG}_{n,k} = \sum_\lambda \frac{\mathbb{E}_{S_n}[\prod_a F_{k,\lambda_a}(\sigma)]}{z_\lambda} p_\lambda. \tag{506}$$

88.2 Coefficientwise limit

Fix a finite set of exponents. Every variable in (505) is a polynomial in finitely many cycle counts. We use the following elementary factorial-moment lemma. For fixed nonnegative integers $r_1, \ldots, r_R$,

$$\mathbb{E} \prod_{j=1}^{R} (A_j)_{r_j} = \prod_{j=1}^{R} j^{-r_j} \quad \text{whenever} \quad n \geq \sum_{j=1}^{R} j r_j.$$

Indeed, the falling-factorial product counts ordered selections of pairwise disjoint distinguished cycles, with r_j cycles of length j. Choosing their supports and cyclic orders and then permuting the remaining labels gives $n! \prod_j j^{-r_j}/n!$. These are exactly the joint factorial moments of independent $Z_j \sim \mathrm{Poisson}(1/j)$. Hence the finite vector of cycle counts converges jointly, and every factorial moment required by a fixed coefficient is eventually equal to its limiting value. Thus

$$(A_1, \ldots, A_R) \Longrightarrow (Z_1, \ldots, Z_R), \qquad Z_j \text{ independent}, \quad Z_j \sim \mathrm{Poisson}(1/j).$$

Thus

$$F_{k,r}(\sigma) \Longrightarrow F_{k,r}^{(\infty)} := [u^k] \prod_{j=1}^{kr} (1 + u^{j/\gcd(j,r)})^{\gcd(j,r) Z_j},$$

jointly for every finite collection of r's. Moment stabilization permits expectations in (506) to pass to the limit, proving the coefficientwise Λ-limit

$$\boxed{\mathrm{SMG}_\infty^{(k)} = \sum_\lambda \frac{\mathbb{E}[\prod_a F_{k,\lambda_a}^{(\infty)}]}{z_\lambda} p_\lambda.}$$

For $r = 1$, $F_{k,1}^{(\infty)}$ has mean one. The action has rank $k+1$ once $n \geq 2k$, hence its limiting second moment is $k+1$ and its variance is k. A Poisson variable of mean one has variance one; consequently the limit is non-Poisson for $k \geq 2$. For example

$$F_{2,1}^{(\infty)} = \binom{Z_1}{2} + Z_2.$$

88.3 Exact finite formula checks

For each $(n, k) = (4, 2), (5, 2), (6, 2), (6, 3)$, the verifier enumerates all $n!$ permutations, induces their action on lexicographically ordered k-subsets, and materializes the corresponding zero-one matrix. Distinct byte keys prove that the constructed matrices form a faithful copy of S_n. It then compares the integer matrix traces $\mathrm{Tr}(M^r)$, $1 \le r \le 4$, with (505). Six joint products indexed by $(1), (2), (3), (1, 1), (2, 1), (1, 1, 1)$ are averaged independently on both sides.

n	k	dimension	matrices	trace checks	$(\mathbb{E}F, \operatorname{Var} F)$
4	2	6	24	96	$(1, 2)$
5	2	10	120	480	$(1, 2)$
6	2	15	720	2880	$(1, 2)$
6	3	20	720	2880	$(1, 3)$

All 6,336 trace comparisons and all 24 joint-average comparisons pass exactly. These exhaustive matrices verify the finite fixed-point identity, not the asymptotic convergence; convergence is proved by the factorial-moment lemma above.

88.4 Reproduction and reference

```
.venv/bin/python for_this_guy/limiting_lambda_claims/
  symmetric_k_subsets/verification.py
.venv/bin/marimo edit marimo_notebooks/symmetric_k_subsets.py
```

Join each wrapped command onto one line. For the fixed-point and moment background, see P. Diaconis, J. Fulman, and R. Guralnick, *On fixed points of permutations*, https://arxiv.org/abs/0708.2643.

89 Rooted-tree automorphism groups

Abstract

We prove the exact recursive sigma–MGF law for automorphisms of any finite rooted tree. Explicit leaf-permutation matrix groups for complete binary trees through height three and ternary trees through height two give 30 exact mixed-coefficient comparisons. The document separates this proved finite recursion from the still-open general random-tree Λ-limit.

89.1 Wreath-product lemma

Let a finite permutation group H act on X, and let $H \wr S_m$ act on $X \times [m]$. Define

$$\mathrm{SMG}_H(\mathbf{t}) = \frac{1}{|H|} \sum_{h \in H} \prod_i \det(I - t_i h)^{-1}.$$

Write an element of the wreath product as $(h_1, \ldots, h_m; \sigma)$. On a top cycle $c = (i_1 \cdots i_\ell)$, one circuit accumulates $h_c = h_{i_\ell} \cdots h_{i_1}$ and block elimination gives

$$\det(I - tg|_c) = \det(I - t^\ell h_c).$$

Uniform independent h_i make the products h_c uniform and independent over the top cycles. Averaging over S_m and using its cycle index proves

$$\boxed{\sum_{m \geq 0} u^m \,\mathrm{SMG}_{H \wr S_m}(\mathbf{t}) = \exp\left(\sum_{\ell \geq 1} \frac{u^\ell}{\ell} \,\mathrm{SMG}_H(t_1^\ell, t_2^\ell, \ldots)\right).} \tag{507}$$

89.2 Arbitrary rooted trees

If the root branches have isomorphism types A with multiplicities m_A, then internal automorphisms and permutations of equal branches give

$$\mathrm{Aut}(T) \cong \prod_A (\mathrm{Aut}(A) \wr S_{m_A}).$$

The leaf representation is the corresponding direct sum. Determinants multiply on direct sums, so (507) yields

$$\boxed{\mathrm{SMG}_T(\mathbf{t}) = \prod_A [u_A^{m_A}] \exp\left(\sum_{\ell \geq 1} \frac{u_A^\ell}{\ell} \,\mathrm{SMG}_A(\mathbf{t}^\ell)\right).} \tag{508}$$

For the complete b-ary tree, $G_h = G_{h-1} \wr S_b$, starting with the trivial action on one leaf.

89.3 Exact computational verification

The verifier recursively enumerates every wreath-product permutation and materializes its zero-one matrix. Distinct matrix keys and the identity

$$|G_h| = b! \, |G_{h-1}|^b$$

independently check each closure. For a matrix M, the direct symmetric characters are obtained over exact rationals from Newton's identity

$$d h_d(M) = \sum_{r=1}^{d} \mathrm{Tr}(M^r) h_{d-r}(M).$$

The other side is computed from the exact cycle-index average in (507), with truncated multivariate polynomial arithmetic. Thus the two routes share neither matrices nor coefficient extraction.

b	height	leaves	base order	group order
2	1	2	1	2
2	2	4	2	8
2	3	8	8	128
3	1	3	1	6
3	2	9	6	1296

For every row the program compares coefficients indexed by $(1), (2), (3), (1, 1), (2, 1), (2, 2)$; all 30 comparisons pass exactly.

89.4 What is and is not proved

Equation (508) is an exact theorem for each finite tree. For a Galton–Watson tree it becomes a distributional recursion after conditioning on branch-type counts. Also,

$$\log|\operatorname{Aut}(T)| = \sum_i \log|\operatorname{Aut}(T_i)| + \sum_A \log(m_A!)$$

is an additive tree functional; Gaussian-type asymptotics for this scalar quantity are known. Neither fact alone proves convergence of the full random series $\mathrm{SMG}_T(\mathbf{t})$. A general limiting rooted-tree Λ-distribution therefore remains open, and the verifier makes no such claim.

89.5 Reproduction and reference

```
.venv/bin/python for_this_guy/limiting_lambda_claims/
  rooted_tree_automorphisms/verification.py
.venv/bin/marimo edit marimo_notebooks/rooted_tree_automorphisms.py
```

Join wrapped commands onto one line. For automorphism-count asymptotics, see M. Olsson and S. Wagner, *The distribution of the number of automorphisms of random trees*, https://arxiv.org/abs/2211.12801.

89.6 Level-transitive leaf actions and the universal no-limit obstruction

Let $L \leq S_d$ be transitive, put $G_0 = 1$, and define $G_{h+1} = G_h \wr L$ in the natural action on the d^{h+1} leaves of the rooted d-ary tree. If F_h is the number of fixed leaves of a uniform element of G_h, and if Y is the number of fixed letters of a uniform element of L, then

$$F_{h+1} \stackrel{d}{=} \sum_{j=1}^{Y} F_h^{(j)}, \tag{509}$$

where the summands are independent copies of F_h. Transitivity and Burnside's lemma give $\mathbb{E}Y = 1$. Writing $r_L = \#(L\backslash[d]^2) = \mathbb{E}(Y^2)$ and conditioning in the preceding recursion yields

$$\mathbb{E}F_h = 1, \qquad \mathbb{E}(F_h^2) = 1 + h(r_L - 1). \tag{510}$$

Consequently

$$[m_{(1,1)}]\,\mathrm{SMG}_{G_h,\mathbb{C}[X_h]} = 1 + h(r_L - 1),$$

so every nontrivial transitive local action fails to have a raw coefficientwise limit.

There is also a recurrence-free form. For any group acting transitively on the leaves at height h, the depth of the lowest common ancestor of an ordered pair is invariant. Equality and the h possible divergence depths therefore give

$$[m_{(1,1)}]\,\mathrm{SMG}_{G_h,\mathbb{C}[X_h]} = \#(G_h\backslash X_h^2) \geq h + 1. \tag{511}$$

This proves failure of raw moment convergence for every level-transitive family, not only for iterated full wreath products. It does not rule out convergence in probability or a normalized theory.

89.7 Exact audit of the iterated examples

Direct leaf permutations, truncated cycle-index recurrences, fixed-point squares, and pair-orbit searches give the same values:

local action	height	leaves	group order	$\mathbb{E}F_h^2$	pair orbits
C_2	1	2	2	2	2
C_2	2	4	8	3	3
C_2	3	8	128	4	4
C_3	1	3	3	3	3
C_3	2	9	81	5	5
S_3	1	3	6	2	2
S_3	2	9	1296	3	3

The six coefficients indexed by $(1), (2), (1,1), (3), (2,1), (2,2)$ agree in every row. The finite recursion for arbitrary rooted trees remains exact, whereas an infinite-height random-tree law is model-dependent.

A Self-contained build note

This edition is a single-source artifact. Every proof body is present in this file, and compilation does not read any local TeX source, bibliography, graphic, code, data, or manifest. File names and commands printed inside the proofs are human-readable provenance notes only; they are not build inputs. The only required dependencies are the standard LaTeX packages declared in the preamble.

Part IX

Provisional claims and unresolved proof obligations

B Local-ring generating-tuple transitivity

Status: Open lemma in this edition. The exact finite formulas in Section 44 do not depend on this lemma; only the advertised uniform rank-stability bound does.

Let R_a be a finite chain ring and L a finite R_a-module generated by at most n elements. The stability argument would follow from:

$\mathrm{GT}(n, L)$: any two surjections $R_a^n \twoheadrightarrow L$ differ by precomposition with an element of $\mathrm{GL}_n(R_a)$.

Reducing modulo the maximal ideal proves transitivity on minimal generating sets over the residue field, and elementary column operations lift changes of basis through $I + \pi M_n(R_a)$. The remaining issue is to prove that the kernels of arbitrary nonminimal surjections can always be matched by such a lift for every module $L \le R_a^{dD}$. Until this module-theoretic statement is cited or proved, equation (210) is a conditional theorem, not an unconditional all-chain-ring result.

C Formed-space extension over finite chain rings

The bounded-support argument for symplectic and orthogonal towers requires a precise theorem saying that an isometry between supported split, unimodular formed submodules extends to the ambient group in the stated rank range. In residue characteristic two the quadratic refinement must be part of the data. A complete proof must specify the ring class, form type, nondegeneracy and split hypotheses, and the exact unused-rank requirement. Without these hypotheses the universal fixed-point formula remains exact, but the eventual-stability conclusion is only conditional.

D Fixed-tail wreath character polynomials

Equation (40) requires a wreath-product character-polynomial theorem with an explicit threshold $N_{\boldsymbol{\alpha}}(D)$ that remains valid after the class-power maps used for every $r \le D$. Once such a theorem is supplied, the marked Poisson moment argument is finite-dimensional and rigorous. This edition does not claim a universal threshold for arbitrary multipartition tails.

E Generic braid and TQFT image families

For a specified finite represented image, or for a specified compact closure with normalized Haar measure, the Molien and balanced formulas in Section 51 are exact. A generic Jones or TQFT family additionally requires a theorem identifying the image or compact closure and its represented character/power data. The concrete Ising, Heisenberg, Weil, and Clifford examples do not by themselves establish such a family theorem.

F Exact computational certificates still required

The following upgrades are evidentiary rather than changes to the universal formulas:

- reconstruct H_3 over $\mathbb{Q}(\sqrt{5})$ or use a certified character table;
- attach versioned class sizes, power maps, frame shapes, and hashes for M_{24}, Co_0, lattice, and sporadic examples;
- for every numerical rank, report the largest singular value declared zero and the smallest declared positive;
- archive the scripts, lock files, exact outputs, and data tables referenced by the reproduction commands.